

# DETERMINANTAL FLAGS AND PRIMARY BOUNDARIES IN MULTIPLICATION FAILURE

QIAN QI

**ABSTRACT.** We determine embedded primary structures in multiplication failure for finite algebras. For universal quadratic multiplication tensors of Hilbert function  $(1, e, q)$ , with  $q \leq \binom{e}{2}$ , we give an irredundant primary decomposition on the projection-corank-one open where products with the projected hyperplane span the quadratic layer. Every embedded support is a quadratic restriction-rank stratum; we determine its symbolic thickness, normality, a projective resolution, and the exact local nilradical index. The proof identifies the full Fitting ideal with a weighted determinantal ideal and computes its primary flag coefficient by coefficient. A family of Hilbert function  $(1, 3, 3)$  has a smooth embedded support which is a rank-four bundle over an elliptic curve times a projective plane. Its nilradical is an invertible sheaf on that support with square zero, while the reduced failure divisor stays fixed and the elliptic  $j$ -invariant varies. We also retain the complete primary classification for Hilbert function  $(1, 2, 2)$ , including its extreme-corank locus, and relate the primary strata to conductor kernels. For general finite locally free algebras, products of a unital codimension- $r$  subbundle over a  $\mathbb{Q}$ -scheme stabilize in degree  $r + 1$ , sharply in every codimension and generating rank. Relative conductor comparisons identify finite-algebra and global section-multiplication cokernels under explicit regularity hypotheses, with arbitrary base change in their stated range.

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## 1. PRIMARY FLAGS AND ELLIPTIC BOUNDARIES

An embedded component of a multiplication-failure scheme can carry moduli which disappear after taking the reduced support. We prove this by a uniform primary decomposition over the parameter space of quadratic multiplication tensors, and by a family in which the embedded support is a vector bundle over an elliptic curve times a projective plane. The reduced failure divisor is independent of the elliptic parameter.

Here and in Sections 2–4, the ground field is  $\mathbb{C}$ . Let  $V, S$  have dimensions  $e, q$ , and let  $\gamma : \text{Sym}^2 V \twoheadrightarrow S$ . The augmented algebra  $B_\gamma = \mathbb{C} \oplus V \oplus S$  has multiplication  $\gamma$  in degree two and zero products involving  $S$  and  $V \oplus S$ . For  $K \in \text{Gr}(e, V \oplus S)$  set  $W = \mathbb{C} \oplus K$ . The scheme  $D_\gamma$  is defined by the zeroth Fitting ideal of the cokernel of  $\text{Sym}^m W \rightarrow B_\gamma$ ; it is independent of  $m \geq 2$ . These are unital planes. They contain a unit and hence generate the corresponding sheaf, but need not generate  $B_\gamma$  as an algebra. This distinction is essential at a failure point.

Put  $b = e - 1$ ,  $p = \binom{e}{2}$ , and suppose  $1 \leq q \leq p$ . Allow  $\gamma$  to vary over all surjections. On the projection-corank-one locus write  $H = \text{im}(K \rightarrow V)$  and impose the open condition  $\gamma(HV) = S$ . If  $\Delta$  denotes the projection determinant divisor and  $Z_j \subset \Delta$  is defined by  $\text{rank}(\gamma|_{\text{Sym}^2 H}) < j$ , our primary flag theorem gives

$$\mathcal{I}_D = \mathcal{I}_\Delta \cap \bigcap_{j=\max\{1, q-e+2\}}^q \mathcal{I}_{Z_j}^{(q-j+2)}. \quad (1.1)$$

The intersection is irredundant; these are exactly its associated supports. The superscript denotes a symbolic power in the smooth ambient parameter space. The reduced divisor has multiplicity one. At a point where the restricted quadratic map has rank  $a$ , the nilradical has exact index  $q - a + 1$ . The supports  $Z_j$  are normal, and their natural Grassmannian incidences are projective resolutions with explicitly determined fibres (Theorems 3.1 and 3.2). Thus the result is a primary classification on a specified universal open set, in arbitrary embedding dimension; it is not an assertion about unspecified specializations or about the omitted higher projection-corank strata.

The underlying algebra is particularly simple. For a generic  $q$  by  $p$  matrix  $A$ , introduce one independent parameter  $t$ . We prove

$$t \sum_{i=0}^q t^{q-i} I_i(A) = (t) \cap \bigcap_{j=1}^q (t, I_j(A))^{(q-j+2)}. \quad (1.2)$$

A coefficientwise symbolic-order argument gives both equality and explicit witnesses to irredundancy. The geometric input identifies the full multiplication ideal with the left side, not merely with its radical. The normality of generic determinantal supports is a classical input, not a new assertion of this paper.

For  $e = q = 3$ , the family

$$B_\lambda = \mathbb{C}[x, y, z] / ((x, y, z)^3, xy - \lambda z^2, yz - \lambda x^2, zx - \lambda y^2)$$

has Hilbert function  $(1, 3, 3)$ . For an explicit nonempty open set of parameters, its full failure ideal on the entire projection-corank-at-most-one open is

$$\mathcal{I}_{D_\lambda} = \mathcal{I}_\Delta \cap \mathcal{I}_{E_\lambda}^2, \quad 0 \longrightarrow \mathcal{O}_{E_\lambda}(-\Delta) \longrightarrow \mathcal{O}_{D_\lambda} \longrightarrow \mathcal{O}_\Delta \longrightarrow 0.$$

Here  $E_\lambda$  is smooth and normal, and is the total space of a rank-four bundle over  $C_\lambda \times \mathbb{P}^2$ , where

$$C_\lambda : \quad \lambda(a^3 + b^3 + c^3) + (1 - 4\lambda^3)abc = 0.$$

The curves  $C_\lambda$  have nonconstant  $j$ -invariant. In particular, this is a positive-moduli family of algebras and embedded primary supports, not two orbit types presented with a redundant parameter. The nilradical is an invertible sheaf on  $E_\lambda$  and has square zero. A smooth projective compactification of  $E_\lambda$  has Albanese variety  $C_\lambda$ . The conductor has constant rank one on this boundary, while the quadratic restriction retains the elliptic moduli (Theorem 4.1).

There are three distinct scopes in the article. The coefficient-ring stabilization and conductor comparison apply to general finite locally free algebras. The quadratic factorization applies to split augmented algebras with cube-zero augmentation ideal. The primary flag theorem classifies their universal first-order-spanning corank-one open; the  $(1, 2, 2)$  theorem, by contrast, treats that entire fixed Hilbert-function class, including its extreme-corank locus. We retain the complete latter theorem and the higher-contact, relative-incidence, and global section-multiplication results. Their hypotheses are not replaced by the new open conditions.

The article is self-contained as an algebraic-geometric contribution. No theorem from a paper numbered A1, or an unverified application in a later paper, is part of its significance argument. Literature inputs and the precise remaining full-text comparison are separated in Section 22.3.

The accompanying complete edition retains all complementary geometry and statistical proofs; these are not premises of the primary flag theorem.

## 2. A WEIGHTED DETERMINANTAL PRIMARY DECOMPOSITION

For an ideal  $J$ , write  $I_i(A)$  for the ideal of  $i$ -minors of  $A$ , with  $I_0(A) = (1)$ . For a prime  $P$  in a noetherian ring  $R$ , put  $P^{(s)} = P^s R_P \cap R$  for  $s \geq 1$ , and  $P^{(s)} = R$  for  $s \leq 0$ .

**Lemma 2.1** (Coefficientwise symbolic orders). *Let  $R_0 = \mathbb{C}[x_1, \dots, x_n]$ , let  $J \subset R_0$  be prime, and put  $P = (t, J) \subset R_0[t]$ . For  $m \geq 1$  and  $f = \sum_{u \geq 0} t^u f_u$ , one has*

$$f \in P^{(m)} \iff f_u \in J^{(m-u)} \text{ for every } 0 \leq u < m.$$

*Proof.* Localize  $R_0$  at  $J$ , and denote its maximal ideal by  $\mathfrak{m}$ . In  $R_{0,J}[t]$  the quotient by  $(\mathfrak{m}, t)^m$  is an Artinian local ring: it is a finite module over  $R_{0,J}/\mathfrak{m}^m$  and  $t$  is nilpotent. Hence  $(\mathfrak{m}, t)^m$  is  $(\mathfrak{m}, t)$ -primary, and contracting it from the localization at  $(\mathfrak{m}, t)$  does not enlarge it. This localization is  $R_0[t]_P$ . Polynomial expansion gives

$$(\mathfrak{m}, t)^m = \bigoplus_{u \geq 0} t^u \mathfrak{m}^{\max\{m-u, 0\}}$$

as an  $R_{0,J}$ -submodule. Contracting coefficient by coefficient to  $R_0$  proves the assertion. No assumption about ordinary and symbolic powers of  $J$  is involved.  $\square$

**Lemma 2.2** (Order of a minor at a rank stratum). *Let  $A$  be a generic  $q$  by  $p$  matrix, where  $p \geq q$ . For  $1 \leq j \leq k \leq q$ ,*

$$I_k(A) \subset I_j(A)^{(k-j+1)}.$$

*The ideal  $I_j(A)$  is prime and its quotient is normal.*

*Proof.* Primality and normality are the classical generic determinantal results; for instance [39, Theorem 3.5(c)], with  $r = j - 1$ , gives normality and the domain assertion (the case  $j = 1$  is a polynomial quotient). To prove the order statement, invert a  $(j - 1)$ -minor not in  $I_j(A)$ ; for  $j = 1$  no inversion is needed. Invertible row and column operations replace  $A$  by  $\text{diag}(I_{j-1}, A')$ . At the prime  $I_j(A)$ , the entries of  $A'$  generate its extension. A  $k$ -minor must use at least  $k - j + 1$  entries from  $A'$ . This proves the inclusion after localization at  $I_j(A)$ , and contraction proves the symbolic inclusion.  $\square$

**Theorem 2.3** (Weighted determinantal primary flag). *Let  $A$  be a generic  $q$  by  $p$  matrix with  $p \geq q \geq 1$ , let  $R = \mathbb{C}[A, t]$ , and set*

$$F = t \sum_{i=0}^q t^{q-i} I_i(A), \quad P_j = (t, I_j(A)).$$

*Then an irredundant primary decomposition is*

$$F = (t) \cap \bigcap_{j=1}^q P_j^{(q-j+2)}. \tag{2.1}$$

*In particular*

$$\text{Ass}_R(R/F) = \{(t), P_1, \dots, P_q\}, \quad \sqrt{F} = (t), \quad \text{codim } P_j = 1 + (q - j + 1)(p - j + 1).$$

*The reduced divisor has generic multiplicity one. At a point of  $t = 0$  where  $A$  has rank  $a$ , the nilradical of the local failure ring has exact index  $q - a + 1$ . In particular its maximal index is  $q + 1$ .*

*Proof.* Every displayed component is primary:  $P_j$  is prime and a symbolic power is the contraction of a power of the maximal ideal of  $R_{P_j}$ . Membership in  $(t)$  requires  $f_0 = 0$ . For  $u \geq 1$  put  $k = q + 1 - u$ . If  $1 \leq k \leq q$ , Lemma 2.1 says that the condition supplied by the  $j$ th component is

$$f_u \in I_j(A)^{(k-j+1)}.$$

The condition with  $j = k$  is exactly  $f_u \in I_k(A)$ . It implies every condition with  $j < k$  by Lemma 2.2; those with  $j > k$  are void. For  $u \geq q + 1$  all these conditions are void. Consequently the intersection has coefficient ideal  $I_{q+1-u}(A)$  in degrees  $1 \leq u \leq q$ , coefficient ideal (1) in degrees  $u \geq q + 1$ , and zero constant coefficient. These are exactly the coefficient ideals of  $F$ , proving (2.1).

Here are explicit irredundancy witnesses. Choose any nonzero  $(j - 1)$ -minor  $d_{j-1}$ , with  $d_0 = 1$ . The polynomial

$$g_j = t^{q-j+1} d_{j-1}^2$$

does not belong to  $P_j^{(q-j+2)}$ , since  $d_{j-1}$  is a unit at  $I_j(A)$ . It belongs to  $(t)$ . For  $h > j$  its  $t$ -order alone is at least  $q - h + 2$ . For  $h < j$ , its order at  $P_h$  is at least

$$q - j + 1 + 2(j - h) \geq q - h + 2$$

by Lemma 2.2. It therefore belongs to all the other primary components. Finally, the square of a nonzero  $q$ -minor belongs to every  $P_j^{(q-j+2)}$ , since  $2(q - j + 1) \geq q - j + 2$ , but not to  $(t)$ . Thus no component can be removed, and the standard associated-prime characterization of an irredundant primary decomposition gives the asserted set. The codimension formula is the usual rank-matrix dimension count: rank  $j - 1$  matrices have dimension  $(j - 1)(q + p - j + 1)$ .

At a rank- $a$  point invert an  $a$ -minor and perform block elimination. The ideal generated by  $A$  and  $tI_q$  has a free rank- $a$  image; a presentation of its cokernel is  $[A', tI_{q-a}]$ , with  $A'$  the  $(q - a)$  by  $(p - a)$  Schur complement. Since

$$I_q([A, tI_q]) = \sum_{i=0}^q t^{q-i} I_i(A),$$

the local failure ideal becomes

$$t \sum_{i=0}^{q-a} t^{q-a-i} I_i(A'). \tag{2.2}$$

All entries of  $A'$  vanish at the point. Its coefficient ring is a localized polynomial ring independent of  $t$ . Formula (2.2) contains  $t^{q-a+1}$  but not  $t^{q-a}$  when  $q - a > 0$ : the latter membership would require  $1 \in I_1(A')$ . The same nonmembership persists in the local ring at the point, since a denominator not vanishing there has invertible constant term modulo  $I_1(A')$ . For  $a = q$  the local ideal is  $(t)$  and the nilradical is zero, whose index we take to be one. This proves both the exact local indices and generic multiplicity.  $\square$

*Remark 2.4* (What is and is not specialized). The ideal  $tI_q([A, tI_q])$  and its coefficient formula can be substituted in any coefficient ring. The primary decomposition (2.1) uses independent matrix entries and is not asserted to remain a primary decomposition under an arbitrary substitution. Polynomial extension and localization do preserve the argument. The geometric application below is first made on the universal tensor space, where independence holds; the elliptic specialization is proved separately by local parameters.

## 3. UNIVERSAL QUADRATIC FLAGS AND THEIR RESOLUTIONS

Fix  $e \geq 2$ ,  $b = e - 1$ ,  $p = \binom{e}{2}$ , and  $1 \leq q \leq p$ . Let  $V, S$  be fixed vector spaces of dimensions  $e, q$ , and let

$$\Gamma = \text{Surj}(\text{Sym}^2 V, S), \quad \mathcal{X} = \Gamma \times \text{Gr}(e, V \oplus S).$$

The algebra  $\mathcal{B} = \mathcal{O} \oplus V \oplus S$  over  $\Gamma$  has the universal quadratic multiplication  $\gamma$  and  $S(V \oplus S) = 0$ . Write  $\mathcal{K}$  for the tautological bundle and  $D$  for the failure scheme of  $\mathcal{O} \oplus \mathcal{K}$  in any degree  $m \geq 2$ .

Delete the locus  $\text{rank}(\mathcal{K} \rightarrow V) \leq e - 2$ . On its determinant divisor  $\Delta$  there are bundles

$$H = \text{im}(\mathcal{K} \rightarrow V), \quad \Lambda = \mathcal{K} \cap S, \quad \text{rank } H = b, \quad \text{rank } \Lambda = 1.$$

The subspace  $HV \subset \text{Sym}^2 V$  is generated by products with one factor in  $H$ . Delete also the closed subset of this divisor on which  $\gamma(HV) \neq S$ . Denote the resulting open scheme by  $\mathcal{U}$ , and continue to write  $D, \Delta$  for their restrictions to it. For  $1 \leq j \leq q$ , define the closed subscheme

$$Z_j = \{(\gamma, K) \in \Delta : \text{rank}(\gamma|_{\text{Sym}^2 H}) < j\} \quad (3.1)$$

using the  $j$ -minor ideal on  $\Delta$ . As a subscheme of  $\mathcal{U}$  its ideal includes  $\mathcal{I}_\Delta$ .

**Theorem 3.1** (Universal first-order-spanning primary classification). *Put  $j_0 = \max\{1, q - b + 1\}$ . The schemes  $Z_j$  with  $j < j_0$  are empty; each  $Z_j$  with  $j_0 \leq j \leq q$  is nonempty, integral and normal. On the whole of  $\mathcal{U}$ ,*

$$\mathcal{I}_D = \mathcal{I}_\Delta \cap \bigcap_{j=j_0}^q \mathcal{I}_{Z_j}^{(q-j+2)} \quad (3.2)$$

*is an irredundant primary decomposition. Its associated points are exactly the generic point of  $\Delta$  and the generic points of these  $Z_j$ . In particular there are no other associated supports on  $\mathcal{U}$ .*

*The divisor  $\Delta$  is smooth, and its generic multiplicity in  $D$  is one. At a point of  $\Delta$  where  $\text{rank}(\gamma|_{\text{Sym}^2 H}) = a$ , the nilradical has exact index  $q - a + 1$ . Its maximal index on  $\mathcal{U}$  is  $\min\{q, b\} + 1$ . The codimension of  $Z_j$  in  $\mathcal{U}$  is  $1 + (q - j + 1)(p - j + 1)$ .*

*Proof.* Work on a frame chart of  $\mathcal{K}$ . Write its inclusion as a stacked matrix with upper block  $M : \mathcal{K} \rightarrow V$ . The block-minor argument of Theorem 13.1 gives, over the universal coefficient ring,

$$\mathcal{I}_D = (\det M) I_q(\gamma \text{Sym}^2 M).$$

Indeed all products of degree at least three in  $K$  vanish, and the image is  $\mathcal{O} + K + K^2$ . Every maximal minor of the full presentation must use all  $e$  linear columns. This supplies the factor  $\det M$  before any quadratic minor is taken.

Near a corank-one point, invert a  $b$ -minor of  $M$ . Block elimination and invertible changes of the source and target bases give  $M = \text{diag}(I_b, t)$ , with  $t$  a local equation for  $\Delta$ . With respect to  $V = H \oplus \langle v \rangle$ , let  $A, B, c$  be the blocks of  $\gamma$  on  $\text{Sym}^2 H, Hv, v^2$ . They have sizes  $q \times p$ ,  $q \times b$ , and  $q \times 1$ . Up to harmless nonzero scalar factors in the symmetric basis,

$$\gamma \text{Sym}^2 M = [A, tB, t^2c].$$

The condition on  $\mathcal{U}$  says that  $[A, B]$  is surjective at  $t = 0$ . Shrink the neighborhood so that it has a right inverse. This gives an equality of column modules over the local coefficient ring:

$$\text{im}[A, tB, t^2c] = \text{im}[A, tI_q]. \quad (3.3)$$

For the nontrivial inclusion, write  $I_q = AU + BV$  using the right inverse and multiply by  $t$ . The reverse inclusion follows because  $tB$  and  $t^2c$  are generated by  $tI_q$ . Thus

$$\mathcal{I}_D = t \sum_{i=0}^q t^{q-i} I_i(A). \quad (3.4)$$

The entries of  $A, B, c$  are independent coordinates of the universal tensor space after the indicated invertible changes of basis; the other frame coordinates add polynomial variables and localizations. Theorem 2.3 therefore applies. Its prime  $(t, I_j(A))$  is exactly the intrinsic ideal in (3.1). On the spanning open, a rank- $(j-1)$  choice of  $A$  is possible exactly when  $q-j+1 \leq b$ . It is obtained by completing its image with the  $b$  columns of  $B$ . Hence exactly the primes indexed by  $j \geq j_0$  survive. Their generic strata also meet the full-frame open: prescribe  $M = \text{diag}(I_b, 0)$  and give its last column a nonzero  $S$  component. The witnesses in Theorem 2.3, or its local rank-eliminated form at each surviving generic point, show that the surviving decomposition is irredundant.

For completeness, the global geometry of each support can be read without a frame. The divisor of corank-one planes is the total space of  $\text{Hom}(H, S/\Lambda)$  over  $\text{Gr}(b, V) \times \mathbb{P}(S)$ : a plane with projection  $H$  and kernel  $\Lambda$  is exactly the inverse graph of a map  $H \rightarrow S/\Lambda$ . Over this smooth base the restriction map from all tensors  $\gamma$  to  $\text{Hom}(\text{Sym}^2 H, S)$  is a surjective vector-bundle map. Thus  $Z_j$  is, locally, a generic determinantal scheme times an affine space, restricted to the nonempty spanning open. The determinantal rank stratum is irreducible; the vector-bundle description shows that these local models form an integral global scheme. Lemma 2.2 gives its normality.

Symbolic powers localize, so the local intersections glue. Alternatively, apply Lemma 5.1 to the intrinsic ideals just defined. It also identifies the associated points on  $\mathcal{U}$ , rather than only on its frames. Off  $\Delta$ ,  $M$  is invertible, so all ideals in the decomposition are unit ideals and  $D$  is empty. The local nilpotency calculation is Theorem 2.3. Since  $a \geq \max\{0, q-b\}$  on the spanning locus and that lower rank occurs, its maximal index is as stated. The same local models give the codimension and generic multiplicity assertions.  $\square$

**Theorem 3.2** (Resolutions of the embedded supports). *For  $j_0 \leq j \leq q$ , let  $\tilde{Z}_j$  parametrize triples  $((\gamma, K), T)$  with  $(\gamma, K) \in \Delta$  and*

$$T \in \text{Gr}(j-1, S), \quad \gamma(\text{Sym}^2 H) \subset T.$$

*Then  $\tilde{Z}_j$  is smooth and integral, and the forgetful map  $\rho_j : \tilde{Z}_j \rightarrow Z_j$  is a projective birational resolution. At a point where  $\gamma|_{\text{Sym}^2 H}$  has rank  $r$ , its scheme-theoretic fibre is*

$$\text{Gr}(j-1-r, S/\gamma(\text{Sym}^2 H)) \simeq \text{Gr}(j-1-r, q-r). \quad (3.5)$$

*These constructions are equivariant for changes of bases in  $V, S$  and for frame changes of  $K$ .*

*Proof.* Start with  $\text{Gr}(b, V) \times \mathbb{P}(S) \times \text{Gr}(j-1, S)$ , carrying the bundles  $H, \Lambda, T$ . The condition  $\gamma(\text{Sym}^2 H) \subset T$  is the kernel of the surjective bundle map

$$\text{Hom}(\text{Sym}^2 V, S) \longrightarrow \text{Hom}(\text{Sym}^2 H, S/T).$$

Consequently its solution space is a vector bundle. Adjoin the independent bundle  $\text{Hom}(H, S/\Lambda)$  recording  $K$ , and restrict to  $\gamma(HV) = S$ . This is precisely  $\tilde{Z}_j$ . It is a nonempty open of a vector bundle over an integral smooth base, hence is smooth and integral. The projection is projective, since it is a closed incidence in the relative Grassmannian over  $\Delta$  restricted to  $Z_j$ . It is surjective and has one-point fibre when the image of  $\text{Sym}^2 H$  has dimension  $j-1$ ; over that open the inverse sends a matrix to its image subbundle and is a morphism. This proves birationality. Over a point of rank  $r$ , choosing  $T$  is exactly choosing a  $(j-1-r)$ -subspace of the quotient by that image; the same statement for test schemes proves the fibre formula with its scheme structure. All defining maps are intrinsic, proving equivariance.  $\square$

**Proposition 3.3** (Conductor and restriction flags). *On the corank-one locus of any fixed  $B_\gamma$ , put  $W = \mathbb{C} \oplus K$ ,  $H = \text{im}(K \rightarrow V)$  and  $\Lambda = K \cap S$ . Then  $W$  is a subalgebra exactly when  $\gamma(\text{Sym}^2 H) \subset \Lambda$ . The largest ideal of  $B_\gamma$  contained in  $W$  is*

$$\text{cond}_{B_\gamma}(W) = \ker [K \longrightarrow \text{Hom}(V, S/\Lambda)], \quad k \longmapsto (v \mapsto \gamma(\bar{k}, v)). \quad (3.6)$$

*On geometric fibres its dimension is  $1 + \dim\{h \in H : \gamma(h, V) \subset \Lambda\}$ . The kernel commutes with base change on every exact-rank stratum of the displayed map. Thus the restriction-rank flag defining the primary supports and the conductor flag are computed by two specified maps of the same quadratic tensor; neither is replaced by a numerical conductor dimension.*

*Proof.* The multiplication image is  $\mathbb{C} + K + \gamma(\text{Sym}^2 H)$ , proving the closure assertion. A conductor element with nonzero constant coefficient would be a unit of the local algebra and would force  $B_\gamma \subset W$ , which is impossible. Over coefficient rings the same exclusion follows when  $q > 1$  by multiplying with the locally free quotient  $S/\Lambda$ ; when  $q = 1$ , multiply with  $V/H$ , which is a free line and receives the scalar coefficient of multiplication by a lift of  $V/H$ . For  $k \in K$ , multiplication by  $S$  is zero and multiplication by a lift of  $v$  is  $\gamma(\bar{k}, v)$ . Such a product is in  $K$  exactly when it is in  $\Lambda$ . This proves the kernel formula, including over coefficient rings. Finally  $K \rightarrow H$  has kernel  $\Lambda$ , and on a scheme-theoretic exact-rank locus the indicated matrix has locally split kernel and image. The dimension and base-change assertions follow.  $\square$

#### 4. AN ELLIPTIC FAMILY OF EMBEDDED PRIMARY BOUNDARIES

The universal theorem has a global consequence in a non-isotrivial family of algebras of Hilbert function  $(1, 3, 3)$ . In this section the parameter tensor is specialized, so its primary decomposition is proved directly; no compatibility of primary components with arbitrary base change is presumed.

Let

$$B_\lambda = \mathbb{C}[x, y, z]/((x, y, z)^3, xy - \lambda z^2, yz - \lambda x^2, zx - \lambda y^2). \quad (4.1)$$

The elements  $1, x, y, z, x^2, y^2, z^2$  are a basis, also over the polynomial parameter ring. Thus this is a finite free rank-seven family. Write  $V = \langle x, y, z \rangle$  and  $S = \langle x^2, y^2, z^2 \rangle$ . Work over the nonempty open

$$\Omega = \{\lambda : \lambda \neq 0, \quad \lambda^3 \neq 1, \quad (4\lambda^3 - 1)^3 \neq 27\lambda^3\}. \quad (4.2)$$

For example  $\lambda = 1/2$  belongs to  $\Omega$ . Let  $X = \text{Gr}(3, V \oplus S)$  and  $U = \{K : \text{rank}(K \rightarrow V) \geq 2\} \subset X$ . Its determinant divisor  $\Delta$  has projection rank exactly two and is independent of  $\lambda$ . On  $\mathbb{P}(V^*)$  put

$$C_\lambda : F_\lambda(a, b, c) = \lambda(a^3 + b^3 + c^3) + (1 - 4\lambda^3)abc = 0. \quad (4.3)$$

Denote by  $E_\lambda \subset \Delta$  its inverse image under  $K \mapsto H = \text{im}(K \rightarrow V)$ .

**Theorem 4.1** (Elliptic embedded boundary). *For every  $\lambda \in \Omega$  the curve  $C_\lambda$  is smooth and has genus one. For all  $m \geq 2$ , the full failure scheme  $D_\lambda$  of  $\text{Sym}^m(\mathbb{C} \oplus K) \rightarrow B_\lambda$  on all of  $U$  has the irredundant primary decomposition*

$$\mathcal{I}_{D_\lambda} = \mathcal{I}_\Delta \mathcal{I}_{E_\lambda} = \mathcal{I}_\Delta \cap \mathcal{I}_{E_\lambda}^2. \quad (4.4)$$

*Its only associated points are the generic points of  $\Delta$  and  $E_\lambda$ . The reduced divisor is smooth and has generic multiplicity one. The embedded support is smooth and normal, with*

$$E_\lambda \simeq \text{Tot}_{C_\lambda \times \mathbb{P}(S)} \text{Hom}(H, S/\Lambda), \quad (4.5)$$

*where  $H$  is the tautological hyperplane bundle and  $\Lambda$  is the tautological line of  $\mathbb{P}(S)$ . This is a rank-four bundle, so  $\dim E_\lambda = 7$  and  $\text{codim}_U E_\lambda = 2$ .*

*The nilradical has square zero and is the line bundle  $\mathcal{O}_{E_\lambda}(-\Delta)$ , viewed as an  $\mathcal{O}_{D_\lambda}$ -module supported on  $E_\lambda$ . More precisely there is an exact sequence*

$$0 \rightarrow \mathcal{O}_{E_\lambda}(-\Delta) \rightarrow \mathcal{O}_{D_\lambda} \rightarrow \mathcal{O}_\Delta \rightarrow 0. \quad (4.6)$$

*For every  $K \in \Delta$ , the conductor of  $\mathbb{C} \oplus K$  in  $B_\lambda$  is exactly  $\Lambda = K \cap S$ .*

*The family has genuine moduli: the elliptic curves have nonconstant  $j$ -invariant, and distinct  $j$ -values give nonisomorphic algebras  $B_\lambda$ . A smooth projective compactification of  $E_\lambda$  is a projective bundle over  $C_\lambda \times \mathbb{P}^2$  and has Albanese variety  $C_\lambda$ .*

*Proof.* A hyperplane with equation  $ax + by + cz = 0$ , on the chart  $c \neq 0$ , has basis  $u = (c, 0, -a)$  and  $v = (0, c, -b)$ . The restriction of the quadratic multiplication, in the product basis  $u^2, uv, v^2$ , is

$$A_\lambda = \begin{pmatrix} c^2 & -\lambda ac & -2\lambda bc \\ -2\lambda ac & -\lambda bc & c^2 \\ a^2 & ab + \lambda c^2 & b^2 \end{pmatrix}, \quad \det A_\lambda = -c^3 F_\lambda(a, b, c). \quad (4.7)$$

Expansion along the first row proves the determinant identity. Cyclic charts give the same equation up to the transition unit of  $\det(\text{Sym}^2 H)^*$ . Since  $\det(\text{Sym}^2 H) = (\det H)^3$ , this determinant is a section of  $\mathcal{O}_{\mathbb{P}(V^*)}(3)$ . Thus (4.3) defines its global degeneracy curve, not a component seen only on one chart.

Divide  $F_\lambda$  by  $\lambda$  and set

$$k = \frac{4\lambda^3 - 1}{3\lambda}.$$

This gives the Hesse cubic  $a^3 + b^3 + c^3 - 3kabc = 0$ . Its three partial derivatives vanish simultaneously at a projective point exactly when  $k^3 = 1$ : no coordinate of such a point is zero, and multiplication of  $a^2 = kbc$ ,  $b^2 = kac$ ,  $c^2 = kab$  forces  $k^3 = 1$ ; the converse follows by solving these equations. Condition (4.2) therefore makes  $C_\lambda$  smooth. It is integral, since distinct components of a reducible plane cubic intersect and give singularities. Adjunction gives genus one, and  $[1 : -1 : 0]$  is a point on every fibre. Moreover  $A_\lambda$  has rank exactly two along the curve. If its rank were at most one, all its cofactors, hence all derivatives of its determinant in the local hyperplane coordinates, would vanish, contrary to smoothness.

We next check the first-order-spanning hypothesis at every hyperplane, not only at a general point of the curve. A nonzero functional of  $S$  annihilating  $\gamma(HV)$  would give a quadratic form in the dual net vanishing on  $HV$ . Such a form must be a square of the equation of  $H$ , up to a scalar. The dual net consists of

$$\alpha(X^2 + 2\lambda YZ) + \beta(Y^2 + 2\lambda ZX) + \chi(Z^2 + 2\lambda XY).$$

A square  $(aX + bY + cZ)^2$  belongs to it only if

$$ab = \lambda c^2, \quad bc = \lambda a^2, \quad ca = \lambda b^2.$$

When  $\lambda \neq 0$ , a nonzero solution has all coordinates nonzero. Their product gives  $\lambda^3 = 1$ , excluded on  $\Omega$ . It follows that  $\gamma(HV) = S$  for every  $H$ .

Now fix  $K \in \Delta$ . The block elimination of (3.4) is valid for this particular  $\gamma$ . If  $H \notin C_\lambda$ , the restriction has rank three, and the local ideal is  $(t)$ . If  $H \in C_\lambda$ , invert a nonzero two-minor of  $A_\lambda$ . Eliminating it leaves a one-row presentation, so the local ideal is

$$(t^2, tf), \quad (4.8)$$

where  $t$  is a local equation for  $\Delta$  and  $f$  is the remaining Schur entry, a unit times  $F_\lambda$  in local hyperplane coordinates. Since  $C_\lambda$  is smooth and the map  $\Delta \rightarrow \mathbb{P}(V^*)$  is smooth,  $t, f$  are part of a regular system of parameters. Off  $\Delta$ ,  $M$  is invertible and the failure scheme is empty. This proves the ideal calculation on all of  $U$  without specializing a universal primary decomposition.

The intrinsic graph description of corank-one planes gives (4.5), so  $E_\lambda$  is smooth and integral. Let  $P = (t, f)$  locally. Its associated graded ring is a polynomial ring in two variables over the integral ring of  $E_\lambda$ . Consequently  $P^2$  is  $P$ -primary and  $(P^2 : t) = P$ . It follows that  $(t) \cap P^2 = tP$ , proving (4.4) and its irredundancy. The two intrinsic ideals glue, and Lemma 5.1 supplies an alternative frame-to-scheme verification. The ideal  $\mathcal{I}_\Delta$  is invertible, and therefore

$$\mathcal{I}_\Delta / (\mathcal{I}_\Delta \mathcal{I}_{E_\lambda}) \simeq \mathcal{I}_\Delta \otimes \mathcal{O}_{E_\lambda}.$$

This is the kernel in (4.6); its square vanishes because  $\mathcal{I}_\Delta \subset \mathcal{I}_{E_\lambda}$ . It is nonzero along  $E_\lambda$ . This proves all nilradical and multiplicity assertions.

For the conductor, multiplication by a vector  $(x, y, z) \in V$  has matrix

$$T(x, y, z) = \begin{pmatrix} x & \lambda z & \lambda y \\ \lambda z & y & \lambda x \\ \lambda y & \lambda x & z \end{pmatrix}.$$

For nonzero  $(x, y, z)$  this matrix has rank at least two on  $\Omega$ . Indeed its principal two-minors imply, if its rank is at most one,  $xy = \lambda^2 z^2$ ,  $yz = \lambda^2 x^2$ , and  $zx = \lambda^2 y^2$ . No coordinate can then vanish. The minor in rows 1, 2 and columns 1, 3 also gives  $x^2 = \lambda yz$ . Combining the latter equality with  $yz = \lambda^2 x^2$  gives  $\lambda^3 = 1$ , a contradiction. Thus no nonzero element of  $H$  multiplies all of  $V$  into a line  $\Lambda$ . Proposition 3.3 gives  $\text{cond}(W) = \Lambda$ . Notice also that  $W$  is not a subalgebra on this locus, since  $\text{rank } \gamma(\text{Sym}^2 H) \geq 2$ . Here the conductor means the largest ideal contained in  $W$ ; it does not presuppose that  $W$  is a subalgebra.

Finally the Hesse Weierstrass calculation gives

$$j(C_\lambda) = \frac{27k^3(k^3 + 8)^3}{(k^3 - 1)^3}, \quad k = \frac{4\lambda^3 - 1}{3\lambda}. \quad (4.9)$$

For normalization, substitute  $(u_0, u_1) = (1, -k/2)$  in [40, Section 2, equations (4)–(5)] and use  $j = 1728 4A^3 / (4A^3 + 27B^2)$  for its Weierstrass coefficients. This yields (4.9). It is a nonconstant rational function of  $\lambda$ . An isomorphism of local algebras induces isomorphisms of  $\mathfrak{m}/\mathfrak{m}^2$  and  $\mathfrak{m}^2$  intertwining  $\gamma$ . It therefore carries the hyperplane restriction degeneracy curve projectively to the corresponding curve. Distinct  $j$ -values give nonisomorphic algebras.

The compactification

$$\overline{E}_\lambda = \mathbb{P}_{C_\lambda \times \mathbb{P}(S)}(\mathcal{O} \oplus \text{Hom}(H, S/\Lambda))$$

contains  $E_\lambda$  as the affine-bundle chart. It is smooth and projective. A morphism from a projective space to an abelian variety is constant; applying this first on its projective-bundle fibres and then on the  $\mathbb{P}(S)$  fibres shows that its Albanese is the Jacobian of  $C_\lambda$ , identified with  $C_\lambda$  by the specified point. Equivalently, projective-bundle cohomology gives  $h^1(\mathcal{O}_{\overline{E}_\lambda}) = 1$ . This gives a global geometric invariant carried by the embedded support although the reduced determinant divisor stays fixed.  $\square$

**Corollary 4.2** (A non-isotrivial family of full failure schemes). *Over  $\Omega$ , the construction above gives relative schemes  $\mathcal{E} \subset \Omega \times \Delta$  and  $\mathcal{D} \subset \Omega \times U$  with*

$$\mathcal{I}_{\mathcal{D}} = \mathcal{I}_{\Omega \times \Delta} \mathcal{I}_{\mathcal{E}}, \quad 0 \rightarrow \mathcal{O}_{\mathcal{E}}(-\Delta) \rightarrow \mathcal{O}_{\mathcal{D}} \rightarrow \mathcal{O}_{\Omega \times \Delta} \rightarrow 0.$$

*The morphisms  $\mathcal{E} \rightarrow \Omega$  and  $\mathcal{D} \rightarrow \Omega$  are flat; the former is smooth. Each geometric fibre has exactly the two associated supports in Theorem 4.1.*

*Proof.* The cubic equation defines a smooth proper family of plane curves on  $\Omega$ , since every fibre has nonvanishing relative gradient. The graph construction is a vector bundle over its product with  $\mathbb{P}(S)$ , hence  $\mathcal{E} \rightarrow \Omega$  is smooth. The local proof of (4.8) works over the parameter ring: its two-minor is invertible and the residual equation  $f$  cuts out that smooth family. It gives the displayed ideal and exact sequence before specialization. Both outer terms of the sequence are flat over  $\Omega$ , so the middle term is flat. Tensoring the sequence with any residue field is exact, and the fibre proof above gives the stated primary decomposition. Flatness here is a conclusion of this specific family, not a general assertion about Fitting schemes.  $\square$

## 5. PRIMARY DESCENT FROM FRAMES

We record explicitly the descent used in the global primary statements. Its hypotheses are verified on Zariski frame charts; no claim about arbitrary regular-fibre morphisms is needed.

**Lemma 5.1** (Frame descent of primary decompositions). *Let  $X$  be a noetherian scheme, let  $\mathcal{K}$  be a vector bundle, and let  $f : F = \text{Fr}(\mathcal{K}) \rightarrow X$  be its frame torsor. Suppose  $\mathcal{I}, \mathcal{Q}_1, \dots, \mathcal{Q}_s$  are coherent ideals of  $\mathcal{O}_X$  and  $Y_1, \dots, Y_s \subset X$  are distinct integral closed subschemes. Assume that on  $F$  the pulled-back ideals give a primary decomposition*

$$f^*\mathcal{I} = \bigcap_i f^*\mathcal{Q}_i, \quad \sqrt{f^*\mathcal{Q}_i} = \mathcal{I}_{f^{-1}Y_i},$$

*with each nonempty local component primary. Then  $\mathcal{I} = \bigcap_i \mathcal{Q}_i$  on  $X$ , each  $\mathcal{Q}_i$  is primary along  $Y_i$ , and the associated points of  $\mathcal{O}_X/\mathcal{I}$  are the generic points of the components in its irredundant decomposition. The intersection on  $X$  is irredundant if and only if the intersection on  $F$  is irredundant.*

*The hypotheses can be checked by polynomial extension and localization of affine primary decompositions, provided every claimed associated stratum meets the frame open. For an invariant symbolic-power ideal, its intrinsic description on  $X$  is its symbolic power at the generic point of the indicated integral support.*

*Proof.* On an affine trivializing open  $U = \text{Spec } A$ , one has

$$f^{-1}U = \text{Spec } A[z_{ij}, \det(z)^{-1}].$$

This ring is faithfully flat over  $A$ , and the inclusion has the retraction  $z_{ij} \mapsto \delta_{ij}$ . Flatness commutes with a finite intersection of ideals, as follows by tensoring the kernel sequence defining the intersection. Faithful flatness detects ideal equality, so the displayed intersection contracts to the asserted one on  $U$ .

If  $ab \in \mathcal{Q}_i(U)$  and  $b \notin \mathcal{Q}_i(U)$ , faithfulness implies that  $b$  remains outside its extended ideal. Primarity upstairs therefore implies that a power of  $a$  lies in the extended ideal, and contraction puts that power in  $\mathcal{Q}_i(U)$ . The radical contracts to the ideal of  $Y_i$ . When a trivializing open requires several affine subcharts of  $F$ , the same argument is local: an element nonzero in the quotient remains nonzero on some such chart; the prime support specifies its unique generic point. This proves the local primary assertion. Conversely polynomial extension of a primary ideal is primary, and localization preserves primarity unless the ideal becomes the unit ideal. One may verify the polynomial assertion by the usual zero-divisor criterion for polynomials over a primary ring, or by contracting from the localization at the extended prime.

If a component can be omitted upstairs, faithful flatness contracts the resulting equality. If it can be omitted downstairs, flatness extends the equality. This proves the irredundancy equivalence. On an affine chart the associated primes of an irredundant primary decomposition with distinct radicals are exactly those radicals; applying this statement and gluing gives the associated points on  $X$ . In particular the assertion is not deduced solely from a statement about fibre regularity.

For symbolic powers, polynomial extension followed by localization at the extended prime commutes with taking the contraction of a power of the maximal ideal. Equivalently apply the defining membership test at the generic point on each trivialization. Thus the ideals described this way agree on overlaps with the intrinsic symbolic-power sheaves. All assertions are compatible with further Zariski localization.  $\square$

**Corollary 5.2** (The global ideals in Hilbert function  $(1, 2, 2)$ ). *In Theorem 13.2, let  $\mathcal{D} = \mathcal{I}_\Delta$  and let  $\mathcal{P}, \mathcal{Q}$  denote the ideals of the factor surfaces  $Y_L$  (one surface in the square case, two in the distinct factor case). In the latter case put*

$$\mathcal{N} = \mathcal{P} + \mathcal{Q} = \mathcal{I}_{\{K=S_2\}}, \quad \mathcal{J} = (\mathcal{I}_D : \mathcal{D}^2),$$

$$\mathcal{Q}_0 = \mathcal{J} + \mathcal{P}^2 + \mathcal{Q}^2, \quad \mathcal{Q}_* = \mathcal{D}^2 \mathcal{Q}_0 + \mathcal{N}^7.$$

*These are global coherent ideals on  $\mathrm{Gr}(2, V \oplus S_2)$ . The square decomposition is  $\mathcal{D}^2 \cap \mathcal{P}^4$ ; the distinct-factor decomposition is  $\mathcal{D}^2 \cap \mathcal{P}^3 \cap \mathcal{Q}^3 \cap \mathcal{Q}_*$ . Both are irredundant globally and their associated points are exactly those stated in Theorem 13.2.*

*Proof.* The surface ideal is the zero ideal of the bundle map  $K \rightarrow V/L$ , and the point ideal is the zero ideal of  $K \rightarrow V$ . Thus the ideals  $\mathcal{P}, \mathcal{Q}, \mathcal{N}$  are intrinsic; on frames they are  $(a, c), (b, d), (a, b, c, d)$  after choosing the factor lines as axes. The determinant ideal is intrinsic as well. The ideal quotient by its square commutes with flat pullback: locally that square is generated by a nonzerodivisor, and the quotient is the kernel of multiplication by that generator on  $\mathcal{O}_X/\mathcal{I}_D$ . This identifies its pullback with the ideal  $J$  used in the affine calculation. The stated sums and products therefore pull back to  $Q_0, Q_*$  exactly.

The affine primary equalities are proved in Theorem 13.2. Polynomial extension by the lower frame block and restriction to the full-frame open preserve them. Every associated stratum survives this restriction: the lower block can be the identity matrix, independently of the upper block. The strata  $\Delta, Y_L, \{K = S_2\}$  are integral, and their inverse images under a frame torsor are integral. Apply Lemma 5.1 to obtain equality, primarity, irredundancy and the exact associated-point set on the Grassmannian. Permuting the two factor lines permutes the two surface ideals and leaves the unordered decomposition unchanged. Thus no labelled choice is needed for the asserted geometric collection.  $\square$

## 6. FINITE-ALGEBRA AND GLOBAL CONDUCTOR FRAMEWORK

The multiplication map of a linear series can fail to be surjective with a scheme structure which changes even when its support does not. We study the actual multiplication cokernel on generating Grassmannians of finite algebras, and transport it to global sections of projective schemes. The central link is a codimension bound over the coefficient ring, followed by a conductor incidence which retains the local rings of the failure scheme.

Let  $W \subset B$  contain 1 and have locally free quotient of rank  $r$ . Theorem 9.1 proves  $W^{r+1} = \mathcal{O}[W]$  over every  $\mathbb{Q}$ -algebra. The proof compresses the commuting multiplication operators to  $B/W$ . Their commutators land in the quadratic residual image; polarized Cayley–Hamilton then bounds the residual filtration without discarding nilpotents on the base. In codimension two the entire stable failure scheme is therefore defined by two by two minors of one degree-three residual matrix (Theorem 12.1). Its entry ideal defines a different scheme: the extreme-corank subalgebra locus.

Off that locus the full failure scheme is an open relative Grassmannian over  $\mathrm{Hilb}^2(B)$ . This is an isomorphism of schemes, not only a parameterization of geometric points. The proper incidence extending it has fibres controlled by the conductor of the codimension-two subalgebra. When the conductor action has rank one, the fibre is the length-two quotient scheme of a rank-three algebra, expressed by a binary cubic. When the action has rank two, it is the spectrum of a quadratic algebra (Proposition 12.2). The rank-three and rank-four alternatives of Theorem 11.4 thus govern one incidence over the full multiplication failure scheme; they do not by themselves classify every primary component on its extreme-corank locus.

The incidence yields a systematic non-Cartier family. For  $B_h = \mathbb{C}[x, y]/(x^2, xy, y^h)$ ,  $h \geq 5$ , the full failure scheme of unital  $(h-1)$ -planes has local rings which are localized polynomial extensions of

$$\mathbb{C}[b, u, v]/(F_h, vF_{h-1}, bv, b^2u), \quad F_1 = 1, \quad F_{j+1} = uF_j + vF_{j-1}.$$

Their exact primary decomposition is

$$(u, v) \cap (b^2, bv, F_h, vF_{h-1}).$$

The minimal support has codimension  $h-3$ , an embedded component lies over  $b = u = v = 0$ , and the nilradical has exact index  $2h-3$  (Theorem 14.1). The mixed equations  $bv$  and  $b^2u$  couple

the extra tangent direction to the higher contact coefficients. Corollary 14.2 also supplies genuine three-generator examples. The unit-cover description places these local rings on smooth covers of the full generating Grassmannians, rather than on selected linear slices.

A general conductor theorem completes the passage to global geometry. If  $I_Z$  is the saturated ideal of any zero-dimensional scheme in  $\mathbb{P}^e$ , then  $n \geq 2 \operatorname{reg} I_Z - 1$  suffices to identify the global and finite multiplication cokernels in every degree  $m \geq 2$  (Theorems 15.1 and 15.2). It applies to nonmonomial local schemes, unions, and finite flat families with a uniform regularity bound. The proof fills the conormal layer and the ordinary ideal-square layer. The square-regularity bound is the classical input of [35, Theorem 1.8]; the exact cokernel comparison is proved here. Corollary 15.3 transports the preceding embedded components to actual section multiplication on  $\mathbb{P}^2$ .

These results extend, rather than replace, the preceding calculations. The stable generated algebra still carries its representation on the multiplication defect; its exact-action-rank schemes define conductor flags  $C \subset Q$  compatible with base change on each stratum (Theorem 11.1). Their rank transitions can themselves be nonreduced. The following determinant-power family provides a complementary, pure-primary case of the theory.

Let  $Z$  be the order- $h$  fat point of  $\mathbb{P}^e$ , with  $e \geq 2$  and  $h \geq 3$ . For every  $n \geq 2h - 1$ , take all series containing  $H^0(\mathcal{I}_Z(n))$  and restricting to a generating  $(e + 1)$ -plane on  $Z$ . In symmetric degrees  $m \geq h - 1$  their failure ideal is

$$\mathcal{I}_\Delta^{\binom{e+h-1}{e+1}}, \quad (6.1)$$

where  $\Delta$  is the integral determinant divisor of the map from the augmentation kernel of the restricted series to  $\mathfrak{m}/\mathfrak{m}^2$ . This is the full primary ideal, and all its powers remain primary (Corollary 17.3). The proof separates a filtered substitution determinant from a triangular global conductor argument. It gives three-plane results on every  $\mathbb{C}[x, y]/(x, y)^h$  and works in arbitrary embedding dimension; no diagonal-ideal theorem is used. The global range depends on the fat-point order, rather than its binomially growing length.

Let  $D \subset \mathbb{P}^1$  have length  $d$ , put  $E_{n,D} = H^0(\mathcal{O}_D(n))$  and  $W_D = H^0(\mathcal{O}(n)(-D))$ , and let  $A \subset E_{n,D}$  be a generating  $k$ -plane. Write  $U_A = \operatorname{res}_D^{-1}(A)$ . For  $1 \leq k < d$  our conductor range is  $n \geq 2d - k$ . In every symmetric degree  $m \geq 2$  it gives

$$\operatorname{coker}(\operatorname{Sym}^m U_A \rightarrow V_{mn}) \simeq \operatorname{coker}(\operatorname{Sym}^m A \rightarrow E_{mn,D}). \quad (6.2)$$

The isomorphism holds on the entire relative generating Grassmannian and after arbitrary base change. The range is sharp uniformly in  $D, A, m$ : at  $n = 2d - k - 1$  an explicit monomial series already violates it in degree two. The exact fibrewise obstruction is the first cohomology of an evaluation kernel. On a curve of genus  $g$ , intrinsic cohomological hypotheses give the same transport; the uniform sufficient bound is  $\deg L \geq 2d + 2g - 1$ .

The defect-one structural theorem concerns generating hyperplanes in an arbitrary finite locally free algebra, not only pencils or powers of one element. Its failure scheme in all degrees  $m \geq 2$  is the space of rank-two quotient algebras with a generating line. The identification is an isomorphism of functors on arbitrary, including nonreduced, test schemes. It follows from the conductor of a unital hyperplane subalgebra and a cyclic presentation of the multiplication cokernel (Theorem 10.2).

For  $D = \sum_i d_i p_i$  this gives a complete primary classification of all generating  $(d - 1)$ -planes. The components supported on  $2p_i$  have transverse algebra

$$G_{d_i} = \mathbb{C}[u, v]/(F_{d_i}, F_{d_i+1}), \quad F_0 = 0, \quad F_1 = 1, \quad F_{j+1} = uF_j + vF_{j-1};$$

those supported on  $p_i + p_j$  have transverse algebra  $\mathbb{C}[x, y]/(x^{d_i}, y^{d_j})$ . Their lengths are  $\binom{d_i}{2}$  and  $d_i d_j$ , but their exact nilpotency indices are  $2d_i - 3$  and  $d_i + d_j - 1$ . The reduced components are disjoint smooth curves on the generating open. The moving scheme is smooth and is a

generating open of a projective-line bundle over  $\text{Sym}^2 C \times \text{Sym}^{d-2} C$ . These results apply to global multiplication on  $\mathbb{P}^1$  for  $n \geq d + 1$ . They classify the whole contact-hyperplane family for every multiplicity partition, not a collection of transverse examples.

There is also a complete stable description for generating three-planes on a reduced divisor. On unit frames its ideal is

$$\bigcap_{i < j} (x_i - x_j, y_i - y_j).$$

The equality, and the analogous equality for every power of the ideal, use Haiman's diagonal-ideal theorem explicitly. The step specific to multiplication is an exact total-degree truncation and the conductor comparison; the radical-ideal theorem is a classical input, not a new assertion here.

**The pencil family and the distinction between fibres and total spaces.** The earlier contact-pencil classification remains part of the article, now in the improved range. We record its complete statement to fix the bases of its primary assertions.

**Theorem 6.1** (Contact-pencil failure schemes). *Let  $d \geq 4$  and  $n \geq 2d - 2$ . On the generating open  $G_D^\circ \subset \text{Gr}(2, E_{n,D})$ , let  $\mathcal{J}_m$  be the zeroth Fitting ideal of the global multiplication cokernel. For  $2 \leq m < d - 1$ , this ideal is zero: the failure scheme is the whole parameter space. For  $m \geq d - 1$  it is a Cartier divisor, and, on a fixed divisor  $D = \sum_i d_i p_i$ , its complete primary decomposition is*

$$\mathcal{J}_m = \bigcap_{i: d_i \geq 2} \mathcal{I}_{T_i}^{\binom{d_i}{2}} \cap \bigcap_{i < j} \mathcal{I}_{E_{ij}}^{d_i d_j}. \quad (6.3)$$

Here  $T_i$  and  $E_{ij}$  are failure to separate  $2p_i$  and  $p_i + p_j$ . These are distinct smooth irreducible divisors; there are no embedded associated points. The exact local equations, all coranks and infinitesimal layers are given in Theorems 19.2 and 19.3. Over a moving multiplicity stratum, primary descent is governed by Lemma 18.1.

With  $D$  unrestricted, the pencil failure divisor is integral, and its normalization is the smooth finite incidence marking a scalar length-two subdivisor. The normalization and its singular-support test are given in Theorem 20.2 and Proposition 20.3.

On a unit pencil frame, with  $D = (g)$  and ratio  $v$ , the determinant is the classical index form  $\Delta(g, v) = \det[1, v, \dots, v^{d-1}]$ . Its relation to the characteristic polynomial  $\chi_v$  is

$$\text{disc}(\chi_v) = \text{disc}(g)\Delta(g, v)^2.$$

The equation of failure is  $\Delta = 0$ , not the discriminant on the left: the latter also vanishes when  $g$  has repeated roots, even if  $v$  generates the algebra. The quotient and index-form constructions are related to the schemes of generators in [29]. The Grassmannian cohomology presentation of  $G_r$  is likewise classical. What must be proved here is the exact identification with the multiplication schemes, including primary fibres and base change.

For the new non-Cartier results the dependence is

$$\begin{aligned} \text{codimension stabilization} &\implies \text{full degree-three Fitting scheme,} \\ \text{hyperplane conductor incidence} &\implies \text{embedded multigenerator local rings,} \\ \text{regularity and the conormal layer} &\implies \text{global embedded failure schemes.} \end{aligned}$$

The first arrow is valid over nonreduced coefficient rings, the second is a scheme isomorphism on the specified open, and the last is a coherent-cokernel isomorphism. These are different kinds of assertions.

The dependence of the retained arguments is

$$\begin{aligned}
& \text{evaluation kernel} \implies \text{conductor cokernel}, \\
& \text{unital hyperplane conductor} \implies \text{rank-two quotients}, \\
& \text{rank-two quotients} \implies \text{primary fibres and smooth total scheme}, \\
& \text{index form} \implies \text{pencil primary fibres and normalization}, \\
& \text{polarized Cayley-Hamilton} + \text{Haiman} \implies \text{three-plane schemes}.
\end{aligned}$$

None of these arguments depends on the statistical interpretation, the quadratic excess classification, or the unrestricted hyperplane embedded structure. Those developments, including all their proofs and hypotheses, are retained in the full complementary manuscript. The present family is not identified with the unrestricted ambient Grassmannian. Section 22 gives the precise classical inputs and the remaining limitation in the nearest-source comparison.

**Primary geometry across the conductor boundary.** The principal boundary theorem is a classification of an entire Hilbert-function class, rather than a further isolated local example. Every complex local algebra with Hilbert function  $(1, 2, 2)$  is isomorphic to  $\mathbb{C}[x, y]/((x, y)^3, y^2)$  or  $\mathbb{C}[x, y]/((x, y)^3, xy)$ . For all its unital three-planes, in every degree at least two, Theorem 13.2 determines the full zeroth-Fitting scheme and every associated stratum, including the extreme-corank boundary. On a smooth frame cover put  $\delta = ad - bc$ ,  $P = (a, c)$  and  $Q = (b, d)$ . In the square case the ideal is  $\delta^2 P^2 = (\delta^2) \cap P^4$ . In the two-factor case it is  $\delta^2(ab, ad + bc, cd)$ , with embedded primes  $P, Q, P + Q$  and the explicit primary decomposition (13.7). Both schemes have a normal reduced determinant divisor, generic multiplicity two, and nilradical index four.

The factor lines of the quadratic relation index the embedded surfaces. A double factor gives a surface contained entirely in the subalgebra locus; distinct factors give two surfaces meeting that locus and an additional embedded point at the quadratic layer. Theorem 13.3 computes the conductor kernel on these strata, over their actual coefficient rings. This is the boundary information not supplied by the corank-one incidence isomorphism alone. Proposition 13.4 places both types in one finite flat algebra family, with an exact formula for the full Fitting ideal before specialization.

Three levels of statement should be distinguished. For an arbitrary finite algebra, Theorem 12.1 is a global presentation and an incidence theorem. For cube-zero augmentation ideals, Theorem 13.1 factors the full multiplication ideal through the quadratic multiplication tensor. For Hilbert function  $(1, 2, 2)$ , Theorem 13.2 converts this factorization into a complete primary classification. The connected higher-contact family of Theorem 14.1 supplies a complementary phenomenon: unbounded nilpotent complexity. Corollary 13.5 transports the complete length-five classification to section multiplication on the projective plane.

Finally, Proposition 9.2 proves optimality of the stabilization bound in every codimension and every generating rank at least two. Its elementary field-level interpretation is separated from the coefficient-ring assertion. The arguments in Sections 12.1 and 15.1 give the full functorial and cohomological details needed for arbitrary, possibly nonreduced, base change.

## 7. SHARP CONDUCTOR REDUCTION AND A CURVE CRITERION

All schemes in this article are over  $\mathbb{C}$ . Put  $V_r = H^0(\mathbb{P}^1, \mathcal{O}(r))$ , with  $V_r = 0$  for  $r < 0$ . For a subspace  $U$  of sections,  $U^j$  denotes its multiplication image, not its abstract symmetric power.

**Lemma 7.1** (The evaluation-kernel bound). *Let  $U \subset V_n$  be base-point-free, of codimension  $c$ , and write  $M_U = \ker(U \otimes \mathcal{O} \rightarrow \mathcal{O}(n))$ . There are integers  $a_i \geq 1$ , with  $\sum_i a_i = n$ , such that  $M_U = \bigoplus_{i=1}^{n-c} \mathcal{O}(-a_i)$ . If  $a = \max_i a_i$ , then for  $t \geq 0$ ,*

$$\text{coker}(U \otimes V_t \rightarrow V_{n+t}) \simeq H^1(M_U(t)), \quad UV_t = V_{n+t} \iff t \geq a - 1.$$

In particular  $a \leq c + 1$ , so  $UV_t = V_{n+t}$  for  $t \geq c$ .

*Proof.* The splitting theorem for vector bundles on  $\mathbb{P}^1$  applies to the locally free evaluation kernel. Its degree is  $-n$  and its rank is  $n - c$ . Moreover  $H^0(M_U) = 0$ , because  $U \rightarrow H^0(\mathcal{O}(n))$  is injective. Thus all its summands have negative degree. Twisting the evaluation sequence by  $\mathcal{O}(t)$  and using  $H^1(\mathcal{O}(t)) = 0$  gives the asserted cokernel. The formula for  $H^1(\mathcal{O}(t - a_i))$  gives the equivalence. Finally the other  $n - c - 1$  positive integers have sum at least  $n - c - 1$ , whence  $a \leq n - (n - c - 1) = c + 1$ .  $\square$

**Theorem 7.2** (Sharp uniform conductor range). *Let  $D \subset \mathbb{P}^1$  have length  $d \geq 2$ , let  $1 \leq k < d$ , and suppose*

$$n \geq 2d - k. \quad (7.1)$$

*Let  $A \subset H^0(\mathcal{O}_D(n))$  be a generating  $k$ -plane, and put  $W = H^0(\mathcal{O}(n)(-D))$  and  $U = \text{res}_D^{-1}(A)$ . For every  $m \geq 2$ ,*

$$WU^{m-1} = H^0(\mathcal{O}(mn)(-D)). \quad (7.2)$$

*On the entire relative generating Grassmannian over  $\text{Hilb}^d(\mathbb{P}^1)$ , restriction gives*

$$\text{coker}(\text{Sym}^m \mathcal{U} \rightarrow V_{mn} \otimes \mathcal{O}) \simeq \text{coker}(\text{Sym}^m \mathcal{A} \rightarrow \pi_* \mathcal{O}_D(mn)). \quad (7.3)$$

*These are isomorphisms of coherent sheaves after arbitrary base change; all their Fitting ideals agree. For a particular geometric fibre, the weaker condition  $n - d \geq \max_i a_i(U) - 1$  suffices for the same conclusion. Write  $X_{d,k}^\circ$  for this relative generating Grassmannian. The bound (7.1) is sharp as a bound uniform in  $D, A$  and simultaneously in all  $m \geq 2$ . For  $k = d$  the cokernel assertion is immediate from  $U = V_n$  in the range  $n \geq d - 1$ ; the saturation assertion in that case holds for  $n \geq d$ .*

*Proof.* Write  $D = (g)$ , so  $W = gV_{n-d}$ . The series  $U$  is base-point-free:  $W$  generates away from  $D$  and  $A$  generates on  $D$ . Its codimension in  $V_n$  is  $c = d - k$ . Lemma 7.1 and  $n - d \geq c$  give  $WU = gV_{2n-d}$ . Further multiplication by  $U$  uses input degrees  $2n - d, 3n - d, \dots$ , all at least  $c$ , and proves (7.2). The same argument uses  $a - 1$  in place of  $c$  on a particular fibre.

All restriction targets and kernels are vector bundles: their first cohomology groups vanish in this range. The map  $\mathcal{W} \otimes \text{Sym}^{m-1} \mathcal{U} \rightarrow \pi_* I_D(mn)$  is onto on every geometric fibre, hence is a surjection of vector bundles. It remains onto after every base change. Locally split  $0 \rightarrow \mathcal{W} \rightarrow \mathcal{U} \rightarrow \mathcal{A} \rightarrow 0$ . The kernel of the map on symmetric powers is the image of  $\mathcal{W} \otimes \text{Sym}^{m-1} \mathcal{U}$ . Quotienting the restriction square by these kernels proves (7.3). Fitting base change now applies [25].

For sharpness take  $n = 2d - k - 1$ ,  $D = d\{0\}$ , and, in the affine coordinate  $t$ , take the degree- $n$  homogenizations of

$$U = \langle 1, t, \dots, t^{k-1}, t^d, t^{d+1}, \dots, t^n \rangle.$$

This is base-point-free and is the inverse image of a generating  $k$ -plane on  $D$ . Exponents of its quadratic products lie in

$$[0, 2k - 2] \cup [d, n + k - 1] \cup [2d, 2n].$$

None is  $2d - 1$ . Thus  $t^{2d-1} \notin U^2$ , although it belongs to  $H^0(I_D(2n))$ . Its nonzero cokernel class restricts to zero on  $D$ , so even the cokernel comparison fails at this boundary in degree two. This does not assert a sharp threshold separately for each  $m > 2$ .  $\square$

The numerical bound can be replaced by a cohomological hypothesis on a curve. Here and below an ideal such as  $I_D$  is the ideal on the curve, whereas a Fitting ideal is on the parameter space.

**Proposition 7.3** (An exact cohomological test). *Let  $C$  be a smooth projective curve, let  $L$  be a line bundle, and let  $U \subset H^0(L)$  be base-point-free. Put  $W = H^0(L(-D))$ . If  $H^1(L(-D)) = 0$ , then*

$$\operatorname{coker}(W \otimes U \rightarrow H^0(L^2(-D))) \simeq H^1(M_U \otimes L(-D)).$$

*More generally, for  $B_j = U^j \subset H^0(L^j)$  and its evaluation kernel  $M_{B_j}$ , the obstruction to  $WB_j = H^0(L^{j+1}(-D))$  is exactly  $H^1(M_{B_j} \otimes L(-D))$ .*

*Proof.* Tensor the evaluation sequence for  $B_j$  by  $L(-D)$  and take cohomology. The next term is  $B_j \otimes H^1(L(-D)) = 0$ .  $\square$

**Theorem 7.4** (Conductor transport on curves). *Let  $C$  have genus  $g$ , and let  $D$  be effective of length  $d \geq 2$ . Suppose  $L(-D)$  is globally generated,  $\operatorname{Sym}^2 H^0(L(-D)) \rightarrow H^0(L^2(-2D))$  is surjective, and*

$$H^1(L^j(-2D)) = 0 \quad (j \geq 1). \quad (7.4)$$

*For every generating  $A \subset H^0(L|_D)$  and every  $m \geq 2$ ,  $U = \operatorname{res}_D^{-1}(A)$  satisfies*

$$WU^{m-1} = H^0(L^m(-D)), \quad \operatorname{coker}(\operatorname{Sym}^m U \rightarrow H^0(L^m)) \simeq \operatorname{coker}(\operatorname{Sym}^m A \rightarrow H^0(L^m|_D)).$$

*In particular these assertions hold if*

$$\deg L \geq 2d + 2g - 1. \quad (7.5)$$

*For the relative assertion, let  $f : C \rightarrow S$  be smooth and projective of relative dimension one, with  $S$  locally Noetherian over  $\mathbb{C}$ , let  $D$  be a relative effective Cartier divisor finite flat of degree  $d$ , and let  $L$  be invertible. Assume the preceding hypotheses on every geometric fibre. For each  $j \geq 1$  and  $b = 0, 1, 2$ , put  $E_{j,b} = f_* L^j(-bD)$ . These are locally free,  $R^1 f_* L^j(-bD) = 0$ , and their formation commutes with arbitrary base change. The same base-change property holds for the finite direct images of  $L^j|_D$  and  $L^j(-D)|_D$ . On the relative generating Grassmannian form  $\mathcal{U}$  as the inverse image of  $\mathcal{A}$  in  $E_{1,0}$ . The displayed saturation and cokernel assertions then hold as maps of these bundles and sheaves, after every base change.*

*Proof.* The hypotheses imply  $H^1(L^j(-D)) = 0$  by the exact sequence with quotient on  $D$ . They make  $U$  base-point-free. It contains a pencil without base points, obtained by avoiding the finitely many zeros of one section. The pencil sequence, tensored by  $L^j(-2D)$ , shows that multiplication onto  $H^0(L^{j+1}(-2D))$  is surjective for  $j \geq 2$ : its obstruction is  $H^1(L^{j-1}(-2D))$ . Starting with the assumed surjectivity for  $W^2$ , this proves  $W^2 U^{m-2} = H^0(L^m(-2D))$ .

The map  $W \rightarrow H^0(L(-D)|_D)$  is onto by (7.4). Choose  $a \in A$  a unit on  $D$  and lift it to  $u \in U$ . Multiplication by  $u^{m-1}$  identifies the first normal quotients in these degrees. Consequently  $Wu^{m-1}$  fills that quotient in  $H^0(L^m(-D))$ . Together with the double-vanishing space just proved, this gives the assertion about  $WU^{m-1}$ . The locally split restriction square gives the cokernel isomorphism as before.

Under (7.5), the line bundle  $L(-D)$  has degree at least  $2g + 1$ . The classical normal-generation theorem in this degree range gives the required quadratic surjectivity and global generation; see [14, p. 73, Introduction]. Serre duality gives (7.4), since  $\deg L(-2D) \geq 2g - 1$ . For the relative assertion, the assumed fibrewise  $H^1$ -vanishing for  $b = 2$  implies the same vanishing for  $b = 1, 0$  by restriction to  $D$ . Cohomology and base change for these flat line bundles on a proper flat relative curve gives the named direct images, with no higher cohomology; finite flat pushforward gives the two restriction bundles. In particular  $0 \rightarrow E_{1,1} \rightarrow E_{1,0} \rightarrow f_*(L|_D) \rightarrow 0$  is a locally split exact sequence. The quadratic map  $\operatorname{Sym}^2 E_{1,1} \rightarrow E_{2,2}$  and each map  $E_{1,1} \otimes \operatorname{Sym}^{m-1} \mathcal{U} \rightarrow E_{m,1}$  are surjective on every geometric fibre by the fixed-curve proof. They are thus surjections of vector bundles and remain so under arbitrary pullback. The same locally split restriction square proves the relative cokernel assertion; no base-change assumption on an unflattened image  $U^j$  is needed.  $\square$

For example on any elliptic curve, a line bundle of degree at least  $2d + 1$  transports all the finite-contact schemes below. The curve statement is not a sharpness assertion in positive genus. Its intrinsic hypotheses, and the exact obstruction in Proposition 7.3, distinguish a failure of regularity from a failure of the finite-contact multiplication.

## 8. GENERATING OPENS AND FINITE PRESENTATIONS

Here “generating” refers to generation as an ideal or invertible module. It does not assert generation as a unital algebra.

**Lemma 8.1** (The relative generating open). *Let  $B/S$  be a finite locally free commutative algebra over a  $\mathbb{C}$ -scheme, and let  $A \subset B$  be a subbundle. The following conditions are equivalent:  $B \otimes A \rightarrow B$  is surjective; on every geometric fibre  $A$  has nonzero image in every residue field of  $B$ ; and the open  $A^\times$  of sections of  $A$  which are units of  $B$  is smooth and surjective over  $S$ . These conditions define an open in the Grassmannian. The assertions have the same form for a subbundle of an invertible  $B$ -module, using generating sections in place of units.*

*Proof.* Surjectivity is equivalent to its geometric-fibre version by Nakayama’s lemma. On a finite algebra over an algebraically closed field, an ideal is the unit ideal exactly when its images in all residue fields are nonzero. These evaluations cut out finitely many proper hyperplanes in  $A$ . Their complement is nonempty over an infinite field and consists precisely of units. On the total space of the vector bundle  $A$ , invertibility is the open condition  $\det(b \mapsto ab) \neq 0$ . This open is smooth over  $S$ , and the preceding fibre calculation proves surjectivity. Conversely a unit section after a faithfully flat cover generates the unit ideal, which descends. The Grassmannian open is the complement of the support of the cokernel of  $B \otimes A \rightarrow B$ . Local trivializations of an invertible module prove the last assertion.  $\square$

**Lemma 8.2** (Total-degree truncation). *Let  $B$  be a rank- $d$  finite locally free algebra over a  $\mathbb{C}$ -algebra  $R$ , and let  $b_1, \dots, b_q \in B$ . The  $R$ -subalgebra they generate is spanned as an  $R$ -module by the monomials of total degree at most  $d - 1$ . This assertion commutes with base change, with polynomial reduction coefficients on a free chart.*

*Proof.* Set  $w = \sum_j t_j b_j$  in  $B[t_1, \dots, t_q]$  and apply Cayley–Hamilton to multiplication by  $w$ . The coefficient of  $t^\alpha$ , for  $|\alpha| = d$ , in  $w^d$  is  $\binom{d}{\alpha} b^\alpha$ . All other terms have  $B$ -monomial degree less than  $d$ : coefficients of the characteristic polynomial are scalars in  $R[t]$ . The multinomial coefficient is invertible in  $R$ . This expresses each degree- $d$  monomial using lower degrees. Multiplying these identities and inducting reduces every higher monomial. The adjugate identity for Cayley–Hamilton makes the coefficients polynomial in the multiplication table.  $\square$

## 9. A CODIMENSION BOUND OVER THE COEFFICIENT RING

The degree needed to generate a finite algebra can be bounded by the codimension of the initial subspace, rather than by the length of the algebra. A fibrewise dimension argument suggests this bound but does not prove the assertion over a nonreduced parameter scheme. The following argument proves an equality of submodules over the coefficient ring.

**Theorem 9.1** (Codimension stabilization). *Let  $B$  be a commutative finite locally free algebra over a  $\mathbb{Q}$ -scheme, and let  $W \subset B$  be a subbundle containing 1, with  $B/W$  locally free of rank  $r$ . Write  $W^j$  for the image of  $\text{Sym}^j W \rightarrow B$ . Then*

$$W^{r+1} = \mathcal{O}[W], \quad W^j = W^{r+1} \quad (j \geq r + 1). \quad (9.1)$$

*These equalities remain true after every base change, without a flatness hypothesis on  $W^j$ .*

For a generating rank- $k$  subbundle  $A$  of a rank- $d$  algebra over a  $\mathbb{C}$ -scheme, all Fitting ideals of the multiplication cokernel are consequently independent of the degree  $j \geq d - k + 1$ . The generating opens and normalization by units are those of Lemma 8.1. This is a statement about ideal equalities; the unnormalized cokernels need not have a preferred isomorphism in different degrees.

*Proof.* The case  $r = 0$  is immediate. Work otherwise on a free chart and choose a splitting  $B = W \oplus M$ , where  $\text{rank } M = r$ . For  $w \in W$  write multiplication by  $w$  in blocks

$$L_w = \begin{pmatrix} A_w & D_w \\ \beta_w & C_w \end{pmatrix}.$$

Thus  $\beta_w : W \rightarrow M$  and  $C_w : M \rightarrow M$ . Let  $N_0 = \sum_{w \in W} \text{im } \beta_w$ , and, for  $j \geq 0$ , let  $N_j$  be the sum of the images of  $N_0$  under all words of length at most  $j$  in the operators  $C_w$ . A finite local basis of  $W$  suffices in these definitions. Since  $1 \in W$ , every  $W^m$  contains  $W$ . Projection to  $M$  and subtraction of its  $W$  component therefore give, as submodules of the chosen splitting,

$$W^{j+2} = W \oplus N_j. \quad (9.2)$$

Indeed the initial case is the definition of  $N_0$ , and multiplication by  $W$  changes the residual module  $N$  into  $N_0 + \sum_w C_w N$ .

Commutativity of the full multiplication operators gives the identity

$$[C_w, C_v] = \beta_v D_w - \beta_w D_v, \quad \text{im}[C_w, C_v] \subset N_0. \quad (9.3)$$

Consider a word of length  $r$  applied to  $N_0$ . Interchanging two adjacent factors changes it by a word with a commutator in those positions. The commutator sends its entire argument into  $N_0$ ; at most  $r - 2$  factors remain to its left. The change thus belongs to  $N_{r-2}$ , and in particular to  $N_{r-1}$ . For  $r = 1$  no interchange is needed. All permutations of a length- $r$  word therefore agree modulo  $N_{r-1}$  when applied to  $N_0$ .

Apply Cayley–Hamilton to the  $r$  by  $r$  matrix  $C(t) = \sum_{i=1}^r t_i C_{w_i}$  and extract the coefficient of  $t_1 \cdots t_r$ . The leading term is the sum of the  $r!$  ordered products of the  $C_{w_i}$ . Every other term is a scalar linear combination of words of length at most  $r - 1$ . Applied to  $N_0$  these other terms lie in  $N_{r-1}$ . The preceding interchange argument and the invertibility of  $r!$  now show that every length- $r$  word sends  $N_0$  into  $N_{r-1}$ . Hence  $N_r = N_{r-1}$ . This equality says that  $N_{r-1}$  is invariant under all  $C_w$ , so the residual filtration is constant thereafter. Equation (9.2) proves (9.1).

This calculation used no division except by  $r!$  and no passage to residue fields. It is valid over every base-changed coefficient ring; in particular, it does not assert that an unsplit image is a vector bundle. Finally choose a unit  $a \in A$  smoothly locally and put  $W = a^{-1}A$ . Multiplication by  $a^j$  identifies  $W^j$  with  $A^j$ , and is an automorphism of  $B$ . The Fitting ideals are thus constant in the stated range on a faithfully flat cover and hence on the original parameter scheme.  $\square$

Combining this theorem with Lemma 8.2 gives the stable range  $j \geq \min\{d - 1, d - k + 1\}$  for  $d \geq 2$ . The improvement is largest when the starting subspace has small codimension. In particular, a codimension-two multiplication problem is determined in degree three, however large  $d$  is. Section 12 uses this ring-level assertion to retain the entire failure scheme, including the closed locus where the defect is two.

**9.1. Sharpness and the algebra-length problem.** For a unital subspace  $W$  over a field, let  $\ell(W)$  be the least  $j$  such that  $W^j$  is the algebra generated by  $W$ . This is the usual length of a generating subspace. Algebra-length bounds are studied in the literature of Pappacena and Markova; see [36, 37] and the more recent discussion in [38, Sections 1 and 6]. No novelty is claimed here for the following field-level dimension argument, which works even without commutativity. If  $W^j = W^{j+1}$ , multiplication by  $W$  stabilizes that space and the sequence is constant thereafter.

Until then its dimension strictly increases. Starting with dimension  $k$  in an algebra of dimension  $d$  gives  $\ell(W) \leq d - k + 1$ . Theorem 9.1, by contrast, proves an equality of submodules over a coefficient ring without assuming the intervening images flat. A dimension count on residue fields does not establish that equality. Its application here is stability of all Fitting ideals after arbitrary base change.

**Proposition 9.2** (Sharpness in every codimension). *For every pair of integers  $r \geq 1$ ,  $k \geq 2$ , there is a connected commutative local algebra  $B$  of dimension  $r + k$  and a unital  $k$ -plane  $W \subset B$  such that*

$$W^r \neq W^{r+1} = B.$$

*Thus  $r + 1$  is the optimal universal bound for codimension  $r$ , even when the generating rank  $k$  is prescribed.*

*Proof.* Take

$$B = \mathbb{C}[z, \epsilon_1, \dots, \epsilon_{k-2}] / (z^{r+2}, z\epsilon_i, \epsilon_i\epsilon_j : 1 \leq i, j \leq k-2)$$

and  $W = \text{span}(1, z, \epsilon_1, \dots, \epsilon_{k-2})$ . When  $k = 2$  the lists involving  $\epsilon_i$  are empty. A basis of  $B$  is  $1, z, \dots, z^{r+1}, \epsilon_1, \dots, \epsilon_{k-2}$ . For  $1 \leq j \leq r + 1$ , a basis of  $W^j$  is  $1, z, \dots, z^j, \epsilon_1, \dots, \epsilon_{k-2}$ . The assertions follow, with  $z^{r+1}$  witnessing strictness.  $\square$

The result is an optimal bound depending on the initial codimension, not a claim to improve every bound involving the minimal-polynomial degrees or the internal structure of a particular algebra. The relative proof uses commutativity of the full multiplication operators; it does not assert that their compressed blocks commute.

## 10. GENERATING HYPERPLANES AND LENGTH-TWO QUOTIENTS

We now solve a multigenerator problem without assuming that the finite algebra is monogenic, reduced, or constant. If  $B$  is a finite locally free commutative algebra of rank  $d$  over a scheme  $S$ , let  $\text{Hilb}^2(B/S)$  denote the functor of quotient algebras  $B_T \rightarrow Q$  that are locally free of rank two over  $T$ . It is the relative Hilbert scheme of two points of  $\text{Spec}_S B$ . A generating line in  $Q$  means a line subbundle whose  $Q$ -span is  $Q$ .

**Lemma 10.1** (The conductor of a unital hyperplane). *Unital subalgebras  $S_0 \subset B_T$  which are hyperplane subbundles are naturally equivalent to rank-two quotient algebras of  $B_T$ . The maps are*

$$S_0 \longmapsto B_T/J, \quad J = \{b \in B_T : bB_T \subset S_0\}, \quad (B_T \rightarrow Q) \longmapsto \text{preimage}(\mathcal{O}_T \cdot 1).$$

*The conductor  $J$  is an ideal subbundle of rank  $d - 2$ , and its formation commutes with arbitrary base change.*

*Proof.* Work locally with  $T = \text{Spec } R$ , trivialize  $B/S_0$ , and write it as the target of  $\phi : B \rightarrow R$ . Choose  $e$  with  $\phi(e) = 1$ . Since  $1 \in S_0$ , the bilinear form  $\beta(b, c) = \phi(bc)$  has  $\beta|_{S_0 \times S_0} = 0$  and  $\beta(1, e) = 1$ . An element of its radical first lies in  $S_0$ , by pairing with 1; it then lies in the kernel of the split surjection

$$S_0 \longrightarrow R, \quad s \longmapsto \phi(se),$$

which sends 1 to 1. Thus the radical has rank  $d - 2$ , and  $B/J$  has basis the classes of  $1, e$ . Associativity shows that the radical is an ideal:  $\beta(jb, c) = \beta(j, bc) = 0$ . It is precisely the displayed conductor. This split-kernel description proves base change and gives  $S_0/J = R \cdot 1$ .

Conversely, in a rank-two locally free algebra  $Q$ , the unit is a unimodular vector, so  $Q/R \cdot 1$  is a line bundle. The inverse image of the scalars is a unital hyperplane subalgebra. No nonzero ideal of  $Q$  can be contained in  $R \cdot 1$ : in a basis  $1, e$ , if  $c \cdot 1$  belongs to such an ideal, then  $ce$  must be scalar, so  $c = 0$ . Its conductor is therefore exactly the original kernel. The two constructions are inverse and are independent of all trivializations.  $\square$

**Theorem 10.2** (The universal hyperplane failure scheme). *Let  $B/S$  be finite locally free of rank  $d \geq 3$  over a  $\mathbb{C}$ -scheme. On  $X = \text{Gr}_S(d-1, B)^\circ$ , the generating open, let  $\mathcal{A}$  be the tautological hyperplane and put*

$$\mathcal{C}_m = \text{coker}(\text{Sym}^m \mathcal{A} \rightarrow B), \quad Y_m = V(\text{Fitt}_0 \mathcal{C}_m).$$

*For every  $m \geq 2$  the ideal of  $Y_m$  is the same. The common scheme  $Y$  is naturally isomorphic to*

$$\{(B \rightarrow Q, \ell) : \text{rank } Q = 2, \ell \subset Q \text{ a generating line}\}. \quad (10.1)$$

*The map to  $X$  sends  $(Q, \ell)$  to the inverse image of  $\ell$ . If  $i : Y \hookrightarrow X$  and  $B_Y \twoheadrightarrow Q \supset \ell$  is the universal quotient with its generating line, the canonical formula is*

$$\mathcal{C}_m \simeq i_*(Q/\ell^{(m)}), \quad \ell^{(m)} = \text{im}(\ell^{\otimes m} \rightarrow Q). \quad (10.2)$$

*Both  $\ell^{(m)}$  and its quotient are line bundles on  $Y$ . In particular all positive-index Fitting ideals are the unit ideal. The isomorphism of schemes and all these assertions commute with arbitrary base change. The same statement holds for an invertible  $B$ -module in place of  $B$ , with target its  $m$ th tensor power.*

*Proof.* Locally for the smooth topology, choose a unit  $a \in \mathcal{A}$ . Such choices cover  $X$ : on each geometric fibre avoid the finitely many evaluation hyperplanes. Replace  $\mathcal{A}$  by  $A' = a^{-1}\mathcal{A}$ , and the target by multiplication by  $a^{-m}$ . Now  $1 \in A'$ . Split  $B = A' \oplus Re$  locally. Since  $(A')^m$  contains  $A'$ , elementary row and column operations present the cokernel by a single row. For  $m = 2$  its entries are exactly the coefficients of the products  $ss'$  in  $B/A'$ , for  $s, s' \in A'$ . Call their ideal  $I$ . Thus  $Y_2$  represents the condition that  $A'$  is a subalgebra, with its full scheme structure, not just its support. Modulo  $I$  all higher products lie in  $A'$ . Conversely every quadratic product is an  $m$ -fold product by inserting 1's. Hence the residual ideals equal  $I$  for every  $m \geq 2$ , and the cokernel is locally  $R/I$ .

Lemma 10.1 supplies the unique rank-two quotient on  $Y$ . Its kernel  $J$  also lies in  $\mathcal{A} = aA'$ , since  $J$  is an ideal and  $aJ = J$ . The image of  $\mathcal{A}$  is a generating line in  $Q$ . The construction does not depend on  $a$ . Indeed another unit  $b \in \mathcal{A}$  gives  $b/a \in A'$ , and its inverse belongs to  $A'$  by Cayley–Hamilton in  $B$  (the constant coefficient is a unit). Thus  $(b/a)^{-1}A' = A'$  and the conductor is unchanged. This also proves descent from the smooth cover. Conversely the inverse image of a generating line in a rank-two quotient becomes the inverse image of the scalars after division by a unit, so it satisfies precisely the residual equations. These constructions are inverse on every test scheme, proving the scheme isomorphism. On  $Y$ , the normalized subalgebra  $A'$  has conductor  $J$ , and  $J \subset a^m A' = (\mathcal{A})^m$  since  $a$  is a unit and  $J$  an ideal. Passing to  $Q$  identifies  $(\mathcal{A})^m/J$  with  $\ell^{(m)}$ . This gives the canonical map in (10.2); the local cyclic presentations identify it as an isomorphism. Multiplication by a local unit spanning  $\ell$  makes  $\ell^{(m)}$  a direct-summand line, so the quotient is indeed a line bundle. For an invertible  $B$ -module  $P$ , put  $P_Q = P \otimes_B Q$ ; the same formula is  $i_*(P_Q^{\otimes m}/\ell^{(m)})$ , where now  $\ell \subset P_Q$ . Trivialization of  $P$  gives the local proof and this intrinsic expression supplies its gluing.  $\square$

**10.1. Every fixed multiplicity partition.** For  $r \geq 2$  define polynomials

$$F_0 = 0, \quad F_1 = 1, \quad F_2 = u, \quad F_{j+1} = uF_j + vF_{j-1}, \quad G_r = \mathbb{C}[u, v]/(F_r, F_{r+1}). \quad (10.3)$$

For positive  $r, s$  put  $H_{r,s} = \mathbb{C}[x, y]/(x^r, y^s)$ .

**Theorem 10.3** (Complete primary schemes for contact hyperplanes). *Let  $D = \sum_i d_i p_i$  have length  $d \geq 3$ . Consider the entire generating open  $\text{Gr}(d-1, H^0(\mathcal{O}_D))^\circ$ . For every  $m \geq 2$ , its multiplication failure scheme is the disjoint union*

$$Y_D \simeq \coprod_{i:d_i \geq 2} (\mathbb{A}^1 \times \text{Spec } G_{d_i}) \amalg \coprod_{i < j} (\mathbb{G}_m \times \text{Spec } H_{d_i, d_j}). \quad (10.4)$$

The identifications use local coordinates on  $D$  and trivializations of the restricted line bundle. Each displayed summand is a primary component on this fixed-divisor parameter space. There are no embedded associated points. All reduced components are smooth and pairwise disjoint. Their transverse lengths and exact nilradical indices are, respectively,

| length-two support | length           | nilpotency index  |
|--------------------|------------------|-------------------|
| $2p_i$             | $\binom{d_i}{2}$ | $2d_i - 3$        |
| $p_i + p_j$        | $d_i d_j$        | $d_i + d_j - 1$ . |

The convention for a reduced component is index one. Every failure fibre has cokernel dimension one. On binary inverse-image subseries these are exactly the multiplication failure schemes whenever  $n \geq d + 1$ ; on curves they are the same schemes under Theorem 7.4.

*Proof.* A degree-two quotient of the product algebra  $\prod_i \mathbb{C}[z_i]/(z_i^{d_i})$  decomposes according to the ranks of its idempotent factors. These ranks are locally constant even on a nonreduced base. Its possibilities are rank two at a single factor, or rank one at each of two distinct factors. This is a decomposition into open and closed Hilbert subschemes, not merely a decomposition of closed points.

For rank two in  $\mathbb{C}[z]/(z^r)$ , the elements  $1, z$  form a basis in every geometric quotient, hence on every family by Nakayama. Write  $z^2 = uz + v$ . Then

$$z^j = F_j(u, v)z + vF_{j-1}(u, v).$$

The relation  $z^r = 0$  is equivalent to  $F_r = vF_{r-1} = 0$ , equivalently  $F_r = F_{r+1} = 0$ . Thus this Hilbert scheme is  $\text{Spec } G_r$ . A rank-one quotient at a factor of length  $r$  is  $\text{Spec } \mathbb{C}[x]/(x^r)$ , so the two-factor case is  $\text{Spec } H_{r,s}$ . A generating line in the first rank-two algebra is uniquely spanned by  $1 + \tau z$ ; in the split two-factor algebra it is uniquely spanned by  $(1, t)$ , with  $t$  a unit. Theorem 10.2 now proves (10.4) over arbitrary test schemes.

The algebra  $G_r$  is Artin local: if  $v \neq 0$ , two consecutive zero values in the recurrence force  $F_1 = 0$ , and if  $v = 0$  they force  $u = 0$ . The two equations form a regular sequence in  $\mathbb{C}[u, v]$ . With  $\deg u = 1$ ,  $\deg v = 2$ , its Hilbert series is

$$\frac{(1 - t^{r-1})(1 - t^r)}{(1 - t)(1 - t^2)}. \quad (10.5)$$

It has length  $\binom{r}{2}$  and weighted socle degree  $2r - 4$ . To see the exact ordinary nilpotency, identify  $u$  and  $-v$  with the elementary symmetric functions in two variables. Then  $F_j = h_{j-1}$  and

$$G_r \simeq H^{2*}(\text{Gr}(2, r), \mathbb{C}), \quad (10.6)$$

where the cohomological degree is divided by two. This is the classical presentation by  $h_{r-1}, h_r$ , not a new presentation of Grassmannian cohomology; the rectangular Schur basis is recalled in [31, Theorem 2.7 and the following discussion]. Successive multiplication by  $u = s_{(1)}$  adds one box. The only partition of size  $2r - 4$  in the 2 by  $r - 2$  rectangle is  $(r - 2, r - 2)$ ; the ballot-path count of its standard tableaux is  $\frac{1}{r-1} \binom{2r-4}{r-2}$ . Consequently

$$u^{2r-4} = \frac{1}{r-1} \binom{2r-4}{r-2} (-v)^{r-2} \neq 0.$$

Every monomial of ordinary degree  $2r - 3$  has weighted degree at least  $2r - 3$  and vanishes. This proves the index  $2r - 3$ , including  $r = 2$ . In  $H_{r,s}$  the top monomial is  $x^{r-1}y^{s-1}$ , giving length  $rs$  and index  $r + s - 1$ .

Each summand is a complete intersection with reduced support a smooth integral curve, so it is primary and has no embedded points. Equivalently, over a local Artin coefficient algebra, a polynomial with nonzero reduction is a nonzerodivisor, as one checks on the successive powers

of the maximal ideal. The same holds after inverting  $t$ . This proves the primary assertions also globally on the indicated affine curves. The cokernel assertion follows from Theorem 10.2. The two transports follow from Theorems 7.2 and 7.4.  $\square$

**Theorem 10.4** (The smooth moving scheme). *Let  $d \geq 3$ . On the relative generating Grassmannian of  $(d-1)$ -planes over  $\text{Hilb}^d(C)$ , for a smooth connected projective curve  $C$ , the finite-contact failure scheme is a generating open of a projective-line bundle over*

$$\text{Sym}^2 C \times \text{Sym}^{d-2} C, \quad (Z, Q) \mapsto D = Z + Q.$$

*It is smooth, irreducible, and of dimension  $d+1$ . Its projection to  $\text{Sym}^d C$  is flat of relative dimension one. The finite map of divisor spaces has degree  $\binom{d}{2}$ ; the primary lengths in Theorem 10.3 sum to this degree on every fixed fibre. Under either conductor theorem, these are statements about the original global multiplication scheme.*

*Proof.* A nested pair  $Z \leq D$  of effective divisors on a smooth curve is uniquely  $D = Z + Q$ . This holds in families: a finite flat subscheme of a smooth relative curve is a relative Cartier divisor, and inclusions of their invertible ideals give the residual divisor. The incidence in Theorem 10.2 is therefore exactly the following canonical open projective bundle. On  $T = \text{Sym}^2 C \times \text{Sym}^{d-2} C$  let  $\mathcal{Z} \subset C \times T$  be the universal degree-two divisor and set  $F = (\pi_{\mathcal{Z}})_* \mathcal{O}_{\mathcal{Z}}$ . Use the projectivization  $\mathbb{P}_T^{\text{lines}}(F)$  parametrizing line subbundles of  $F$ , and retain the open where the tautological line  $\ell$  satisfies  $F\ell = F$ . This is the claimed bundle, not an unspecified local form. For a restricted line bundle replace  $F$  by  $(\pi_{\mathcal{Z}})_*(L|_{\mathcal{Z}})$  and use generation as an  $F$ -module. The open is nonempty on every fibre of its smooth irreducible base, and hence is smooth and irreducible of dimension  $d+1$ .

The map of divisor spaces is finite (proper with finite geometric fibres). Both spaces are smooth of dimension  $d$ ; the finite source is Cohen–Macaulay and the target regular, so the map is flat. Over a reduced divisor it has  $\binom{d}{2}$  points. This proves the degree and flatness assertions; alternatively, the fibre lengths computed above give  $\sum_i \binom{d_i}{2} + \sum_{i < j} d_i d_j = \binom{d}{2}$ . The open projective bundle is smooth of relative dimension one, so its composite projection is flat of that relative dimension.  $\square$

**10.2. A direct residual presentation.** For comparison with the multiplication matrix, normalize a unit and write, on the unital hyperplane chart in  $\mathbb{C}[z]/(z^r)$ ,

$$A' = \langle 1, z^i - c_i z : 2 \leq i \leq r-1 \rangle.$$

Put  $c_1 = 1$  and  $c_j = 0$  for  $j \geq r$ . After eliminating the pivot columns, the remaining entries of the quadratic matrix are

$$c_{i+j} - c_i c_{j+1} - c_j c_{i+1} + c_2 c_i c_j \quad (2 \leq i \leq j \leq r-1). \quad (10.7)$$

For  $r \geq 4$ , the relations with  $i = 2$  give  $c_{j+2} = u c_{j+1} + v c_j$ , with  $u = c_2$ ,  $v = c_3 - c_2^2$ . They eliminate  $c_4, \dots, c_{r-1}$  and leave  $F_r, F_{r+1}$ . Every other relation follows from the recurrence identity

$$F_{i+j} = F_i F_{j+1} + F_j F_{i+1} - u F_i F_j,$$

proved by induction in  $j$ . For  $r = 3$  the single residual entry is  $c_2^3$ , agreeing with  $G_3 \simeq \mathbb{C}[u]/(u^3)$ . For  $r = 4$  one obtains

$$G_4 = \mathbb{C}[u, v]/(u^3 + 2uv, u^2v + v^2), \quad \text{length } G_4 = 6, \quad \mathfrak{m}^5 = 0 \neq \mathfrak{m}^4.$$

Thus the multigenerator transverse scheme is not a powered smooth divisor, even though its length equals the corresponding pencil weight. The smooth moving scheme has these nonreduced fibres; nonreducedness of a fibre is not an embedded component of the total scheme. Nor does this determine the embedded primary ideal along the evaluation curve in the unrestricted, non-generating hyperplane problem retained in the complementary material.

**10.3. A noncurvilinear algebra and an embedded component.** The general finite-algebra theorem is not restricted to the primary patterns arising from divisors on a smooth curve.

**Proposition 10.5** (An exact embedded chart). *Let  $B = \mathbb{C}[x, y]/(x^2, y^2)$ . On the unit-normalized hyperplane chart*

$$A' = \langle 1, y - ax, xy - bx \rangle \subset B \otimes \mathbb{C}[a, b],$$

*the multiplication failure ideal, for every  $m \geq 2$ , is*

$$I = (ab, b^2) = (b) \cap (a, b)^2.$$

*Its associated primes are  $(b)$  and  $(a, b)$ . Its nilradical is a one-dimensional square-zero module supported at the embedded point. These equations give the pullback of the full failure scheme on the generating Grassmannian, not only its reduced support.*

*Proof.* Use the quotient functional  $\phi(1) = 0$ ,  $\phi(x) = 1$ ,  $\phi(y) = a$ ,  $\phi(xy) = b$ . Products involving 1 are pivots. The other quadratic products have residual values  $-2ab, -b^2, 0$ . This gives  $I$ ; Theorem 10.2 gives every higher degree. The displayed primary decomposition follows by intersecting the monomial ideals. The nilradical is generated by  $b$ , whose annihilator modulo  $I$  is  $(a, b)$ . The normalization by a unit is the same smooth-cover calculation used in that theorem. In particular adding the freely chosen unit does not remove the embedded associated prime.  $\square$

Thus the quotient description distinguishes smooth moving contact schemes from finite-algebra families with embedded structure. The absence of embedded points in Theorem 10.3 is a consequence of the curvilinear contact algebra and its explicit Hilbert schemes, not a general assertion about finite algebras.

## 11. STABLE IMAGES AND HIGHER-DEFECT CONDUCTOR STRATA

The rank-two quotient in Theorem 10.2 is specific to a hyperplane. In larger codimension the appropriate object is a quotient algebra together with a subalgebra acting faithfully on the quotient module. The rank of this action is an additional stratification invariant. We keep its scheme structure throughout.

For a map  $u$  of vector bundles write  $I_j(u)$  for its ideal of  $j$ -minors. Its *exact-rank  $s$  locus* means  $V(I_{s+1}(u)) \setminus V(I_s(u))$ , with this locally closed scheme structure. On it the image and cokernel are vector bundles and commute with arbitrary base change: after inverting a pivot, elementary operations reduce the matrix to  $\text{diag}(I_s, 0)$ . This is stronger than requiring constant rank on geometric points.

**Theorem 11.1** (Stable image and conductor stratification). *Let  $B$  be a commutative finite locally free algebra of rank  $d \geq 2$  over a locally Noetherian  $\mathbb{C}$ -scheme  $S$ , and let  $\mathcal{A}$  be the universal bundle on  $X = \text{Gr}_S(k, B)^\circ$ , with  $1 \leq k \leq d$ . Put  $\mu_m : \text{Sym}^m \mathcal{A} \rightarrow B_X$ . All Fitting ideals of  $\text{coker } \mu_m$  are independent of  $m \geq d - 1$ ; Theorem 9.1 also gives stability for  $m \geq d - k + 1$ . We use  $d - 1$  below to retain a uniform notation for the stable-image strata. On the exact-rank  $s$  locus  $X_s$  of  $\mu_{d-1}$  there is a canonical unital subalgebra bundle  $E \subset B_{X_s}$  of rank  $s$ . There is an invertible  $E$ -module  $L \subset B_{X_s}$ , also a subbundle over  $X_s$ , such that*

$$E = \mathcal{O}[a^{-1}\mathcal{A}], \quad L = E\mathcal{A} = aE, \quad \text{im } \mu_m = L^m \quad (m \geq d - 1) \quad (11.1)$$

*on any local unit cover  $a \in \mathcal{A} \cap B^\times$ .*

*Suppose  $r = d - s > 0$  and put  $M = B/E$ . The action*

$$\theta : E \longrightarrow \mathcal{E}nd_{\mathcal{O}}(M)$$

defines exact-rank strata  $X_{s,t}$ , with  $1 \leq t \leq r^2$ . On such a stratum the conductor and its quotient flag are

$$J = \ker \theta = \{b \in B : bB \subset E\}, \quad C = E/J \subset Q = B/J, \quad \text{rank } C = t, \quad \text{rank } Q = r + t. \quad (11.2)$$

Here  $J$  is an ideal subbundle of  $B$ , and  $C \rightarrow \mathcal{E}nd(Q/C)$  is a subbundle injection. Each of these constructions commutes with arbitrary base change within the specified stratum.

More generally, the functor of unital codimension- $r$  subalgebra bundles with action of exact rank  $t$  is equivalent to the functor of flags  $B \twoheadrightarrow Q \supset C$  of these ranks, where the first map is a surjective morphism of unital commutative algebras,  $Q$  is locally free of rank  $r + t$ ,  $C \subset Q$  is a unital subalgebra subbundle of rank  $t$ ,  $Q/C$  is locally free of rank  $r$ , and the specified multiplication action  $C \rightarrow \mathcal{E}nd(Q/C)$  is a locally split subbundle injection. The inverse sends the flag to the preimage of  $C$ .

*Proof.* Choose a unit  $a$  smoothly locally, as in Lemma 8.1, and set  $W = a^{-1}\mathcal{A}$ . Then  $1 \in W$ . Lemma 8.2 says that  $W^{d-1} = \mathcal{O}[W]$  and that  $W^m = W^{d-1}$  for  $m \geq d-1$ , over the coefficient ring itself. Multiplication by  $a^m$  identifies the original image with  $a^m\mathcal{O}[W]$ . The resulting cokernels are locally isomorphic in these degrees, proving equality of every Fitting ideal before any rank restriction.

On  $X_s$ , the pivot description makes  $E = \mathcal{O}[W]$  a subbundle with locally free quotient. If  $b$  is a second unit in  $\mathcal{A}$ , then  $b/a \in E$  is a unit of  $B$ . Cayley–Hamilton for multiplication by  $b/a$ , whose constant coefficient is a unit, puts its inverse in  $E$ . Thus  $\mathcal{O}[b^{-1}\mathcal{A}] \subset E$ ; reversing  $a, b$  gives equality. This proves descent of  $E$ . The image  $EA$  is locally  $aE$  and therefore descends to the stated invertible  $E$ -module and subbundle  $L$ . It also proves (11.1) universally.

Because  $E$  is a subalgebra, it acts on  $M = B/E$ . Its action kernel consists of the  $e \in E$  with  $eB \subset E$ . An element  $b$  of the displayed conductor already lies in  $E$ , by multiplication by 1; hence the two kernels agree. If  $jB \subset E$ , then for  $b, c \in B$  one has  $(jb)c = j(bc) \in E$ , so  $J$  is an ideal of  $B$ . On an exact-rank  $t$  stratum, pivot reduction for  $\theta$  makes  $J$  locally free of rank  $s - t$  and  $C$  a subbundle of  $\mathcal{E}nd(M)$ . The exact sequence  $0 \rightarrow C \rightarrow B/J \rightarrow M \rightarrow 0$  proves that  $Q$  is locally free of rank  $r + t$ . All the kernels and images just used are split as  $\mathcal{O}$ -modules locally; this proves their base-change assertions. The unit acts as the identity on  $M$ , so  $t \geq 1$ ; the other bound comes from  $\text{rank } \mathcal{E}nd(M) = r^2$ .

Conversely, given a flag as in the statement, its inverse image  $E$  is a subalgebra bundle and  $B/E \simeq Q/C$ . The kernel of its action is precisely the kernel of  $B \rightarrow Q$ : a larger kernel would give an element of  $C$  annihilating  $Q/C$ , contrary to the subbundle injection. Thus the conductor construction recovers the flag, including over nonreduced test schemes. These mutually inverse constructions respect all pullbacks within the exact-rank functors.  $\square$

The word “stratum” cannot be removed from the base-change assertion for  $J$ . The multiplication cokernel is a universally defined coherent sheaf on  $X$ ; its conductor need not be a bundle on  $X$ . Example 11.5 below exhibits the distinction.

**Proposition 11.2** (The extreme-corank failure scheme). *On the Grassmannian of rank- $(d - r)$  subbundles  $W \subset B$  containing 1, with  $1 \leq r < d$ , the unital subalgebra scheme is*

$$V(\text{Fitt}_{r-1} \text{coker}(\text{Sym}^m W \rightarrow B)), \quad m \geq 2.$$

*The defining ideal is independent of  $m$ , including its nonreduced structure. Its conductor exact-rank  $t$  subschemes are the quotient-flag functors of Theorem 11.1.*

*Proof.* Since  $1 \in W$ , the image  $W^m$  contains  $W$ . Split  $B = W \oplus M$  locally and remove the identity pivot columns. The residual presentation has  $r$  rows, so its  $(r - 1)$ st Fitting ideal is the ideal of all entries. In degree two these entries are precisely the coefficients of  $W \cdot W$  in

$B/W$ . They vanish if and only if  $W$  is a subalgebra, on every test scheme. Modulo their ideal, every higher product remains in  $W$ . Conversely each quadratic product is an  $m$ -fold product by insertion of 1's. The entry ideals are equal. The last assertion is the functorial equivalence just proved.  $\square$

For  $r = 1$  this is the usual zeroth-Fitting failure scheme. For  $r > 1$  it is the *extreme-corank* locus, not the whole locus where multiplication is nonsurjective. This distinction prevents a false extension of the hyperplane theorem by changing only the quotient rank.

### 11.1. The two quotient mechanisms in codimension two.

**Lemma 11.3** (Relative cyclic vectors). *Let  $M$  be a rank-two vector bundle, and let  $C \subset \mathcal{E}nd(M)$  be a unital rank-two algebra subbundle over a  $\mathbb{C}$ -scheme. Then  $M$  is an invertible  $C$ -module locally on the base. More explicitly, write  $C = \langle 1, T \rangle$  and  $T = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$  after trivializing. The opens*

$$D(c), \quad D(b), \quad D(c + d - a - b)$$

*cover the base and have cyclic vectors  $(1, 0)$ ,  $(0, 1)$ , and  $(1, 1)$ , respectively. These opens and the resulting module isomorphisms commute with base change.*

*Proof.* On every geometric fibre, the subbundle injection makes  $T$  nonscalar. If all three displayed functions vanish, then  $b = c = d - a = 0$ , contradicting that assertion. Their principal opens therefore cover. The determinant of the map  $C \rightarrow M$ ,  $q \mapsto qv$ , is respectively  $c$ ,  $-b$ , and  $c + d - a - b$  for the three chosen vectors. On the relevant open it is a unit, so the map is a  $C$ -linear isomorphism. The same invertible matrices prove the assertion after any pullback.  $\square$

**Theorem 11.4** (Codimension-two quotient flags). *Let  $E \subset B$  be a unital subalgebra bundle with  $M = B/E$  of rank two over a locally Noetherian  $\mathbb{C}$ -scheme. For its action  $\theta$  one has  $I_3(\theta) = 0$  scheme-theoretically. The exact-rank-one closed locus and the exact-rank-two open locus have the following functorial descriptions.*

*On the first,  $E$  is the inverse image of the scalar subalgebra in a locally free rank-three quotient algebra of  $B$ . On the second,  $E$  is the inverse image of  $C$  in a quotient  $B \rightarrow Q$ , where  $C$  is a rank-two algebra over the base and  $Q$  is a locally free rank-two algebra over  $C$ . In particular  $Q$  has rank four over the base. In both cases the kernel of  $B \rightarrow Q$  is the conductor, and these descriptions commute with arbitrary base change on their respective strata.*

*Proof.* Locally trivialize  $M$ . Since 2 is invertible, write every matrix as a scalar plus a trace-zero matrix. The trace-zero parts of the commuting matrices  $\theta(E)$  still commute. The bracket map

$$\bigwedge^2 \mathfrak{sl}_2 \longrightarrow \mathfrak{sl}_2$$

is an isomorphism over  $\mathbb{C}$ : for the usual basis  $H, U, V$  its values are  $2U, -2V, H$ . Thus the wedge of any two trace-zero columns is zero over the whole coefficient ring, not just over its residue fields. Adjoining the scalar identity column proves  $I_3(\theta) = 0$ . The identity also gives  $I_1(\theta) = \mathcal{O}$ .

On the exact-rank-one locus the image is the scalar algebra. Theorem 11.1 gives  $C = \mathcal{O}$  and  $\text{rank } Q = 3$ . Conversely the scalar line in any rank-three algebra has faithful action on its rank-two quotient module: the action is ordinary scalar multiplication. Its inverse image has exactly this conductor.

On the exact-rank-two locus,  $C \subset \mathcal{E}nd(M)$  is a rank-two subalgebra bundle. Locally write  $C = \langle 1, T \rangle$ . Lemma 11.3 gives an explicit cyclic-vector open cover and proves that  $M$  is locally free of rank one as a  $C$ -module. The exact sequence of  $C$ -modules  $0 \rightarrow C \rightarrow Q \rightarrow M \rightarrow 0$  consequently splits locally, so  $Q$  is locally free of rank two over  $C$ .

Conversely, if  $Q$  is a locally free rank-two  $C$ -algebra, its unit is unimodular as a  $C$ -section. Hence  $Q/C$  is locally free of rank one over  $C$ . Its regular  $C$ -action is faithful and is a subbundle

injection into its endomorphism bundle over the original base (check on geometric fibres and use a pivot). The quotient-flag equivalence therefore gives exactly the second stratum. All arguments are functorial and use split modules on the indicated strata.  $\square$

*Remark 11.5.* Consider  $R = \mathbb{C}[s]$ ,  $B = R[z]/(z^5)$ , and the direct summand

$$E = \langle 1, sz^2 + z^3, z^4 \rangle_R.$$

It is a subalgebra since  $(sz^2 + z^3)^2 = s^2 z^4$ . Modulo  $E$  take the basis  $\bar{z}, \bar{z}^2$ ; then  $\bar{z}^3 = -s\bar{z}^2$ . The action of  $sz^2 + z^3$  sends  $\bar{z}$  to  $-s^2\bar{z}^2$  and  $\bar{z}^2$  to zero, while  $z^4$  acts as zero. Thus the exact-rank-one subscheme is  $s^2 = 0$ , not merely  $s = 0$ . Over  $R$  the conductor is  $Rz^4$ ; at the reduced fibre  $s = 0$  it is  $\langle z^3, z^4 \rangle$ . Quotient rank drops from four to three. This simultaneously records the nonreduced rank stratum and the failure of conductor formation to commute with unrestricted specialization. The fibre subalgebras belong to the classical codimension-two polynomial-algebra classification of [34, Theorem 39]; the calculation here retains the relative action and its determinantal thickening.

## 12. THE FULL CODIMENSION-TWO FAILURE SCHEME

Fix a finite locally free commutative algebra  $B/S$  of rank  $d \geq 4$  over a locally Noetherian  $\mathbb{C}$ -scheme. Let

$$X = \text{Gr}_S(d-3, B/\mathcal{O} \cdot 1),$$

viewed as the Grassmannian of rank- $(d-2)$  subbundles  $W \subset B$  containing 1. This is a unital parameter space, not an open subset of the Grassmannian of all generating planes. We explain the passage to the latter at the end of the section.

Write  $\mu_j : \text{Sym}^j W \rightarrow B_X$ , and let  $D$  denote the stable zeroth-Fitting failure scheme. Let  $T \subset X$  denote the unital subalgebra scheme. In general  $T$  is a proper closed subscheme of  $D$ ; its ideal is the first, rather than the zeroth, Fitting ideal.

**Theorem 12.1** (Equations and the corank-one incidence). *The stable scheme  $D$  is defined already by  $\mu_3$ . On a splitting  $B_X = W \oplus M$ , with  $\text{rank } M = 2$ , use the block notation of (9.3). Let  $\beta$  be the matrix with columns  $\beta_{w_i}(w_j)$ , for a local basis  $(w_i)$  of  $W$ , and set*

$$\mathcal{K} = [\beta \mid C_{w_1}\beta \mid \cdots \mid C_{w_{d-2}}\beta]. \quad (12.1)$$

For every  $m \geq 3$  there are equalities

$$\begin{aligned} \text{coker } \mu_m &\simeq \text{coker } \mathcal{K}, \\ \mathcal{I}_D &= I_2(\mathcal{K}), \\ \mathcal{I}_T &= I_1(\beta) = I_1(\mathcal{K}) = \text{Fitt}_1 \text{coker } \mu_m. \end{aligned} \quad (12.2)$$

The cokernel isomorphism uses the chosen splitting; the ideals glue canonically. These are equations for the full schemes, not just for their radicals or their exact-rank strata.

Let  $H = \text{Hilb}_S^2(B)$  be the scheme of locally free rank-two quotient algebras. Over  $H$ , let  $E_H \subset B_H$  be the preimage of the scalar subalgebra in the universal quotient. Put

$$Z = \text{Gr}_H(d-3, E_H/\mathcal{O} \cdot 1), \quad Z^\circ = \{W : W^2 \longrightarrow E_H \text{ is onto}\}.$$

Then the forgetful morphism induces an isomorphism

$$Z^\circ \xrightarrow{\sim} D \setminus |T|. \quad (12.3)$$

It holds on arbitrary test schemes, including nonreduced ones. On this open the stable generated algebra is the hyperplane  $E_H$ , and the normalized multiplication cokernel is its quotient line  $B/E_H$ . The proper morphism  $Z \rightarrow X$  factors through  $D$  and is surjective on geometric points of  $D$ .

*Proof.* Theorem 9.1 gives  $W^3 = \mathcal{O}[W]$ . By (9.2), its residual image in  $M$  is  $N_1 = \text{im } \mathcal{K}$ . The identity columns supplied by  $W \subset W^3$  may be removed from a presentation of  $B/W^3$ . This proves the first two assertions in (12.2). Every entry of  $C_{w_i}\beta$  is a linear combination of entries of  $\beta$ , so their entry ideals are equal. Vanishing of these entries is exactly  $W^2 \subset W$ , which is the subalgebra functor. This proves the last line.

Work now where a column  $v$  of  $\beta$  is unimodular in  $M$ . Its lift is a quadratic product modulo  $W$ , and  $E = W + \mathcal{O}v$  is a hyperplane subbundle of  $B$ . Pivot on the unit coordinate of  $v$ . The two by two minors of  $\mathcal{K}$  then say precisely that all its columns belong to  $\mathcal{O}v$ . In the quotient ring by those minors,  $W^2 = E$  and  $WE \subset E$ . Moreover  $v \in W^2$ , so multiplication by  $v$  preserves  $E$ : write  $v$  as a sum of products of elements of  $W$  and use  $WE \subset E$  twice. Thus  $E$  is a unital subalgebra. Equivalently its equality with the stable image follows directly from Theorem 9.1.

The conductor equivalence of Lemma 10.1 associates to  $E$  a unique rank-two quotient of  $B$ . It recovers a point of  $Z^\circ$ . Conversely, a point of  $Z^\circ$  has stable image  $E_H$ , since  $W^2 = E_H$  and  $E_H$  is an algebra. Its residual multiplication rank is exactly one, so it factors through  $D \setminus |T|$ . The two constructions are inverse on every test scheme and are defined by split pivots and the hyperplane conductor. They therefore glue to (12.3), including its nilpotent structure. The same description gives the cokernel line.

The relative Hilbert scheme is projective: it is the closed ideal-quotient subscheme of the Grassmannian of rank-two module quotients of  $B$ . Hence  $Z \rightarrow X$  is proper. Every multiplication image on  $Z$  lies in the hyperplane algebra  $E_H$ , so its maximal minors vanish and the map factors through  $D$ . Surjectivity away from  $T$  is already proved. Surjectivity on geometric points of  $T$  follows from the two fibre descriptions in Proposition 12.2 below.  $\square$

The theorem gives both an equation for  $D$  along  $T$  and a complete functorial normal form off  $T$ . In particular, it is not obtained by replacing the ideal  $I_2(\mathcal{K})$  with the entry ideal  $I_1(\beta)$ . Nor does it assert that the corank-one open is dense in every component: for a particular algebra a component may be contained in  $T$ .

**Proposition 12.2** (Fibres over the conductor strata). *Restrict  $T$  to an exact-action-rank stratum and write  $J = \text{cond}_B(W)$ ,  $C = W/J$ , and  $Q = B/J$ . The fibre product  $Z \times_X T$  has the following descriptions, with arbitrary base change within the indicated stratum.*

*If the action rank is one,  $C = \mathcal{O}$  and  $Q$  has rank three. The fibre product is  $\text{Hilb}^2(Q)$ . More explicitly, on  $\mathbb{P}(Q/\mathcal{O})$ , in the convention of lines, it is the zero scheme of the canonical binary cubic*

$$\ell^{\otimes 3} \longrightarrow \bigwedge^2(Q/\mathcal{O}), \quad v^{\otimes 3} \longmapsto \bar{v} \wedge \overline{v^2}. \quad (12.4)$$

*The cubic may vanish identically; no relative degree-three flatness is asserted without a nonvanishing hypothesis.*

*If the action rank is two,  $C$  has rank two and  $Q$  is a rank-two  $C$ -algebra. The fibre product is canonically  $\text{Spec } C$ , a finite locally free double cover of the stratum, possibly nonreduced.*

*Proof.* A point of the fibre product is a hyperplane algebra  $E \subset B$  containing  $W$ . Since  $J$  is an ideal of  $B$  contained in  $W$ , it is contained in the conductor of  $E$ . Thus the problem is exactly to choose a unital hyperplane subalgebra of  $Q$  containing  $C$ . Conversely its inverse image supplies  $E$ .

In the first case this is the unital-hyperplane functor of a rank-three algebra, hence  $\text{Hilb}^2(Q)$  by Lemma 10.1. A line  $\ell \subset Q/\mathcal{O}$  has an algebra as its preimage if and only if its square remains in that preimage. This is exactly (12.4). Changing a lift  $v$  by a scalar changes  $v^2$  by a scalar and a multiple of  $v$ , so the displayed wedge is independent of the lift. This also proves its scheme-theoretic meaning over a nonreduced base.

In the second case put  $M = Q/C$ . By Lemma 11.3,  $M$  is locally an invertible  $C$ -module. A hyperplane extension gives a  $C$ -stable subbundle  $\ell \subset M$  of rank one over the original base. The induced action on the quotient line  $M/\ell$  is a character  $C \rightarrow \mathcal{O}$ . Conversely such a character, with kernel  $I$ , gives  $\ell = IM$ ; it is a subbundle and the quotient is a line. These constructions are inverse, as can be checked after trivializing  $M$  over  $C$ . They are the functor of  $\mathrm{Spec} C$ . There is no additional multiplication equation. Indeed locally write  $Q = C \oplus Cz$ , with  $z^2 = \alpha + \gamma z$ ,  $\alpha, \gamma \in C$ . The extension is  $C \oplus Iz$ , which is an algebra because  $I$  is an ideal of  $C$ . This proves the assertion over every test scheme.

For completeness both fibres are nonempty over an algebraically closed field. A rank-three commutative algebra has a quotient of length two, by an ideal composition series; a rank-two algebra has a character. This proves the geometric surjectivity used above.  $\square$

For example, the rank-three algebra  $\mathbb{C}^3$  gives three points in (12.4), whereas  $\mathbb{C}[u, v]/(u, v)^2$  gives the whole projective line. Mere invariance under the scalar action would incorrectly give a projective line in both cases. The square condition in (12.4) is essential. In contrast, on the rank-two action stratum it follows from the quadratic-algebra tower. The conductor stratification thus controls the exceptional fibres of one proper incidence over the full failure scheme. We do not identify this morphism with a normalization or a blow-up.

*Remark 12.3* (Unit covers and unrestricted generating planes). Let  $G = \mathrm{Gr}_S(d-2, B)^\circ$  be the full generating Grassmannian, and let  $\tilde{G}$  parametrize pairs  $(A, a)$  with  $a \in A \cap B^\times$ . The morphism  $\tilde{G} \rightarrow G$  is smooth and surjective, and

$$\tilde{G} \simeq B^\times \times_S X, \quad (A, a) \longmapsto (a, a^{-1}A).$$

Multiplication by  $a^{-m}$  identifies the pulled-back cokernel with the unital one. All the ideal identities above therefore give the full failure scheme on this smooth cover of  $G$  and descend. Over  $\mathbb{C}$ , local primary structures discussed below may equally be read on this cover; the assertion is not that a special linear slice of  $G$  determines its ambient scheme structure.

**12.1. The quadratic incidence fibre as a functor.** In Theorem 12.1, the notation  $D \setminus |T|$  means the open complement of the support of the possibly nonreduced closed scheme  $T$ . Nothing is subtracted from its structure sheaf. We supply a full functorial description of the rank-two action fibre in Proposition 12.2.

Let  $C \subset Q$  be the conductor flag on that exact-action-rank stratum. Then  $C$  is a rank-two algebra,  $Q$  is a rank-two locally free  $C$ -algebra, and  $M = Q/C$  is an invertible  $C$ -module. The incidence fibre parametrizes rank-three subalgebras  $E' \subset Q$  containing  $C$ , with  $Q/E'$  a line bundle over the base.

A morphism  $U \rightarrow \mathrm{Spec} C$  is an algebra map  $\chi : C_U \rightarrow \mathcal{O}_U$ . Write  $I = \ker \chi$ ; it is a line subbundle of  $C_U$ , since the unit splits  $\chi$ . Tensoring  $0 \rightarrow I \rightarrow C_U \rightarrow \mathcal{O}_U \rightarrow 0$  with the invertible  $C_U$ -module  $M_U$  gives

$$0 \longrightarrow IM_U \longrightarrow M_U \longrightarrow M_U \otimes_{C_U, \chi} \mathcal{O}_U \longrightarrow 0.$$

The last term is a line bundle over  $U$ . The middle term has rank two over  $U$ , so  $IM_U$  is a line subbundle and this sequence is locally split over  $\mathcal{O}_U$ . Let  $E'_\chi$  be its inverse image in  $Q_U$ . It is a rank-three subbundle containing  $C_U$ . To check multiplicative closure, one may work faithfully flat locally where  $M_U$  has a  $C_U$ -generator lifted to  $e \in Q_U$ . Then  $Q_U = C_U \oplus C_U e$ ,  $e^2 = \alpha + \beta e$ , and  $E'_\chi = C_U + Ie$ . Products stay in this submodule because  $I$  is an ideal and  $I^2 \subset I$ . The closure condition descends, and the definition of  $E'_\chi$  did not use the generator.

Conversely let  $E'$  be such a subalgebra, and put  $N = E'/C_U$ . The exact sequence

$$0 \longrightarrow N \longrightarrow M_U \longrightarrow Q_U/E' \longrightarrow 0$$

exhibits  $N$  as a line subbundle. Since  $C_U \subset E'$ , the submodule  $N$  is  $C_U$ -stable. Its quotient line  $L' = Q_U/E'$  therefore defines an algebra map

$$\chi : C_U \longrightarrow \text{End}_{\mathcal{O}_U}(L') = \mathcal{O}_U.$$

The equality on the right is canonical for an invertible module. Let  $I = \ker \chi$ . The quotient map factors through  $M_U \otimes_{C_U, \chi} \mathcal{O}_U \rightarrow L'$ . This is a surjective map of line bundles, hence an isomorphism. Thus  $N = IM_U$  and  $E' = E'_\chi$ . Conversely the action on the quotient constructed from  $\chi$  recovers  $\chi$  itself. All the quotient sequences are locally split as sequences of base modules, and tensoring an invertible  $C$ -module is exact. The two constructions therefore commute with every pullback and are inverse on every test scheme, not just on geometric points. This proves the asserted fibre isomorphism with  $\text{Spec } C$  by the functor of points.

### 13. PRIMARY GEOMETRY AT A CONDUCTOR BOUNDARY

The distinction between a presentation of the full failure scheme and its primary geometry is essential along the subalgebra locus. We now resolve that geometry for every complex local algebra with Hilbert function  $(1, 2, 2)$ . The proof begins with a factorization valid for all algebras whose augmentation ideal has cube zero. In the classified case the quadratic relation determines the embedded associated strata, including a stratum supported entirely at the extreme-corank boundary. No restriction to the quadratic-generation open is made.

**13.1. The quadratic-layer factorization.** Let  $B$  be a finite local  $\mathbb{C}$ -algebra, with maximal ideal  $\mathfrak{m}$  satisfying  $\mathfrak{m}^3 = 0$ . Put  $V = \mathfrak{m}/\mathfrak{m}^2$ ,  $S_2 = \mathfrak{m}^2$ , and write  $e = \dim V$ ,  $q = \dim S_2$ . Multiplication induces a surjection

$$\gamma : \text{Sym}^2 V \longrightarrow S_2.$$

Choose a splitting  $\mathfrak{m} = V \oplus S_2$ . The unital parameter space of  $(e+1)$ -planes is  $X = \text{Gr}(e, \mathfrak{m})$ : its tautological plane is  $W = \mathcal{O} \cdot 1 \oplus K$ . On its frame bundle write

$$\begin{pmatrix} M \\ H \end{pmatrix} : \mathcal{O}^e \longrightarrow V \oplus S_2, \quad M \in \text{Mat}_{e \times e}, \quad H \in \text{Mat}_{q \times e}. \quad (13.1)$$

The stacked matrix has rank  $e$ . Its entries, not just the entries of  $M$ , are the frame coordinates.

**Theorem 13.1** (Quadratic-layer factorization). *For every degree  $m \geq 2$ , the full zeroth-Fitting ideal of  $\text{Sym}^m W \rightarrow B$  pulls back to*

$$\mathcal{I}_D = (\det M) I_q(\gamma \text{Sym}^2 M). \quad (13.2)$$

*The identity holds on the whole frame bundle and after arbitrary base change, including substitution of coefficients in a nonreduced ring. In particular it also gives equations at  $M = 0$  whenever that stratum meets the frame bundle. The variables  $H$  are unrestricted except for the frame condition; the formula is not a transverse slice.*

*Proof.* Products of three elements of  $K$  vanish. Because  $1 \in W$ , the multiplication image in every degree  $m \geq 2$  is  $\mathcal{O} \cdot 1 + K + K^2$ . After deleting zero and repeated columns, a presentation of its cokernel is given by

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & M & 0 \\ 0 & H & \gamma \text{Sym}^2 M \end{pmatrix}.$$

Every nonzero maximal minor must use the first column and all  $e$  columns of the middle block: no other column has an entry in the  $V$  rows. Expansion in the first  $e+1$  rows leaves exactly  $\det M$  times a  $q$  by  $q$  minor of the last block. These are all the maximal minors. This is a polynomial identity in the universal coefficient ring. It proves the assertion before taking a radical, localizing at a tangent minor, or passing to residue fields.  $\square$

The same calculation applies to an augmented locally free algebra  $\mathcal{O} \oplus V \oplus S_2$  over a base scheme, with locally free layers,  $S_2(V \oplus S_2) = 0$ , and multiplication  $\gamma$ . Formula (13.2) therefore describes families of quadratic relations as well as families of generating planes.

**13.2. The complete class of Hilbert function  $(1, 2, 2)$ .** Every local algebra in this class has  $\mathfrak{m}^3 = 0$ . The kernel of  $\gamma : \text{Sym}^2 V \rightarrow S_2$  is a line of binary quadratic forms. Over  $\mathbb{C}$  its generator is either a square or a product of two independent linear forms. Thus the complete list of isomorphism types is

$$\begin{aligned} B_{\text{sq}} &= \mathbb{C}[x, y]/((x, y)^3, y^2), \\ B_{\text{tf}} &= \mathbb{C}[x, y]/((x, y)^3, xy). \end{aligned} \quad (13.3)$$

Indeed a linear splitting of  $\mathfrak{m} \rightarrow V$  identifies  $B$  with  $\mathbb{C} \oplus V \oplus S_2$ , with multiplication determined entirely by  $\gamma$ ; there are no further extension coefficients when  $\mathfrak{m}^3 = 0$ .

In both cases  $X = \text{Gr}(2, \mathfrak{m})$  parametrizes unital three-planes. On its frame bundle put

$$M = \begin{pmatrix} a & c \\ b & d \end{pmatrix}, \quad \delta = ad - bc, \quad R = \mathbb{C}[a, b, c, d], \quad P = (a, c), \quad Q = (b, d), \quad \mathfrak{n} = (a, b, c, d).$$

The four entries of  $H$  are independent additional variables. All the following ideals are extended to  $R[H]$  and restricted to the open where the stacked matrix (13.1) has rank two. A component absent from an individual affine chart is simply omitted there; the decomposition is irredundant on the whole frame bundle.

**Theorem 13.2** (Complete primary decomposition for Hilbert function  $(1, 2, 2)$ ). *For every  $m \geq 2$  the full multiplication failure scheme has the following primary decompositions.*

For  $B_{\text{sq}}$ ,

$$I_{\text{sq}} = \delta^2 P^2 = (\delta^2) \cap P^4, \quad \text{Ass}(R/I_{\text{sq}}) = \{(\delta), P\}. \quad (13.4)$$

The prime  $P$  is embedded. More generally, for every  $j \geq 1$ ,

$$I_{\text{sq}}^j = (\delta^{2j}) \cap P^{4j}. \quad (13.5)$$

For  $B_{\text{tf}}$ , set

$$\begin{aligned} J &= (ab, ad + bc, cd), \\ Q_0 &= J + P^2 + Q^2, \\ Q_* &= \delta^2 Q_0 + \mathfrak{n}^7. \end{aligned} \quad (13.6)$$

Then  $Q_0$  and  $Q_*$  are  $\mathfrak{n}$ -primary and

$$\begin{aligned} I_{\text{tf}} &= \delta^2 J = (\delta^2) \cap P^3 \cap Q^3 \cap Q_*, \\ \text{Ass}(R/I_{\text{tf}}) &= \{(\delta), P, Q, \mathfrak{n}\}. \end{aligned} \quad (13.7)$$

The last three primes are embedded. The displayed  $Q_*$  is a specified primary representative, not a canonical choice of an embedded primary component.

These identities describe the entire schemes on  $X$ , including the extreme-corank locus  $T$ . Intrinsically, an embedded surface is indexed by a linear factor  $L \subset V$  of the quadratic relation: its support is

$$Y_L = \{K \subset \mathfrak{m} : K \rightarrow V/L \text{ is zero}\} = \text{Gr}(2, S_2 + L) \simeq \mathbb{P}^2. \quad (13.8)$$

There is one such surface for a square and two for distinct factors. In the two-factor case there is also the embedded point  $K = S_2$ . There are no further associated strata.

The reduced support in both cases is the same integral normal divisor

$$\Delta = \{\text{rank}(K \rightarrow V) \leq 1\} \subset X.$$

*Its singular locus is the point  $K = S_2$ . The generic multiplicity of  $D$  along  $\Delta$  is two, whereas the nilradical of  $\mathcal{O}_D$  has exact nilpotency index four. Neither full failure scheme is Cartier.*

*Proof.* Choose bases in the quadratic layer, and rescale the middle symmetric basis vector where necessary. The two quadratic product matrices are

$$L_{\text{sq}} = \begin{pmatrix} a^2 & 2ac & c^2 \\ ab & ad + bc & cd \end{pmatrix}, \quad L_{\text{tf}} = \begin{pmatrix} a^2 & 2ac & c^2 \\ b^2 & 2bd & d^2 \end{pmatrix}.$$

For the first matrix the target basis may be taken to be  $x^2, 2xy$ ; for the second it is  $x^2, y^2$ . Their two-minor ideals are, respectively,  $\delta P^2$  and  $\delta J$ . Theorem 13.1 gives the asserted full failure ideals. Notice the extra factor  $\delta$  supplied by the linear layer; dropping that layer would produce a different primary problem.

For the square case, the  $P$ -adic associated graded ring of  $R$  is  $\mathbb{C}[b, d][a, c]$ , a domain. The  $P$ -adic order of  $\delta$  is one. Consequently

$$(P^t : \delta^u) = P^{\max\{t-u, 0\}} \quad (t, u \geq 0).$$

It follows that  $(\delta^2) \cap P^4 = \delta^2 P^2$ , and the same calculation with  $2j, 4j$  proves (13.5). The polynomial  $\delta$  is prime, and powers of  $P$  are primary. The two components are irredundant:  $\delta^2 \notin P^4$  and  $a^4 \notin (\delta^2)$ . This proves the associated-prime assertion.

For the second case we first prove

$$J = P \cap Q \cap Q_0. \quad (13.9)$$

Since  $P \cap Q = PQ$ , its quotient by  $J$  is spanned by the class of  $\delta$ : the relations set  $ab = cd = 0$  and  $ad = -bc$ . Every element of  $\mathfrak{n}$  annihilates this class, and the class is nonzero by linear independence in degree two. In  $R/Q_0$  a basis is

$$1, a, b, c, d, ad,$$

with  $bc = -ad$  and all other quadratic monomials zero. Thus  $\mathfrak{n}^3 \subset Q_0$ ,  $Q_0$  is  $\mathfrak{n}$ -primary, and  $\delta = 2ad$  remains nonzero in this quotient. The natural map  $PQ/J \rightarrow R/Q_0$  is therefore injective, proving (13.9). It is irredundant; the embedded prime  $\mathfrak{n}$  is also witnessed directly by  $\text{ann}_{R/J}(\bar{\delta}) = \mathfrak{n}$ .

The same associated-graded argument as before gives

$$(P^3 : \delta^2) = P, \quad (Q^3 : \delta^2) = Q.$$

We claim that  $(Q_* : \delta^2) = Q_0$ . For a homogeneous polynomial of degree  $j$ , membership of its product with  $\delta^2$  in  $\mathfrak{n}^7$  is equivalent to  $j \geq 3$ . More generally the homogeneous decomposition, and injectivity of multiplication by  $\delta^2$  in  $R$ , give

$$(\delta^2 Q_0 + \mathfrak{n}^7 : \delta^2) = Q_0 + \mathfrak{n}^3 = Q_0.$$

Intersecting with  $(\delta^2)$  and using (13.9) proves (13.7). Since  $Q_* \subset \mathfrak{n}$  and contains  $\mathfrak{n}^7$ , it is  $\mathfrak{n}$ -primary. For completeness multiplication by  $\delta^2$  gives an injection  $R/J \hookrightarrow R/I_{\text{tf}}$ . Hence its three associated primes  $P, Q, \mathfrak{n}$  remain associated. The prime  $(\delta)$  is the unique minimal prime of  $I_{\text{tf}}$ . This proves irredundance and excludes additional associated primes.

All these decompositions survive polynomial extension by  $H$  and localization, and every associated stratum meets the frame open by taking  $H$  to be the identity matrix. The global ideals, irredundancy, and associated points are established formally in Corollary 5.2, using Lemma 5.1. That corollary uses only the affine primary equalities just proved; it does not assume the global conclusion of the present theorem. In particular no specialization assertion or unexpanded regular-fibre argument is needed here.

The reduced equation on the frame cover is  $\delta = 0$ . It is an integral hypersurface whose singular locus is  $M = 0$ , of codimension three within that hypersurface. A hypersurface is Cohen–Macaulay;

regularity in codimension one and Serre's criterion give normality. Integrality on  $X$  also follows from the surjection  $\mathrm{Gr}_{\mathbb{P}(V)}(2, S_2 + \mathcal{L}) \rightarrow \Delta$  from an integral scheme. At a general point of  $\Delta$  the residual ideal  $P^2$  or  $J$  is a unit ideal, so the generic multiplicity is two.

In either case write  $I = \delta^2 J'$ , where  $J' = P^2$  or  $J$ . One has  $\delta \notin J'$  and  $\delta^2 \in J'$ ; in the latter case use  $\delta^2 = (ad + bc)^2 - 4abcd$ . Therefore  $\delta^4 \in I$  and  $\delta^3 \notin I$ . Since  $\sqrt{I} = (\delta)$ , the nilradical has exact index four. Nonvanishing is retained on the chart where  $H$  is invertible. Finally a nonzero principal ideal in the regular scheme  $X$  cannot have these embedded associated primes. Thus  $D$  is not Cartier.  $\square$

**13.3. Conductor equations on the embedded strata.** The preceding theorem computes local rings across  $T$ , not just the fibres of the incidence over  $T$ . We now locate the conductor strata inside these primary supports. This also specifies which embedded components lie entirely in the extreme-corank locus.

**Theorem 13.3** (Conductor position of the primary strata). *On the exact rank-one locus of  $K \rightarrow V$ , write  $L = \mathrm{im}(K \rightarrow V)$  and  $H_0 = K \cap S_2$ . Both are line bundles. The subalgebra locus  $T$  there is precisely the zero scheme of*

$$\gamma|_{\mathrm{Sym}^2 L} : \mathrm{Sym}^2 L \longrightarrow S_2/H_0. \quad (13.10)$$

*On that scheme the conductor is the actual kernel*

$$\mathrm{cond}_B(W) = \ker(K \longrightarrow \mathrm{Hom}(V, S_2/H_0)), \quad k \longmapsto (v \longmapsto \gamma(\bar{k}, v)). \quad (13.11)$$

*If the induced map  $L \rightarrow \mathrm{Hom}(V, S_2/H_0)$  has exact rank zero, the conductor is  $K$  and the conductor action has rank one. If it has exact rank one, the conductor is  $H_0$  and the action has rank two. These kernel statements commute with base change on the specified scheme-theoretic rank strata.*

*For  $B_{\mathrm{sq}}$  the embedded surface  $Y_{\langle y \rangle}$  is contained in  $T$ . On its rank-one part the action rank is one exactly when  $H_0 = \langle xy \rangle$ , and is two otherwise. For  $B_{\mathrm{tf}}$  the two embedded surfaces are  $Y_{\langle x \rangle}$  and  $Y_{\langle y \rangle}$ . Their rank-one intersections with  $T$  are, respectively,  $H_0 = \langle x^2 \rangle$  and  $H_0 = \langle y^2 \rangle$ ; the conductor action has rank one on both intersections. Elsewhere on the rank-one part of  $T$  it has rank two.*

*At  $K = S_2$ , in both types,  $W = \mathbb{C} \oplus S_2$  has conductor  $S_2$  and quotient  $B/S_2 = \mathbb{C} \oplus V$  with square-zero augmentation ideal. The incidence fibre there is  $\mathbb{P}(V)$ , whereas the action-rank-two fibres above are spectra of the dual numbers. The extra embedded point at  $K = S_2$  occurs precisely in the two-factor type. Thus numerical action rank alone does not determine associated primes: the quadratic relation in the conductor data is indispensable.*

*Proof.* Since  $S_2 \mathfrak{m} = 0$ , one has  $W^2 = \mathcal{O} \cdot 1 + K + \gamma(L^2)$ . The quotient  $S_2/H_0$  is a line bundle, so closure under multiplication is exactly (13.10), including its nonreduced zero-scheme structure.

An element  $\lambda \cdot 1 + k \in W$  in the conductor must have  $\lambda = 0$ : multiplication by  $S_2$  and projection to the free line  $S_2/H_0$  is scalar multiplication by  $\lambda$ . For  $k \in K$ , multiplication by  $S_2$  is zero, and multiplication by a lift of  $v \in V$  is  $\gamma(\bar{k}, v)$ . It belongs to  $W$  exactly when it belongs to  $H_0$ . This proves (13.11) over every coefficient ring on the indicated rank locus. Its restriction to the exact-rank strata is a kernel of constant rank, so the sequences split locally and pull back. The conductor has rank two or one accordingly; as  $\mathrm{rank} W = 3$ , the action ranks are one or two.

In the square type,  $\gamma(y^2) = 0$  and  $\gamma(y, V) = \langle xy \rangle$ . This proves the assertion about the entire embedded surface and its action-rank jump. If  $L \neq \langle y \rangle$ , then  $\gamma(L, V) = S_2$ , so action rank one cannot occur on its rank-one subalgebra locus. In the two-factor type, multiplication by  $x$  has image  $\langle x^2 \rangle$  and multiplication by  $y$  has image  $\langle y^2 \rangle$ . For any other line both quadratic basis vectors occur and the image is  $S_2$ . These observations prove all the rank-one assertions. They are equations in the universal line and quotient bundles, rather than an appeal to continuity of ranks.

At  $K = S_2$  the conductor computation is immediate. The action-rank-one incidence is the scheme of length-two quotients of  $\mathbb{C} \oplus V$ , hence the projective line: its binary cubic square obstruction vanishes identically. On the action-rank-two locus the rank-two conductor algebra is local with square-zero augmentation line, since every quadratic product in  $K$  lies in  $H_0$ . Proposition 12.2 therefore gives the spectrum of the dual numbers on geometric fibres. The assertion about the embedded point follows from Theorem 13.2, not from the incidence fibre list.  $\square$

On the rank-one, action-rank-one loci just described, the residual rank-three algebra is curvilinear: the surviving tangent vector has nonzero square. Thus its length-two quotient scheme has length three. The change from these cubic fibres, or the quadratic fibres, to the projective line at  $K = S_2$  is accompanied by the local equations (13.4) and (13.7). Those equations, rather than the dimensions of the fibres, determine the nilpotent boundary.

**Proposition 13.4** (Collision of the quadratic factors). *The algebras*

$$\mathcal{B}_\tau = \mathbb{C}[\tau, x, y] / ((x, y)^3, y^2 - \tau xy)$$

*form a finite free rank-five family over  $\mathbb{C}[\tau]$ . On the full unital three-plane frame space its failure ideal in every degree  $m \geq 2$  is*

$$\delta^2(a(a + \tau b), 2ac + \tau(ad + bc), c(c + \tau d)). \quad (13.12)$$

*Its fibre at  $\tau = 0$  is the square-type scheme, and every fibre at  $\tau \neq 0$  is the two-factor-type scheme. Thus the primary strata in Theorem 13.2 occur in one family of full Fitting schemes with fixed algebra length and Hilbert function. No flatness of that failure family, or compatibility of a chosen primary decomposition with specialization, is asserted.*

*Proof.* The elements  $1, x, y, x^2, xy$  are a free basis over  $\mathbb{C}[\tau]$ ; the displayed relation is the only elimination in degree two and all terms of degree at least three vanish. The quadratic matrix, in the product basis without rescaling the middle column, is

$$\begin{pmatrix} a^2 & ac & c^2 \\ 2ab + \tau b^2 & ad + bc + \tau bd & 2cd + \tau d^2 \end{pmatrix}.$$

Its minors are  $\delta$  times the three displayed generators. Theorem 13.1 proves the ideal identity over  $\mathbb{C}[\tau]$  and after specialization. The quadratic relation is  $y(y - \tau x)$ , proving the assertions about the fibres.  $\square$

**Corollary 13.5** (Projective realization of the complete boundary). *Let  $Z \subset \mathbb{P}^2$  have homogeneous ideal  $((X, Y)^3, Y^2)$  or  $((X, Y)^3, XY)$  in  $\mathbb{C}[T, X, Y]$ . For every  $n \geq 5$ , the inverse-image linear series restricting to all generating three-planes on  $Z$  have, in each symmetric degree  $m \geq 2$ , exactly the primary schemes of Theorem 13.2, on their smooth unit covers. In particular the embedded conductor-boundary components are components of actual global section-multiplication failure.*

*Proof.* The ideals are saturated because  $T$  is a nonzerodivisor on the quotient. The affine algebras have top graded degree two, so their homogeneous ideals have regularity three. Apply Theorem 15.1, using  $n \geq 2 \cdot 3 - 1$ , and trivialize  $\mathcal{O}_Z(n)$  by  $T^n$ . The cokernel isomorphism transfers the full Fitting ideal. Finally Remark 12.3 transfers the equations from unital planes to a smooth cover of the unrestricted generating Grassmannian. No boundary is removed in either step.  $\square$

#### 14. EMBEDDED PRIMARY STRATA FOR CONNECTED LOCAL ALGEBRAS

The incidence of Theorem 12.1 transfers more than the support of a quotient scheme. On its quadratic-generation open it transfers its local rings, and hence its primary structure. We apply

it to a connected noncurvilinear family in which the full multiplication failure scheme has an embedded component.

For  $h \geq 5$  set

$$B_h = \mathbb{C}[x, y]/(x^2, xy, y^h), \quad \dim_{\mathbb{C}} B_h = h + 1. \quad (14.1)$$

Let  $F_0 = 0$ ,  $F_1 = 1$ , and  $F_{j+1} = uF_j + vF_{j-1}$ , and put

$$\begin{aligned} J_h &= (F_h, vF_{h-1}) \subset \mathbb{C}[u, v], \\ I_h &= (F_h, vF_{h-1}, bv, b^2u) \subset \mathbb{C}[b, u, v], \\ P_h &= (u, v), \quad Q_h = (b^2, bv, F_h, vF_{h-1}). \end{aligned} \quad (14.2)$$

The equality  $J_h = (F_h, F_{h+1})$  follows from the recurrence.

**Theorem 14.1** (A connected embedded-primary family). *Consider the unital Grassmannian of  $(h-1)$ -planes in  $B_h$ , and its full zeroth-Fitting multiplication failure scheme  $D_h$  in any degree  $m \geq 3$ . At the plane*

$$W_0 = \langle 1, x, y^2, y^3, y^5, \dots, y^{h-1} \rangle \quad (\text{the term } y^4 \text{ is omitted}), \quad (14.3)$$

*there is a Zariski neighbourhood in  $D_h$  isomorphic to*

$$\text{Spec}((\mathbb{C}[b, u, v]/I_h)[t_1, \dots, t_{h-2}])_f, \quad f(0) = 1. \quad (14.4)$$

*Here the variables  $t_i$  are actual Grassmannian coordinates, and  $f$  is the coefficient of a specified quadratic product in the one-dimensional quotient of the conductor hyperplane by  $W$ . This is a local description of the full failure scheme, not of a transverse test family.*

*The ideal in (14.2) has the irredundant primary decomposition*

$$I_h = P_h \cap Q_h. \quad (14.5)$$

*Its associated primes are  $(u, v)$  and  $(b, u, v)$ . The first is the minimal prime and the second is embedded. The same decomposition, with the ideals extended and localized, describes (14.4). The reduced support has codimension  $h-3 \geq 2$  in the unital Grassmannian, so this failure scheme is not Cartier. Its nilradical has exact nilpotency index  $2h-3$ . For the displayed choice of embedded primary ideal,*

$$\dim_{\mathbb{C}} \mathbb{C}[b, u, v]/Q_h = \binom{h}{2} + h - 1, \quad \dim_{\mathbb{C}} \sqrt{I_h}/I_h = \binom{h}{2} + h - 3. \quad (14.6)$$

*The second formula concerns the coefficient ring before adjoining the Grassmannian variables. It records a nilradical supported at the embedded point, rather than multiplicity along the generic point of a Cartier divisor.*

*Proof.* We first compute an entire open chart of  $\text{Hilb}^2(B_h)$ . Choose the chart on which  $1, y$  is a basis of the universal quotient. Write  $y^2 = uy + v$  and  $x = a + by$ . The relation  $xy = 0$  gives  $a = -bu$  and  $bv = 0$ . After this substitution,  $x^2 = 0$  is equivalent to  $b^2u = 0$ : its constant equation follows from  $bv = 0$  and  $b^2u = 0$ . Finally

$$y^h = F_h y + vF_{h-1}$$

shows that  $y^h = 0$  is equivalent to the two generators of  $J_h$ . These calculations give a quotient algebra free on  $1, y$  over  $\mathbb{C}[b, u, v]/I_h$ , and conversely every such quotient has these coefficients. They prove an isomorphism of the chart functor with  $\text{Spec } \mathbb{C}[b, u, v]/I_h$  over arbitrary coefficient rings. No radical has been taken.

Here is a direct primary calculation. Write  $A_h = \mathbb{C}[u, v]/J_h$  and  $\mathfrak{n} = (u, v)A_h$ . It is an Artin local algebra, with residue field  $\mathbb{C}$ . In  $A_h[b]$ , coefficient comparison in the variable  $b$  gives

$$\mathfrak{n}A_h[b] \cap (b^2, bv) = (bv, b^2\mathfrak{n}) = (bv, b^2u).$$

Lifting to  $\mathbb{C}[b, u, v]$  proves (14.5). The ideal  $P_h$  is prime; the quotient by  $Q_h$  is an Artin local ring supported at  $(b, u, v)$ , so  $Q_h$  is primary. The decomposition is irredundant:  $b^2 \in Q_h \setminus P_h$ , whereas  $u \in P_h \setminus Q_h$ . The asserted associated primes follow. Polynomial extension and localization at a function nonzero at the specified point retain both components.

For clarity, the numerical invariants can be read without any primary-decomposition algorithm. The ring  $A_h$  is the contact algebra  $G_h$  of Theorem 10.3. It has dimension  $\binom{h}{2}$ , and

$$A_h/(v) \simeq \mathbb{C}[u]/(u^{h-1}), \quad \mathfrak{n}^{2h-3} = 0, \quad u^{2h-4} \neq 0.$$

These last assertions also follow from its two-row Schur basis:  $\deg u = 1$ ,  $\deg v = 2$ , the top weighted degree is  $2h - 4$ , and the coefficient of the top Schur class in  $u^{2h-4}$  is nonzero. This is the classical basis input recorded in [31]. As vector spaces the quotient by  $Q_h$  is  $A_h \oplus b(A_h/(v))$ , proving the first length formula. In the quotient by  $I_h$ , the nilradical is

$$\mathfrak{n} \oplus b(u\mathbb{C}[u]/(u^{h-1})).$$

All coefficients of  $b^j$  for  $j \geq 2$  are scalars. This proves the second length formula. The inclusion of the constant coefficient ring  $A_h$  is injective, so the upper bound  $2h - 3$  on nilpotency is attained. Localization at  $f(0) = 1$  does not kill its nonzero top class: it is supported at the retained maximal ideal.

It remains to prove that this chart is realized by the full multigenerator failure scheme. At  $b = u = v = 0$ , the conductor hyperplane is

$$E_0 = \langle 1, x, y^2, \dots, y^{h-1} \rangle.$$

The plane  $W_0$  in (14.3) is a hyperplane in  $E_0$ , and  $W_0^2 = E_0$ , since  $(y^2)^2 = y^4$ . In the universal quotient chart the conductor hyperplane has the basis

$$1, \quad x - by, \quad y^j - F_j y \quad (2 \leq j \leq h-1).$$

Take the Grassmannian chart in which the vector  $y^4 - F_4 y$  is the omitted direction, and add its multiples  $t_i$  to each of the other nonconstant basis vectors. This gives  $h - 2$  free coordinates on  $\text{Gr}(h - 2, E/\mathcal{O})$ . Let  $q_2$  be the resulting lift of the  $y^2$  basis vector. Its square belongs to  $E$ , because  $E$  is an algebra. The coefficient  $f$  of  $q_2^2$  in  $E/W$ , in the omitted basis direction, equals 1 at the origin. On  $f \neq 0$  one has  $W^2 = E$ . Theorem 12.1 now identifies this localized polynomial chart with an open neighbourhood of  $W_0$  in  $D_h$ , proving (14.4).

The unital Grassmannian has dimension  $2(h - 2)$ , whereas the support of (14.4) has dimension  $1 + (h - 2) = h - 1$ . Their difference is  $h - 3$ . This proves the non-Cartier assertion and finishes the proof.  $\square$

For  $h = 5$  the theorem gives four-planes in a connected length-six algebra. Its coefficient ideal is explicitly

$$(u^4 + 3u^2v + v^2, u^3v + 2uv^2, bv, b^2u).$$

The reduced support is the  $b$ -axis, the chosen embedded primary quotient has length 14, and the nilradical has dimension 12 and index 7. Increasing  $h$  changes both the higher-jet equations and the embedded nilpotent algebra; it does not merely change the exponent of a fixed determinant divisor. The parameter  $b$  couples the extra square-zero tangent direction  $x$  to the higher contact coefficients  $u, v$  through  $bv$  and  $b^2u$ .

**Corollary 14.2** (A three-generator member after adjoining a reduced factor). *For every  $s \geq 1$ , let  $B = (\mathbb{C}[x, y]/(x^2, y^2)) \times \mathbb{C}^s$ . The full stable failure scheme of unital codimension-two planes has a chart, meeting an embedded associated stratum, of the form*

$$\text{Spec}((\mathbb{C}[a, b]/(ab, b^2))[t_1, \dots, t_{s+1}])_g, \quad g(0) = 1.$$

*Its primary decomposition is the extension of  $(ab, b^2) = (b) \cap (a, b)^2$ . In particular  $s = 1$  gives three multiplication generators and an embedded component on the full failure scheme, rather than on a selected slice.*

*Proof.* Use the open-and-closed part of  $\text{Hilb}^2(B)$  supported in the first factor. Proposition 10.5 gives the coefficient chart  $\mathbb{C}[a, b]/(ab, b^2)$  there. At its origin the conductor hyperplane is  $\langle 1, y, xy \rangle \times \mathbb{C}^s$ . Let  $e_i$  denote the standard idempotent of the  $i$ th reduced factor. The plane generated by

$$1_B, \quad (y, e_1), \quad (xy, 0), \quad e_2, \dots, e_s$$

has rank  $s + 2 = \dim B - 2$ . Its square contains  $e_1 = (y, e_1)^2$ , so its quadratic image is the entire conductor hyperplane. The relative Grassmannian chart has  $s + 1$  parameters. Localize at the quadratic-generation minor nonzero at this point and apply Theorem 12.1. The displayed primary identity is retained under this polynomial extension and localization.  $\square$

The connected family of Theorem 14.1 and the three-generator corollary serve different purposes. The first gives unbounded connected local complexity; the second supplies a three-generator instance of the same incidence mechanism. Both also describe smooth local covers of the unrestricted generating Grassmannians by Remark 12.3. Neither uses a claim that primary decomposition survives arbitrary specialization.

## 15. CONDUCTOR TRANSPORT FOR ARBITRARY FINITE PROJECTIVE SCHEMES

The finite-algebra results apply to arbitrary zero-dimensional projective schemes once the conductor kernel is filled. This can be done by a normal-layer argument in which Castelnuovo–Mumford regularity replaces the monomial calculation for a homogeneous fat point. We distinguish the regularity of a homogeneous ideal from that of its associated sheaf; the distinction is essential for its ordinary square.

**Theorem 15.1** (Regularity-controlled conductor transport). *Let  $Z \subset \mathbb{P}_{\mathbb{C}}^e$ ,  $e \geq 1$ , be a nonempty zero-dimensional closed subscheme, and let  $I \subset R = \mathbb{C}[X_0, \dots, X_e]$  be its saturated homogeneous ideal. Put*

$$r = \text{reg } I, \quad s = \text{reg}(I^2),$$

*where  $I^2$  is the ordinary homogeneous ideal square. If*

$$n \geq \max\{r, s - 1\}, \tag{15.1}$$

*then every generating subspace  $A \subset H^0(Z, \mathcal{O}_Z(n))$ , with  $U = \text{res}_Z^{-1} A$  and  $K_t = H^0(\mathbb{P}^e, \mathcal{I}_Z(t))$ , satisfies*

$$K_n U^{m-1} = K_{mn} \quad (m \geq 2). \tag{15.2}$$

*Consequently restriction induces a canonical isomorphism*

$$\text{coker}(\text{Sym}^m \mathcal{U} \rightarrow H^0(\mathbb{P}^e, \mathcal{O}(mn)) \otimes \mathcal{O}) \simeq \text{coker}(\text{Sym}^m \mathcal{A} \rightarrow H^0(Z, \mathcal{O}_Z(mn)) \otimes \mathcal{O}) \tag{15.3}$$

*on the entire generating Grassmannian, for every  $m \geq 2$ . In particular all Fitting ideals agree.*

*The simpler sufficient range*

$$n \geq 2 \text{reg } I - 1 \tag{15.4}$$

*always implies (15.1). It has no restriction on the local embedding dimension, the support partition, or a monomial presentation of  $Z$ .*

*Proof.* We use the following classical regularity facts. An ideal of regularity  $r$  is generated in degrees at most  $r$ . If an ideal has regularity  $s$ , then it agrees with its saturation in degrees at least  $s$ , and the first cohomology of its associated sheaf vanishes in degrees at least  $s - 1$ ; see [35, Theorem 1.3]. Moreover

$$\text{reg}(I^2) \leq 2 \text{reg } I \tag{15.5}$$

for a zero-dimensional projective scheme, by [35, Theorem 1.8]. This is a bound for the ordinary ideal square, not merely its sheafification. It proves the implication from (15.4). We prove the sharper assertion under (15.1).

Choose  $g \in U$  whose restriction generates  $\mathcal{O}_Z(n)$  as an invertible  $\mathcal{O}_Z$ -module. Such a section exists by Lemma 8.1. Let  $t \geq n$ . Since  $t \geq s - 1$ , the conormal sequence gives a surjection

$$K_t \longrightarrow H^0(Z, (\mathcal{I}_Z/\mathcal{I}_Z^2)(t)). \quad (15.6)$$

Multiplication by  $g$  is an isomorphism between the conormal twists  $t$  and  $t + n$ , since its restriction is a unit after trivializing  $\mathcal{O}_Z(n)$ . Thus, for any  $f \in K_{t+n}$ , one can choose  $q \in K_t$  for which  $f - gq$  lies in  $H^0(\mathbb{P}^e, \mathcal{I}_Z^2(t + n))$ .

The ideal  $I$  is generated in degrees at most  $r$ , and  $n, t \geq r$ . Factoring coefficient monomials between two chosen ideal generators therefore gives the elementary homogeneous equality

$$(I^2)_{t+n} = I_n I_t. \quad (15.7)$$

Indeed for generators of degrees  $a, b \leq r$ , a coefficient monomial of degree  $t + n - a - b$  can be split into factors of degrees  $n - a$  and  $t - b$ . This proves both containments. Since  $s \geq 2$  and  $n \geq s - 1$ , one has  $t + n \geq s$ . Agreement with saturation in that degree identifies the preceding sheaf section space with  $(I^2)_{t+n}$ . Also  $K_t = I_t$ , because  $I$  is saturated. We have proved

$$K_{t+n} = gK_t + K_n K_t = UK_t. \quad (15.8)$$

The first equality means equality of the sum with the whole kernel; the last follows from  $g, K_n \subset U$ , with the reverse inclusion clear from vanishing on  $Z$ . Iterating for  $t = n, 2n, \dots$  proves (15.2).

Restriction in degrees  $n$  and  $mn$  is surjective by regularity. The restriction sequence  $0 \rightarrow K_n \rightarrow U \rightarrow A \rightarrow 0$  is locally split on the parameter Grassmannian. The kernel on symmetric powers is generated by  $K_n \otimes \text{Sym}^{m-1} U$ , whose multiplication image is the full target restriction kernel by (15.2). Taking cokernels of the restriction square proves (15.3). Its map is restriction itself, so the local choices of  $g$  disappear from the result. The relative version below supplies, in particular, the asserted scheme-level statement over this whole Grassmannian.  $\square$

**Theorem 15.2** (Relative finite schemes and base change). *Let  $S$  be a locally Noetherian  $\mathbb{C}$ -scheme and  $\mathcal{Z} \subset \mathbb{P}_S^e$  a finite locally free family. Suppose the saturated homogeneous ideal  $I_z$  of every geometric fibre has regularity at most  $r$ . For  $n \geq 2r - 1$ , the kernel identity and cokernel comparison of Theorem 15.1 hold on each relative generating Grassmannian over  $S$  and commute with arbitrary base change. The same conclusion holds under uniform bounds  $\text{reg } I_z \leq r$ ,  $\text{reg}(I_z^2) \leq s$  and  $n \geq \max\{r, s - 1\}$ .*

*No hypothesis that formation of a conductor ideal or of a primary decomposition commutes with specialization is used. The assertion is about the actual coherent multiplication cokernels.*

*Proof.* Let  $p : \mathbb{P}_S^e \rightarrow S$ . For  $t \geq n$ , regularity on geometric fibres and cohomology and base change give split exact sequences locally on  $S$ ,

$$0 \longrightarrow p_* \mathcal{I}_Z(t) \longrightarrow p_* \mathcal{O}(t) \longrightarrow p_* \mathcal{O}_Z(t) \longrightarrow 0.$$

Their terms are vector bundles and their formation commutes with base change. One may also obtain this directly from fibrewise surjectivity of restriction, the local freeness of its two outer bundles, and Nakayama's lemma. The resulting kernel has the required fibre by the exact sequence on each fibre.

Over the relative Grassmannian, form the multiplication morphism

$$p_* \mathcal{I}_Z(n) \otimes \text{Sym}^{m-1} \mathcal{U} \longrightarrow p_* \mathcal{I}_Z(mn).$$

It is surjective on every geometric fibre by the preceding theorem, using (15.5) if only the bound  $r$  is specified. Its coherent cokernel therefore vanishes by Nakayama's lemma. The same symmetric-power restriction square proves the relative cokernel isomorphism. The restriction bundles and

their split sequences pull back, and a surjective morphism remains surjective after every pullback. Hence the construction proves arbitrary base change, not just flat base change. Fitting ideals commute with base change for finitely presented modules. This argument does not require flatness of the first infinitesimal neighbourhood of  $\mathcal{Z}$  or base change for a selected primary component.  $\square$

The bound recovers the fat-point range  $n \geq 2h - 1$ , since the homogeneous ideal of an order- $h$  fat point has regularity  $h$ . The direct triangular proof in Theorem 17.2 is retained, as is its sharp boundary example. The present proof also applies to nonmonomial local schemes, unions with unequal local lengths, and finite flat deformations with a uniform regularity bound. A uniform bound is available on a fixed projective Hilbert scheme by boundedness; sharpness of  $2r - 1$  is not asserted for each individual finite scheme or each generating rank.

**Corollary 15.3** (Global embedded multiplication failure). *Let  $h \geq 5$  and let  $Z_h \subset \mathbb{P}^2$  have saturated ideal*

$$I = (X^2, XY, Y^h) \subset \mathbb{C}[T, X, Y].$$

*For every  $n \geq 2h - 1$ , take all inverse-image series of generating  $(h - 1)$ -planes on  $Z_h$ . In every degree  $m \geq 3$ , their global multiplication failure scheme has, on a smooth unit cover near the plane (14.3), the local primary description (14.4)–(14.5). It has the embedded component and exact nilradical index stated there. The coherent cokernel comparison persists in any finite flat family with the stated regularity bound.*

*Proof.* The variable  $T$  is a nonzerodivisor on  $\mathbb{C}[T, X, Y]/I$ , so  $I$  is saturated and the scheme is supported at  $[1 : 0 : 0]$ . Its affine algebra is  $B_h$ . The standard monomials in  $\mathbb{C}[X, Y]/I$  are  $1, X, Y, \dots, Y^{h-1}$ , and its top degree is  $h - 1$ ; thus  $\text{reg } I = h$ . For comparison, the minimal monomial generators of  $I^2$  are

$$X^4, X^3Y, X^2Y^2, XY^{h+1}, Y^{2h},$$

and its top quotient degree is  $2h - 1$ , so  $\text{reg}(I^2) = 2h$ . The conductor theorem applies at  $n \geq 2h - 1$ . Trivialize  $\mathcal{O}_{Z_h}(n)$  by  $T^n$  and apply Theorem 14.1 and the unit-cover construction. The equality of coherent cokernels gives the same Fitting scheme globally. The final assertion is the relative theorem, not a claim that the displayed embedded primary ideal persists unchanged under every deformation.  $\square$

**15.1. Cohomology and arbitrary base change in conductor transport.** We make explicit the bundle assertions used in Theorem 15.2. Write  $p : \mathbb{P}_S^e \rightarrow S$  and  $q : \mathcal{Z} \rightarrow S$ . The sheaf  $\mathcal{I}_{\mathcal{Z}}$  is  $S$ -flat: it is the kernel of a surjection between  $S$ -flat sheaves  $\mathcal{O}$  and  $\mathcal{O}_{\mathcal{Z}}$ . For  $t \geq \max\{0, r - 1\}$  the two modules

$$E_t = p_*\mathcal{O}(t), \quad F_t = q_*\mathcal{O}_{\mathcal{Z}}(t)$$

are locally free and commute with arbitrary base change. The first is the familiar homogeneous polynomial bundle; for the second use that  $q$  is finite locally free and  $\mathcal{O}_{\mathcal{Z}}(t)$  is invertible. The fibre regularity bound gives  $H^1(\mathbb{P}_S^e, \mathcal{I}_{\mathcal{Z}_s}(t)) = 0$  in this range. Hence the restriction map  $E_t \rightarrow F_t$  is surjective on each geometric fibre and, by Nakayama's lemma, on  $S$ .

Its kernel  $K_t$  is locally free because the quotient  $F_t$  is locally free, so

$$0 \longrightarrow K_t \longrightarrow E_t \longrightarrow F_t \longrightarrow 0$$

is locally split. The long exact sequence of direct images identifies  $K_t = p_*\mathcal{I}_{\mathcal{Z}}(t)$  and gives  $R^1p_*\mathcal{I}_{\mathcal{Z}}(t) = 0$ . In fact all higher direct images vanish in this range:  $\mathcal{Z}$  is finite over  $S$ , and  $\mathcal{O}(t)$  on projective space has no higher direct images for  $t \geq 0$ .

After any base change  $S' \rightarrow S$  the displayed split sequence stays exact. The corresponding sequence of sheaves on projective space also stays exact because the original family is flat. Pushforward identifies the new kernel with  $p'_*\mathcal{I}_{\mathcal{Z}'}(t)$ ; restriction is still surjective and its first

higher direct image is still zero. This proves the arbitrary base-change assertion directly, without assuming  $S'$  reduced, flat over  $S$ , or Noetherian.

Now form the relative generating Grassmannian and let  $\mathcal{U}$  be the inverse image of its tautological restricted series. For  $m \geq 2$  the map

$$K_n \otimes \mathrm{Sym}^{m-1} \mathcal{U} \longrightarrow K_{mn}$$

is a morphism of locally free sheaves. If the uniform bounds are  $\mathrm{reg} I_{\bar{s}} \leq r$  and  $\mathrm{reg}(I_{\bar{s}}^2) \leq s$ , the single-fibre conductor proof applies for  $n \geq \max\{r, s-1\}$ . If only  $r$  is given, Sidman's ordinary-square bound supplies  $s = 2r$ . These bounds are used on the geometric fibres only: no flatness or base change is required for  $\mathcal{I}_{\mathbb{Z}}^2$  or its conormal quotient. Thus the displayed morphism is fibrewise surjective, and its finitely presented cokernel is zero. It remains surjective after every pullback.

Finally the locally split sequence  $0 \rightarrow K_n \rightarrow \mathcal{U} \rightarrow \mathcal{A} \rightarrow 0$  shows that the kernel of  $\mathrm{Sym}^m \mathcal{U} \rightarrow \mathrm{Sym}^m \mathcal{A}$  is generated by  $K_n \otimes \mathrm{Sym}^{m-1} \mathcal{U}$ . Its multiplication image is exactly the target restriction kernel  $K_{mn}$ . The induced isomorphism of multiplication cokernels is therefore canonical restriction and commutes with arbitrary base change. This explains all relative assertions without replacing a coherent cokernel by its fibre dimensions.

## 16. THREE GENERATORS ON A REDUCED CONTACT DIVISOR

This theorem realizes Haiman's diagonal ideal as the stable Fitting ideal of an actual multiplication map. The all-powers identity is Haiman's input, not a theorem deduced here from multiplication. The resulting global-section failure schemes and their coherent cokernels are obtained using the conductor comparison.

**Theorem 16.1** (The complete stable three-plane ideal). *Let  $D = p_1 + \cdots + p_d$  be reduced,  $d \geq 4$ , and let  $A$  vary on the entire generating open  $\mathrm{Gr}(3, H^0(\mathcal{O}_D))^\circ$ . On the smooth unit-frame cover write  $A = \langle a, ax, ay \rangle$ ; thus the matrix with rows  $(1, x_i, y_i)$  has rank three. For  $m \geq d-1$ , the zeroth Fitting ideal  $J_m$  is independent of  $m$ , and*

$$J_m = \bigcap_{i < j} P_{ij}, \quad P_{ij} = (x_i - x_j, y_i - y_j). \quad (16.1)$$

For every  $q \geq 1$  one has the stronger identity

$$J_m^q = \bigcap_{i < j} P_{ij}^q. \quad (16.2)$$

*These are complete primary decompositions, including all component intersections. Thus  $J_m$  is radical, all its components are smooth of codimension two on this cover and on the fixed- $D$  Grassmannian, and neither it nor any of its powers has embedded associated points. The corank is  $d$  minus the number of distinct pairs  $(x_i, y_i)$ . For global binary subseries the same statements hold for  $n \geq 2d-3$ .*

*Proof.* After dividing the target by  $a^m$ , the multiplication matrix has as columns all monomials in  $x, y$  of total degree at most  $m$ , evaluated at the  $d$  labelled points. Lemma 8.2 identifies, over the entire coefficient ring, the ideal of its  $d$ -minors for  $m \geq d-1$  with the ideal generated by determinants of arbitrary monomial columns. Those determinants span the alternating polynomials in the  $d$  pairs of variables: antisymmetrize a monomial and use linearity. Haiman's theorem [30, Corollary 3.8.3] states that the ideal generated by these alternants has every power equal to the intersection of the corresponding powers of the pair-diagonal ideals. It gives (16.1) and (16.2) before any localization; hence they hold on the rank-three open. In that corollary Haiman's  $n$  is our number  $d$  of labelled points, and Haiman's exponent  $d$  is our power  $q$ . These

two indices are independent; no restriction on  $q$  is imposed. This use of the diagonal-ideal theorem is essential.

Each  $P_{ij}$  is a smooth prime ideal of codimension two; its powers are primary (use two normal coordinates in a regular local ring). The intersections therefore have exactly the indicated associated primes. Smooth faithful flat descent gives the fixed- $D$  assertions. Polynomial functions separate any finite set of distinct affine points, so their image dimension is the number of distinct pairs. The truncation lemma gives the stated corank in this finite degree. Finally apply Theorem 7.2 with  $k = 3$ .  $\square$

The stable moving scheme over the reduced-divisor stratum is integral, though it need not be smooth or unibranched. Its normalization marks one equal pair and is smooth and finite. Indeed the ordered configuration cover is etale and the permutation group is transitive on pairs. Upstairs the reduced scheme is the union in (16.1); its normalization is the disjoint union of the smooth pair diagonals. This construction descends. A point is smooth precisely when only one pair coincides; two or more failed pairs give distinct local branches after etale base change. No assertion about nonreduced divisors is inferred from this etale calculation; those have their own quotient-algebra structure in Section 10 when  $k = d - 1$ .

## 17. NONREDUCED MULTIGENERATOR SCHEMES AND FAT-POINT TRANSPORT

We next determine a multiplication failure scheme with arbitrarily large embedding dimension and nilpotence order. It is not a diagonal arrangement and its proof does not use Haiman's theorem. Fix integers  $e \geq 2$  and  $h \geq 3$ , and put

$$B_{e,h} = \mathbb{C}[z_1, \dots, z_e]/(z_1, \dots, z_e)^h, \quad d = \binom{e+h-1}{e}, \quad N = \binom{e+h-1}{e+1}. \quad (17.1)$$

Write  $\mathfrak{m}$  for its augmentation ideal. On  $X = \text{Gr}(e+1, B_{e,h})^\circ$  the augmentation of the universal bundle  $\mathcal{A}$  is surjective. Its kernel  $K$  has rank  $e$ . Let  $\Delta \subset X$  be the determinant divisor of the canonical map

$$K \longrightarrow (\mathfrak{m}/\mathfrak{m}^2) \otimes \mathcal{O}_X. \quad (17.2)$$

Thus this divisor is defined without a choice of unit or coordinates.

**Theorem 17.1** (Exact primary law for truncated local algebras). *For every  $m \geq h - 1$ , on the entire generating Grassmannian  $X$ ,*

$$\text{Fitt}_0 \text{ coker}(\text{Sym}^m \mathcal{A} \rightarrow B_{e,h}) = \mathcal{I}_\Delta^N. \quad (17.3)$$

*The divisor  $\Delta$  is nonempty, integral, and Cartier. The displayed ideal is primary, its exact nilradical index is  $N$ , and its generic transverse length is  $N$ . For every  $q \geq 1$  its  $q$ th ordinary power has the complete primary description  $\mathcal{I}_\Delta^{Nq}$ ; there are no embedded associated points. On a smooth unit-frame cover,  $\Delta$  is a generic  $e$  by  $e$  determinant, and its singular locus is given by matrix rank at most  $e - 2$ .*

*More precisely, choose a unit  $a \in \mathcal{A}$  and write  $\mathcal{A} = \langle a, ax_1, \dots, ax_e \rangle$ , with  $x_i \in \mathfrak{m}$ . After multiplying the target by  $a^{-m}$ , the cokernel in (17.3) is the cokernel of the square substitution map*

$$T_x : B_{e,h} \longrightarrow B_{e,h}, \quad z_i \longmapsto x_i.$$

*If  $M$  is the matrix of linear coefficients of the  $x_i$ , then  $\det T_x = (\det M)^N$ . These presentation and ideal identities hold after arbitrary base change. The primary assertions refer to the universal parameter space  $X$  over  $\mathbb{C}$ , not to every specialization. For  $m < h - 1$  the zeroth Fitting ideal is zero: the failure scheme is all of  $X$ , not the empty scheme. Thus the stated stable threshold is sharp for this family.*

*Proof.* A unit exists locally because the augmentation of  $\mathcal{A}$  is onto; subtracting constants from the remaining normalized basis vectors puts all  $x_i$  in  $\mathfrak{m}$ . The normalized multiplication matrix consists of the columns  $x^\alpha$  with  $|\alpha| \leq m$ . Columns of total degree at least  $h$  vanish. When  $m \geq h - 1$  the remaining columns are exactly the  $d$  columns of  $T_x$ ; all other columns are zero. This proves the cokernel assertion, over any coefficient algebra.

The map  $T_x$  preserves the  $\mathfrak{m}$ -adic filtration and its block on  $\mathfrak{m}^j/\mathfrak{m}^{j+1}$  is  $\text{Sym}^j M$ . Hence

$$\det T_x = \prod_{j=1}^{h-1} \det(\text{Sym}^j M) = (\det M)^{\sum_{j=1}^{h-1} \binom{e+j-1}{e}} = (\det M)^N.$$

For completeness, the middle identity follows by taking  $M$  diagonal: among all degree- $j$  monomials each diagonal entry occurs with total exponent  $\binom{e+j-1}{e-1} = \binom{e+j-1}{e}$ . The identity holds for arbitrary matrices because both sides are polynomial functions and diagonalizable matrices with distinct eigenvalues form a dense open over  $\mathbb{C}$ . It is consequently a polynomial identity and remains valid on nonreduced bases. No condition on the coefficients of degree at least two was used.

The determinant of (17.2) differs from  $\det M$  by a unit: the vectors  $ax_i$  form a basis of  $K$ , and their linear terms are the linear terms of  $x_i$  multiplied by the nonzero augmentation of  $a$ . Thus the ideal equality descends canonically. To check its geometric structure, send  $A$  to  $K = A \cap \mathfrak{m}$ . This identifies  $X$  with the total space of the quotient bundle  $\mathfrak{m}/K$  over  $\text{Gr}(e, \mathfrak{m})$ : a lift of the augmented line has the unique form  $1 + v$  modulo  $K$ . The projection of its universal  $e$ -plane to  $\mathfrak{m}/\mathfrak{m}^2$  vanishes in determinant precisely on the standard Schubert divisor

$$\Sigma = \{K \in \text{Gr}(e, \mathfrak{m}) : K \cap \mathfrak{m}^2 \neq 0\}.$$

This is the integral Schubert divisor for the fixed projection with kernel  $\mathfrak{m}^2$ , and  $\Delta$  is its pullback along the quotient-bundle projection. This identifies the divisor globally. For an elementary verification of integrality, its frame cover has independent linear coefficients  $M$  and independent higher coefficients, subject only to full column rank of the combined matrix. The generic determinant is irreducible. Its zero scheme remains nonempty on the frame open, as witnessed by

$$x_1 = z_1, \dots, x_{e-1} = z_{e-1}, \quad x_e = z_1^2.$$

The standard integral determinant hypersurface on that open descends to  $\Sigma$ , and its vector-bundle pullback is integral and Cartier.

Locally its equation is a prime element in a regular local ring. Every positive power is primary; alternatively, a hypersurface in a regular scheme has no embedded associated points and its support here has a single generic point. The radical of  $(\delta^N)$  is  $(\delta)$ , and  $\delta^{N-1}$  is nonzero modulo  $\delta^N$  at its generic point. This proves the exact nilradical index and length, and the same argument proves the assertion for all powers. The partial derivatives of a determinant are its cofactors, giving the stated singular locus. Finally  $\text{rank Sym}^m \mathcal{A} = \binom{e+m}{e} < d$  if  $m < h - 1$ ; there are no maximal minors. For  $A = \langle 1, z_1, \dots, z_e \rangle$  the stable map is surjective, so this threshold cannot be lowered.  $\square$

For  $e = 2$  this is a genuine three-plane theorem for every  $\mathbb{C}[x, y]/(x, y)^h$ : the contact length is  $\binom{h+1}{2}$  and the failure multiplicity is  $\binom{h+1}{3}$ . Already  $h = 3$  gives a length-six, embedding-dimension-two algebra whose failure divisor has multiplicity four. These are schemes on a full generating Grassmannian, not selected transverse slices. They are distinct from the curvilinear hyperplane schemes and the reduced pair-diagonal ideal.

**17.1. A global conductor theorem in arbitrary projective dimension.** The preceding algebra is not the algebra of a divisor on a curve. To apply its scheme calculation to global sections, we require a separate conductor theorem. Its range depends on the order  $h$  of the fat point, not on its length  $d$ .

**Theorem 17.2** (Fat-point conductor transport). *Let  $e \geq 1$ ,  $h \geq 2$ , let  $p \in \mathbb{P}^e$ , and let  $Z$  be the fat point with ideal  $\mathcal{I}_p^h$ . Suppose  $n \geq 2h - 1$ . For any generating  $k$ -plane  $A \subset H^0(Z, \mathcal{O}_Z(n))$ , let*

$$W = H^0(\mathbb{P}^e, \mathcal{I}_p^h(n)), \quad U = \text{res}_Z^{-1}(A).$$

*For every  $m \geq 2$  one has*

$$WU^{m-1} = H^0(\mathbb{P}^e, \mathcal{I}_p^h(mn)). \quad (17.4)$$

*Restriction induces an isomorphism of coherent cokernels on the entire relative generating Grassmannian, also with  $p$  varying,*

$$\text{coker}(\text{Sym}^m \mathcal{U} \rightarrow H^0(\mathcal{O}(mn)) \otimes \mathcal{O}) \simeq \text{coker}(\text{Sym}^m \mathcal{A} \rightarrow \pi_* \mathcal{O}_Z(mn)). \quad (17.5)$$

*It and all its Fitting ideal equalities commute with arbitrary base change. The bound is sharp uniformly over all generating dimensions  $k$  and all  $m \geq 2$ ; no rank-specific sharpness is asserted.*

*Proof.* First fix  $p = [1 : 0 : \cdots : 0]$  and work on its affine chart with coordinates  $z_1, \dots, z_e$ . A section of  $\mathcal{O}(n)$  is a polynomial of degree at most  $n$ , and  $W$  has basis the monomials of total degree between  $h$  and  $n$ . Choose  $u \in U$  with constant coefficient  $u_0$  a unit. This is possible over a local parameter base because the augmentation of  $A$  and restriction onto  $A$  are surjective.

For every integer  $t$  with  $h \leq t \leq mn$ , one can choose  $1 \leq j \leq m$  with  $jh \leq t \leq jn$ . Indeed the integral intervals  $[jh, jn]$  have no gaps:  $jn + 1 \geq (j+1)h$  follows from  $n \geq 2h - 1$ . Every monomial  $z^\alpha$  of total degree  $t$  can be factored into  $j$  monomials of degrees between  $h$  and  $n$ . Their product, multiplied by  $u^{m-j}$ , belongs to  $WU^{m-1}$  and has the form

$$u_0^{m-j} z^\alpha + \text{monomials of total degree strictly greater than } t.$$

Its degree is at most  $mn$ . Choosing this column for every monomial of degree between  $h$  and  $mn$  gives a square triangular matrix with units on the diagonal. It is invertible over the coefficient ring. These columns are thus a basis of the whole space on the right of (17.4). The reverse containment is immediate.

Restriction in degrees  $n$  and  $mn$  is surjective, since those degrees are at least  $h - 1$ . Its kernels and targets are vector bundles, with monomial bases on the current chart. In the locally split sequence  $0 \rightarrow W \rightarrow U \rightarrow A \rightarrow 0$ , the kernel on symmetric powers is the image of  $W \otimes \text{Sym}^{m-1} U$ . Equation (17.4) identifies its multiplication image with the kernel of target restriction. Taking the quotient of this restriction square proves (17.5). The triangular proof works over every coefficient ring once  $u_0$  is a unit. For moving  $p$ , choose projective coordinates smoothly locally; restriction and the cokernel isomorphism are intrinsic, so they descend. All the split restriction sequences and surjective maps remain so after arbitrary pullback.

At the boundary  $n = 2h - 2$ , take  $A = \langle 1 \rangle$ . Then  $U = \langle 1 \rangle + W$ , and its quadratic monomials have degrees in  $\{0\} \cup [h, 2h - 2] \cup [2h, 4h - 4]$ . They omit  $z_1^{2h-1}$ , although it lies in the conductor kernel in degree  $2n$ . Its nonzero global cokernel class restricts to zero on  $Z$ . Thus the cokernel comparison already fails in degree two.  $\square$

**Corollary 17.3** (Global noncurvilinear primary schemes). *For  $e \geq 2$ ,  $h \geq 3$ , and  $n \geq 2h - 1$ , consider all inverse-image series  $U$  of generating  $(e+1)$ -planes on  $Z = (\mathcal{I}_p^h)$ . For every  $m \geq h - 1$ , their global multiplication failure scheme has the exact primary ideal  $\mathcal{I}_\Delta^{\binom{e+h-1}{e+1}}$ . Every power is primary and there are no embedded associated points on this fixed- $p$  generating parameter space. In degrees  $2 \leq m < h - 1$  the failure ideal is zero, so the whole parameter space fails. The same primary assertions hold on the relative parameter space with  $p \in \mathbb{P}^e$  varying; the coherent cokernel comparison itself holds after every base change.*

*Proof.* Apply Theorems 17.1 and 17.2. For moving  $p$  there is the following canonical construction. On  $P \times P$ ,  $P = \mathbb{P}^e$ , let  $\mathcal{D}_h$  be the subscheme defined by the  $h$ th power of the diagonal ideal, and let  $p_1, p_2$  be the projections. Set

$$\mathcal{B}_h = (p_1)_* \mathcal{O}_{\mathcal{D}_h}, \quad \mathcal{L}_n = (p_1)_* (\mathcal{O}_{\mathcal{D}_h} \otimes p_2^* \mathcal{O}_P(n)).$$

The first is the rank- $d$  jet algebra; the second is an invertible  $\mathcal{B}_h$ -module, locally free of rank  $d$  over  $P$ . Let  $\mathcal{M}_n$  be the kernel of  $\mathcal{L}_n \rightarrow \mathcal{O}_P(n)$ . Its first normal quotient is  $\Omega_P^1 \otimes \mathcal{O}_P(n)$ . On the relative generating open of  $\mathrm{Gr}_P(e+1, \mathcal{L}_n)$ , the universal augmentation kernel  $K$  therefore has the determinant line map

$$\det K \longrightarrow \det(\Omega_P^1 \otimes \mathcal{O}_P(n)).$$

Its zero divisor is canonical. The map  $A \mapsto K$  to  $\mathrm{Gr}_P(e, \mathcal{M}_n)$  is an affine-bundle torsor, locally the space of lifts of  $\mathcal{O}_P(n)$  in  $\mathcal{L}_n/K$ . The divisor is the pullback of the relative Schubert divisor for  $\mathcal{M}_n \twoheadrightarrow \Omega_P^1 \otimes \mathcal{O}_P(n)$ . Local splittings of these vector-bundle sequences exhibit the latter as the product with the standard integral Schubert divisor. Since  $P$  is integral and the fibres are nonempty, this divisor and its affine-bundle pullback are integral and Cartier. The total generating parameter space is smooth, so the same primary-power argument applies. No primary-decomposition assertion is inferred for arbitrary nonreduced specializations of this parameter space.  $\square$

## 18. PRIMARY DESCENT OVER MULTIPLICITY STRATA

The following descent statement concerns ideals, not just the permutation action on their generic points.

**Lemma 18.1** (Etale descent of weighted prime divisors). *Let  $X$  be a regular integral Noetherian  $\mathbb{C}$ -scheme and  $p: \tilde{X} \rightarrow X$  a finite etale Galois cover, with group  $G$ . Let  $\{D_\alpha\}$  be a finite  $G$ -stable collection of distinct integral effective Cartier divisors on  $\tilde{X}$ . Work componentwise if the cover is disconnected. Suppose the positive integers  $b_\alpha$  are constant on  $G$ -orbits. For an orbit  $O$ , there is a unique integral effective Cartier divisor  $D_O$  on  $X$  whose ideal satisfies*

$$p^* \mathcal{I}_{D_O} = \prod_{\alpha \in O} \mathcal{I}_{D_\alpha} = \bigcap_{\alpha \in O} \mathcal{I}_{D_\alpha}.$$

*If a descended ideal  $\mathcal{J}$  pulls back to  $\prod_{\alpha} \mathcal{I}_{D_\alpha}^{b_\alpha}$ , then its complete primary decomposition is*

$$\mathcal{J} = \prod_O \mathcal{I}_{D_O}^{b_O} = \bigcap_O \mathcal{I}_{D_O}^{b_O}. \quad (18.1)$$

*There are no embedded associated points, and the formula holds at intersections as well as generically. The descended prime divisor need not be smooth or geometrically unibranched.*

*Proof.* In each regular local ring upstairs, equations for distinct prime Cartier divisors are distinct prime factors. Unique factorization gives product equal to intersection, with or without positive powers. The reduced orbit union has an invariant invertible ideal subsheaf of  $\mathcal{O}_{\tilde{X}}$  with its canonical descent data. Faithfully flat descent gives an invertible ideal subsheaf of  $\mathcal{O}_X$ . It defines an effective Cartier divisor. Its support is the image of any one member of the orbit, hence is irreducible, since the map is finite. Reducedness descends faithfully flat, so this divisor is integral.

The same factorization upstairs, and flatness for finite intersections, identify the pullbacks of all three ideals in (18.1). Faithfulness makes them equal downstairs. Locally downstairs each  $\mathcal{I}_{D_O}$  is generated by a prime element of a regular local ring; its  $b_O$ th power is primary. Their intersection is irredundant after discarding components absent from the chart, so it is the complete primary decomposition and has no embedded primes. No normal-crossing hypothesis was used. An orbit can have several members through a point upstairs, giving several etale branches of the integral descended divisor. This explains the final distinction.  $\square$

Apply the lemma to the relative generating pencil Grassmannian over a multiplicity stratum. Ordering the distinct support points gives a finite étale cover with permutations among equal multiplicities. On it the divisors  $T_i, E_{ij}$  are the relative rank-one Schubert divisors. Their ideals are prime Cartier, and their weights are  $\binom{d_i}{2}, d_i d_j$ . Formula (18.1) supplies the whole descended primary decomposition. In particular it remains valid at a braid intersection where several values agree. The fixed labelled divisors are smooth; the descended orbit divisors are asserted integral, not automatically smooth or unbranched.

This argument applies to the earlier pencil calculation. For the contact-hyperplane family, Theorem 10.2 gives a scheme isomorphism with the quotient incidence and does not require a conjectural primary descent rule.

### 19. CONTACT PENCILS: EQUATIONS, PRIMARY COMPONENTS, AND INFINITESIMAL LAYERS

Fix  $d \geq 4$  and  $n \geq 2d - 2$ . In this section the divisor  $D$  is fixed, although it need not be reduced. Set  $G_D^\circ = \text{Gr}(2, E_{n,D})^\circ$ . It is a smooth irreducible variety of dimension  $2d - 4$ . The conductor construction embeds it as the open generating locus in the Grassmannian of subseries containing  $W_D$ . Let  $\mathcal{J}_m$  be the zeroth Fitting ideal of  $\text{coker}(\text{Sym}^m \mathcal{U} \rightarrow V_{mn})$  on this variety.

Trivialize  $\mathcal{O}_D(n)$ , and put  $R = H^0(\mathcal{O}_D)$ . There is a smooth surjective frame map

$$\mathcal{F}_D = \{(a, v) : a \in R^\times, v \in R \setminus \mathbb{C}\} \longrightarrow G_D^\circ, \quad (a, v) \longmapsto \langle a, av \rangle. \quad (19.1)$$

It is the open part of the frame bundle where the first basis vector is a unit. Its relative dimension is four. All calculations on  $\mathcal{F}_D$  below are therefore exact faithfully flat local calculations for  $G_D^\circ$ , not transverse slices inferred from pointwise ranks.

**Lemma 19.1** (The power-basis presentation). *On  $\mathcal{F}_D$  the multiplication cokernel is presented by*

$$K_m(v) = [1, v, v^2, \dots, v^m] : \mathcal{O}^{m+1} \longrightarrow R \otimes \mathcal{O}. \quad (19.2)$$

For  $m \geq d - 1$ , every later column is an  $\mathcal{O}$ -linear combination of the first  $d$  columns, and

$$\mathcal{J}_m \mathcal{O}_{\mathcal{F}_D} = (\Delta_D(v)), \quad \Delta_D(v) = \det[1, v, \dots, v^{d-1}]. \quad (19.3)$$

For  $2 \leq m < d - 1$ ,  $\mathcal{J}_m = 0$ .

*Proof.* By Theorem 7.2, it suffices to multiply sections in the pencil  $\langle a, av \rangle$  on  $D$ . Its image is generated by  $a^m v^j$  for  $0 \leq j \leq m$ . Multiplication by  $a^{-m}$  is an invertible target operation and gives (19.2). Multiplication by  $v$  is an endomorphism of the rank- $d$  free module  $R \otimes \mathcal{O}$ . Its monic characteristic polynomial annihilates it over the coefficient ring, by the adjugate proof of Cayley–Hamilton. Thus  $v^d$  and every higher power lie in the span of the first  $d$  powers, with regular coefficients. Column operations reduce the presentation to  $[K_{d-1}(v), 0]$ . This proves (19.3), as an identity of ideals and not merely of radicals. If  $m + 1 < d$ , all  $d$ -minors are zero. Faithful flatness descends this assertion.  $\square$

Write  $D = \sum_{i=1}^s d_i p_i$ , with distinct  $p_i$ . Choose local parameters  $z_i$  and the Chinese-remainder identification

$$R = \prod_{i=1}^s \mathbb{C}[z_i]/(z_i^{d_i}), \quad v_i = \lambda_i + \sum_{r=1}^{d_i-1} c_{ir} z_i^r.$$

All  $\lambda_i, c_{ir}$  are independent affine coordinates before removing the scalar locus. For  $d_i \geq 2$  define  $T_i$  by failure of  $A$  to separate  $2p_i$ ; for  $i < j$  define  $E_{ij}$  by failure to separate  $p_i + p_j$ . On the frame space they have equations  $c_{i1} = 0$  and  $\lambda_j - \lambda_i = 0$ , respectively.

**Theorem 19.2** (Full fixed-contact classification). *For every  $m \geq d - 1$ , in the ordered jet basis  $1, z_i, \dots, z_i^{d_i-1}$ ,*

$$\Delta_D(v) = \prod_{i: d_i \geq 2} c_{i1}^{\binom{d_i}{2}} \prod_{i < j} (\lambda_j - \lambda_i)^{d_i d_j}. \quad (19.4)$$

*The complete primary decomposition on  $G_D^\circ$  is*

$$\mathcal{J}_m = \bigcap_{i: d_i \geq 2} \mathcal{I}_{T_i}^{\binom{d_i}{2}} \cap \bigcap_{i < j} \mathcal{I}_{E_{ij}}^{d_i d_j}. \quad (19.5)$$

*All displayed divisors are nonempty, smooth, irreducible, and distinct. They are precisely the associated subvarieties; no embedded associated points occur. The failure divisor has dimension  $2d - 5$  and divisor class  $\binom{d}{2}$  times the restricted Pluecker class.*

*Every multiplication corank, for every  $m \geq 2$ , is determined as follows. Put  $e_i = \text{ord}_{z_i}(v_i - \lambda_i)$ , with  $e_i = d_i$  for a constant  $v_i$ , and put  $\ell_i = \lceil d_i/e_i \rceil$ . For each distinct value  $\lambda$  among the  $\lambda_i$ , set  $L_\lambda = \max_{i: \lambda_i = \lambda} \ell_i$ , and let  $s(v) = \sum_\lambda L_\lambda$ . Then*

$$\text{corank}(\text{Sym}^m U_A \rightarrow V_{mn}) = d - \min\{m + 1, s(v)\}. \quad (19.6)$$

*On the frame space every higher Fitting ideal is, exactly, the corresponding minor ideal of (19.2).*

*Proof.* For a polynomial  $P$  of degree at most  $d - 1$ , Taylor expansion gives

$$P(v_i(z_i)) = \sum_{r=0}^{d_i-1} \frac{P^{(r)}(\lambda_i)}{r!} (v_i(z_i) - \lambda_i)^r \pmod{z_i^{d_i}}.$$

The linear change from the divided derivatives of  $P$  at  $\lambda_i$  to the coefficients of  $P(v_i(z_i))$  is triangular, with diagonal  $1, c_{i1}, \dots, c_{i1}^{d_i-1}$ . Its determinant is  $c_{i1}^{\binom{d_i}{2}}$ , including when this coefficient is zero. The determinant of the divided-derivative evaluation map

$$P \mapsto \left( \frac{P^{(r)}(\lambda_i)}{r!} \right)_{i, 0 \leq r < d_i}$$

in the basis  $1, T, \dots, T^{d-1}$  is the confluent Vandermonde  $\prod_{i < j} (\lambda_j - \lambda_i)^{d_i d_j}$ . Here is an explicit normalization of that identity. Start with  $d$  distinct variables, cluster  $d_i$  of them near  $\lambda_i$ , and in each cluster replace successive evaluation rows by divided-difference rows. The ordinary Vandermonde determinant is divided by the product of the differences *within* each cluster. In the cluster limit the rows are the divided derivatives above, while the remaining cross-cluster factors tend to the displayed product. This proves the identity on a dense open set and hence as a polynomial identity, including all collisions. Combining the two determinants proves (19.4) without inverting any of its factors.

For any length-two  $Z \leq D$ , restriction  $E_{n,D} \rightarrow E_{n,Z}$  is surjective. The locus in  $\text{Gr}(2, E_{n,D})$  where its restriction to  $A$  has rank at most one is the corresponding irreducible Schubert divisor. Its rank-zero locus is excluded by generation on  $D$ . Its rank-one locus is smooth: choose a basis of  $A$  with one nonzero image, and vary the second basis vector in a complementary target direction; the determinant then has a nonzero linear derivative. For irreducibility one may also mark the image line in  $E_{n,Z}$ ; the resulting incidence is a Grassmannian bundle over  $\mathbb{P}(E_{n,Z})$ , and projects birationally onto the rank-one locus. The generating open is nonempty for each divisor when  $d \geq 4$ . On frames this follows directly by choosing  $a = 1$  and a nonscalar  $v$  with the single prescribed linear equality. The same coordinates show that the divisors are distinct.

On each affine frame chart the coordinate ring is a localization of a polynomial ring. The distinct linear factors in (19.4) generate distinct prime ideals there. Unique factorization gives  $(\prod f_\alpha^{b_\alpha}) = \bigcap (f_\alpha^{b_\alpha})$ ; each ideal  $(f_\alpha^{b_\alpha})$  is primary. These identities descend by faithful flatness to

(19.5). Equivalently, at each point of the smooth base the local ring is regular, each displayed divisor is smooth, and its powers are primary. This also proves absence of embedded primes. The dimension follows since this is a nonzero Cartier divisor. Each component is a Pluecker hyperplane section, and

$$\sum_i \binom{d_i}{2} + \sum_{i < j} d_i d_j = \binom{d}{2},$$

which proves the class formula.

Finally, in  $\mathbb{C}[z_i]/(z_i^{d_i})$  the element  $v_i - \lambda_i$  has nilpotency index  $\ell_i$ . Its minimal polynomial is  $(T - \lambda_i)^{\ell_i}$ . In the product algebra the minimal polynomial is therefore

$$P_v(T) = \prod_{\lambda} (T - \lambda)^{L_{\lambda}}. \quad (19.7)$$

No polynomial of smaller degree can annihilate  $v$ , and division by this monic polynomial shows that  $1, v, \dots, v^{s(v)-1}$  form a basis of  $\mathbb{C}[v]$ . Consequently the rank of  $K_m(v)$  is  $\min\{m+1, s(v)\}$ . The cokernel isomorphism of Theorem 7.2 gives (19.6) and the assertion about Fitting ideals.  $\square$

**Theorem 19.3** (Exact infinitesimal layers and singularities). *Let  $m \geq d-1$ . At a point of a smooth frame chart discard the nonvanishing factors of (19.4). Write the remaining distinct linear factors as  $f_{\alpha}$ , with multiplicities  $b_{\alpha}$ , and put*

$$F = \prod_{\alpha} f_{\alpha}^{b_{\alpha}}, \quad P = \prod_{\alpha} f_{\alpha}.$$

*In the regular local ring  $B$  of that chart the failure ring is  $B/(F)$  and its nilradical is  $\mathcal{N} = (P)/(F)$ . Its exact nilpotency index is  $\max_{\alpha} b_{\alpha}$ , with index one for a reduced local ring. For every  $q \geq 1$ , there is an isomorphism of  $B$ -modules, with indicated generator  $P^q$ ,*

$$\mathcal{N}^q / \mathcal{N}^{q+1} \simeq B / \left( P, \prod_{\alpha} f_{\alpha}^{(b_{\alpha}-q)+} \right). \quad (19.8)$$

*The completed local equation is the same weighted linear arrangement, with unused frame and higher-jet coordinates as smooth parameters. Its tangent cone is defined by the homogeneous product of the active linear factors with these same multiplicities. Its singular scheme on this chart has ideal  $(F, \partial F)$ , and its singular support is the union of the components with  $b_{\alpha} \geq 2$  and the pairwise intersections of distinct components. These descriptions descend smoothly to  $G_D^{\circ}$ .*

*Proof.* The local ring in question is a localization of a polynomial ring in the unit coordinates of  $a$  and the independent jet coordinates of  $v$ . The factors  $f_{\alpha}$  are prime, distinct, and vanish at the selected point. Unique factorization shows that  $P^q \in (F)$  if and only if  $q \geq b_{\alpha}$  for every  $\alpha$ . This proves the nilpotency assertion, including at multiple intersections.

The homomorphism  $B \rightarrow \mathcal{N}^q / \mathcal{N}^{q+1}$  sending 1 to the class of  $P^q$  is onto. Its kernel is

$$((P^{q+1}, F) : P^q) = (P) + (F : P^q) = \left( P, \prod_{\alpha} f_{\alpha}^{(b_{\alpha}-q)+} \right).$$

For the middle equality, if  $rP^q = sP^{q+1} + tF$ , subtract  $sP$  from  $r$ ; cancellation gives membership in  $(F : P^q)$ . The last colon is computed by prime-factor exponents. This proves (19.8). Globally the chosen generator carries the corresponding line-bundle trivialization; the displayed isomorphism is a local one and makes no claim of a preferred global generator.

At the chosen point every active factor is linear in centered coordinates. Every discarded factor is a unit, so the completed ideal is exactly the displayed product, not a truncation of it. The initial ideal is generated by the same homogeneous product, which proves the tangent-cone statement. The hypersurface cotangent presentation gives  $(F, \partial F)$ . At a point on exactly one component of

multiplicity one the differential is a unit times  $df_\alpha \neq 0$ . At every point on a multiple component or on two distinct components it is zero. This proves the singular support assertion. Smooth faithfully flat pullback preserves these local statements; in particular the four frame parameters do not alter the nilpotency index or the singular-support criterion.  $\square$

For illustration, length four already gives subseries of codimension two, in every  $V_n$  with  $n \geq 6$ . For every  $m \geq 3$  their fixed-contact primary equations are as follows. Put  $\delta_{ij} = \lambda_j - \lambda_i$ ; only first-jet coefficients of multiple points are displayed.

| Multiplicities of $D$ | Exact equation on frames                         | Largest local nilpotency index |
|-----------------------|--------------------------------------------------|--------------------------------|
| $(1, 1, 1, 1)$        | $\prod_{i < j} \delta_{ij}$                      | 1                              |
| $(2, 1, 1)$           | $c_{11} \delta_{12}^2 \delta_{13}^2 \delta_{23}$ | 2                              |
| $(2, 2)$              | $c_{11} c_{21} \delta_{12}^4$                    | 4                              |
| $(3, 1)$              | $c_{11}^3 \delta_{12}^3$                         | 3                              |
| $(4)$                 | $c_{11}^6$                                       | 6                              |

The reduced arrangement need not have normal crossings: three coincident values give three braid hyperplanes in a two-dimensional normal space. The equation, rather than a normal-crossing label, records that singularity. Formula (19.6) also distinguishes points with the same support of failure but different higher-jet vanishing orders.

## 20. MOVING CONTACTS AND THE NORMALIZATION OF THE FAILURE DIVISOR

Let  $X_d^\circ = X_{d,2}^\circ$  be the relative generating Grassmannian in Theorem 7.2. It is smooth and irreducible of dimension  $3d - 4$ . The fixed-divisor schemes of the preceding section are its scheme-theoretic fibres, not its irreducible components. For  $m \geq d - 1$  denote its failure scheme by  $Y_m$ .

**20.1. Components over every multiplicity stratum.** For a partition  $\mathbf{d} = (d_1, \dots, d_s)$  of  $d$ , let  $B_{\mathbf{d}} \subset \text{Sym}^d \mathbb{P}^1$  be the locally closed stratum of divisors with exactly this multiplicity partition, and restrict  $X_d^\circ$  to it. Let  $M$  be the set of distinct multiplicities, and let  $r_e = \#\{i : d_i = e\}$  for  $e \in M$ .

**Corollary 20.1** (Complete component descent on multiplicity strata). *For  $m \geq d - 1$ , the failure scheme over  $B_{\mathbf{d}}$  has*

$$\#\{e \in M : e \geq 2\} + \binom{\#M}{2} + \#\{e \in M : r_e \geq 2\} \quad (20.1)$$

*irreducible components, all of dimension  $s + 2d - 5$ , and no embedded associated points. A ramification component is indexed by each  $e \geq 2$  in  $M$  and has primary multiplicity  $\binom{e}{2}$ . An identification component is indexed by each unordered pair of multiplicities  $e, f$  that occurs at distinct support points and has primary multiplicity  $ef$ . The ideal of the scheme is the intersection of these prime-divisor ideals raised to the stated powers. On the labelled etale cover its full local equations and nilradical layers are exactly those of Theorem 19.3.*

*Proof.* The ordered configuration space of  $s$  distinct points of  $\mathbb{P}^1$  is an integral finite etale cover of  $B_{\mathbf{d}}$ , with deck group  $\prod_{e \in M} \mathfrak{S}_{r_e}$ . The relative first-jet and two-point restriction maps are surjective over this cover. Consequently each labelled  $T_i$  and  $E_{ij}$  is an irreducible relative rank-one Schubert divisor. The local factorization (19.4) holds with the support coordinates free, so there are no further components. The deck group permutes  $T_i$  exactly according to the value  $d_i$ , and permutes  $E_{ij}$  exactly according to the unordered pair  $\{d_i, d_j\}$ . The images of the orbit unions are precisely the irreducible components downstairs: each is the image of an integral labelled component, and two such images agree exactly for components in the same orbit. This gives (20.1).

Lemma 18.1 applies to this regular ambient space and its invariant weighted divisors, including all branch intersections. It proves that each component downstairs is a prime Cartier divisor; irreducibility here does not assert smoothness or geometric unibranchness. Pulling its reduced ideal back to the labelled cover gives the reduced product of its labelled branch equations. Their multiplicity is constant on an orbit, so the primary ideal identity in (19.5) descends faithfully flatly. This also proves absence of embedded points and gives the stated local layers. The base stratum has dimension  $s$ , giving the dimension formula. A descended component can have intersecting local branches; smoothness of each labelled divisor is not a claim that every descended component is smooth.  $\square$

**Theorem 20.2** (An integral failure divisor with smooth normalization). *For  $d \geq 4$ ,  $n \geq 2d - 2$ , and  $m \geq d - 1$ , the schemes  $Y_m$  are all the same integral Cartier divisor  $Y$  of dimension  $3d - 5$ . In particular  $Y$  has no embedded associated points. Its normalization is*

$$\tilde{Y} = \{(D, A, Z) : (D, A) \in X_d^\circ, Z \leq D, \deg Z = 2, \text{rank}(A \rightarrow H^0(\mathcal{O}_Z(n))) = 1\}. \quad (20.2)$$

The forgetful map  $\tilde{Y} \rightarrow Y$  is finite and birational, and  $\tilde{Y}$  is smooth and irreducible.

More explicitly, on an affine support chart write  $D = (g(t))$  with  $g$  monic of degree  $d$ , and on a frame chart put  $v = b/a \in \mathbb{C}[t]/(g)$ , represented by a polynomial of degree less than  $d$ . The failure equation is

$$\Delta(g, v) = \det[1, v, \dots, v^{d-1}]_{\mathbb{C}[t]/(g)} = 0. \quad (20.3)$$

The normalization on this chart is parametrized by

$$g = hq, \quad v = \lambda + hw, \quad (20.4)$$

where  $h, q$  are monic of degrees  $2, d - 2$ , respectively,  $\lambda \in \mathbb{C}$ , and  $\deg w \leq d - 3$ . The scalar-pencil locus is omitted; the unit  $a$  is a free smooth parameter.

*Proof.* Theorem 7.2 applies on the whole relative space. The proof of Lemma 19.1 applies over the polynomial coefficient ring of a varying monic  $g$ , not only on its fibres. Thus every  $Y_m$  has the identical ideal (20.3) on a smooth covering by frame charts. This determinant is not identically zero:  $v = t$  makes its matrix the identity for every  $g$ .

The space of pairs  $Z \leq D$  with lengths  $2, d$  is  $\text{Sym}^2 \mathbb{P}^1 \times \text{Sym}^{d-2} \mathbb{P}^1$ ; the map sends  $(Z, Q)$  to  $D = Z + Q$ . It is finite. Indeed it is proper, and a fixed effective divisor has only finitely many degree-two subdivisors, obtained by choosing multiplicities at its finitely many support points. Restriction  $E_{n,D} \rightarrow E_{n,Z}$  is a surjection of vector bundles. The rank-at-most-one condition on a pencil is a relative Schubert divisor; its rank-zero part is excluded by the generating hypothesis. This makes (20.2) a closed subscheme of the finite base change to  $X_d^\circ$ , so its projection is finite.

Smoothness and irreducibility can be checked explicitly. A divisor with finite support is contained in an affine coordinate chart. On that chart  $Z$  is defined by a unique monic quadratic  $h$ , and  $Z \leq D$  means  $g = hq$ . Since  $a$  is a unit, rank one on  $Z$  is equivalent to  $v \bmod h$  being a scalar  $\lambda$ . Division by the monic  $h$  gives uniquely  $v = \lambda + hw$ , with the stated degree bound. Thus the framed incidence is an open part of affine space with coordinates the coefficients of  $h, q, w$ , the scalar  $\lambda$ , and the unit  $a$ . This proves smoothness. Globally irreducibility also follows from the relative rank-one Schubert description over the integral pair-divisor base; each fibre is irreducible, and its generating open is nonempty. These charts describe the same incidence, so no extra component appears at infinity.

An element  $v$  fails to generate the length- $d$  algebra  $R = H^0(\mathcal{O}_D)$  precisely when it either takes the same value at two distinct support points or has zero first derivative at a multiple support point. This follows at once from the minimal polynomial (19.7): its degree is  $d$  if and only if the support values are distinct and each local  $\ell_i = d_i$ . These two failure mechanisms are precisely

the existence of a length-two  $Z \leq D$  on which  $v$  is constant. Therefore the image of the finite incidence is the whole support of (20.3). That support is irreducible.

To determine its multiplicity, work over divisors with distinct support, label their roots after an étale base change, and use evaluation coordinates for  $v$ . The fixed-fibre formula becomes a product of the distinct differences  $\lambda_j - \lambda_i$ . At a point with exactly one equal pair the determinant has a nonzero linear derivative. Hence its multiplicity along its unique irreducible support is one. Since the ambient space is smooth, a Cartier hypersurface has no embedded associated points; a generically reduced such hypersurface is reduced. One may see the last assertions locally from unique factorization in the regular local ring, or from the Cohen–Macaulay and  $R_0 + S_1$  criterion [24, Tag 031R]. Thus  $Y$  is integral.

Over the dense open subset with reduced  $D$  and precisely one equal pair there is exactly one possible  $Z$ . The finite projection is therefore birational in characteristic zero. Its source is smooth and hence normal, so it is the normalization. The affine parametrization is exactly (20.4), as already proved. The dimension of the divisor is  $3d - 5$ .  $\square$

**Proposition 20.3** (Exact singular-support test across collisions). *At a closed point  $(D, A) \in Y$ , call a length-two  $Z \leq D$  scalar if  $A$  has rank one on  $Z$ . The point is smooth on  $Y$  if and only if there is precisely one scalar  $Z$  and the following linear system has no nonzero solution. In the associated factorization (20.4), the unknown is  $r(t)$  of degree less than two, and the system is*

$$h \mid qr, \quad wr \bmod h \text{ is constant.} \quad (20.5)$$

*In particular, this criterion is effective at nonreduced  $D$  and is not merely the distinct-support pair-count criterion. The full singular scheme on a frame chart has ideal*

$$(\Delta, \partial\Delta/\partial g_0, \dots, \partial\Delta/\partial g_{d-1}, \partial\Delta/\partial v_0, \dots, \partial\Delta/\partial v_{d-1}). \quad (20.6)$$

*Proof.* If the normalization has more than one point above  $(D, A)$ , the local ring of  $Y$  cannot be regular. Suppose there is just one. We first justify the differential criterion for a finite normalization  $\nu : N \rightarrow Y$  with  $N$  smooth. A regular local ring is normal, so a smooth point of  $Y$  has an isomorphic normalization near it. Conversely let  $A_0 \subset B_0$  be the local finite extension at a point with a unique preimage and common residue field  $\mathbb{C}$ . Injectivity of  $d\nu$  means

$$\mathfrak{m}_{B_0}/(\mathfrak{m}_{B_0}^2 + \mathfrak{m}_{A_0}B_0) = 0.$$

Nakayama’s lemma first gives  $\mathfrak{m}_{B_0} = \mathfrak{m}_{A_0}B_0$ . It then gives, for the finite  $A_0$ -module  $B_0$ , that 1 generates  $B_0$ , since the quotient by  $\mathfrak{m}_{A_0}$  is  $\mathbb{C}$ . The injective map  $A_0 \rightarrow B_0$  is consequently surjective. Thus  $A_0 = B_0$  is regular. Local finiteness here follows by localizing the finite morphism; uniqueness of the preimage makes the resulting semilocal ring local.

In the affine normalization coordinates, a tangent vector with  $\dot{g} = \dot{v} = 0$  satisfies

$$0 = q\dot{h} + h\dot{q}, \quad 0 = \dot{\lambda} + w\dot{h} + h\dot{w}.$$

Set  $r = \dot{h}$ , so  $\deg r < 2$  because  $h$  is monic. The first equation is soluble exactly when  $h \mid qr$ , and then determines  $\dot{q} = -qr/h$ . The second is soluble exactly when the remainder of  $wr$  modulo  $h$  is constant; it determines  $\dot{\lambda}$  as the negative of that constant and determines  $\dot{w}$  by division. The resulting degrees satisfy the required bounds. If  $r = 0$  all four variations are zero. Thus the differential has nonzero kernel exactly when (20.5) has a nonzero solution. The unit frame parameter has zero variation in a kernel and creates no additional one. Smooth frame descent proves the criterion on  $Y$ . Finally (20.6) is the hypersurface Jacobian presentation in its smooth affine ambient space.  $\square$

*Remark 20.4* (A smooth total point with a triple primary fibre). Take  $d = 4$ ,  $g = t^3(t - 1)$ , and  $A = \langle 1, t^2 \rangle$  on  $D = (g)$ . The only scalar length-two subdivisor is  $h = t^2$ . In (20.4) one has  $q = t(t - 1)$  and  $w = 1$ . For  $r = r_0 + r_1t$ , divisibility  $h \mid qr$  forces  $r_0 = 0$ , while constancy of

$wr \bmod h$  forces  $r_1 = 0$ . Hence the total divisor is smooth here. On the fixed-divisor fibre, the two support values are distinct and (19.4) is a unit times  $c_{11}^3$ . Its exact local nilpotency index is three. Thus even a smooth point of the integral total divisor can have a nonreduced primary fibre.

**Proposition 20.5** (Discriminants and specialization). *Let  $\chi_v(T)$  be the characteristic polynomial of multiplication by  $v$  in  $\mathbb{C}[t]/(g)$ . With the usual monic polynomial normalization of discriminants,*

$$\text{disc}(\chi_v) = \text{disc}(g) \Delta(g, v)^2. \quad (20.7)$$

*This is a polynomial identity on the entire coefficient space. The equation  $\Delta = 0$ , not  $\text{disc}(\chi_v) = 0$ , defines the failure divisor, including when  $g$  has multiple roots. Its fibre over each fixed  $D$  is exactly the primary scheme in (19.5).*

*Proof.* When  $g$  has distinct roots  $t_1, \dots, t_d$ , the evaluation matrix changes the basis  $1, t, \dots, t^{d-1}$  to point values. Therefore

$$\Delta(g, v) = \frac{\prod_{i < j} (v(t_j) - v(t_i))}{\prod_{i < j} (t_j - t_i)}.$$

Squaring gives (20.7), because  $\chi_v(T) = \prod_i (T - v(t_i))$  on this open set. Both sides are polynomial in the coefficients, so the identity holds everywhere. For  $v = t$  the determinant is one even when  $g$  is nonreduced; this also shows why the factor  $\text{disc}(g)$  must not be included in the failure equation. The scheme-theoretic fibre statement follows from Fitting base change in Theorem 7.2, or by specializing the polynomial determinant and applying (19.4).  $\square$

*Proof of Theorem 6.1.* The fixed-contact assertions are Theorems 19.2 and 19.3. The moving-contact assertions are Theorem 20.2 and Proposition 20.3. All four concern the original multiplication map, by the coherent cokernel isomorphism of Theorem 7.2.  $\square$

## 21. PENCIL COLLISIONS AND COMPARISON OF LOCAL TYPES

The normalization criterion above separates phenomena which a set-theoretic discriminant calculation conflates. In the affine normalization write  $g = hq$  and  $v = \lambda + hw$ , with  $h$  monic of degree two. The kernel test is

$$h \mid qr, \quad wr \bmod h \text{ constant}, \quad \deg r < 2.$$

The following three cases give concrete local distinctions.

If  $g$  has distinct roots, the normalization is unramified; a unique equal value pair gives a smooth point, and two disjoint equal pairs give an ordinary double crossing. Three equal values give the three braid factors  $\delta_{12}\delta_{13}(\delta_{13} - \delta_{12})$ , not a normal-crossing divisor in a two-dimensional normal space. These are exact equations because root labelling is étale here.

For  $g = t^3(t - 1)$  and  $v = t^2$ , the scalar subdivisor  $h = t^2$  is unique. One has  $q = t(t - 1)$  and  $w = 1$ . Divisibility forces the constant term of  $r$  to vanish and the second condition forces its linear term to vanish. The total pencil divisor is smooth, although its fixed- $D$  fibre has primary equation a unit times  $c_{11}^3$ .

For  $g = t^4$  and  $v = t^2$ , again  $h = t^2$  is the unique scalar subdivisor. Now  $q = t^2$  and  $w = 1$ . The constant  $r = 1$  satisfies the kernel test. Thus the total pencil divisor is singular and its normalization is ramified at the unique preimage. This last example is asserted to have a unique normalization point, not an unproved analytic cusp normal form. In contrast, the moving contact-hyperplane scheme is smooth at all these divisor collisions by Theorem 10.4. Its fixed fourfold-contact transverse algebra is  $G_4$ , with length six and nilpotency index five, rather than the pencil's sixth power of a prime divisor.

## 22. RELATION TO EARLIER CONSTRUCTIONS

The determinant  $\det[1, v, \dots, v^{d-1}]$  is a classical local index form. Arpin, Bozlee, Herr and Smith construct schemes of monogenerators and polygenerators for finite locally free algebras [29]. Their Proposition 3.6 and Definition 3.12 identify the index-form open and its closed complement; Proposition 3.14 gives the polygenerator open by minors. Thus neither the generator functor nor its index determinant is claimed as new here. Our use concerns the coherent cokernel of a global section-multiplication map and, for generating hyperplanes, its exact closed incidence with a rank-two quotient. Lemma 10.1 and Theorem 10.2 prove that closed-scheme identification on nonreduced bases, rather than deriving it from agreement of generator opens.

The algebra  $G_r$  is the classical two-row Grassmannian cohomology ring; [31, Theorem 2.7] provides the rectangular Schur basis in a more general quotient. Its role here is an exact transverse algebra of multiplication failure, determined by a rank-two quotient of  $\mathbb{C}[z]/(z^r)$ . The explicit residual equations (10.7) verify this identification without a tangent-cone approximation. The diagonal-ideal and all-powers identities in the three-plane result are precisely [30, Corollary 3.8.3]; the argument here supplies the finite total-degree presentation and conductor transport, not a new proof of that theorem.

Normal generation in sufficiently positive degree on a curve is classical [14]. The curve conductor theorem uses it only for  $L(-D)$ , and then proves the precise quotient comparison by filling the double-vanishing space and its first normal layer. On  $\mathbb{P}^1$  the evaluation-kernel argument gives the sharper rank-dependent bound and its boundary example.

For pencils, confluent Vandermonde factorization and the discriminant identity are classical algebra. The moving normalization is the explicit scalar-pair incidence of that index-form divisor. Its finite map and differential test are proved in Section 20; the normalization is not offered as a construction independent of index-form geometry. The contact-hyperplane incidence is different: it is the failure scheme itself, not just its normalization, and its total space is smooth even at collisions.

**Subalgebra varieties, quotient flags, and the additional comparisons.** Iovanov and Sistko [32, Theorem 0.1] classify maximal subalgebras of finite-dimensional algebras over fields under their Schur hypothesis. In the commutative algebraically closed case, the alternatives include identifying two residue factors and imposing a first-order condition at one factor. These closed-point mechanisms are classical and are not claimed here as new. Sistko [33, Definition 3.1, Theorems 1.1 and 4.1] constructs subalgebra varieties and computes the maximal-subalgebra vanishing ideal for basic algebras. The distinction needed here is between such a variety and the full relative quadratic-equation functor: for  $\mathbb{C}[z]/(z^r)$  the latter already has the transverse algebra  $G_r$ , of length  $\binom{r}{2}$ . Agreement on the reduced variety does not identify this thickening.

The Hilbert functor used in Section 10 is the standard functor of locally free quotient algebras; its representability is not a contribution of this article. Likewise, nested effective divisors on a curve are classically expressed by a divisor and its residual divisor. What is established here is the identification of these functors with specified multiplication Fitting schemes, the canonical cokernel line (10.2), and the global-section transports. These are separate assertions from the classical classification of geometric hyperplanes.

The faithful-action viewpoint is also classical; compare [33, Proposition 3.4] for subalgebras of matrix algebras. Our conductor flags record the *specified inclusion* into  $B$ , the action on  $B/E$ , and its exact-rank scheme structure. A conductor rank bound alone is not the flag equivalence, nor does either imply unrestricted base change across an action-rank jump. The one-variable codimension-two classification is already given in [34, Theorem 39]. We use it to locate the fibres of Example 11.5, not as a new classification of polynomial subalgebras. The quotient-tower statement concerns arbitrary finite  $B$  and its relative exact-rank functors.

In [29, Example 4.16], the two-generator jet calculation has curvilinear target  $k[\varepsilon]/\varepsilon^{m+1}$ . Our target in (17.1) has embedding dimension  $e$ . The symmetric-power determinant identity in the proof is classical linear algebra. Its role is to determine the full primary power of the tangent determinant, uniformly in  $e, h$ , and then identify that power with global section multiplication on  $\mathbb{P}^e$  using Theorem 17.2. Merely knowing the polygenerator open would not determine this exponent or the global degree range. These comparisons identify the inputs and additional claims; they are not a certificate of exhaustive priority against all literature.

**Codimension, embedded incidences, and regularity.** The elementary dimension bound for an increasing sequence of subspaces is not a scheme-theoretic stabilization theorem. The proof of Theorem 9.1 retains the compressed commutator relation over the coefficient ring before using polarized Cayley–Hamilton. The polygenerator minors of [29] remain the classical ambient construction. The assertions added here are the codimension-sensitive degree bound, the full codimension-two matrix, and its conductor incidence on nonreduced bases. We do not claim an exhaustive priority comparison for every possible length bound.

The binary cubic for a rank-three algebra is the explicit square condition for a line to lift to a subalgebra. No independent novelty claim is made for that classical cubic-algebra calculation. Its role in Proposition 12.2 is to describe an exceptional fibre of the same incidence which transfers the embedded local rings in Theorem 14.1. The contact algebra  $G_h$  and its Schur basis are classical inputs; the mixed equations  $bv, b^2u$ , their displayed primary intersection, and their realization as local rings of full multiplication failure are proved separately.

Sidman’s Theorems 1.3 and 1.8 in [35] supply the relationship between regularity and saturation, and the ordinary-square estimate  $\text{reg}(I^2) \leq 2 \text{reg } I$  for a finite projective scheme. Sidman also records the earlier power bounds of Chandler and of Geramita–Gimigliano–Pitteloud. None of these regularity inequalities is claimed as new. Their use here is the normal-layer identity (15.8) and the resulting isomorphism of actual multiplication cokernels, including relative families. This separation is particularly important: bounding the regularity of an ideal is not by itself a theorem identifying the Fitting scheme of a prescribed incomplete linear series.

Ballico’s failure-locus work [4, 5] is a necessary nearest-source comparison. The accessible statements of [5, Theorem 0.2, Proposition 2.2, Remark 2.3] concern failure on finite linear sections of a fixed completely embedded variety. The complete theorem pages of [4] have not been obtained in this revision, so their exact relation to the present scheme statements remains unverified. We make no inference of non-anticipation from that access limitation. The results here are submitted with complete proofs and explicit classical inputs; priority of the resulting package is a separate question for source comparison and editorial assessment.

**22.1. Generators, algebra length, and the boundary classification.** The scheme-theoretic study of generators is an established input. Arpin–Bozlee–Herr–Smith [29] construct the index-form non-monogenerator scheme (Definition 3.12) and describe the polygenerator open by maximal minors (Proposition 3.14); the numbering here refers to the identified preprint version. Thus neither representability, index forms, nor the use of maximal minors is a novelty claim of this paper. In their construction Cayley–Hamilton bounds powers of each individual generator. The additional tasks here are the codimension-sensitive equality of image modules, descent to the generating Grassmannian, the conductor incidence, and the complete primary decompositions of the full boundary in the stated classes.

The field-level length inequality is separated from the relative assertion in Section 9.1. We do not use a special numerical estimate from the algebra-length literature as an unproved premise. The elementary dimension argument and the sharpness family are proved in the text. The coefficient-ring proof supplies the Fitting-ideal consequence needed when the base is nonreduced.

Theorems 13.1–13.3 identify the present boundary contribution precisely: a full multiplication ideal, all its associated strata for Hilbert function  $(1, 2, 2)$ , their primary representatives, and their conductor position. They do not assert a primary classification for every finite algebra. The higher-contact and complete contact-hyperplane results remain separate theorems with their original scope. The discussion of [4] in Section 22 remains subject to the same source-access limitation; no exhaustive priority comparison with that article is asserted on the basis of its title or bibliographic record.

**22.2. Logical dependencies.** The finite-algebra core is self-contained. It does not take a theorem from a manuscript labelled A1 as an unstated premise. The evaluation and conductor comparisons proved here are the interfaces through which a global linear-series problem becomes a finite-algebra problem. The outputs available to further work are canonical multiplication cokernels and their Fitting ideals, exact-action conductor flags, and the primary boundary classification. A downstream application must verify the stated generating and regularity hypotheses and retain the scheme structure; the determinant support alone loses the embedded strata. The historical polar, residual and statistical sections are not prerequisites for this core. This logical dependency statement does not certify that any particular version of a later manuscript already invokes these results.

**22.3. Determinantal inputs and the elliptic family.** The primality and normality of a generic minor ideal are classical; we use [39, Theorem 3.5(c)]. We also use the elementary rank-stratum codimension count and prove the needed symbolic-order inclusion by a Schur complement. The weighted ideal in Theorem 2.3 is not offered as a new theory of generic determinantal varieties. Its contribution here is the explicit primary intersection for the full multiplication ideal, including the extra linear determinant factor, and its use for the universal restriction flag. The incidence resolution is the usual image-subspace construction, applied to the specific embedded supports and with its fibres proved in Theorem 3.2.

The geometry of the Hesse pencil is also classical. We use the Weierstrass coefficients in [40, Section 2] to normalize the  $j$ -invariant. We do not claim a new classification of plane cubics. The additional statements are the explicit finite algebra (4.1), the full failure ideal on  $U$ , the square-zero line module (4.6), and the realization of the varying elliptic curves as the bases of its embedded supports. Those statements are derived from multiplication in the algebra and not from a resemblance between reduced equations.

The distinction from generator representability and the elementary field-level algebra-length bound is explained in Section 22.1. In particular a field-level dimension bound does not assert equality of image submodules over a nonreduced coefficient ring or identify their Fitting ideals there; the coefficient-ring argument and sharpness construction are provided in full in this article.

The full text of [4] has not been obtained for a theorem-by-theorem comparison. Its bibliographic record and publisher reference listing are not sufficient evidence to affirm or deny anticipation of any theorem here. No exhaustive priority conclusion with respect to that article is asserted. This documentary question is separate from the proofs above and remains explicitly recorded in the accompanying response and literature audit.

## COMPLEMENTARY GEOMETRY: FULL STATEMENTS AND PROOFS

The following sections retain the quadratic, polar, residual, wall and unrestricted hyperplane developments with their original hypotheses. They are not premises of the finite-quotient and boundary theorems above. The first section retains the earlier two-layer conductor proof in its original sufficient range.

## 23. EXACT CONDUCTOR REDUCTION IN FAMILIES

We work over  $\mathbb{C}$ . For a divisor  $D \subset \mathbb{P}^1$  of length  $d$  write  $I_D = \mathcal{O}(-D)$ . The notation  $U^m$  means the image of  $\text{Sym}^m U$  under multiplication, not an abstract symmetric power. For a finite scheme  $D$ , a subspace of  $H^0(\mathcal{O}_D(n))$  is called generating if its evaluation map onto  $\mathcal{O}_D(n)$  is surjective. This condition is open in a family. In relative notation,  $\pi$  denotes the projection from  $\mathbb{P}^1$  times the parameter space; all bundles are pulled back to the indicated relative Grassmannian when needed.

**Lemma 23.1** (Pencil multiplication). *If  $U \subset V_n$  is base-point-free, then  $UV_t = V_{n+t}$  for  $t \geq n-1$ .*

*Proof.* Choose  $0 \neq f \in U$ . For each of its finitely many zeros the sections of  $U$  vanishing there form a proper hyperplane. Choose  $g \in U$  outside their union. The binary forms  $f, g$  are coprime. The kernel of  $(A, B) \mapsto fA + gB$  on  $V_t \oplus V_t$  consists of  $(gh, -fh)$  with  $h \in V_{t-n}$ , where this space is zero for  $t < n$ . For  $t = n-1$  the map is between two  $2n$ -dimensional spaces and is injective. For  $t \geq n$  its rank is  $2(t+1) - (t-n+1) = n+t+1$ . In either case its image is  $V_{n+t}$ .  $\square$

**Theorem 23.2** (Relative conductor reduction). *Suppose  $n \geq 2d-1$ ,  $m \geq 2$ , and  $1 \leq k \leq d$ . On  $\text{Hilb}^d(\mathbb{P}^1) = \mathbb{P}^d$  let  $\mathcal{D}$  be the universal divisor and  $\mathcal{E}_j = \pi_* \mathcal{O}_{\mathcal{D}}(j)$ , of rank  $d$ . Let  $X_{d,k}^\circ$  be the open relative Grassmannian of generating  $k$ -planes  $\mathcal{A} \subset \mathcal{E}_n$ , and define*

$$\mathcal{W} = \pi_* I_{\mathcal{D}}(n), \quad \mathcal{U} = \ker(V_n \otimes \mathcal{O} \longrightarrow \mathcal{E}_n / \mathcal{A}).$$

*Then  $\mathcal{U}$  has rank  $n+1-d+k$ , and*

$$\text{im}(\mathcal{W} \otimes \text{Sym}^{m-1} \mathcal{U} \longrightarrow V_{mn} \otimes \mathcal{O}) = \pi_* I_{\mathcal{D}}(mn). \quad (23.1)$$

*Restriction induces an isomorphism*

$$\text{coker}(\text{Sym}^m \mathcal{U} \longrightarrow V_{mn} \otimes \mathcal{O}) \xrightarrow{\sim} \text{coker}(\text{Sym}^m \mathcal{A} \longrightarrow \mathcal{E}_{mn}). \quad (23.2)$$

*Both assertions remain true after arbitrary base change to  $X_{d,k}^\circ$ . In particular the two cokernels have identical Fitting ideals in every index, including on nonreduced bases.*

*Proof.* We first prove (23.1) on a geometric fibre. Write  $D = (g)$ , with  $g$  a binary form of degree  $d$ , so  $W = gV_{n-d}$ . The inverse-image subseries  $U$  is base-point-free: outside  $D$  this follows from  $W$ , and on  $D$  it follows from the generating hypothesis on  $A$ .

Products of two elements of  $W$  fill  $g^2 V_{2n-2d}$ . Since  $2n-2d \geq n-1$ , repeated applications of Lemma 23.1 give

$$W^2 U^{m-2} = g^2 V_{mn-2d} = H^0(I_D^2(mn)). \quad (23.3)$$

For  $m=2$  no application of the lemma is needed. For each further factor the input degree is  $2n-2d+jn \geq n-1$ .

The restriction of  $W$  to the first normal layer is surjective:

$$H^0(I_D(n)) \longrightarrow H^0((I_D/I_D^2)(n)).$$

Indeed the obstruction is  $H^1(I_D^2(n)) = H^1(\mathcal{O}(n-2d))$ , which is zero because  $n-2d \geq -1$ . A generating subspace  $A$  contains a section nonzero at all support points of  $D$ : avoid the finitely many evaluation hyperplanes. Such a section is a generator of the invertible  $\mathcal{O}_D$ -module  $\mathcal{O}_D(n)$ . Lift it to  $u \in U$ . Multiplication by  $u^{m-1}$  identifies the first normal layer in degree  $n$  with that in degree  $mn$ . Consequently  $Wu^{m-1}$  surjects onto  $H^0((I_D/I_D^2)(mn))$ . This quotient is also the quotient of  $H^0(I_D(mn))$  by  $H^0(I_D^2(mn))$ , since  $mn-2d \geq -1$ . Combining this fact with (23.3) proves (23.1) on every geometric fibre.

All the targets just used are vector bundles on the parameter space: the requisite first cohomology groups vanish uniformly. The map in (23.1) is therefore a surjection of vector bundles by the fibrewise criterion and Nakayama's lemma. It stays surjective under every base change.

The exact sequence  $0 \rightarrow \mathcal{W} \rightarrow \mathcal{U} \rightarrow \mathcal{A} \rightarrow 0$  is locally split; hence the kernel of  $\mathrm{Sym}^m \mathcal{U} \rightarrow \mathrm{Sym}^m \mathcal{A}$  is the image of  $\mathcal{W} \otimes \mathrm{Sym}^{m-1} \mathcal{U}$ . Restrict the multiplication diagram to  $\mathcal{D}$  and divide both source and target by these kernels. The surjectivity just proved identifies the two cokernels, giving (23.2). This argument is compatible with base change. Presentation independence and base change for Fitting ideals [25, Lemma 15.8.4, Tag 07ZA] finish the proof.  $\square$

*Remark 23.3* (What the reduction preserves). The theorem identifies coherent modules, not only the ranks of their fibres. Thus it preserves the equations of every corank scheme and all primary structure of its zeroth Fitting scheme on the stated base. It does not assert a primary decomposition for an arbitrary finite-dimensional multiplication system. Such a decomposition is proved below for  $k = 2$ . The inequality  $n \geq 2d - 1$  is a sufficient uniform bound; no assertion that it is necessary is used.

#### 24. THE QUADRATIC FAMILY AND THE POLAR RESIDUAL FRAMEWORK

Let  $V_n = H^0(\mathbb{P}^1, \mathcal{O}(n))$  and  $k = n + 1$ . Multiplication of sections defines, on the product of Grassmannians  $X_L = \mathrm{Gr}(c, V_n)^L$ , the bundle map

$$\mu : \bigoplus_{\nu=1}^L \mathrm{Sym}^2 \mathcal{U}_\nu \longrightarrow V_{2n} \otimes \mathcal{O}_{X_L}. \quad (24.1)$$

There are no products between different factors. We study the scheme  $\mathcal{D}_L$  of maximal minors of this map, rather than only the rank at a general tuple. Put

$$q = \binom{c+1}{2}, \quad e = Lq, \quad p = 2k - 1, \quad M = Lc(k - c), \quad a = e - p + 1, \quad b = Lc - 3,$$

and assume  $e \geq p$ . The competition between  $a$  and  $b$  reflects two geometrically different kinds of annihilator. A nondegenerate Hankel form leads to the expected determinantal codimension  $a$ ; a rank-two form leads to signed secant families of codimension  $b$ . This competition is measured by multiplication on a smaller residual space and governs the singularities of the failure scheme.

The reduced support  $\Delta^{(L)} = |\mathcal{D}_L|$  is the set of tuples for which  $\sum_\nu U_\nu^2 \neq V_{2n}$ . The basic codimension formula holds in the entire range  $4 \leq c < k$ .

**Theorem 24.1** (Reduced failure locus and independent contacts). *Suppose  $4 \leq c < k$ ,  $L \geq 1$ , and  $Lq \geq p$ . Then*

$$\mathrm{codim}_{X_L} \Delta^{(L)} = \min\{Lq - p + 1, Lc - 3\}. \quad (24.2)$$

*There is an irreducible common-secant subvariety  $\Sigma_L \subset \Delta^{(L)}$  of codimension  $Lc - 3$ . If  $Lc - 3 \leq Lq - p + 1$ , it is a maximal-dimensional irreducible component of  $\Delta^{(L)}$ . Consequently a general  $L$ -tuple is informative if and only if  $Lq \geq p$ .*

Theorem 30.1 refines this formula to a component classification in all these dimensions. For  $a \leq b$  it determines all components and proves reducedness and Cohen–Macaulayness; for  $b < a$  it determines all maximal-dimensional components. Exactly five additional equality incidences occur outside the two principal rank families. In each, the plane contains the full radical and is maximal isotropic in the quotient. The orientation covers in Section 29 determine their irreducibility.

The principal deformation result is independent of this binary classification. Given any map  $\beta : \mathrm{Sym}^m V \rightarrow F$ , the plane derivative at an annihilator defines a polar residual space  $Q$  and a map  $\tau : Q \rightarrow \ker \ell$ . Theorem 25.1 expresses the tangent excess of the incidence, and of its corank-one image, as  $\dim \ker \tau$ . In degree two,

$$Q = \bigoplus_\nu \mathrm{Sym}^2(U_\nu \cap \mathrm{rad} H_\ell),$$

so the obstruction is a smaller multiplication problem. The formulation also applies to products of sections on arbitrary projective varieties and to higher symmetric powers.

This tangent calculation has a scheme-theoretic global form. Theorem 25.2 determines the full singular support in expected codimension: all points of original corank at least two are singular, and a corank-one point is singular exactly when  $\tau$  is not injective. On every constant-rank polar stratum, the Fitting ideals of the cotangent sheaf are the indicated minors of  $\tau$ . These identities include the higher tangent-excess strata and glue with exact Schur–Jacobian presentations at larger corank. Corollary 25.3 consequently handles all annihilator ranks of the quadratic binary problem, including support collisions. Proposition 25.5 gives the corresponding local criterion in excess codimension.

The distinction between restricted ideals and neighbourhoods is addressed by Theorem 26.2. At arbitrary original corank, the joint polar relations are the obstruction to a transverse generic matrix chart. Even when that obstruction is nonzero, the theorem gives an exact matrix of residual convergent equations, retaining all higher terms and all nested rank ideals. In corank one, Theorem 26.5 reduces these to  $\delta = \dim \ker \tau$  equations in  $M - a + \delta$  parameters and identifies their quadratic obstruction. In expected codimension they form a regular sequence. Proposition 26.7 constructs transverse examples in every symmetric degree; the cubic example in Section 26 also exhibits a nontransverse cuspidal failure germ. These are exact local statements, not a claim that the binary higher-product component problem follows from the two-point calculation.

At a rank-two annihilator with distinct support, its radical is  $gV_{n-2}$  for the vanishing quadratic  $g$ . Writing  $U_\nu \cap \text{rad } H_\ell = gK_\nu$ , the wall  $a = b$  is exactly

$$L\binom{c}{2} = 2k - 5,$$

the square threshold for the residual multiplication of the  $(c-1)$ -planes  $K_\nu$ . Its entire determinantal rank stratification is the pullback of the corresponding cotangent Fitting stratification on the original corank-one locus (Corollary 25.4). On a nonempty dense open subset of its integral divisor, the analytic equation is  $uv = 0$  (Theorem 31.9). The normal-bundle factorization, relative determinant line and simultaneous nonemptiness needed for this equation are proved separately in Section 31.3. We also give all same-annihilator signed intersections over the split rank-two base and an explicit primary decomposition of the fixed-annihilator isotropy fibre. These describe different objects: fibre embedded structure is not being asserted for the global scheme.

For higher symmetric powers the general polar theorem remains applicable. Section 32 additionally computes the two-point annihilator sectors by algebraic dihedral descent. This explicit family is not identified with all components of the higher-product failure scheme. The arbitrary- $\beta$  theorem, rather than such an identification, is the general extension of the quadratic residual mechanism.

A complete higher-product family is provided by Theorem 33.1. For all hyperplanes  $U \subset V_n$ ,  $n \geq 4$ , and every  $m \geq 2$ , the entire reduced failure locus of  $\text{Sym}^m U \rightarrow V_{mn}$  is the same secant threefold. We determine every corank, including the jump to  $m$  on the evaluation curve. In degree at least three we also determine all associated points of the projective failure scheme: the secant generic point and the embedded evaluation-curve generic point. A lower bound  $\lceil m/2 \rceil$  for the nilpotency index shows that the high-degree scheme structure is not recovered from this fixed reduced support. This supplies a global excess-codimension consequence in every higher degree, beyond merely inserting a multiplication map into the general polar formalism.

The geometry arises naturally in recovery of calibrated information metrics: second-order distance contacts give the moment functional on the within-contact products  $U_\nu^2$ . A short application is given in Section 35. The appendices give the exact-contact, native-realization, boundary-experiment and finite-precision arguments, with their distinct hypotheses. None is

used as an assumption in the algebraic proofs. Classical inputs and the precise extent of the primary-source comparison are recorded in Section 34.

## 25. POLAR RESIDUAL SPACES AND GLOBAL SINGULARITY STRATA

The deformation mechanism does not require binary forms. Let  $V, F$  be finite-dimensional complex vector spaces,  $\dim F = p$ , and fix a linear map

$$\beta : \operatorname{Sym}^m V \longrightarrow F, \quad m \geq 2.$$

Choose integers  $0 < c_\nu < \dim V$ , put  $X = \prod_\nu \operatorname{Gr}(c_\nu, V)$ , and write  $M = \dim X$ . The tautological multiplication map, its source rank and its expected failure codimension are

$$\mu : \mathcal{E} = \bigoplus_\nu \operatorname{Sym}^m \mathcal{U}_\nu \longrightarrow F \otimes \mathcal{O}_X, \quad e = \operatorname{rank} \mathcal{E}, \quad a = e - p + 1.$$

Assume  $e \geq p$ , and let  $D = V(I_p(\mu))$ . No assumption on the codimension of  $D$  is made until explicitly stated. The incidence  $\mathcal{I} \subset \mathbb{P}(F^*) \times X$  is the zero scheme of  $\ell\mu = 0$ . Write  $D^{(1)}$  for the locus where  $\mu$  has corank one.

At a point  $z = ([\ell], U)$  of this incidence, choose a representative  $\ell$  and define the plane derivative

$$\begin{aligned} B_z : T_U X &\longrightarrow \mathcal{E}_U^*, \\ B_z(A)(u_1 \cdots u_m) &= \sum_{i=1}^m \ell \beta(u_1 \cdots A_\nu u_i \cdots u_m), \quad u_i \in U_\nu. \end{aligned} \quad (25.1)$$

This is well defined for  $A_\nu : U_\nu \rightarrow V/U_\nu$ : changing a lift adds a product annihilated by  $\ell$ . Define the *polar residual space* and its multiplication map by

$$Q_z = \ker(B_z^* : \mathcal{E}_U \longrightarrow T_U^* X), \quad \tau_z = \mu_U|_{Q_z} : Q_z \longrightarrow \ker \ell \subset F. \quad (25.2)$$

The kernel  $Q_z$  is independent of the chosen nonzero representative of  $\ell$ . On a constant-rank stratum of  $B$ , these spaces are vector bundles and  $\tau$  is a bundle map. Equivalently, before trivializing the tautological line  $\mathcal{L} = \mathcal{O}_{\mathbb{P}(F^*)}(-1)$ , the plane derivative has target  $\mathcal{E}^* \otimes \mathcal{L}^*$ .

**Theorem 25.1** (Polar residual tangent sequence). *At every point  $z = ([\ell], U) \in \mathcal{I}$  there is an exact sequence*

$$0 \longrightarrow \ker B_z \longrightarrow T_z \mathcal{I} \longrightarrow (\mu_U Q_z)^\perp / \mathcal{C}\ell \longrightarrow 0. \quad (25.3)$$

*In particular*

$$\dim T_z \mathcal{I} = M - a + \dim \ker \tau_z. \quad (25.4)$$

*The projection  $\mathcal{I} \rightarrow D$  is a scheme isomorphism over  $D^{(1)}$ , so the same formula computes  $\dim T_U D$  there. For  $m = 2$ , set  $H_\ell(v, w) = \ell \beta(vw)$  and  $W_\nu = U_\nu \cap \operatorname{rad} H_\ell$ . Then*

$$Q_z = \bigoplus_\nu \operatorname{Sym}^2 W_\nu, \quad \tau_z((q_\nu)_\nu) = \sum_\nu \beta(q_\nu). \quad (25.5)$$

*Thus the quadratic obstruction is an actual smaller multiplication problem for every symmetric bilinear system, not only for binary forms.*

*Proof.* Differentiating the incidence equations gives  $\dot{\ell}\mu_U + B_z(\dot{U}) = 0$ . A vector in  $\mathcal{E}_U^*$  belongs to  $\operatorname{im} B_z$  exactly when it annihilates  $Q_z$ . The equation is therefore soluble in  $\dot{U}$  exactly when  $\dot{\ell}$  annihilates  $\mu_U Q_z$ . Its set of solutions is a torsor under  $\ker B_z$ . Taking  $\dot{\ell}$  modulo  $\mathcal{C}\ell$  proves (25.3). If  $h = \dim Q_z$ , then  $\operatorname{rank} B_z = e - h$ , so its middle dimension is

$$M - (e - h) + p - 1 - \operatorname{rank} \tau_z = M - a + \dim \ker \tau_z.$$

On a chart with an invertible  $(p-1)$ -minor, invertible row and column operations put  $\mu$  in the form  $\operatorname{diag}(I_{p-1}, S)$ , where  $S$  is a single row. The incidence equations first solve the normalized

left kernel vector uniquely and then impose  $S = 0$ . This proves the scheme isomorphism, not just a bijection on points.

For the quadratic assertion, the map  $V/U_\nu \rightarrow U_\nu^*$  induced by  $H_\ell$  has image  $(U_\nu/W_\nu)^*$ . Its transpose has kernel  $W_\nu$ , and it descends through  $U_\nu$  because  $U_\nu$  is isotropic. Choose  $U_\nu = W_\nu \oplus Z_\nu$ . Varying vectors of  $W_\nu$  in lifts dual to  $Z_\nu$  prescribes every mixed symmetric entry; varying the vectors of  $Z_\nu$  prescribes their symmetric block, using division by two. Consequently  $\text{im } B_z$  is exactly the space of forms vanishing on every  $\text{Sym}^2 W_\nu$ . Taking annihilators gives (25.5).  $\square$

We use  $I_s(A)$  for the ideal of  $s$ -minors of a matrix, with  $I_s = (1)$  for  $s \leq 0$  and  $I_s = (0)$  if the requested minors are too large. For a finite module with presentation  $R^q \rightarrow R^N \rightarrow C \rightarrow 0$ ,  $\text{Fitt}_i C$  is generated by the  $(N - i)$ -minors of the presentation matrix. This definition is independent of the presentation. All statements about restrictions of these ideals below mean pullback of the ideals of  $D$ ; they do not mean the intrinsic singular scheme of a locally closed stratum.

**Theorem 25.2** (Global singularities and residual Fitting strata). *Suppose  $D$  is nonempty and of pure expected codimension  $a > 0$  in  $X$ , and put  $d = M - a$ . Then its full singular support is*

$$\text{Sing}(D)_{\text{red}} = \{U : \text{corank } \mu_U \geq 2\}_{\text{red}} \cup \{U \in D^{(1)} : \ker \tau_{([\ell_U], U)} \neq 0\}_{\text{red}}. \quad (25.6)$$

*The union denotes the indicated subsets of  $D$ , with their reduced closed union; the second set is locally closed in  $D$ . On any locally closed constant-rank stratum  $T \subset D^{(1)}$  of  $B$ , with  $h = \text{rank } Q$ , one has, for every integer  $j \geq 1$ ,*

$$\text{Fitt}_{d+j-1}(\Omega_{D/\mathbb{C}})\mathcal{O}_T = I_{h-j+1}(\tau). \quad (25.7)$$

*In particular the restriction of the singular scheme  $V(\text{Fitt}_d \Omega_{D/\mathbb{C}})$  is the residual noninjectivity scheme  $V(I_h(\tau))$ , including its determinantal scheme structure. Formula (25.7) detects all tangent-excess strata, including points outside a generic node locus and points whose annihilator has colliding support.*

*At a point of original corank  $\rho \geq 1$ , write  $S = D_0 - CA^{-1}B_0$  for the Schur complement of an invertible  $(p - \rho)$ -block of  $\mu$ . Let  $f_\alpha$  be all its  $\rho$ -minors and choose smooth local coordinates  $x_1, \dots, x_M$  on  $X$ . On this chart the singular scheme has the exact ideal*

$$I_\rho(S) + I_a((\partial f_\alpha / \partial x_i)_{\alpha, i}) \quad \text{in } \mathcal{O}_X. \quad (25.8)$$

*The Schur–Jacobian presentations hold on open charts for every  $\rho \geq 1$ , so they cover the whole failure scheme and are compatible on overlaps. The residual-minor simplification is a pullback to the stated strata, not an assertion of a product neighbourhood transverse to those strata.*

*Proof.* Over  $D^{(1)}$  use the incidence isomorphism of Theorem 25.1. In local coordinates on  $\mathbb{P}(F^*) \times X$ , the differential of its  $e$  equations is the map from a free module of rank  $M + p - 1$  to one of rank  $e$  obtained by adjoining  $\ell\mu$  to  $B$ . Its transpose presents the cotangent module of the incidence. Restrict to  $T$  and split off an invertible  $(e - h)$ -minor of  $B$ . Constant rank means that its remaining Schur block is identically zero on  $T$ , not just at a chosen point. After elementary operations the remaining nonzero block is the map from  $F^*/C\ell$  to  $Q^*$  induced by  $\ell\mu$ ; it is  $\tau^*$ , with the tautological-line twist if no representative of  $\ell$  was chosen. Fitting ideals commute with base change. The minors defining  $\text{Fitt}_{d+j-1}$  have size

$$(M + p - 1) - (d + j - 1) = e - j + 1.$$

After removing the identity block their size is  $h - j + 1$ . This proves (25.7). Change of frames or of a representative of  $\ell$  multiplies the matrices by invertible matrices and hence leaves these ideals unchanged.

Since the local dimension on  $D$  is  $d$ , the tangent formula gives regularity at a corank-one point exactly when  $\tau$  is injective. Over  $\mathbb{C}$ , regularity is smoothness. At a point of corank  $\rho \geq 2$ , Schur elimination identifies the defining ideal with  $I_\rho(S)$ , where every entry of  $S$  vanishes at the point.

Thus this ideal is contained in the square of the maximal ideal. All its first differentials vanish, and  $T_U D = T_U X$  has dimension  $M > d$ . This proves (25.6). Finally the usual exact cotangent presentation for the quotient by  $(f_\alpha)_\alpha$  gives (25.8) directly from the definition of  $\text{Fitt}_d$ . The presentation independence of Fitting ideals proves compatibility with every other Schur chart and with (25.7).  $\square$

**Corollary 25.3** (Every annihilator rank for quadratic binary subseries). *In the notation of Theorem 24.1, suppose  $4 \leq c < k$ ,  $e \geq p$ , and  $a \leq b$ . Theorem 25.2 applies to  $\mathcal{D}_L$ . On its corank-one locus, put  $W_\nu = U_\nu \cap \text{rad } H_{\ell_U}$  and  $h = \sum_\nu \binom{\dim W_\nu + 1}{2}$ . Then*

$$U \text{ is smooth} \iff \bigoplus_\nu \text{Sym}^2 W_\nu \longrightarrow V_{2n} \text{ is injective.} \quad (25.9)$$

*On each fixed-radical-intersection-dimension stratum, the singular ideal and all higher tangent-excess ideals are the minors in (25.7) of this smaller multiplication map. No assumption on distinct support, maximal annihilator rank, or membership in just one signed component is required.*

*Proof.* Theorem 24.1 and Proposition 28.6 give pure expected codimension. Equation (25.5) identifies the residual map. Constant dimensions of the intersections give vector bundles after stratification. The preceding theorem applies.  $\square$

**Corollary 25.4** (The entire residual wall stratification). *At  $a = b$ , on a signed rank-two stratum with original multiplication of corank one and  $U_\nu \not\subset \text{rad } H_\ell$ , the pullback of the singular scheme is exactly the residual determinant scheme of  $\mu'_K$ . More generally its tangent-excess-at-least- $j$  ideal is  $I_{h-j+1}(\mu'_K)$ , where  $h = L(\binom{c}{2}) = 2k - 5$ . This includes every residual corank, not only the open ordinary-double-crossing locus.*

*Proof.* Here  $Q = \bigoplus \text{Sym}^2(gK_\nu)$ , and multiplication factors as  $\mu'_K$  followed by the injective multiplication by  $g^2$ . The source and residual target both have dimension  $h$ . Apply (25.7); multiplication by  $g^2$  is a subbundle inclusion on the split rank-two base.  $\square$

**Proposition 25.5** (Excess-codimension form of the residual criterion). *Without the expected-codimension hypothesis, let  $T \subset D^{(1)}$  be a constant-rank stratum of  $B$ . For any integer  $t$ , locally*

$$\text{Fitt}_{M-t}(\Omega_{D/\mathbb{C}})\mathcal{O}_T = I_{h-a+t}(\tau). \quad (25.10)$$

*At a point where  $\dim_U D = M - t$ , smoothness is equivalent to  $\dim \ker \tau = a - t$ . For quadratic binary subseries in the excess regime  $b < a$ , this gives  $t = b$  at every corank-one point lying on a maximal-dimensional component. It is a statement about the full local failure scheme there, not only its selected component.*

*Proof.* The same cotangent presentation has  $M + p - 1$  generators, so the requested minors have size  $p - 1 + t = e - a + t$ . Remove the  $(e - h)$ -identity block. This gives (25.10). Compare (25.4) with the stated local dimension. In the last assertion the global dimension bound is  $M - b$ , and the component through the point attains it.  $\square$

These results separate a full singular-locus calculation from a classification of local analytic types or of associated primes. Equation (25.8) retains nonlinear Schur entries: replacing it by minors of their linear parts would lose information. Neither the global criterion nor the excess formula asserts that every germ is a normal crossing, or that every excess scheme is unmixed.

The generality in  $\beta$  has concrete instances. For a projective variety  $Y$  and a line bundle  $A$ , one may take  $V = H^0(Y, A)$ ,  $F = H^0(Y, A^m)$  and  $\beta$  to be multiplication of sections; the same construction applies to multivariate polynomial spaces and their linear quotients. Theorems 25.1 and 25.2 use precisely the hypotheses stated above in each case. They do not require a secant parameterization or a classification of catalecticant ranks. For  $m > 2$  the polar residual  $Q$  need

not be a symmetric power of a radical, which is why the general statement uses the kernel of the plane derivative instead of imposing a quadratic description.

## 26. INTRINSIC RESIDUAL EQUATIONS AT EVERY MULTIPLICATION CORANK

The residual Fitting identities describe restrictions of intrinsic ideals. To pass from these restrictions to a neighbourhood, one must retain the nonlinear equations. We give such a presentation at every point. The linear algebra in this section applies to a map of vector bundles on any smooth complex variety; the polar formula identifies its intrinsic obstruction for the multiplication systems of Section 25.

The matrix normal derivative and the transverse determinantal model used below are classical [10, Remark 1.18; Section 1.3.2]. The point here is their explicit expression in multiplication polars, including the joint relations between distinct annihilators, and the nonlinear equations left when transversality fails.

**26.1. The joint polar obstruction.** Fix  $U \in D$ , and write

$$K_U = \ker \mu_U, \quad C_U = \operatorname{coker} \mu_U, \quad \rho = \dim C_U, \quad s = \dim K_U = e - p + \rho.$$

Here  $C_U^* \subset F^*$  is the entire space of annihilators, not a choice of one projective annihilator. Differentiating  $\mu$  and then restricting to its kernel and projecting to its cokernel defines

$$\partial_U : T_U X \longrightarrow \operatorname{Hom}(K_U, C_U), \quad A \longmapsto [\dot{\mu}_U(A)|_{K_U}]. \quad (26.1)$$

This map is intrinsic. Indeed a change of source or target frame changes  $\dot{\mu}$  by terms of the form  $P\mu_U + \mu_U Q$ ; these vanish after restriction to  $K_U$  and projection to  $C_U$ .

**Proposition 26.1** (Joint polar relations). *For a symmetric multiplication system, the transpose of (26.1) is*

$$\begin{aligned} \partial_U^* : K_U \otimes C_U^* &\longrightarrow T_U^* X, \\ \sum_i q_i \otimes \ell_i &\longmapsto \sum_i B_{([\ell_i], U)}^*(q_i), \end{aligned} \quad (26.2)$$

where the representative  $\ell_i$  is used in the definition of  $B$ ; the expression is linear in that representative and extends to  $\ell_i = 0$ . Put

$$\mathcal{R}_U = \ker \partial_U^*, \quad t = \operatorname{rank} \partial_U, \quad b_U = \dim \mathcal{R}_U = \rho s - t.$$

If  $\rho = 1$ , a choice of nonzero annihilator identifies  $\mathcal{R}_U$  with  $\ker \tau_{([\ell], U)}$ . If  $\rho > 1$ , the kernel in (26.2) includes relations between distinct annihilators; it is not in general the direct sum of the single-annihilator kernels.

*Proof.* For  $q \in K_U$ ,  $\ell \in C_U^*$  and  $A \in T_U X$ , the natural trace pairing gives

$$\langle q \otimes \ell, \partial_U(A) \rangle = \ell \dot{\mu}_U(A)(q) = B_{([\ell], U)}(A)(q).$$

Summing proves the formula. If  $C_U^* = \mathbb{C}\ell$ , its kernel is  $K_U \cap \ker B^* = K_U \cap Q = \ker(\mu_U|_Q)$ . All identifications are compatible with changes of frame by the intrinsic construction of  $\partial_U$ .  $\square$

**26.2. An exact residual presentation.** Let  $H_U = \operatorname{Hom}(K_U, C_U)$  and  $H_0 = \operatorname{im} \partial_U$ . Choose a vector-space complement  $H_1$  so that  $H_U = H_0 \oplus H_1$ , and choose an isomorphism  $L : \mathbb{C}^t \rightarrow H_0$ . Such choices specify coordinates, not additional hypotheses. In particular  $H_1$  identifies with  $\operatorname{coker} \partial_U$  and has dimension  $b_U$ .

**Theorem 26.2** (Residual matrix germ). *At every  $U \in D$  there are analytic coordinates  $(y_1, \dots, y_t, x_1, \dots, x_{M-t})$  centred at  $U$  and an  $H_1$ -valued convergent power series  $\Psi(y, x)$  with  $\Psi(0) = d\Psi(0) = 0$  such that*

$$\hat{\mathcal{O}}_{D,U} \simeq \mathbb{C}[[y_1, \dots, y_t, x_1, \dots, x_{M-t}]]/I_\rho(L(y) + \Psi(y, x)). \quad (26.3)$$

*The same presentation holds with convergent power series for the analytic local ring. Moreover, for  $1 \leq j \leq \rho$ , the ideal of the locus where the original multiplication has rank at most  $p - \rho + j - 1$  is, in this presentation,*

$$I_j(L(y) + \Psi(y, x)). \quad (26.4)$$

*These are equalities of ideals after analytic change of coordinates, not merely tangent-space or set-theoretic assertions. No expected-codimension assumption is required.*

*Proof.* Put  $r = p - \rho$ . Trivialize source and target and choose an invertible  $r$ -minor at  $U$ . After constant changes of frame the matrix is

$$\mu = \begin{pmatrix} A & B_0 \\ C & D_0 \end{pmatrix}, \quad A(U) = I_r, \quad B_0(U) = C(U) = D_0(U) = 0.$$

Invertible analytic row and column operations reduce it to  $\text{diag}(I_r, S)$ , where  $S = D_0 - CA^{-1}B_0$  is a  $\rho \times s$  matrix vanishing at  $U$ . If  $r = 0$ , this means  $S = \mu$ , with no inverse block. The differential of  $S$  at  $U$  is precisely (26.1); the correction  $CA^{-1}B_0$  starts in degree two.

Let  $P_0 : H_U \rightarrow H_0$  and  $P_1 : H_U \rightarrow H_1$  be the projections. The map  $L^{-1}P_0S$  has differential of rank  $t$ . Complete its  $t$  coordinate functions to analytic local coordinates  $(y, x)$  on  $X$  using the inverse function theorem. Then  $P_0S = L(y)$  exactly, whereas  $\Psi = P_1S$  has no constant or linear term. This proves the claimed matrix expression. For a block diagonal matrix with an invertible  $r$ -block,

$$I_{r+j}(\text{diag}(I_r, S)) = I_j(S) \quad (1 \leq j \leq \rho).$$

To see both inclusions, a  $j$ -minor of  $S$  extends by the identity block to an  $(r+j)$ -minor; all other terms are minors of  $S$  of size at least  $j$  and belong to  $I_j(S)$ . This proves (26.4) and, with  $j = \rho$ , the analytic presentation of  $D$ . Completion gives (26.3).  $\square$

*Remark 26.3.* The power series  $\Psi$  is not a canonical function: complements, frames and coordinates enter its construction. The completed local ring and the nested determinantal ideals are canonical. Their compatibility on overlaps follows because each presentation is obtained from the original ideals  $I_{r+j}(\mu)$  by invertible operations. The theorem therefore supplies local equations also at support collisions and at points with several annihilators. It does not replace  $\Psi$  by its quadratic term or assert that a constant polar-rank stratum has a product neighbourhood.

The first nonzero residual terms can be computed without solving all the equations. For example, along a curve with parameter  $h$ , write

$$\mu(h) = \begin{pmatrix} A_0 + hA_1 + O(h^2) & hB_1 + h^2B_2 + O(h^3) \\ hC_1 + h^2C_2 + O(h^3) & hD_1 + h^2D_2 + O(h^3) \end{pmatrix}, \quad A_0 \text{ invertible.}$$

Expansion of the inverse gives

$$S(h) = hD_1 + h^2(D_2 - C_1A_0^{-1}B_1) + O(h^3). \quad (26.5)$$

For a velocity in  $\ker \partial_U$ , the class of the second coefficient in  $H_U/H_0$  is independent of the acceleration of the curve. An acceleration changes that coefficient by an element of  $\text{im } \partial_U$ . The same argument, using the chain rule and  $D_1 = 0$ , proves independence of coordinates and frames. Thus (26.5), paired with  $\mathcal{R}_U = (H_U/H_0)^*$ , is the intrinsic quadratic obstruction on  $\ker \partial_U$ . It describes the quadratic part of  $\Psi(0, x)$ ; it is not by itself the initial ideal of all the maximal minors.

**Corollary 26.4** (Transverse determinantal neighbourhoods). *If  $\mathcal{R}_U = 0$ , then  $M \geq \rho s$  and*

$$\widehat{\mathcal{O}}_{D,U} \simeq \mathbb{C}[[z_1, \dots, z_{M-\rho s}, (y_{ij})_{1 \leq i \leq \rho, 1 \leq j \leq s}]]/I_\rho((y_{ij})). \quad (26.6)$$

*This is also an analytic product decomposition. The germ is reduced, Cohen–Macaulay and of dimension  $M - a$ . If  $\rho = 1$  it is smooth; if  $\rho \geq 2$  its singular support is the product of the smooth factor with the rank-at-most- $(\rho - 2)$  generic matrix locus. This last assertion concerns reduced support, not equality of the singular-scheme ideal with  $I_{\rho-1}((y_{ij}))$ .*

*Proof.* Here  $t = \rho s$ ,  $H_1 = 0$ , and (26.3) has no nonlinear residual part. Choosing the entries of  $S$  as coordinates proves (26.6). For completeness, the generic rank-at-most- $(\rho - 1)$  matrix variety is irreducible: it is the closure of products of a  $\rho \times (\rho - 1)$  matrix and a  $(\rho - 1) \times s$  matrix. Its rank- $(\rho - 1)$  stratum has dimension  $(\rho - 1)(s + 1)$ , as one sees by choosing its image and a surjection onto it. Its codimension is  $s - \rho + 1 = a$ . The expected-grade maximal-minor resolution [8] consequently gives a Cohen–Macaulay quotient. At rank  $\rho - 1$ , Schur elimination leaves  $a$  independent entries, so the quotient is generically smooth and hence generically reduced. Cohen–Macaulayness gives  $S_1$ ;  $R_0 + S_1$  implies reducedness [24]. At rank at most  $\rho - 2$ , every first derivative of every maximal minor vanishes. The tangent dimension is the ambient dimension, strictly larger than the local dimension. These and only these points are singular. For  $\rho = 1$  the ideal consists of independent coordinates. The dimension and all product assertions follow.  $\square$

**26.3. Corank-one equations and their quadratic obstruction.** Corank one admits a smaller presentation than the general matrix normal form: all independent linear equations may be eliminated.

**Theorem 26.5** (Minimal residual equations in expected codimension). *Suppose  $\mu_U$  has corank one, put  $z = ([\ell], U)$  and  $\delta = \dim \ker \tau_z$ , and let  $d = M - a$ . There is an exact presentation*

$$\widehat{\mathcal{O}}_{D,U} \simeq \mathbb{C}[[x_1, \dots, x_{d+\delta}]]/(f_1, \dots, f_\delta), \quad f_i \in (x_1, \dots, x_{d+\delta})^2, \quad (26.7)$$

*and an analogous convergent presentation. If  $D$  has codimension  $a$  at  $U$ , these  $\delta$  equations form a regular sequence. In that case the displayed embedding dimension and number of equations are minimal. Without that codimension hypothesis the exact presentation still holds, but regularity of the sequence is not asserted.*

*The quadratic obstruction takes values in  $(\ker \tau_z)^*$ . For a tangent vector  $(\lambda, A) \in T_z \mathcal{I}$ , choose graph lifts of the maps  $A_\nu : U_\nu \rightarrow V/U_\nu$ . On a monomial  $u_1 \cdots u_m \in \text{Sym}^m U_\nu$  define*

$$\begin{aligned} \mathcal{H}_{\lambda,A}(u_1 \cdots u_m) &= \sum_{i < j} \ell \beta(u_1 \cdots A_\nu u_i \cdots A_\nu u_j \cdots u_m) \\ &\quad + \sum_i \lambda \beta(u_1 \cdots A_\nu u_i \cdots u_m). \end{aligned} \quad (26.8)$$

*Extend linearly to  $\mathcal{E}_U$ . Its restriction to  $\ker \tau_z$  is the intrinsic quadratic obstruction and equals the degree-two part of the residual equations, after identifying their coordinate space with  $(\ker \tau_z)^*$ .*

*Proof.* The incidence is the zero scheme of a rank- $e$  section on the smooth  $(M + p - 1)$ -dimensional variety  $\mathbb{P}(F^*) \times X$ . Its differential is

$$T_{[\ell]} \mathbb{P}(F^*) \oplus T_U X \longrightarrow \mathcal{E}_U^*, \quad (\lambda, A) \longmapsto \lambda \mu_U + B_z(A).$$

The annihilator of its image is  $\ker \mu_U \cap \ker B_z^* = \ker \tau_z$ ; thus its rank is  $e - \delta$ . Choose that many independent equations and eliminate them, together with the corresponding coordinates, by the analytic implicit function theorem. The resulting smooth space has dimension  $M + p - 1 - (e - \delta) = d + \delta$ . Subtracting appropriate linear combinations of the eliminated equations leaves  $\delta$  equations with zero differential. Projection from the incidence is a scheme isomorphism here by Theorem 25.1; this proves (26.7).

If the local codimension is  $a$ , this local ring has dimension  $d$ . In the regular local ring with  $d + \delta$  parameters its ideal, generated by  $\delta$  elements, has height  $\delta$ . Such a sequence is regular: a regular local ring is Cohen–Macaulay, and a height-equal-to-length generating sequence is a partial system of parameters, hence a regular sequence. Since all  $f_i$  are quadratic or higher, no further smooth coordinate can be eliminated, and a regular sequence of this length minimally generates the ideal.

To compute the quadratic terms, trivialize the projective line near  $[\ell]$  and use the graph path  $(\ell + h\lambda, u_i + hA_\nu u_i)$ . Expanding the incidence section, its coefficient of  $h^2$  is exactly (26.8). The order- $h$  coefficient vanishes because  $(\lambda, A)$  is tangent. A second-order change of path changes the coefficient of  $h^2$  by the image of the section's differential, which is annihilated by  $\ker \tau_z$ . Changes of graph lifts, projective trivialization or frames have precisely the same effect, by the chain rule and the vanishing first-order section. Thus the restriction is well defined. The nonlinear corrections used in eliminating the independent equations are again second-order changes of path and have zero pairing with this kernel. This proves the asserted identification with the degree-two residual equations.  $\square$

*Remark 26.6.* Theorem 26.5 identifies an actual complete-intersection germ in expected codimension, not just its tangent dimension. Its quadratic forms need not generate the full initial ideal. Higher-order terms in the  $f_i$  or in  $\Psi$  must be retained unless an additional normal-form argument removes them. The simple-normal-determinant argument below supplies such an argument on the residual wall. In the excess regime the same exact equations remain available without silently imposing expected grade.

**26.4. Realization in every symmetric degree.** The transverse class is not confined to two-point annihilators or to quadratic binary multiplication. The following explicit model realizes it for every symmetric degree and every possible multiplication corank.

**Proposition 26.7** (Realization of transverse matrix germs). *Let  $m \geq 2$ ,  $e \geq p \geq 1$  and  $1 \leq \rho \leq p$  be integers. There exist a vector space  $V$ , a surjection  $\beta : \text{Sym}^m V \rightarrow \mathbb{C}^p$ , and  $e$  one-dimensional contacts in  $X = \text{Gr}(1, V)^e$  for which the multiplication has corank  $\rho$  and  $\mathcal{R}_U = 0$ . Consequently its failure germ is the full generic  $\rho \times (e - p + \rho)$  maximal-minor germ, times a smooth factor, for every  $m$ .*

*Proof.* Put  $r = p - \rho$  and take  $V$  with basis  $x_1, \dots, x_e, w_1, \dots, w_\rho$ . Let  $f_1, \dots, f_p$  be the target basis. Define  $\beta$  on monomials by

$$\begin{aligned} \beta(x_i^m) &= f_i & (1 \leq i \leq r), \\ \beta(x_j^{m-1} w_\alpha) &= \frac{1}{m} f_{r+\alpha} & (r < j \leq e, 1 \leq \alpha \leq \rho), \end{aligned} \tag{26.9}$$

and set it equal to zero on all remaining basis monomials. The listed monomials are distinct, including when  $m = 2$ , so this defines a linear map. It is surjective because  $e - r = e - p + \rho \geq \rho > 0$ . At  $U_j = \mathbb{C}x_j$ , the source of the contact multiplication has basis  $(x_j^m)_j$ , its first  $r$  columns are  $f_1, \dots, f_r$ , and its remaining columns are zero. Thus  $K_U$  is spanned by the last  $s = e - r$  source vectors and  $C_U$  by  $f_{r+1}, \dots, f_p$ . Varying only the  $j$ th line from  $x_j$  toward  $w_\alpha$  has derivative

$$\left. \frac{d}{dh} \right|_{h=0} \beta((x_j + hw_\alpha)^m) = f_{r+\alpha}, \quad j > r.$$

These  $\rho s$  independent tangent directions map under  $\partial_U$  to every matrix unit in  $\text{Hom}(K_U, C_U)$ . The map is surjective, and Corollary 26.4 applies.  $\square$

This proposition concerns arbitrary symmetric linear systems. It exhibits genuine higher-degree failure germs and supplies concrete models for the intrinsic theorem. It is separate from, and does

not presuppose, a global classification for  $\text{Sym}^m U \rightarrow V_{mn}$  on the binary Grassmannian. The latter system's two-point sectors remain the subject of Section 32.

**26.5. Two explicit nontransverse residual germs.** These familiar singularities provide direct tests of the distinction between a residual space and the nonlinear residual equations.

A simultaneous-annihilator relation. For the bundle matrix on  $\mathbb{A}^3$

$$\mu = \begin{pmatrix} x & y \\ y & z \end{pmatrix},$$

the origin has  $K = C = \mathbb{C}^2$ . The intrinsic derivative is inclusion of the symmetric matrices into all  $2 \times 2$  matrices. Its transpose has the one-dimensional kernel generated by  $e_1 \otimes e_2^* - e_2 \otimes e_1^*$ . For each fixed nonzero  $\ell \in C^*$ , however, the map  $K \rightarrow T_0^* \mathbb{A}^3$ ,  $q \mapsto \partial^*(q \otimes \ell)$ , is injective: its coefficients are those of the product of two nonzero linear binary forms. Thus checking one annihilator at a time would miss the joint relation. In Theorem 26.2,  $H_0$  is the symmetric subspace and  $\Psi = 0$ ; the exact failure germ is  $xz - y^2 = 0$ . This example is a bundle-matrix example for the intrinsic theorem, not an additional claim about binary subseries.

A cubic multiplication cusp. Let  $V = \mathbb{C}X \oplus \mathbb{C}Y \oplus \mathbb{C}Z$ ,  $F = \mathbb{C}$ , and define  $\beta : \text{Sym}^3 V \rightarrow F$  by  $\beta(Y^2Z) = 1/3$ ,  $\beta(X^3) = -1$ , and zero on the other basis monomials. Take one line contact, and use the chart  $U = \mathbb{C}(Z + xX + yY)$  near  $\mathbb{C}Z$ . Its multiplication matrix is the one-entry matrix

$$\beta((Z + xX + yY)^3) = y^2 - x^3.$$

At the origin  $\delta = 1$  and the quadratic obstruction is  $y^2$ , whereas the exact residual equation is  $y^2 - x^3$ . The map from its completed local ring to  $\mathbb{C}[[t]]$  given by  $x = t^2$ ,  $y = t^3$  is injective: reducing a power series modulo  $y^2 - x^3$  leaves  $A(x) + yB(x)$ , and the even powers in  $A(t^2)$  cannot cancel the odd powers in  $t^3B(t^2)$ . Thus the cusp has one branch, unlike the double crossing. This explicit higher-product system shows why the full residual equation, rather than only its quadratic part, occurs in the theorem.

## 27. THE GLOBAL MULTIPLICATION FAILURE LOCUS

Throughout this section the field is  $\mathbb{C}$ ,  $k = n + 1$ ,  $4 \leq c < k$ ,  $q = \binom{c+1}{2}$ , and  $p = 2k - 1$ . On  $X_L = \text{Gr}(c, V_n)^L$  let  $\mathcal{S}_\nu$  denote the tautological bundles. The failure scheme is defined by the maximal minors of

$$\mu : \bigoplus_{\nu=1}^L \text{Sym}^2 \mathcal{S}_\nu \longrightarrow V_{2n} \otimes \mathcal{O}_{X_L}.$$

Its underlying set is  $\Delta^{(L)}$ . We shall prove Theorem 24.1 without using a maximal-rank theorem for general curves.

**27.1. Annihilating Hankel forms.** For  $\ell \in V_{2n}^*$ , let  $H_\ell(f, g) = \ell(fg)$  on  $V_n$ . The following classical input concerns the whole rank locus, including confluent and limiting forms; it is not an assertion only about sums of distinct point evaluations.

**Lemma 27.1** (Classical Hankel rank strata). *In  $\mathbb{P}(V_{2n}^*)$ , the rank-exactly- $r$  stratum of  $H_\ell$  is nonempty and has dimension*

$$h_r = \begin{cases} 2r - 1, & 1 \leq r < k, \\ 2k - 2, & r = k. \end{cases}$$

*For  $r < k$ , its closure is the variety of sums of at most  $r$  point evaluations, with Zariski closure understood.*

*Identification with the classical statement.* The rank- $\leq r$  equations are the  $(r+1)$ -minors of the  $k \times k$  Hankel matrix in  $2k-1$  independent coordinates. The equality of Hankel minor ideals under changing the rectangular shape, recalled in [7, Section 1], identifies their coordinate ring with that of the maximal minors of an  $(r+1) \times (2k-r-1)$  Hankel matrix. That ring has affine dimension  $2r$ . The secant identification is [7, Section 2, equation (2.0.1)]; thus the projective dimension is  $2r-1$ . The rank- $\leq r-1$  subvariety has smaller dimension, so the exact-rank stratum is nonempty and dense. The full-rank stratum is open in  $\mathbb{P}^{2k-2}$ . These are classical Hankel determinantal facts; no claim about their rational singularities is needed here.  $\square$

**Lemma 27.2** (Isotropic incidence dimensions). *Let a symmetric form on a  $k$ -space have rank  $r$  and radical  $E$  of dimension  $k-r$ . For a totally isotropic  $c$ -plane  $U$ , put*

$$s = \dim((U + E)/E), \quad a = c - s.$$

*Such planes exist precisely for*

$$\max\{0, c - k + r\} \leq s \leq \min\{c, \lfloor r/2 \rfloor\}. \quad (27.1)$$

*The stratum with this value of  $s$  has codimension*

$$D_{r,s} = (c-s)(r-s) + \frac{s(s+1)}{2} \quad (27.2)$$

*in  $\text{Gr}(c, k)$ .*

*Proof.* Choose  $U \cap E$ , an  $a$ -plane in  $E$ , with dimension  $a(k-r-a)$ . The image of  $U$  in the nondegenerate  $r$ -dimensional quotient is an isotropic  $s$ -plane, whose Grassmannian has dimension  $s(r-s) - s(s+1)/2$ . This follows by differentiating the symmetric restriction equations at an isotropic frame: pairing with a complementary dual frame gives all symmetric variations, so the equations are independent. Existence is equivalent to  $2s \leq r$ . Finally the lifts are an affine space of dimension  $s(k-r-a)$ . Subtracting the sum from  $c(k-c)$  gives (27.2). Maximal orthogonal Grassmannians may have two components; both have this same dimension, which is all the count uses.  $\square$

**Lemma 27.3** (Rank optimization). *Assume  $Lq \geq 2k-1$ . For all  $r, s$  satisfying (27.1),*

$$LD_{r,s} - h_r \geq \min\{Lq - 2k + 2, Lc - 3\}. \quad (27.3)$$

*Proof.* Write  $A_0 = Lq - 2k + 2$  and  $B_0 = Lc - 3$ . If  $r = k$ , feasibility forces  $s = c$ , giving exactly  $A_0$ . Suppose  $r < k$ . For  $s = 0$ , the left side is  $(Lc - 2)r + 1 \geq Lc - 1 > B_0$ .

For  $1 \leq s \leq c-2$ , the coefficient of  $r$  in  $LD_{r,s} - 2r + 1$  is  $L(c-s) - 2 \geq 0$ . Since  $r \geq 2s$ , the expression is at least

$$g(s) = 1 + s(Lc - 4) - \frac{Ls(s-1)}{2}.$$

This is concave as a function of  $s$ , and

$$g(1) = B_0, \quad g(c-2) - B_0 = (c-3) \left( \frac{L(c+2)}{2} - 4 \right).$$

The latter is nonnegative except when  $(L, c) = (1, 4)$  or  $(1, 5)$ . In the first exception, saturation forces  $k = 5$  and feasibility forces  $s \leq 1$ . In the second,  $k \leq 8$ ; the values  $g(1), g(2), g(3)$  are 2, 2, 1. The last value is feasible only when  $k = 8$ , in which case  $A_0 = 1$ . Both exceptions therefore satisfy (27.3).

For  $s = c-1$  and  $L = 1$ , the expression decreases with  $r$ , hence is at least its value at  $r = k-1$ , namely  $A_0 + k - 2c + 1$ . Feasibility implies  $k \geq 2c-1$ . For  $s = c-1$  and  $L \geq 2$ , it is minimized at  $r = 2c-2$  and exceeds  $B_0$  by

$$(c-2) \left( \frac{L(c+1)}{2} - 4 \right) \geq 0.$$

Finally, if  $s = c$  and  $r < k$ , it is  $Lq - 2r + 1 \geq A_0 + 1$ . These cases exhaust (27.1).  $\square$

**27.2. The common-secant family.** Let  $C_n \subset \mathbb{P}(V_n^*)$  be the evaluation rational normal curve, and let  $S_n$  be the closure of its secant lines. Since  $n \geq 4$ ,  $S_n$  is irreducible of dimension three. Define

$$\mathfrak{S}_L = \{([h], U_1, \dots, U_L) : [h] \in S_n, U_\nu \subset \ker h\}, \quad \Sigma_L = \text{pr}_{X_L} \mathfrak{S}_L. \quad (27.4)$$

**Proposition 27.4** (A global secant obstruction). *The closed set  $\Sigma_L$  is irreducible, has codimension  $Lc - 3$  in  $X_L$ , and lies in  $\Delta^{(L)}$ .*

*Proof.* The incidence is a relative product of Grassmannians over  $S_n$ , so it is irreducible of dimension  $3 + Lc(k - 1 - c)$  and its projection is closed. For a general  $h = \alpha \text{ev}_x + \beta \text{ev}_y$ , with  $x \neq y$  and  $\alpha\beta \neq 0$ , every  $f \in \ker h$  satisfies  $\alpha f(x) = -\beta f(y)$ . Consequently

$$\ell = \alpha^2 \text{ev}_x^{(2n)} - \beta^2 \text{ev}_y^{(2n)}$$

is nonzero and annihilates all products from each  $U_\nu$ . Closedness extends this conclusion to the whole incidence.

It remains to check the dimension of the image, rather than merely that of the incidence. Fix a general  $h$ . If  $c = k - 1$ , the hyperplane  $U_1$  determines  $h$ . Otherwise, the  $c$ -planes in  $\ker h$  that are also annihilated by some  $h' \in S_n \setminus \{h\}$  form a constructible set of dimension at most

$$3 + c(k - 2 - c) < c(k - 1 - c),$$

because  $c \geq 4$ . Here, for each  $h'$ , the plane lies in the intersection of two distinct hyperplanes. Thus a general  $U_1 \subset \ker h$  admits no such  $h'$ , and the incidence projection is generically one-to-one. Subtraction from  $\dim X_L = Lc(k - c)$  gives  $Lc - 3$ .  $\square$

*Proof of Theorem 24.1.* Failure is equivalent to the existence of a nonzero  $\ell$  annihilating every  $U_\nu^2$ . Consider the closed projective incidence of such pairs  $((U_\nu), [\ell])$ . On the rank- $r$  stratum of  $H_\ell$ , stratify each  $U_\nu$  by  $s_\nu$ . Lemmas 27.1 and 27.2 bound the incidence dimension by

$$\dim X_L + h_r - \sum_\nu D_{r, s_\nu}.$$

Its image cannot have larger dimension. For fixed  $r$ , each term is at least the smallest feasible  $D_{r, s}$ , so Lemma 27.3 gives

$$\text{codim } \Delta^{(L)} \geq \min\{Lq - p + 1, Lc - 3\} \geq 1.$$

This proves, in particular, generic surjectivity without assuming it.

Proposition 27.4 gives the upper bound  $Lc - 3$  and shows that the failure set is nonempty. On a Grassmannian chart the map  $\mu$  is a  $p \times Lq$  matrix over a regular ring. The classical maximal-minor height bound gives codimension at most  $Lq - p + 1$  for its nonempty failure locus; see [9, Introduction], with generic rank  $p$  and minor size  $p$ . The two upper bounds match the lower bound. When  $Lc - 3$  is the minimum, the irreducible closed subvariety  $\Sigma_L$  has the maximal dimension of  $\Delta^{(L)}$ , so it is an irreducible component of its underlying reduced set. If  $Lq < p$ , the dimension of the domain of  $\mu$  makes surjectivity impossible. This proves the final equivalence.  $\square$

**Corollary 27.5** (The base-point locus and the contact count). *For  $L = 1$ , the locus  $\mathcal{B}$  of spaces with a common projective zero is irreducible of codimension  $c - 1$ , has multiplication corank at least two, and is properly contained in  $\Sigma_1$ . It is not an irreducible component of  $\Delta^{(1)}$ . For arbitrary  $L$ , the least number of independently chosen  $c$ -planes with full product span is*

$$L_{\min} = \left\lceil \frac{2k - 1}{\binom{c+1}{2}} \right\rceil.$$

*Proof.* For a fixed linear factor  $l$ , common-base-point spaces form  $\text{Gr}(c, lV_{n-1})$ . Varying  $[l] \in \mathbb{P}^1$  gives an irreducible incidence of dimension  $1 + c(k-1-c)$ , generically one-to-one because a general residual space has no further common factor. The codimension is  $c-1$ . Products are divisible by  $l^2$ , whose multiples have dimension  $p-2$ . The annihilating hyperplane is an evaluation point of  $C_n \subset S_n$ , proving inclusion in  $\Sigma_1$ . The codimensions  $c-1$  and  $c-3$  make that inclusion proper. An irreducible proper subset of the irreducible closed set  $\Sigma_1 \subset \Delta^{(1)}$  cannot itself be a component of  $\Delta^{(1)}$ . The contact count follows from Theorem 24.1 and the dimension obstruction.  $\square$

The theorem determines the exact dimension of the entire bad set and one explicit maximal-dimensional component in the stated regime. It does not assert that every component is a secant component, nor that the determinantal scheme is reduced, equidimensional, or everywhere transverse. In particular, the equality case of the two codimension bounds does not exclude additional components of the same dimension.

## 28. RESIDUAL CONORMALS AND EXACT LOCAL EQUATIONS

We work over  $\mathbb{C}$ . Put  $E_\mu = \bigoplus_\nu \text{Sym}^2 U_\nu$ ,  $M = Lc(k-c)$ ,  $e = L\binom{c+1}{2}$ ,  $p = 2k-1$ , and

$$a = e - p + 1, \quad b = Lc - 3.$$

When  $e \geq p$ , write  $\mathcal{D}_L$  for the scheme of maximal minors of  $\mu : E_\mu \rightarrow V_{2n}$ , and  $\mathfrak{J}$  for the incidence  $\ell\mu = 0$  in  $\mathbb{P}(V_{2n}^*) \times X_L$ . The following result is not restricted to a general point or to a stable range of dimensions.

**Theorem 28.1** (Residual tangent sequence). *Let  $([\ell], U) \in \mathfrak{J}$ , let  $E = \text{rad } H_\ell$ , and put*

$$W_\nu = U_\nu \cap E, \quad d_\nu = \dim W_\nu, \quad R_W = \sum_\nu W_\nu^2 \subset V_{2n}, \quad q = \binom{c+1}{2}.$$

*For each  $\nu$  define*

$$B_\nu : \text{Hom}(U_\nu, V_n/U_\nu) \rightarrow \text{Sym}^2 U_\nu^*, \quad B_\nu(A)(u, v) = H_\ell(Au, v) + H_\ell(u, Av).$$

*Then  $\text{im } B_\nu$  consists exactly of the symmetric forms whose restriction to  $W_\nu \times W_\nu$  is zero. There is an exact sequence*

$$0 \rightarrow \bigoplus_\nu \ker B_\nu \rightarrow T_{([\ell], U)} \mathfrak{J} \rightarrow R_W^\perp / \mathbb{C}\ell \rightarrow 0. \quad (28.1)$$

*In particular*

$$\dim T\mathfrak{J} = M + p - 1 - \sum_\nu \left( q - \binom{d_\nu + 1}{2} \right) - \dim R_W. \quad (28.2)$$

*If  $\mu_U$  has corank one, these are also the tangent space and tangent dimension of  $\mathcal{D}_L$  at  $U$ ; its conormal rank in  $X_L$  is*

$$\sum_\nu \left( q - \binom{d_\nu + 1}{2} \right) - p + 1 + \dim R_W. \quad (28.3)$$

*Proof.* The map  $V_n/U_\nu \rightarrow U_\nu^*$  induced by  $H_\ell$  has image  $(U_\nu/W_\nu)^*$ . Indeed its transpose has kernel  $W_\nu$ ; quotienting by  $U_\nu$  is permitted because  $U_\nu$  is isotropic. Choose a complement  $Z_\nu$  of  $W_\nu$  in  $U_\nu$  and lifts of a dual basis of  $Z_\nu^*$ . The coordinates of  $A|_{W_\nu}$  in these lifts prescribe every  $W_\nu \times Z_\nu$  entry. The coordinates of  $A|_{Z_\nu}$  prescribe every symmetric  $Z_\nu \times Z_\nu$  entry, by taking one half of the desired symmetric matrix. This proves the assertion about  $B_\nu$ , including its rank.

Differentiating  $\ell(uv) = 0$  gives

$$\dot{\ell}(uv) + B_\nu(\dot{U}_\nu)(u, v) = 0.$$

It is solvable for the  $\dot{U}_\nu$  precisely when  $\dot{\ell}$  annihilates all  $W_\nu^2$ . Solutions for different contacts are independent. Taking  $\dot{\ell}$  modulo  $\mathbb{C}\ell$  proves the exact sequence and its dimension. If  $\mu_U$  has corank one, invert a  $(p-1)$ -minor. Row and column operations leave an identity block and one row of  $a$  entries. The normalized left kernel vector is uniquely solved by the identity block; the remaining incidence equations are exactly these  $a$  entries. Hence the incidence projects isomorphically, as a scheme, to  $\mathcal{D}_L$  on this chart. Subtracting the tangent dimension from  $M$  gives the conormal rank.  $\square$

**Proposition 28.2** (Local presentation at every corank). *At a point where  $\mu$  has corank  $\rho \geq 1$ , invert a  $(p-\rho)$ -minor and write its matrix in blocks as  $\begin{pmatrix} A & B \\ C & D \end{pmatrix}$ . On this open chart, and also after completion of its local ring,*

$$I_p(\mu) = I_\rho(D - CA^{-1}B). \quad (28.4)$$

*The matrix on the right has size  $\rho \times (e - p + \rho)$  and vanishes at the given point.*

*Proof.* Multiply by the invertible block matrices eliminating  $C$  and  $B$ , and then by  $A^{-1}$  on the first block. The resulting matrix is  $\text{diag}(I_{p-\rho}, D - CA^{-1}B)$ , in rectangular block notation. Its maximal minors generate the indicated ideal. Invertible row and column operations preserve determinantal ideals, and localization and completion preserve this identity.  $\square$

This is a scheme presentation, not a replacement of its tangent cone by minors of the linear part. Initial ideals can contain additional initial forms obtained from cancellations. Nor does the presentation by itself classify the associated primes of every excess-dimensional germ.

**28.1. The rank-two residual morphism.** Let  $[\ell]$  have rank two and two distinct support points  $x, y$ . Choose the two isotropic hyperplanes  $H_+, H_-$  of  $H_\ell$  and their intersection  $E = gV_{n-2}$ , where  $g$  vanishes at  $x, y$ . Fix a sign pattern  $\varepsilon$  and consider the open incidence on which  $U_\nu \subset H_{\varepsilon_\nu}$  and  $U_\nu \not\subset E$ .

**Proposition 28.3** (Dominance with affine fibres). *For fixed  $[\ell]$  and choices of its hyperplanes, the morphism*

$$(U_\nu)_\nu \longmapsto (g^{-1}(U_\nu \cap E))_\nu \quad \text{to } \text{Gr}(c-1, V_{n-2})^L \quad (28.5)$$

*is smooth and surjective. It is locally an affine-space bundle of relative dimension  $L(k-c-1)$ . The same assertion holds relatively as the distinct support points and their two hyperplanes vary, after trivializing the indicated bundles.*

*Proof.* For a prescribed  $W_\nu = gK_\nu$ , an admissible  $U_\nu$  is the inverse image of a line in  $H_{\varepsilon_\nu}/W_\nu$  projecting isomorphically to  $H_{\varepsilon_\nu}/E$ . Its choices form a torsor under

$$\text{Hom}(H_{\varepsilon_\nu}/E, E/W_\nu),$$

an affine space of dimension  $k-c-1$ . Local choices of a splitting trivialize this torsor. Intersections with  $E$  have constant dimension  $c-1$ , so the assignment is a morphism. Products give the result.  $\square$

**Corollary 28.4** (Exact singularity recursion on a signed stratum). *At a rank-two point as above, suppose that  $\mu_U$  has corank one. Let  $\mu'_K$  be multiplication from  $\bigoplus_\nu \text{Sym}^2 K_\nu$  to  $V_{2n-4}$  and put  $\delta = \dim \text{coker } \mu'_K$ . Then*

$$\dim T_U \mathcal{D}_L = M - b + \delta. \quad (28.6)$$

*If  $4 \leq c < k$ ,  $e \geq p$  and  $b \leq a$ , this point is smooth in  $\mathcal{D}_L$  if and only if  $\mu'_K$  is onto. In this range the singular set on the stated corank-one open stratum is exactly the inverse image of the residual multiplication failure locus.*

*Proof.* Here  $d_\nu = c - 1$  and  $R_W = g^2 \operatorname{im} \mu'_K$ , of dimension  $p - 4 - \delta$ . Equation (28.2) gives the displayed formula. The signed incidence has image of dimension  $M - b$ : a general first  $c$ -plane in its secant hyperplane determines that hyperplane uniquely, since the locus also lying in a second secant hyperplane has dimension at most  $3 + c(k - 2 - c) < c(k - 1 - c)$ . On the corank-one open set the local signed incidence is embedded by the incidence isomorphism. Thus its local dimension is  $M - b$ . Theorem 24.1 bounds the dimension of the entire failure locus by  $M - b$  when  $b \leq a$ . Consequently its local dimension here is  $M - b$ , and equality of tangent and local dimensions is equivalent to  $\delta = 0$ .  $\square$

## 28.2. Projection and commutative algebra.

**Proposition 28.5** (From rank incidences to components). *Stratify the annihilator incidence by the rank  $r$  of  $H_\ell$  and by  $s_\nu = \dim(U_\nu / (U_\nu \cap \operatorname{rad} H_\ell))$ . Every irreducible component  $C$  of the support of  $\mathcal{D}_L$  is dominated by the closure of an irreducible piece of one of these finitely many strata. In particular a stratum of dimension less than  $\dim C$  cannot dominate  $C$ . If a dominating incidence has dimension  $\dim C$ , and its generic fibre contains a nonempty open subset of the projective annihilator space, then that annihilator space is a point and its projection is birational in characteristic zero.*

*Proof.* The projective incidence maps onto the failure locus. Intersect the images of its finitely many locally closed strata and their irreducible pieces with  $C$ . These constructible sets cover  $C$ ; at least one is dense. The dimension theorem gives the first assertion and implies that the generic fibre of an equal-dimensional dominating piece is zero-dimensional. The projective annihilator space is the projectivization of a vector space, not an arbitrary finite collection. A nonempty open subset of a positive-dimensional projective space has positive dimension. Thus the open-fibre hypothesis forces the projective space to be a point. On a corank-one chart the incidence is the graph of its uniquely solved annihilator. This proves birationality, including the scheme-theoretic generic-point assertion.  $\square$

**Proposition 28.6** (Expected grade and absence of embedded primes). *Suppose  $4 \leq c < k$ ,  $e \geq p$ , and  $a \leq b$ . On every nonempty affine Grassmannian chart meeting  $\mathcal{D}_L$ , its maximal-minor ideal has height and grade  $a$ . The Eagon–Northcott complex resolves its quotient, which is Cohen–Macaulay of pure codimension  $a$  and has no embedded associated primes.*

*Proof.* The chart ring is regular. The global codimension theorem gives height at least  $a$ , whereas the maximal-minor height bound gives height at most  $e - p + 1 = a$ . In a regular ring grade equals height. The Eagon–Northcott grade criterion gives a free resolution of length  $a$  and the Cohen–Macaulay conclusion [8, 19]. Localization gives the same conclusion at every prime over the ideal. A Cohen–Macaulay quotient of this expected dimension is unmixed and has no embedded associated primes. No independence assumption on the matrix entries is used.  $\square$

**Proposition 28.7** (The reducedness step). *In the situation of Proposition 28.6, if  $\mathcal{D}_L$  is generically reduced on each of its irreducible components, then it is reduced. If its support is also irreducible, it is integral.*

*Proof.* A Noetherian ring without embedded associated primes satisfies  $S_1$ . Generic reducedness is  $R_0$ . The criterion  $R_0 + S_1$  for reducedness therefore applies [24, Tag 031R]. Alternatively a nonzero nilradical would have an associated prime; without embedded primes this would be a minimal prime, contradicting generic reducedness. The last assertion is the definition of an integral scheme.  $\square$

## 29. ORIENTATION COVERS AND DESCENT OF ISOTROPIC FAMILIES

We give the descent statements used in the component theorem explicitly. They distinguish connectedness of a cover from geometric irreducibility of a fibre, and also distinguish one maximal-isotropic factor from a product of such factors.

**Proposition 29.1** (Relative orthogonal descent). *Let  $S$  be a smooth integral complex variety and let  $(\mathcal{V}, q)$  be a nondegenerate quadratic bundle of rank  $2s$ , with values in a line bundle. Its orientation cover  $\tilde{S} \rightarrow S$  is the degree-two étale cover whose local equation, after choosing frames, is  $z^2 = (-1)^s \det q$ . Changes of frames glue these local covers. Suppose this cover is connected. The relative maximal orthogonal Grassmannian  $\text{OGr}_S(s, \mathcal{V})$  is integral. More generally, the product of  $L$  such Grassmannians for the same quadratic bundle has  $2^{L-1}$  irreducible components, indexed by  $\{+, -\}^L$  modulo simultaneous sign change. Each component is smooth over  $S$ . Adding relative nonmaximal orthogonal Grassmannian factors does not change this count.*

*Proof.* If a frame of  $\mathcal{V}$  changes by  $A$ , its determinant changes by  $(\det A)^2$ . If the frame of the value line changes by  $u$ , the determinant changes by  $u^{-2s}$ . Thus the square-root equations are compatible with the transition functions of  $\det \mathcal{V}^* \otimes (\text{value line})^s$ . Their zero-free right-hand side makes the cover étale. After an étale base change the form has a hyperbolic basis  $e_1, \dots, e_s, f_1, \dots, f_s$ .

Over an algebraically closed field a maximal isotropic plane belongs to one of two families, distinguished by the parity of its intersection dimension with  $\text{span}(e_1, \dots, e_s)$ . A Witt basis sends any such plane to the reference plane by an orthogonal transformation. The transformations of determinant one preserve this parity and act transitively on each family; interchanging  $e_1$  and  $f_1$  reverses it. The special orthogonal group is connected: in a hyperbolic basis its elementary orthogonal root subgroups are affine lines, its diagonal torus is a product of multiplicative groups, and these generate the group by the usual elimination of a Witt basis. Therefore each of its two orbits is irreducible. Equivalently, the graph charts within a family are alternating matrices and are translated under this connected group. The same Witt-basis elimination gives a single irreducible orbit for isotropic planes of dimension strictly less than half the rank.

The two families are consequently locally constant étale labels, with the determinant character describing their interchange. The orientation cover kills this character. Over  $\tilde{S}$  each family is a smooth proper relative variety with geometrically irreducible fibres. It is integral: smoothness gives reducedness, every component dominates the integral base, and the geometric generic fibre has only one component. Here domination follows also from local smoothness and properness: each component is open and closed in the smooth total space, so its nonempty image is open and closed in the connected base. The nontrivial deck transformation exchanges the two families. Their images in the descended space are therefore the same irreducible image, which is the whole space. This proves integrality for one factor.

For  $L$  factors the geometric components over  $\tilde{S}$  are the  $2^L$  products of these integral family bundles. The deck action sends  $\varepsilon$  to  $-\varepsilon$ . The orbit unions descend to the irreducible components: the image of any one integral product is irreducible and equals the image of its entire orbit, and distinct orbits have distinct geometric generic fibres. There are  $2^{L-1}$  orbits. Smoothness descends étale locally. Nonmaximal factors have geometrically irreducible fibres and add no labels.  $\square$

**Proposition 29.2** (The five exceptional orientation bases). *For each exceptional tuple of Theorem 30.1, the incidence containing the full radical and a maximal isotropic quotient is irreducible. For the full-rank case  $(c, k, L) = (7, 14, 1)$  the nondegenerate isotropic incidence is irreducible as well.*

*Proof.* For an exceptional rank  $r = 2s < k$ , work first over the dense ordered support chart

$$T_r = \{(x_1, \dots, x_r, \alpha_1, \dots, \alpha_{r-1}) : x_i \neq x_j, \alpha_i \neq 0\}, \quad \alpha_r = 1.$$

This is a smooth integral open subset of  $\mathbb{A}^r \times \mathbb{G}_m^{r-1}$ . It parameterizes the projective functionals  $[\ell] = [\sum_i \alpha_i \text{ev}_{x_i}]$ . Evaluation identifies the quotient by the radical with  $\mathbb{C}^r$ , and the induced quadratic form is  $\text{diag}(\alpha_1, \dots, \alpha_r)$ . In the function field  $\mathbb{C}(x_1, \dots, x_r, \alpha_1, \dots, \alpha_{r-1})$ , the product  $\prod_{i < r} \alpha_i$  is not a square: its valuation at  $\alpha_1 = 0$  is one. This valuation is a function-field valuation; the boundary  $\alpha_1 = 0$  need not belong to  $T_r$ . The orientation cover obtained by adjoining its square root is therefore integral and connected. Proposition 29.1 applies.

The radical is  $gV_{k-r-1}$  with  $g = \prod_i (t - x_i)$ . Containing this radical identifies the exceptional plane incidence with the maximal orthogonal Grassmannian of the quotient. There is one such factor in each of the five cases. The ordered chart maps dominantly to the corresponding exact-rank base. Its generic fibres merely relabel the support: the radical determines  $g$ , its distinct roots recover the unordered points, and a Vandermonde system recovers the weights up to common scale. Taking the unordered quotient cannot split the irreducible image of this incidence. Its closure is the exceptional incidence claimed in the theorem. Colliding-support points do not create another component of this closure.

In the remaining full-rank case the universal quadratic form has even rank fourteen. The determinant of the universal Hankel matrix has the irreducible corank-one Hankel hypersurface as its zero divisor, with multiplicity one. The irreducibility follows from the Hankel maximal-minor description in Section 27; simplicity can also be checked at a corank-one form represented by thirteen distinct evaluations. Its kernel is generated by a nonzero polynomial  $v$ ; the derivative of the determinant in a Hankel direction  $\ell$  is a nonzero scalar multiple of  $\ell(v^2)$ , which is not identically zero. On an affine projective chart meeting its generic point this simple prime divisor proves nonsquareness in the function field. Scaling the representative of a rank-fourteen form changes the determinant by a fourteenth power, hence by a square, so this is the projective orientation cover. It is connected and Proposition 29.1 gives the assertion.  $\square$

### 30. COMPONENTS IN ALL DIMENSIONS

We retain the entire range  $4 \leq c < k$  and  $Lq \geq p$  of Theorem 24.1. Let  $\Phi_L$  be the closure of the image of the incidence with nonsingular  $H_\ell$ . Let  $\Sigma_\varepsilon$  denote the signed secant image: for distinct  $x, y$  and  $t \neq 0$  its points satisfy

$$U_\nu \subset \ker(\text{ev}_x + \varepsilon_\nu t \text{ev}_y), \quad \varepsilon \in \{+1, -1\}^L / \{\pm 1\}.$$

Only in the five triples in Table 1 do we need an additional notation. For those triples,  $\mathcal{T}_{c,k}$  is the closure of the image of the rank- $r$  incidence indicated in the table. Equivalently, on its dense distinct-support open set,

$$E = \text{rad } H_\ell \subset U, \quad U/E \text{ is a maximal isotropic subspace of } V_n/E.$$

Thus these exceptions have one uniform geometric description; they are not five unrelated numerical accidents.

| $c$ | $k$ | $L$ | $(a, b)$ | $r$ | $\dim E$ | $\dim(U/E)$ |
|-----|-----|-----|----------|-----|----------|-------------|
| 5   | 7   | 1   | (3, 2)   | 4   | 3        | 2           |
| 5   | 8   | 1   | (1, 2)   | 6   | 2        | 3           |
| 6   | 10  | 1   | (3, 3)   | 8   | 2        | 4           |
| 6   | 11  | 1   | (1, 3)   | 10  | 1        | 5           |
| 7   | 13  | 1   | (4, 4)   | 12  | 1        | 6           |

TABLE 1. All nonendpoint equality incidences.

**Theorem 30.1** (All-dimension component theorem). *Suppose  $4 \leq c < k$  and  $Lq \geq p$ .*

- (1) If  $a < b$ , the scheme  $\mathcal{D}_L$  is integral and Cohen–Macaulay of codimension  $a$ . Its support is  $\Phi_L$ , except at  $(c, k, L) = (5, 8, 1), (6, 11, 1)$ , where its support is  $\mathcal{T}_{5,8}, \mathcal{T}_{6,11}$ , respectively.
- (2) If  $a = b$ , the scheme is reduced and Cohen–Macaulay of pure codimension  $a$ . Its components are  $\Phi_L$  and the  $2^{L-1}$  varieties  $\Sigma_\varepsilon$ , except at  $(6, 10, 1), (7, 13, 1)$ , where  $\Phi_L$  is replaced by  $\mathcal{T}_{6,10}, \mathcal{T}_{7,13}$ .
- (3) If  $b < a$ , the maximal-dimensional components are exactly the  $2^{L-1}$  signed secant varieties, except at  $(5, 7, 1)$ , where  $\mathcal{T}_{5,7}$  is one additional maximal-dimensional component. Their codimension is  $b$ .

Every component listed above has general multiplication corank one and is generically smooth in the maximal-minor scheme. Each of its indicated annihilator incidences is birational onto it. The generic Hankel rank is  $k$  on  $\Phi_L$ , two on  $\Sigma_\varepsilon$ , and the rank in Table 1 on  $\mathcal{T}_{c,k}$ . The signed components are rational. In the range  $a \leq b$ ,

$$[\mathcal{D}_L] = c_a \left( \left( \bigoplus_{\nu} \text{Sym}^2 \mathcal{S}_{\nu} \right)^* \right) \quad \text{in } A^a(X_L). \quad (30.1)$$

The third assertion concerns maximal-dimensional components, not all associated primes in the excess range. The theorem removes the stable restriction from the component and expected-codimension scheme assertions; it does not identify distinct notions of completeness.

### 30.1. The complete equality analysis.

**Lemma 30.2** (All endpoint ties). *Among the rank incidences in the proof of Theorem 24.1, an incidence of dimension at least  $M - \min(a, b)$  is either the rank-two incidence with every  $s_{\nu} = 1$ , the full-rank incidence with every  $s_{\nu} = c$ , or one of the five incidences in Table 1. The dimension in each listed case is exactly  $M - \min(a, b)$  when its corresponding endpoint is minimal.*

*Proof.* Use  $D_{r,s}$  and  $h_r$  in Lemmas 27.1 and 27.2. For  $r < k$  and  $s = 0$  the cost is  $(Lc - 2)r + 1 > b$ . For  $1 \leq s \leq c - 2$ , minimization in  $r \geq 2s$  uses the nonnegative coefficient  $L(c - s) - 2$  and gives

$$g(s) = 1 + s(Lc - 4) - Ls(s - 1)/2, \quad g(1) = b, \quad g(c - 2) - b = (c - 3)\{L(c + 2)/2 - 4\}.$$

If  $L \geq 2$ , the last difference is strictly positive, and strict concavity excludes the intervening integers. The coefficient for  $s = 1$  is positive, so equality there requires  $r = 2$ . For  $s = c - 1$ ,  $L \geq 2$ , the cost at its minimum exceeds  $b$  by  $(c - 2)\{L(c + 1)/2 - 4\} > 0$ . For  $s = c$  and  $r < k$ , the cost is at least  $a + 1$ . Thus there are no extra ties for  $L \geq 2$ .

It remains to take  $L = 1$ . Saturation gives  $k \leq (q + 1)/2$ . For  $c = 4$ , this forces  $k = 5$  and feasibility permits only  $s \leq 1$  when  $r < k$ . For  $c = 5$ , the values at  $s = 1, 2, 3$  are 2, 2, 1. At  $s = 2$  equality requires  $r = 4$ , and feasibility together with the minimal endpoint gives  $k = 7$ . At  $s = 3$  the coefficient of  $r$  is zero; feasibility forces  $r = 6, k = 8$ . For  $c = 6$ , the only further lower-endpoint tie is  $s = 4, r = 8, k = 10$ . For  $c \geq 7$ , strict concavity and the displayed endpoint difference exclude every  $2 \leq s \leq c - 2$ .

For  $s = c - 1$  the cost is decreasing in  $r$  and is bounded below by  $a + k - 2c + 1$ . Feasibility requires  $k \geq 2c - 1$ . When  $k \geq 2c$ , this is strictly greater than  $a$ . When  $k = 2c - 1$ , feasibility forces  $r = 2c - 2$ , and

$$a - b = (c - 7)(c - 2)/2.$$

Saturation permits this case for  $c \geq 6$ ; the minimal-endpoint ties are precisely  $c = 6$  and  $c = 7$ , giving  $(6, 11)$  and  $(7, 13)$ . For  $c \geq 8$  the cost  $a$  is larger than  $b$ . The case  $s = c$  was already excluded. This proves the one-contact list.

Finally, if the contacts have different  $s_{\nu}$ , write their total cost as the average of  $LD_{r,s_{\nu}} - h_r$ . Every summand is at least  $\min(a, b)$  by Lemma 27.3; equality of the average requires equality of every summand. The preceding analysis shows that for  $L \geq 2$  only the identical endpoint choices remain. The exceptional cases all have  $L = 1$ .  $\square$

**Lemma 30.3** (Three-dimensional residual subseries). *For  $K \geq 3$  and  $L \geq 1$ , general three-planes in a  $K$ -dimensional binary space have full quadratic product span whenever  $6L \geq 2K - 1$ , provided either  $K = 3$  or  $L \geq 2$ .*

*Proof.* For  $K = 3$  the plane is the entire binary space. Otherwise stratify the annihilator incidence as before, now with  $c = 3$  and  $L \geq 2$ . For  $r < K$ , the costs for  $s = 0, 1, 2, 3$  are respectively

$$(3L - 2)r + 1, \quad (2L - 2)r - L + 1, \quad (L - 2)r + L + 1, \quad 6L - 2r + 1.$$

On their feasible ranges they are positive: use  $r \geq 2$  for  $s = 1$ ,  $r \geq 4$  for  $s = 2$ , and  $r \leq K - 1$  for  $s = 3$ . The full-rank cost, when feasible, is  $6L - 2K + 2 \geq 1$ . The projective incidence therefore cannot dominate the Grassmannian product. This proves generic surjectivity.  $\square$

### 30.2. Irreducibility and the exceptional geometry.

**Lemma 30.4** (Signed incidences in the full range). *For  $c \geq 4$ , the  $2^{L-1}$  signed secant images are distinct, irreducible and rational of codimension  $b$ . A general point of one admits exactly one rank-two projective annihilator.*

*Proof.* A general secant hyperplane has a unique unordered decomposition into two evaluations: evaluations at four distinct points are independent for  $n \geq 3$ , and the limiting assertion follows by Hermite interpolation. Its partner hyperplane and its rank-two quadratic annihilator are therefore determined. The secant parameter space is rational, being birational to a projective-line bundle over  $\text{Sym}^2 \mathbb{P}^1$ . Normalize the first sign to be positive. The incidence is a product of relative Grassmannians of dimension  $3 + Lc(k - 1 - c) = M - b$ , rational over this rational base.

For a fixed general first hyperplane  $h$ , planes admitting a second secant hyperplane have dimension at most  $3 + c(k - 2 - c)$  in  $\text{Gr}(c, \ker h)$ , strictly less than  $c(k - 1 - c)$ . Thus the first plane recovers  $h$  generically. A general other plane lies in only its chosen partner, proving distinctness of the sign sectors and birationality. Every rank-two isotropic plane is contained in one of its isotropic hyperplanes. The exceptional confluent secants have smaller base dimension. These facts also give uniqueness of the general rank-two annihilator.  $\square$

**Lemma 30.5** (Orthogonal incidences and their orientation covers). *In the expected-codimension range, the nondegenerate incidence, when feasible, is irreducible of dimension  $M - a$  and contains points outside all the signed images. Each exceptional incidence in Table 1 is irreducible and its image also contains points outside the signed images.*

*Proof.* If  $2c < k$ , the nondegenerate incidence is smooth over the irreducible open set of nonsingular Hankel forms, with geometrically irreducible fibres  $\text{OGr}(c, k)^L$ . The dimension of each factor is  $c(k - c) - q$ . For a fixed nonsingular form, isotropic  $c$ -planes in a hyperplane have codimension at least  $c$  in this Grassmannian: if the restricted form has a one-dimensional radical, separate planes containing that radical from those avoiding it. Their codimension drops are respectively  $k - c - 1 \geq c$  and  $c$ . Varying a secant hyperplane contributes three parameters, fewer than  $c$ . Hence the first plane can be chosen outside every secant hyperplane.

The inequalities  $a \leq b$  and  $e \geq p$  imply  $2c < k$ , except for the four expected-codimension entries of Table 1 and  $(c, k, L) = (7, 14, 1)$ . This follows at once for  $c \geq 8$  from  $2k \geq L \binom{c}{2} + 5$ ; for  $4 \leq c \leq 7$  substitution gives exactly these five possibilities. In the four table entries the full-rank incidence is empty. For  $(7, 14, 1)$  its maximal orthogonal fibres have two components, but the total incidence is irreducible. Indeed the orientation double cover adjoins a square root of the determinant of the universal Hankel form. This determinant is not a square in the function field: its zero divisor is the irreducible rank-at-most-13 Hankel hypersurface, and it vanishes simply at a general corank-one point. Simplicity also follows by perturbing that point by one further evaluation nonzero on its kernel. The connected orientation cover exchanges the two

maximal orthogonal families. Over it each family has geometrically irreducible fibres, which proves irreducibility of the total incidence. In a maximal orthogonal Grassmannian the condition of containing a fixed isotropic line has codimension  $c - 1 = 6$ ; the same three-parameter secant argument gives a nonsecant first plane.

For an exceptional incidence write  $r = 2(k - c)$  and  $d = 2c - k$ . On the dense distinct-support open set,

$$\ell = \sum_{i=1}^r \alpha_i \text{ev}_{x_i}^{(2n)}, \quad E = gV_{d-1}, \quad V_n/E \simeq \mathbb{C}^r,$$

with quotient form  $\text{diag}(\alpha_i)$  and all  $\alpha_i \neq 0$ . The product  $\prod_i \alpha_i$  is nonsquare in the function field of the ordered support and projective weight parameter space (normalize the last weight to one). Thus its orientation cover is connected. The same argument for maximal orthogonal fibres proves irreducibility of this incidence. Its closure defines the incidence used here; lower-dimensional support collisions do not add another generic family.

To find a point outside the secant images, first suppose  $d \geq 2$ . A secant functional annihilating  $gV_{d-1}$  must have both its support points among the roots of  $g$ , since evaluations of  $V_{d-1}$  at two distinct points are independent. A confluent secant contributes only a pure evaluation, because  $g$  has simple roots. In the nondegenerate quotient an admissible hyperplane must have isotropic dual vector. Each of the finitely many secant lines through support points contains only two such vectors; a pure evaluation has anisotropic dual. A maximal isotropic  $s$ -plane containing any fixed isotropic line is a proper subvariety of codimension  $s - 1 > 0$ . Avoid this finite union. If  $d = 1$ , the secant functionals annihilating  $g$  form a set of dimension at most two. Here  $s = 5$  or  $6$ , and  $s - 1 > 2$ , so the same incidence bound again leaves a nonempty complement. This proves the assertion.  $\square$

**Lemma 30.6** (A corank-one point in the excess exception). *The image  $\mathcal{T}_{5,7}$  has dimension  $M - b$  and general multiplication corank one.*

*Proof.* Dehomogenize and choose a monic square-free polynomial  $g$  of degree four. In  $V_6$  put

$$U = \text{span}\{1, t, g, tg, t^2g\}.$$

For its four roots  $x_i$ , the nonzero barycentric weights  $\alpha_i = 1/g'(x_i)$  satisfy  $\sum_i \alpha_i x_i^j = 0$  for  $j = 0, 1, 2$ . Thus  $\ell = \sum_i \alpha_i \text{ev}_{x_i}^{(12)}$  has Hankel rank four, radical  $gV_2$ , and annihilates  $U^2$ . The twelve products

$$1, t, t^2, g, tg, t^2g, t^3g, g^2, tg^2, t^2g^2, t^3g^2, t^4g^2$$

have distinct degrees, hence are independent. They span a hyperplane of  $V_{12}$ . The annihilator is unique, so the incidence projection is an isomorphism on a neighbourhood in its corank-one open set. Its image has the incidence dimension  $M - b$ . This example is not in a secant hyperplane. Since it contains  $gV_2$ , any secant annihilator would have both support points among the roots of  $g$ , by the argument in Lemma 30.5. Evaluations of  $1, t$  at two distinct such roots are independent; a pure evaluation is also excluded.  $\square$

### 30.3. Proof of the component theorem.

*Proof of Theorem 30.1.* We use Propositions 29.1 and 29.2 for the descent of maximal orthogonal families; in particular a connected orientation cover is not being confused with geometric irreducibility of its fibres.

Assume first  $a \leq b$ . Proposition 28.6 makes  $\mathcal{D}_L$  pure of dimension  $M - a$ . By Proposition 28.5 and Lemma 30.2, its components must be dominated by the listed equal-dimensional incidences. In the strict case only the nondegenerate incidence, or the specified exceptional replacement, can dominate. It is irreducible, so the nonempty failure scheme has just that support. At the wall the

signed images already have the component dimension. The nonsecant point of Lemma 30.5, purity, and the same rank-gap argument force one further component, dominated by the nondegenerate incidence or its exceptional replacement. There can be no others.

For each nondegenerate or exceptional component, its incidence and image have the same dimension. In the nondegenerate case full rank is open in its projective annihilator space. In an exceptional case the rank  $r = 2(k - c)$  is the largest rank permitting an isotropic  $c$ -plane:  $\dim U \leq k - r + \lfloor r/2 \rfloor$  excludes every larger rank. It too is an open rank condition in the annihilator space. A zero-dimensional nonempty open subset of a projective space forces that space to be a point. This proves corank one and birationality.

When  $b < a$ , the same incidence-to-component argument lists all possible maximal-dimensional components. The signed images have the required dimension, and Lemma 30.6 supplies the additional component at  $(5, 7, 1)$ . No other rank incidence has sufficient dimension.

We verify generic corank one on the signed components without assuming it in this dimension argument. At a general signed point the rank-gap lemma excludes all strictly smaller incidences. If its annihilator space had positive dimension and generic rank greater than two, its open maximum-rank part would have positive dimension. The only remaining possible incidences have dimension  $M - b$ , so cannot have a positive-dimensional fibre over a dense subset of a signed component of that same dimension. If the generic rank were two, the entire open part would contradict uniqueness of the rank-two annihilator in Lemma 30.4. These alternatives exhaust a projective space containing its known rank-two point.

The signed residual planes are general by Proposition 28.3. Since  $b \leq a$  is exactly

$$L \binom{c}{2} \geq 2k - 5,$$

their multiplication is onto. If  $k = c + 1$ , each residual plane is the entire residual binary space, whose square spans the target. Otherwise apply the earlier codimension theorem at  $(k - 2, c - 1)$  when  $c \geq 5$ . For  $c = 4$  use Lemma 30.3; if  $L = 1$ , saturation here forces  $k = 5$  and the residual space is the entire three-dimensional binary space. Corollary 28.4 therefore proves generic scheme smoothness on the signed components.

For a nondegenerate component all  $W_\nu$  are zero, so Theorem 28.1 gives tangent dimension  $M - a$ . For an exceptional component  $L = 1$ ,  $W = E = gV_{d-1}$ , and  $\dim W^2 = 2d - 1$ . Its tangent dimension is

$$M - a + \binom{d+1}{2} - (2d - 1) = M - a + (d - 1)(d - 2)/2.$$

For  $d = 1, 2$  this is  $M - a$ . For the excess exception  $d = 3$ , it is  $M - a + 1 = M - b$ . In each case it is exactly the local dimension, proving generic scheme smoothness. Proposition 28.7 now proves reducedness, and integrality in the strict case, of the entire expected-codimension scheme.

Finally apply the Thom–Porteous formula to the dual map with source  $V_{2n}^* \otimes \mathcal{O}$  of rank  $p$  and target  $(\bigoplus_\nu \text{Sym}^2 \mathcal{S}_\nu)^*$  of rank  $e$ , at rank bound  $p - 1$  [16, 11]. Expected codimension has been proved and gives (30.1). Reducedness makes this the sum of the component cycles with multiplicity one. No excess-codimension cycle formula is asserted.  $\square$

### 31. SIGNED INTERSECTIONS AND THE RESIDUAL WALL

This section distinguishes three objects: the incidence over the rank-two base, its reduced normalization, and its image in the full multiplication failure scheme. They agree neither globally nor in all scheme structures.

**31.1. Normalization and the sign incidence complex.** Let  $B_2^\circ$  be the open set of rank-two projective Hankel forms with two distinct support points and nonzero weights. Choosing one of the two isotropic hyperplanes gives a finite étale double cover  $\tilde{B}_2^\circ \rightarrow B_2^\circ$ . On that cover the hyperplane bundles  $H_+, H_-$  and their common radical bundle  $E$  are labelled. Denote by  $\mathfrak{I}_2^\circ$  the restriction of  $\mathfrak{I}$  to this base.

**Theorem 31.1** (Normalization and all same-annihilator signed intersections over the split rank-two base). *After this double-cover base change, the normalization of  $(\mathfrak{I}_2^\circ)_{\text{red}}$  is the disjoint union of the smooth relative varieties*

$$Y_\varepsilon = \prod_{\nu=1}^L \text{Gr}(c, H_{\varepsilon_\nu}), \quad \varepsilon \in \{+, -\}^L.$$

*Descent identifies  $\varepsilon$  with  $-\varepsilon$ , giving  $2^{L-1}$  irreducible components before the cover.*

*For any nonempty collection  $\mathcal{A}$  of sign patterns let  $J$  be the set of contacts at which those patterns do not all agree. Their scheme-theoretic intersection as reduced components over the labelled base is*

$$\bigcap_{\varepsilon \in \mathcal{A}} Y_\varepsilon = \prod_{\nu \in J} \text{Gr}(c, E) \times \prod_{\nu \notin J} \text{Gr}(c, H_{\varepsilon_\nu}). \quad (31.1)$$

*When nonempty it is smooth of dimension  $M - b - c|J|$  and has codimension  $c|J|$  in each  $Y_\varepsilon$ . These are also the local projected intersections wherever the multiplication map has corank one and its annihilator belongs to  $B_2^\circ$ .*

*Proof.* For a split rank-two form its quadratic restriction is a product of two linear functionals. A vector subspace is isotropic exactly when one of these functionals vanishes identically on it: the polynomial ring on that subspace is a domain. The connected cover and its deck action are also established algebraically in Proposition 32.1 with  $m = 2$ . This identifies the reduced incidence as the union of the displayed  $Y_\varepsilon$ . Each is smooth and normal. Their disjoint union maps finitely to the union, is birational on every component, and hence is its normalization. The deck transformation simultaneously interchanges the two hyperplanes; its action on sign patterns has no fixed pattern. The cover is connected, as is also seen from the parameterization by a secant hyperplane and its uniquely recovered partner. Thus its orbits give the stated component count.

The condition  $U \subset H_+$  and  $U \subset H_-$  is exactly  $U \subset E$ . On any frame chart these are independent linear containment equations; adding the ideals of the reduced components therefore gives precisely the Grassmannians in (31.1), with their reduced scheme structures. Replacing  $\text{Gr}(c, k-1)$  by  $\text{Gr}(c, k-2)$  lowers dimension by  $c$  at each differing contact. This proves the formula, including emptiness if a required  $c$ -plane cannot fit in  $E$ . The corank-one incidence isomorphism transfers the local assertion to the projected scheme.  $\square$

The restriction on the last statement is essential. Two signed images can also meet through different annihilators at a higher-corank point, or over a support collision outside  $B_2^\circ$ . Formula (31.1) does not identify such points with a single labelled annihilator.

**Proposition 31.2** (The embedded structure of a rank-two fibre). *Fix a split rank-two form and one contact with  $U \subset E$ . Up to smooth local factors, its full isotropy fibre has coordinate ring*

$$R/J_c, \quad R = \mathbb{C}[a_1, \dots, a_c, b_1, \dots, b_c], \quad J_c = (a_i b_i, a_i b_j + a_j b_i \ (i < j)). \quad (31.2)$$

*Its radical is  $(a_1, \dots, a_c) \cap (b_1, \dots, b_c)$ . For  $c \geq 2$  its nilradical has dimension  $\binom{c}{2}$  as a  $\mathbb{C}$ -vector space, is annihilated by all  $a_i, b_i$ , and squares to zero. Its associated primes are exactly*

$$(a_1, \dots, a_c), \quad (b_1, \dots, b_c), \quad (a_1, \dots, a_c, b_1, \dots, b_c).$$

*For several contacts the fibre is the product of the corresponding isotropy fibres, with independent contact coordinates.*

*Proof.* Choose quotient coordinates so that the rank-two quadratic form is  $2ab$ . In a Grassmannian frame chart at  $U \subset E$ , its two quotient coordinates on the  $i$ th frame vector are  $a_i, b_i$ . The remaining coordinates are unconstrained radical directions. Restricting the bilinear form to all pairs of frame vectors gives exactly  $J_c$ . Its zero set is the union of the two coordinate  $c$ -planes, by the factorization argument above, proving the radical assertion.

Modulo  $J_c$ , the mixed quadratic monomials satisfy  $a_i b_i = 0$  and  $a_i b_j = -a_j b_i$ . Their classes therefore form  $\Lambda^2 \mathbb{C}^c$ , with no further degree-two relations. Every mixed cubic vanishes: for example

$$a_i a_k b_j = -a_i a_j b_k = a_j a_k b_i = -a_k a_i b_j,$$

so it is zero in characteristic zero; the argument for one  $a$  and two  $b$ 's is identical. All mixed monomials of higher degree vanish as well. This proves the assertions about the nilradical  $N$ . In the exact sequence

$$0 \longrightarrow N \longrightarrow R/J_c \longrightarrow R/\sqrt{J_c} \longrightarrow 0,$$

$N$  has only the origin as associated prime, whereas the reduced quotient has its two minimal primes and no embedded prime. The associated-prime containment for a short exact sequence permits no others, and all three occur when  $c \geq 2$ . The equations for different contacts use disjoint variables, proving the product presentation.  $\square$

**Proposition 31.3** (Primary decomposition and Hilbert series of the fibre). *In the notation of (31.2), let  $\mathfrak{a} = (a_1, \dots, a_c)$ ,  $\mathfrak{b} = (b_1, \dots, b_c)$ , and  $\mathfrak{m} = \mathfrak{a} + \mathfrak{b}$ . For  $c \geq 2$  an irredundant primary decomposition is*

$$J_c = \mathfrak{a} \cap \mathfrak{b} \cap (J_c + \mathfrak{m}^3). \quad (31.3)$$

*The last ideal is  $\mathfrak{m}$ -primary, and the Hilbert series is*

$$\text{Hilb}_{R/J_c}(t) = \frac{2}{(1-t)^c} - 1 + \binom{c}{2} t^2. \quad (31.4)$$

*Proof.* The intersection  $\mathfrak{a} \cap \mathfrak{b}$  is generated by all mixed monomials  $a_i b_j$ . Every mixed monomial of degree at least three belongs to  $J_c$ , by the cubic calculation in the preceding proof. In degree two, intersection with  $J_c + \mathfrak{m}^3$  leaves exactly the degree-two part of  $J_c$ . Homogeneity proves (31.3) degree by degree. The quotient by  $J_c + \mathfrak{m}^3$  is an Artinian local ring with maximal ideal the image of  $\mathfrak{m}$ ; every zero divisor is nilpotent. Thus this ideal is  $\mathfrak{m}$ -primary. Neither minimal prime can be omitted:  $a_1^3$  witnesses the need for  $\mathfrak{b}$  and  $b_1^3$  for  $\mathfrak{a}$ . The embedded component cannot be omitted because  $a_1 b_2$  survives modulo  $J_c$  when  $c \geq 2$ . The reduced quotient has in each positive degree the direct sum of the pure  $a$ - and pure  $b$ -monomials, and in degree zero one common constant. The nilradical adds exactly  $\binom{c}{2}$  degree-two classes. This proves (31.4) and verifies algebraically that the three primes listed above are all the associated primes. For  $c = 1$  the third component is redundant and  $J_1 = (a_1 b_1)$  is already reduced.  $\square$

This embedded prime is a statement about a fibre with its annihilator fixed. It does not assert an embedded prime of  $\mathcal{D}_L$ . In particular a reduced total scheme can have these nonreduced fibres.

**31.2. Ordinary double crossings at the residual wall.** For the next theorem suppose

$$c \geq 8, \quad k \geq 2c + 1, \quad a = b. \quad (31.5)$$

The parity in  $L(\binom{c}{2}) = 2k - 5$  in fact forces  $c \geq 10$  in this range. On the signed open set of Proposition 28.3, the residual map is square:

$$\mu'_K : \bigoplus_{\nu} \text{Sym}^2 K_{\nu} \longrightarrow V_{2n-4}, \quad L\left(\binom{c}{2}\right) = 2k - 5.$$

Let  $Z_{\varepsilon}$  be the inverse image of its determinant divisor. It is a divisor in the signed incidence, rather than a new arbitrary choice of a discriminant.

**31.3. Relative determinant and the normal derivative.** In this subsection  $B = \tilde{B}_2^\circ$  is the connected base labelling the two isotropic hyperplanes, and  $\mathcal{R}$  is their common radical bundle. We impose the wall hypotheses (31.5). These relative statements provide the bundle identifications and the nonemptiness arguments for the node normal form.

**Proposition 31.4** (The relative residual divisor). *There is a subbundle  $\mathcal{T} \subset V_{2n} \otimes \mathcal{O}_B$  with fibre  $g^2 V_{2n-4}$  and rank  $h = p - 4$ . On*

$$P = \prod_{\nu=1}^L \text{Gr}_B(c-1, \mathcal{R})$$

*let  $\mathcal{W}_\nu$  denote the tautological bundles. The determinant of residual multiplication*

$$\tau : \bigoplus_{\nu} \text{Sym}^2 \mathcal{W}_\nu \longrightarrow \mathcal{T}$$

*is a section of the line bundle*

$$\det \mathcal{T} \otimes \left( \det \bigoplus_{\nu} \text{Sym}^2 \mathcal{W}_\nu \right)^{-1}. \quad (31.6)$$

*Its zero scheme  $Z_P$  is an integral reduced Cartier divisor in  $P$ . For each sign pattern the signed open incidence  $S_\varepsilon \rightarrow P$  is a smooth affine-space torsor. Its inverse image  $Z_\varepsilon = S_\varepsilon \times_P Z_P$  is likewise an integral reduced divisor. These assertions hold on the entire distinct-support labelled base, not only after fixing the two points.*

*Proof.* The unordered pair of support points gives a line subbundle  $\mathcal{G} \subset V_2 \otimes \mathcal{O}_B$ . Multiplication gives isomorphisms

$$\mathcal{R} = \mathcal{G} \otimes V_{n-2}, \quad \mathcal{T} = \mathcal{G}^{\otimes 2} \otimes V_{2n-4} \subset V_{2n} \otimes \mathcal{O}_B.$$

These formulas are homogeneous and so include a support point at infinity. The source and target of  $\tau$  have the same rank  $h$  at the wall, proving (31.6). Trivializing  $\mathcal{G}$  identifies  $P$  with the product of the base chart and  $\text{Gr}(c-1, V_{n-2})^L$ , and identifies the determinant zero scheme with the same product for the ordinary residual multiplication divisor. By Theorem 30.1 at  $(k', c') = (k-2, c-1)$ , this residual divisor is integral and reduced: its expected codimension is one and  $L(c-1) - 3 > 1$ .

For completeness, these local product descriptions imply the global assertion. Their projections are flat, with geometrically integral fibres. Every irreducible component dominates  $B$ , whereas the geometric generic fibre is irreducible, so there is just one component. The product descriptions are reduced. The section is nonzero and is locally a nonzerodivisor, hence defines a Cartier divisor. The fibres of  $S_\varepsilon \rightarrow P$  are the affine torsors

$$\prod_{\nu} \text{Hom}(H_{\varepsilon_\nu}/\mathcal{R}, \mathcal{R}/\mathcal{W}_\nu).$$

Local splittings trivialize them. Pulling back  $Z_P$  is consequently locally a product with affine space, proving the last assertion by the same generic-fibre and reducedness arguments.  $\square$

**Proposition 31.5** (Intrinsic normal-derivative factorization). *Let  $Y = \mathbb{P}(V_{2n}^*) \times X_L$ , let  $S = S_\varepsilon$  be the signed open incidence over  $B$ , and let  $\mathcal{E} = \bigoplus_{\nu} \text{Sym}^2 \mathcal{U}_\nu$ . The incidence equations are a section  $s$  of  $\mathcal{E}^* \otimes \mathcal{L}^*$ , where  $\mathcal{L}$  is the annihilator line. Their normal derivative is a map of rank- $e$  bundles*

$$d_N s : N_{S/Y} \longrightarrow \mathcal{E}^* \otimes \mathcal{L}^*.$$

*After passing to the labelled etale chart where necessary, the normal bundle has an exact sequence*

$$0 \longrightarrow \bigoplus_{\nu} \text{Hom}(\mathcal{U}_\nu, V_n/H_{\varepsilon_\nu}) \longrightarrow N_{S/Y} \longrightarrow \mathcal{L}^* \otimes \mathcal{T}^* \longrightarrow 0. \quad (31.7)$$

Put  $Q = \bigoplus_{\nu} \text{Sym}^2 \mathcal{W}_{\nu}$ . The first term maps isomorphically onto the kernel of the restriction  $\mathcal{E}^* \otimes \mathcal{L}^* \rightarrow Q^* \otimes \mathcal{L}^*$ . On the remaining associated quotients,  $d_N s$  is  $\text{id}_{\mathcal{L}^*} \otimes \tau^*$ . Consequently its determinant and the residual determinant are identified by an isomorphism of their determinant line bundles. In any local frames this reads

$$\det(d_N s) = u_0 \det(\mu'_K), \quad u_0 \in \mathcal{O}_S^*. \quad (31.8)$$

The unit ambiguity is exactly the change of frames; the equality of zero divisors and their multiplicities is intrinsic.

*Proof.* First restrict  $Y$  to the rank-two base. Relative to this base the normal bundle of  $\text{Gr}(c, H_{\varepsilon_{\nu}})$  in  $\text{Gr}(c, V_n)$  is  $\text{Hom}(\mathcal{U}_{\nu}, V_n/H_{\varepsilon_{\nu}})$ . The normal sequence for  $S \subset Y|_B \subset Y$  gives the first term of (31.7) and the normal bundle of  $B$  in  $\mathbb{P}(V_{2n}^*)$  as quotient.

The affine tangent space of the rank-two cone at  $\ell$  is spanned by the two evaluations and their first derivatives, with the latter weighted by the nonzero coefficients. A polynomial annihilating this four-dimensional space vanishes to order two at both support points. Its space is precisely  $g^2 V_{2n-4}$ . Therefore

$$N_{B/\mathbb{P}(V_{2n}^*)} = \text{Hom}(\mathcal{L}, \mathcal{T}^*) = \mathcal{L}^* \otimes \mathcal{T}^*.$$

This identification is independent of coordinates on  $\mathbb{P}^1$ .

For one contact write  $U = W \oplus \mathbb{C}a$  locally. The restriction of a rank-two form has one nonzero quotient functional in the  $V_n/H_{\varepsilon}$  direction. Varying each vector of  $W$  in this direction prescribes the  $wa$  entries, and varying  $a$  prescribes the  $a^2$  entry, with its nonzero factor two. The  $ww'$  entries remain zero. Thus the plane-normal map is an isomorphism onto the rank- $c$  space of equation forms vanishing on  $\text{Sym}^2 W$ . Doing this for each contact proves the first block assertion.

Restrict the remaining equations to  $Q$ . Variations of the planes contribute zero there. An annihilator-normal variation  $\dot{\ell}$  contributes  $q \mapsto \dot{\ell}(\mu(q))$ , exactly  $\tau^*$  under the preceding normal-bundle identification. Splittings of the two exact sequences therefore give an upper block triangular matrix with an invertible plane block and the block  $\text{id}_{\mathcal{L}^*} \otimes \tau^*$ . Taking determinants proves the assertion on determinant lines. A new splitting alters only an off-diagonal block, and new frames change determinants by units. Finally trivializing  $\mathcal{G}$  cancels the common  $g^2$  factor and identifies  $\tau$  with  $\mu'_K$ , giving (31.8).  $\square$

**Lemma 31.6** (A simultaneous nonempty corank-one wall open). *The integral divisor  $Z_{\varepsilon}$  has a nonempty open subset where the residual map has corank one and nonsingular Hankel annihilator  $\eta$ ,  $K_1 \rightarrow \mathbb{C}^2$  has full evaluation rank at the support points, and the original multiplication has corank one. This open subset is dense in  $Z_{\varepsilon}$ .*

*Proof.* The first two residual conditions hold on a nonempty open subset of the integral residual divisor by Theorem 30.1. We verify that the evaluation-rank condition meets this subset rather than presuming generality of a residual plane.

Write  $c' = c - 1$ ,  $k' = k - 2$ ; here  $k' \geq 2c' + 1$  and  $c' \geq 9$ . For a nondegenerate form on a  $k'$ -space, the locus of isotropic  $c'$ -planes contained in a fixed hyperplane has codimension at least  $c'$  in its orthogonal Grassmannian. For a nonisotropic normal vector this follows by replacing  $k'$  by  $k' - 1$  in the dimension formula. For an isotropic normal vector  $v$ , planes not containing  $v$  are lifts of isotropic  $c'$ -planes in  $v^{\perp}/\mathbb{C}v$ , with  $c'$  affine lift parameters, giving the same bound. Planes containing  $v$  are parametrized by  $\text{OGr}(c' - 1, k' - 2)$  and have codimension  $k' - c' - 1 \geq c'$ . This proves the bound in all cases.

Failure of  $K_1$  to map onto the two evaluation coordinates means containment in one member of their hyperplane pencil. The union has codimension at least  $c' - 1 > 0$ . Over the nondegenerate residual annihilator incidence, its complement is thus nonempty and open. That incidence is irreducible and dominates the residual determinant divisor; its corank-one open is nonempty.

Intersecting these two nonempty opens in an irreducible space proves the simultaneous residual assertion, for each fixed pair of support points.

Choose such a tuple and a nonzero  $k_0 \in K_1 \cap gV_{n-4}$ ; this intersection has dimension at least  $c - 3 > 0$ . Write  $k_0 = gk'_0$  and  $U_1 = gK_1 + \mathbb{C}a_1$ . The already spanned space  $g^2 \operatorname{im} \mu'_K$  has one missing direction in  $g^2V_{2n-4}$ . The product  $(gk_0)a_1 = g^2k'_0a_1$  supplies it if  $\eta(k'_0a_1) \neq 0$ . Changing the lift by  $a_1 \mapsto a_1 + gw$  changes this value by  $H_\eta(k_0, w)$ . This is a nonzero functional because  $\eta$  is nondegenerate. It vanishes on  $K_1$ , since  $K_1$  is isotropic, so it is a well-defined nonconstant affine function on the actual lift torsor, not merely on a space of redundant frames.

After this choice, products from  $(gK_1)a_1$  supply both independent first-derivative coordinates modulo  $g^2V_{2n-4}$ , using full evaluation rank and the nonzero values of  $a_1$  at both support points. The square  $a_1^2$  supplies a further, nonzero value coordinate. The product span therefore has dimension at least  $(p - 4) + 3 = p - 1$ . It is annihilated by  $\ell$ , so its dimension is exactly  $p - 1$ . All the conditions used are open, and we have exhibited choices meeting them. Integrality of  $Z_\varepsilon$  from Proposition 31.4 proves density.  $\square$

**31.4. A simple normal determinant gives an exact double crossing.** The passage from a normal derivative to a local equation is an independent analytic statement. We record the precise hypotheses; first-order corank alone would not suffice.

**Lemma 31.7** (Simple normal determinant). *Let  $Y$  be a smooth complex variety of dimension  $d + e$ , let  $S \subset Y$  be smooth of codimension  $e$ , and let  $s$  be a section of a rank- $e$  vector bundle  $\mathcal{F}$  which vanishes identically on  $S$ . At  $z \in S$ , assume that the normal derivative*

$$d_N s : N_{S/Y} \longrightarrow \mathcal{F}|_S$$

*has rank  $e - 1$ . Suppose its determinant has a simple zero on  $S$  at  $z$ , meaning that a local representative has nonzero differential in  $T_z^*S$ . Then  $d \geq 1$  and the zero scheme  $Z = V(s)$  has analytic local equation  $uv = 0$  in a smooth  $(d + 1)$ -dimensional space. More precisely,*

$$\hat{\mathcal{O}}_{Z,z} \simeq \mathbb{C}[[t_1, \dots, t_{d-1}, u, v]]/(uv), \quad (31.9)$$

*with  $S$  corresponding to the branch  $u = 0$ . No reducedness or component-count assumption on  $Z$  is needed.*

*Proof.* Choose coordinates  $(t_1, \dots, t_d, z_1, \dots, z_e)$  in which  $S = \{z_1 = \dots = z_e = 0\}$ , and trivialize  $\mathcal{F}$ . After invertible changes of the normal coordinates and of the section's components, the first  $e - 1$  components have an invertible Jacobian in  $z_1, \dots, z_{e-1}$  at the point. The implicit function theorem solves these components as  $z_i = h_i(t, u)$ , where  $u = z_e$ . Since the components vanish on  $S$ , uniqueness implies  $h_i(t, 0) = 0$ . Their common zero scheme is a smooth  $(d + 1)$ -dimensional neighbourhood containing  $S = \{u = 0\}$ . For  $e = 1$  this step has no equations to eliminate.

Restrict the last component to this neighbourhood and call it  $f(t, u)$ . It vanishes on  $u = 0$ , so in the convergent power-series ring it factors exactly as  $f = uG(t, u)$ . Along  $S$ , the chain rule for the eliminated equations gives

$$G(t, 0) = \partial_u f(t, 0) = \frac{\det(d_N s)(t)}{\det(\partial(s_1, \dots, s_{e-1})/\partial(z_1, \dots, z_{e-1}))(t, 0)}.$$

For  $e = 1$  the denominator is 1. This is the elementary Schur complement identity for the normal Jacobian. The denominator is a unit. Since the numerator vanishes simply on  $S$ , the differential of  $G(t, 0)$  is nonzero. Consequently  $du$  and  $dG$  are linearly independent at  $z$ . Complete  $u, v = G(t, u)$  to analytic coordinates  $(u, v, t'_1, \dots, t'_{d-1})$  on the neighbourhood. In these coordinates the final equation is exactly  $uv = 0$ . Completion proves (31.9). Changing any trivialization multiplies the determinants by units, so the simple-zero condition and the conclusion are intrinsic.  $\square$

**Corollary 31.8** (Normalization and conductor of the local crossing). *In Lemma 31.7, the normalization of the completed local ring is*

$$\mathbb{C}[[t_1, \dots, t_{d-1}, u]] \oplus \mathbb{C}[[t_1, \dots, t_{d-1}, v]],$$

*and its conductor in that normalization is  $(u) \oplus (v)$ . The conductor in the original ring is  $(u, v)$ , and the scheme-theoretic intersection of the two branches is reduced and smooth of dimension  $d - 1$ .*

*Proof.* The ring in (31.9) embeds into the displayed direct sum as the pairs whose restrictions at  $u = v = 0$  agree. The sum is finite over it: adjoining the idempotent  $(1, 0)$  suffices. Both summands are regular and integrally closed, so this is the normalization. A pair multiplies every pair in the sum back into the original ring precisely when both restrictions at zero vanish. This is  $(u) \oplus (v)$ , whose inverse image is  $(u, v)$ . The quotient by this ideal is  $\mathbb{C}[[t_1, \dots, t_{d-1}]]$ .  $\square$

In the binary wall calculation below,  $Y$  is the annihilator ambient space,  $S$  the chosen signed incidence, and  $s$  its isotropy section. The relative determinant proposition identifies  $\det(d_N s)$ , up to a unit, with the residual multiplication determinant. The relative reduced divisor and the nonemptiness proposition supply its simple-zero locus while the original multiplication still has corank one. Lemma 31.7 then proves the exact local equation used in Theorem 31.9. Thus neither reducedness inferred from a tangent count nor an unproved removal of higher-order terms enters the crossing argument.

### 31.5. The local normal form.

**Theorem 31.9** (The residual double-crossing divisor). *Under (31.5),  $Z_\varepsilon$  is irreducible and reduced. It has a nonempty dense open subset on which the following assertions hold. The original multiplication has corank one, and the residual multiplication has corank one with a nonsingular residual Hankel annihilator. Put  $d = M - a$ . The analytic germ of  $\mathcal{D}_L$  at its projected point has equation  $uv = 0$  in smooth local coordinates  $(t_1, \dots, t_{d-1}, u, v)$ , and its completed local ring is*

$$\mathbb{C}[[t_1, \dots, t_{d-1}, u, v]]/(uv). \quad (31.10)$$

*The two branches are  $\Sigma_\varepsilon$  and  $\Phi_L$ ; no other signed branch passes through this open subset. Their intersection has codimension one in either branch, and codimension  $a + 1$  in  $X_L$ . The tangent cone is the union of two distinct hyperplanes in the  $(d + 1)$ -dimensional tangent space. The germ is seminormal, its normalization separates its two smooth branches, and its conductor in (31.10) is  $(u, v)$ .*

*Proof.* Proposition 31.4 gives the relative integral divisor, Lemma 31.6 supplies a simultaneous dense open, and Proposition 31.5 gives the intrinsic determinant identity. We spell out how they lead to the local equation. At residual dimensions  $k' = k - 2$ ,  $c' = c - 1$  the expected codimension is one and the signed codimension is  $L(c - 1) - 3 > 1$ . Here  $c' \geq 9$  and  $k' \geq 2c' + 1$ . The already proved Theorem 30.1 says that the residual determinant is an integral reduced divisor whose general annihilator  $\eta$  is nonsingular. Pullback by the smooth affine-space bundle in Proposition 28.3 preserves this generic irreducibility and reducedness; bundle trivializations as the support points vary give the same assertion for  $Z_\varepsilon$ .

We check that the original map has corank one on a nonempty open part of this divisor. A general residual tuple has  $K_1$  mapping onto the two evaluation coordinates at  $x, y$ . Indeed failure of this condition places  $K_1$  in one hyperplane of a pencil, and the isotropic Grassmannian incidence count excludes that closed subset. Choose a nonzero

$$k_0 \in K_1 \cap gV_{n-4}, \quad k_0 = gk'_0.$$

Such vectors exist because this intersection has dimension at least  $c - 3 > 0$ . Write  $U_1 = gK_1 + \mathbb{C}a_1$ . The product  $(gk_0)a_1 = g^2k'_0a_1$  fills the missing direction in  $g^2V_{2n-4}$  exactly when  $\eta(k'_0a_1) \neq 0$ .

Varying the lift by  $a_1 \mapsto a_1 + gw$ ,  $w \in V_{n-2}$ , changes this value by  $H_\eta(k_0, w)$ , a nonzero functional because  $H_\eta$  is nonsingular. Thus general lifts fill this direction. The products  $(gK_1)a_1$  give both first-derivative coordinates modulo  $g^2V_{2n-4}$ , since  $K_1$  has full evaluation rank and  $a_1$  has nonzero values at  $x, y$ . The product  $a_1^2$  gives one additional value coordinate. Consequently the original product span has dimension  $(p-4) + 3 = p-1$ . All these conditions are nonempty open conditions.

Work on this corank-one open set in the incidence. The signed incidence  $S$  is smooth of dimension  $d$  in the smooth ambient  $Y = \mathbb{P}(V_{2n}^*) \times X_L$ . Its codimension is

$$(M + p - 1) - d = p - 1 + b = e,$$

and the full incidence is the zero scheme of  $e$  restriction equations. The normal derivative of these equations along  $S$  splits into the  $Lc$  mixed and square equations involving the nonradical vector  $a_\nu$ , and the  $L\binom{c}{2} = p-4$  equations on  $gK_\nu \times gK_\nu$ . The first block is invertible in transverse plane coordinates, by the dual-frame proof of Theorem 28.1. The normal space to the rank-two Hankel base is identified with  $(g^2V_{2n-4})^*$ : its tangent space is the annihilator of  $g^2V_{2n-4}$  modulo  $\mathbb{C}\ell$ , as is seen by differentiating the two evaluations and their weights. In these normal coordinates the remaining derivative block is precisely  $(\mu'_K)^*$ . Hence, up to a nowhere-zero factor,

$$\det(\text{normal derivative along } S) = \det \mu'_K. \quad (31.11)$$

At a general point of  $Z_\varepsilon$  this derivative has corank one and its determinant vanishes simply along  $S$ .

Eliminate  $e-1$  equations and their normal coordinates by the analytic implicit function theorem. The remaining ambient germ is smooth of dimension  $d+1$ , and  $S$  is the smooth hypersurface  $u=0$  in it. The last equation vanishes on  $S$ , so it is  $uG$ . Its normal derivative along  $S$  is  $G|_S$ ; by (31.11) this has a simple zero along the residual divisor. Thus  $dG$  has a nonzero component tangent to  $S$ , and  $u, v = G$  are independent analytic coordinates. This proves (31.10), not only an equality of tangent dimensions. The corank-one projection identifies this germ with the one in  $X_L$ .

Only one sign choice is possible locally because no  $U_\nu$  is contained in the radical and the annihilator is unique. The other branch must therefore be  $\Phi_L$ , by Theorem 30.1. All dimension and tangent-cone assertions follow from the displayed local ring. Its normalization is

$$\mathbb{C}[[t_1, \dots, t_{d-1}, u]] \oplus \mathbb{C}[[t_1, \dots, t_{d-1}, v]],$$

Replacing formal by convergent power series gives the analytic normalization. In either case the original ring consists of pairs with the same restriction at  $u=v=0$ . This identifies the conductor as  $(u, v)$  and proves seminormality: an element of the total fraction ring whose square and cube belong to the original ring lies in each normal branch ring, and its two restrictions agree because their squares and cubes agree.  $\square$

The theorem describes a specified irreducible divisor and its dense open double-crossing locus. It does not claim that every point of  $\Phi_L \cap \Sigma_\varepsilon$  has this type, or that higher-corank and colliding-support intersections are normal crossings.

**Corollary 31.10** (A logarithmic conditioning law at a crossing). *At a real point of the double-crossing locus, with its two real branches, there are real analytic coordinates  $(z_1, \dots, z_{a-1}, u, v, t)$  in  $X_L$  in which*

$$\gamma_{\text{mult}} \asymp \left( \sum_{j=1}^{a-1} z_j^2 + u^2 v^2 \right)^{1/2}. \quad (31.12)$$

On a sufficiently small product box, a probability law whose density is bounded above and below by positive constants satisfies

$$\Pr\{\gamma_{\text{mult}} \leq \epsilon\} \asymp \epsilon^a \log(1/\epsilon), \quad \epsilon \downarrow 0. \quad (31.13)$$

The same law holds in any real loading chart submersing onto this neighbourhood. Constants are local and depend on the chosen chart and density bounds.

*Proof.* At corank one bounded invertible row and column operations compare  $\gamma_{\text{mult}}$  with the norm of the Schur-complement row. The analytic argument in Theorem 31.9, applied in the ambient Grassmannian chart, makes its ideal  $(z_1, \dots, z_{a-1}, uv)$ . Each finite set of generators is a bounded analytic linear combination of the other on a smaller compact chart, so their norms are comparable. This proves (31.12). For the upper bound integrate first over the  $a - 1$  normal coordinates, whose volume is  $O(\epsilon^{a-1})$ . In a fixed square the area of  $|uv| \leq C\epsilon$  is  $O(\epsilon \log(1/\epsilon))$ , by integrating  $\min(r, C\epsilon/|u|)$  in  $u$ . For the lower bound restrict the normal coordinates to a ball of radius  $\epsilon/2$  and take  $|uv| \leq \epsilon/2$  in a smaller fixed square. The same elementary integral is bounded below by a positive constant times  $\epsilon \log(1/\epsilon)$ . Bounded density and Jacobian factors give the assertion. A submersion adds unconstrained smooth coordinates and does not change the exponent or logarithm.  $\square$

This is different from the smooth-stratum law  $\epsilon^a$ . It is a conditioning statement for a specified real design law, not a claim that the exact count-score compression experiment has Fisher information equal to an unwhitened squared singular value. The covariance of that score compression also depends on the design.

### 32. HIGHER PRODUCTS AND DIHEDRAL PHASE SECTORS

The general residual theorem of Section 25 already applies to every symmetric power. Here we calculate explicitly one distinguished family of annihilators within that general setting. The two-hyperplane description has a higher-product counterpart, but its monodromy is no longer exhausted by a simultaneous sign change. For  $m \geq 2$  consider

$$\mu_m : \bigoplus_{\nu=1}^L \text{Sym}^m U_\nu \longrightarrow V_{mn}, \quad p_m = mn + 1, \quad e_m = L \binom{c + m - 1}{m}.$$

Here “two-point annihilator” refers to the family

$$[\ell] = [\text{ev}_x^{(mn)} - t^m \text{ev}_y^{(mn)}], \quad x \neq y, \quad t \neq 0,$$

not to a purported classification of every higher catalecticant rank. Let  $\mu_m$  be the group of  $m$ th roots of unity.

**Proposition 32.1** (Algebraic splitting cover and phase descent). *On the distinct-support two-point base for  $m$ th multiplication, the irreducible components of the reduced isotropic incidence are exactly the orbits of  $\mu_m \rtimes \{1, \sigma\}$  on  $\mu_m^L$ , where simultaneous multiplication acts on all labels and  $\sigma$  inverts all labels. No further splitting occurs for an inversion-fixed relative pattern.*

*Proof.* It suffices first to work over the dense chart where both support points are finite. Its labelled splitting cover is the integral variety

$$T = \{(x, y, t) \in \mathbb{A}^2 \times \mathbb{G}_m : x \neq y\}.$$

The transformations

$$\tau_\zeta(x, y, t) = (x, y, \zeta t), \quad \sigma(x, y, t) = (y, x, t^{-1})$$

satisfy  $\sigma \tau_\zeta \sigma = \tau_{\zeta^{-1}}$ . They form a group of order  $2m$ , acting freely: a nontrivial rotation cannot fix  $t \neq 0$ , and a reflection interchanges the distinct points. The quotient is a finite etale Galois cover of

the two-point base. To identify the quotient with that base, note that the annihilator  $[\text{ev}_x - t^m \text{ev}_y]$  recovers its unordered support and weight ratio. For example the radical of its middle Hankel pairing recovers the degree-two support polynomial, and a two-column Vandermonde system recovers the weights. This also gives algebraic inverses on the usual nonvanishing-minor charts. For the degrees under consideration the pairing has both side dimensions at least two. Thus the displayed transformations are the entire deck group, not just selected monodromies.

Over  $T$  the hyperplanes  $H_\zeta$  are labelled. Their reduced incidences are the integral smooth bundles  $\prod_\nu \text{Gr}(c, H_{\zeta_\nu})$ ; they have geometrically integral fibres. A deck transformation permutes these bundles by the displayed action on the labels (using the inverse rotation gives the same orbit action). The image of any integral bundle under the finite quotient map is irreducible and is exactly the image of its orbit union. Different orbits have distinct generic geometric components. Hence these images are precisely the irreducible components downstairs. An element stabilizing one pattern acts on its integral bundle; its quotient remains integral, since an invariant subring of a domain is a domain. In particular a fixed relative inversion pattern does not split after descent.

Changing affine coordinates on  $\mathbb{P}^1$  only changes trivializations of the evaluation lines. Such charts cover the base, and their nonempty overlaps give the same hyperplane-label cover. Alternatively, each component of the full incidence dominates the base, because its split local models are Grassmannian bundles. Thus it is the closure of a component on this dense chart. No additional component is hidden at infinity. This completes the algebraic descent proof.  $\square$

**Theorem 32.2** (Phase-sector classification for every symmetric power). *Suppose  $n \geq 4$ ,  $4 \leq c < n + 1$ ,  $L \geq 1$ , and  $m \geq 2$ . The reduced incidence of two-point annihilators of  $\mu_m$  has*

$$N_{m,L} = \frac{m^{L-1} + \gcd(m, 2)^{L-1}}{2} \quad (32.1)$$

*irreducible components on the distinct-support open base. The closure of the image of each component in  $X_L$  is irreducible and rational of codimension  $Lc - 3$ . Its incidence projects birationally, and its general point has a unique two-point annihilator. These sectors are indexed by relative phase patterns*

$$(\xi_2, \dots, \xi_L) \in \mu_m^{L-1}, \quad (\xi_2, \dots, \xi_L) \sim (\xi_2^{-1}, \dots, \xi_L^{-1}).$$

*For  $m = 2$  this recovers  $2^{L-1}$ . In general the count is not  $m^{L-1}$ .*

*Proof.* By polarization in characteristic zero,  $\ell$  annihilates  $\text{Sym}^m U$  if and only if the degree- $m$  polynomial  $\ell(f^m)$  vanishes identically for  $f \in U$ . Its factorization is

$$f(x)^m - t^m f(y)^m = \prod_{\zeta \in \mu_m} (f(x) - \zeta t f(y)).$$

Since the polynomial ring on  $U$  is a domain, this holds precisely when  $U$  lies in one of the  $m$  hyperplanes  $H_\zeta = \ker(\text{ev}_x - \zeta t \text{ev}_y)$ . The splitting cover of the two-point base labels all these hyperplanes. Changing the chosen  $m$ th root  $t$  acts by simultaneous cyclic translation of phase labels. Interchanging the two support points acts by inversion after that translation. Proposition 32.1 constructs the finite etale cover, identifies its full deck group, and proves the descent assertion, including sectors with stabilizers. Thus the component orbits are those of the dihedral action on  $\mu_m^L$ .

Quotient first by simultaneous translation, normalizing the first phase to one. The remaining involution inverts every relative phase. Its fixed patterns have each entry of order dividing two; there are  $\gcd(m, 2)^{L-1}$  of them. The two-element orbit formula gives (32.1). Over the splitting cover every phase sector is a smooth irreducible product of relative Grassmannians; monodromy orbits are therefore exactly the irreducible components after descent.

Each incidence has dimension  $3 + Lc(k - 1 - c) = M - (Lc - 3)$ . The first plane determines its general secant hyperplane uniquely by the dimension argument in Lemma 30.4. That hyperplane determines its unordered support points, weight ratio and resulting two-point annihilator, and the other planes determine their phase hyperplanes. This proves generic injectivity and distinctness of the image sectors, hence birationality in characteristic zero.

For rationality, a relative pattern not fixed by inversion chooses an ordering of the support points. Its parameterization uses an open subset of  $\mathbb{P}^1 \times \mathbb{P}^1 \times \mathbb{G}_m$ , with rational Grassmannian fibres trivialized on a dense open set, and is birational. For an inversion-fixed pattern, its first hyperplane instead parameterizes the unordered secant threefold, which is rational; all other hyperplanes are rationally determined from it. The same Grassmannian argument applies. Taking image closures does not change these birational assertions.  $\square$

The theorem classifies the two-point incidence and its image sectors. It does not assert that every such sector is an irreducible component of the entire higher-product failure scheme, or that the full higher-product failure codimension has been determined.

**Theorem 32.3** (Higher-product residual conormal). *On a phase sector suppose no  $U_\nu$  is contained in  $E = gV_{n-2}$ , and write  $U_\nu = gK_\nu + \mathbb{C}a_\nu$ . Put*

$$R_m = \sum_{\nu} (gK_\nu)^2 U_\nu^{m-2} = g^2 \sum_{\nu} K_\nu^2 U_\nu^{m-2} \subset V_{mn}, \quad \delta_m = (p_m - 4) - \dim R_m.$$

*If  $e_m \geq p_m$  and  $\mu_m$  has corank one, then the tangent dimension of its maximal-minor scheme is*

$$M - (Lc - 3) + \delta_m. \quad (32.2)$$

*In particular  $\delta_m = 0$  implies scheme smoothness of codimension  $Lc - 3$  at this point.*

*Proof.* For a fixed contact split  $U = W + \mathbb{C}a$ , with  $W = gK$ . The derivative with respect to the plane vanishes on every monomial having at least two  $W$  factors: after varying one factor another still vanishes at both support points. On the quotient spanned by  $a^m$  and  $wa^{m-1}$ ,  $w \in W$ , it is surjective. Indeed the nonzero quotient linear functional

$$v \mapsto v(x)a(x)^{m-1} - t^m v(y)a(y)^{m-1}$$

allows the coordinates of the variation of each basis vector to solve these  $c$  equations independently; the coefficient for  $a^m$  is the nonzero integer  $m$ . Thus the plane derivative has rank  $c$ . The remaining compatibility equations for  $\dot{\ell}$  are precisely annihilation of  $R_m$ . Independent solutions for the contacts give

$$\dim T\mathfrak{J}_m = M + (p_m - 1) - Lc - \dim R_m,$$

which is (32.2). The corank-one Schur argument identifies this tangent space with that of the maximal-minor scheme. Multiplication by  $g^2$  embeds  $V_{mn-4}$ , of dimension  $p_m - 4$ , so  $\delta_m \geq 0$ . If it is zero, the tangent dimension equals the dimension  $M - (Lc - 3)$  of the local phase incidence, which is embedded by the corank-one projection. Equality of tangent and local dimensions proves smoothness.  $\square$

For  $m = 2$  the residual span is independent of the lifts  $a_\nu$  and reduces to the multiplication of the  $K_\nu$ . For  $m > 2$  it retains  $U_\nu^{m-2}$ . This difference is why a higher-product phase classification alone does not imply the quadratic saturation formula with  $q$  replaced by  $\binom{c+m-1}{m}$ .

### 33. A GLOBAL HIGHER-PRODUCT FAMILY WITH EMBEDDED STRUCTURE

We now apply the residual mechanism to a natural family in every symmetric degree, determining the whole reduced failure locus rather than only an obstruction subfamily. The scheme structure contains a further phenomenon: its support is independent of the degree, but its nilpotent structure is not.

Fix  $n \geq 4$  and put  $V_n = H^0(\mathbb{P}^1, \mathcal{O}(n))$ . A hyperplane  $U \subset V_n$  is the kernel of a unique point  $[h] \in \mathbb{P}(V_n^*)$ . For  $m \geq 2$  let

$$\mu_m : \text{Sym}^m \mathcal{U} \longrightarrow V_{mn} \otimes \mathcal{O}_{\mathbb{P}(V_n^*)}, \quad D_m = V(I_{mn+1}(\mu_m)).$$

Its source rank is  $e_m = \binom{n+m-1}{m} \geq mn + 1$ . Write  $C_n \subset \mathbb{P}(V_n^*)$  for the evaluation rational normal curve, and  $S_n$  for its secant threefold, including tangent secants. Let  $\mathcal{J}_m$  be the ideal sheaf of  $D_m$ , and  $\mathcal{I}_{S_n}, \mathcal{I}_{C_n}$  the reduced ideal sheaves. Associated points below are those of coherent sheaves on projective space; no assertion about an irrelevant-ideal component of an affine homogeneous presentation is implicit.

**Theorem 33.1** (Higher-product hyperplanes: support, ranks and associated points). *For every  $n \geq 4$  and  $m \geq 2$ ,*

$$(D_m)_{\text{red}} = S_n, \quad \text{corank } \mu_{m, \ker h} = \begin{cases} 0, & [h] \notin S_n, \\ 1, & [h] \in S_n \setminus C_n, \\ m, & [h] \in C_n. \end{cases} \quad (33.1)$$

*The scheme  $D_m$  is smooth of dimension three at every point of  $S_n \setminus C_n$ , including the non-evaluation tangent secants. In particular*

$$\mathcal{J}_m : \mathcal{I}_{C_n}^\infty = \mathcal{I}_{S_n}. \quad (33.2)$$

*For every  $m \geq 3$ , the complete set of associated points is*

$$\text{Ass}_{\mathcal{O}_{\mathbb{P}(V_n^*)}}(\mathcal{O}_{D_m}) = \{\eta_{S_n}, \eta_{C_n}\}. \quad (33.3)$$

*Thus there is exactly one reduced irreducible component, and exactly one embedded associated subvariety, the evaluation curve. Its nilradical  $\mathcal{N}_m = \mathcal{I}_{S_n}/\mathcal{J}_m$  satisfies, at every closed point  $x \in C_n$ ,*

$$(\mathcal{N}_m^j)_x \neq 0 \quad \text{whenever } 1 \leq j \text{ and } 2j < m. \quad (33.4)$$

*Consequently the nilpotency index at  $x$  is at least  $\lceil m/2 \rceil$ . The theorem does not assert that this lower bound is the exact index.*

**Lemma 33.2** (Propagation from quadratic surjectivity). *Let  $U \subset V_n$  be base-point-free. Then  $UV_d = V_{n+d}$  for every  $d \geq n - 1$ . In particular,  $U^2 = V_{2n}$  implies  $U^m = V_{mn}$  for every  $m \geq 2$ .*

*Proof.* There are coprime  $f, g \in U$ : fix a nonzero  $f$  and choose  $g$  outside the finitely many proper evaluation hyperplanes at its zeros. The kernel of

$$V_d \oplus V_d \longrightarrow V_{n+d}, \quad (A, B) \longmapsto fA + gB$$

consists exactly of  $(gh, -fh)$  with  $h \in V_{d-n}$ . For  $d = n - 1$  its kernel is zero and both dimensions are  $2n$ . For  $d \geq n$  the rank is  $2(d + 1) - (d - n + 1) = d + n + 1$ . Thus the map is onto in both cases. If  $U^2 = V_{2n}$  then  $U$  is automatically base-point-free; apply the first assertion successively with  $d = 2n, 3n, \dots$   $\square$

**Lemma 33.3** (All higher-product ranks on secants). *For a non-evaluation point  $h \in S_n$ , the image  $U^m$  has codimension one in  $V_{mn}$  for every  $m \geq 2$ . For an evaluation point it has codimension  $m$ .*

*Proof.* First take a distinct-support secant and choose the two points as 0 and  $\infty$ . In the binary monomial basis  $z_j = X^j Y^{n-j}$ , its hyperplane has a basis

$$z_1, \dots, z_{n-1}, \quad a = z_0 + \lambda z_n, \quad \lambda \neq 0.$$

Products of two of the interior monomials give all exponents  $2, \dots, 2n - 2$ . Multiplication of  $a$  by  $z_1$  and  $z_{n-1}$  then supplies exponents 1 and  $2n - 1$ , since the other terms are already in that span. Finally  $a^2$  gives the one allowed combination of the endpoint monomials. Thus  $U^2$  contains every interior monomial and one endpoint combination, and has dimension  $2n$ .

Inductively suppose  $U^m$  contains all monomials of exponents  $1, \dots, mn - 1$ , as well as  $a^m$ . Multiplying its interior monomials by  $z_1, \dots, z_{n-1}$  gives exponents  $2, \dots, (m+1)n - 2$ . Multiplying its two extreme interior monomials by  $a$  supplies the two missing interior exponents. The element  $a^{m+1}$  supplies an endpoint combination. All products satisfy the single endpoint relation inherited from  $f(0) = \lambda^{-1}f(\infty)$  for  $f \in U$ . Hence there can be no second endpoint direction, and the image has dimension  $(m+1)n$ .

At a non-evaluation tangent secant, a change of homogeneous coordinates puts the hyperplane in the form with the first derivative at 0 equal to zero. To justify the normalization, in a local coordinate a change fixing 0 changes a first-jet functional by a nonzero scalar and an arbitrary multiple of evaluation; the latter multiple can therefore be removed. The hyperplane is then

$$U = \text{span}\{z_0, z_2, z_3, \dots, z_n\}.$$

Sums of  $m$  exponents in  $\{0, 2, 3, \dots, n\}$  give exactly  $\{0, 2, 3, \dots, mn\}$ . This follows by induction, since the intervals obtained by adding  $[2, n]$  overlap or are adjacent for  $n \geq 3$ . Thus precisely exponent 1 is missing.

Finally, for  $h$  an evaluation,  $U = lV_{n-1}$  for its vanishing linear form  $l$ . Since products of the complete binary series span,  $U^m = l^m V_{m(n-1)}$ , of dimension  $m(n-1) + 1$ . Its corank is  $m$ .  $\square$

**Lemma 33.4** (Explicit plane-derivative matrix and residual span). *Use the monomials  $z_j = X^j Y^{n-j}$ . At a split secant take*

$$a = z_0 + \lambda z_n, \quad (w_1, \dots, w_{n-1}) = (z_1, \dots, z_{n-1}), \quad v = z_n, \quad \lambda \neq 0,$$

*and let the annihilator be coefficient  $mn$  minus  $\lambda^m$  times coefficient zero. At a tangent secant take  $a = z_0$ ,  $(w_1, \dots, w_{n-1}) = (z_2, \dots, z_n)$ ,  $v = z_1$ , and the coefficient-one annihilator. Set  $u_0 = a$ ,  $u_i = w_i$ , and vary  $u_i$  by  $t_i v$ . On the ordered source quotient*

$$a^m, w_1 a^{m-1}, \dots, w_{n-1} a^{m-1}$$

*the plane-derivative matrix is*

$$\kappa \text{diag}(m, 1, \dots, 1), \quad \kappa = \lambda^{m-1} \text{ or } 1, \quad (33.5)$$

*respectively. It is zero on all monomials with at least two  $w$  factors. Their product image has dimension  $mn - 3$  in both cases.*

*Proof.* Let  $\ell$  be the indicated annihilator. Then  $\ell(va^{m-1}) = \kappa \neq 0$ . The variation of  $a^m$  gives  $mkt_0$ . For  $w_i a^{m-1}$  only variation of  $w_i$  can contribute to  $\ell$ , giving  $\kappa t_i$ ; all terms which retain  $w_i$  vanish. For a monomial with at least two  $w$  factors one such factor remains after every single variation, and the result is annihilated by  $\ell$ . This gives every entry of (33.5).

In the split case  $W = XYV_{n-2}$ , so  $W^2 = (XY)^2 V_{2n-4}$ . The subseries  $U = W + \mathbb{C}a$  is base-point-free:  $W$  has only the two endpoint base points, and  $a$  is nonzero at both. Since  $2n - 4 \geq n - 1$  for  $n \geq 4$ , Lemma 33.2 can be applied successively at degrees  $2n - 4, 3n - 4, \dots$ , proving  $W^2 U^{m-2} = (XY)^2 V_{mn-4}$ .

In the tangent case the exponent set of  $W^2$  is the entire integer interval  $[4, 2n]$ . If the exponent set at degree  $k$  is  $[4, kn]$ , adding the exponent set  $\{0\} \cup [2, n]$  of  $U$  gives  $[4, kn] \cup [6, (k+1)n] = [4, (k+1)n]$ , since  $kn \geq 5$ . Induction from  $k = 2$  proves the assertion in every degree. There are exactly  $mn - 3$  exponents in  $[4, mn]$ .  $\square$

**Lemma 33.5** (Scheme smoothness off the evaluation curve). *At every point of  $S_n \setminus C_n$ , the plane derivative of the annihilator incidence has rank  $n$ , and its polar residual image has dimension  $mn - 3$ . Consequently the tangent space of  $D_m$  has dimension three at each such point.*

*Proof.* For distinct support, let  $g$  be its vanishing quadratic and write  $U = W \oplus \mathbb{C}a$ , with  $W = gV_{n-2}$ . The plane derivative vanishes on monomials with at least two factors in  $W$ , while it

is an isomorphism onto the dual of the  $n$ -dimensional quotient spanned by  $a^m$  and  $wa^{m-1}$ . Indeed the functional on  $V_n/U$  obtained by varying one factor and evaluating against the annihilator is nonzero; the images of the  $n$  basis vectors of  $U$  can be varied independently. This is the computation of Theorem 32.3, now with  $c = n$ . The residual image is

$$W^2U^{m-2} = g^2V_{2n-4}U^{m-2} = g^2V_{mn-4}.$$

The last equality follows from Lemma 33.2, because  $U$  is base-point-free and  $2n - 4 \geq n - 1$ . Its dimension is  $mn - 3$ .

For a tangent secant in the normal form of Lemma 33.3, let  $W = \text{span}\{z_2, \dots, z_n\}$  and  $a = z_0$ . The annihilator in  $V_{mn}^*$  is the coefficient of exponent 1. Since  $V_n/U$  is spanned by  $z_1$ , the derivative independently prescribes the coefficients dual to  $a^m$  and  $wa^{m-1}$ , and has rank  $n$ . Once again its kernel on the dual source consists of monomials with at least two  $W$  factors. Their products span precisely the monomials of exponents  $4, \dots, mn$ :  $W^2$  gives the full interval  $[4, 2n]$ , and successive factors from  $U$  extend it without gaps. This image also has dimension  $mn - 3$ .

The incidence tangent sequence now gives, in both cases,

$$\dim TD_m = \dim T\mathcal{I} = n + (mn + 1) - 1 - n - (mn - 3) = 3,$$

because the original multiplication has corank one by Lemma 33.3.  $\square$

**Lemma 33.6** (The length-two radical, including a double point). *Let  $0 \neq \ell \in V_{2n}^*$  and let  $H_\ell(f, g) = \ell(fg)$  on  $V_n$ . If  $H_\ell$  has rank two, there is a length-two divisor  $Z \subset \mathbb{P}^1$  such that*

$$\ell \in H^0(\mathcal{O}_Z(2n))^* \subset V_{2n}^*, \quad \text{rad } H_\ell = H^0(\mathcal{O}(n)(-Z)).$$

*The induced pairing on  $H^0(\mathcal{O}_Z(n))$  is nondegenerate.*

*Proof.* The rank-at-most-two Hankel variety is the secant variety of the evaluation rational normal curve. The shape-independence of the Hankel minor ideal and the precise secant parametrization are given in [7, Section 1 and Section 2, equation (2.0.1)]; here the minor size is three. The projective bundle of the spaces  $H^0(\mathcal{O}_Z(2n))^*$  over  $\text{Hilb}^2 \mathbb{P}^1$  is proper. Its image contains the distinct-point secant lines and is contained in the rank-at-most-two locus by factorization through a two-dimensional restriction space. Its image is therefore the entire secant variety, including tangent lines. Thus  $\ell$  factors through some  $Z$ .

For two distinct points the pairing on the length-two algebra has matrix  $\text{diag}(\alpha, \beta)$ . Rank two forces  $\alpha\beta \neq 0$ . For  $Z = 2p$ , choose a parameter  $z$  and trivialize the line bundle. Write  $\ell(A + Bz) = \alpha A + \beta B$  on  $\mathbb{C}[z]/(z^2)$ . The pairing matrix in the basis  $1, z$  is  $\begin{pmatrix} \alpha & \beta \\ \beta & 0 \end{pmatrix}$ , which has rank two exactly when  $\beta \neq 0$ . Restriction  $V_n \rightarrow H^0(\mathcal{O}_Z(n))$  is surjective. In either case the kernel of the pulled-back nondegenerate pairing is exactly its restriction kernel  $H^0(\mathcal{O}(n)(-Z))$ .  $\square$

**Lemma 33.7** (Equivariant associated points on a transitive curve). *Let a group act on a Noetherian variety, preserving an ideal  $\mathcal{J}$  with irreducible reduced support  $S$ . Let  $C \subset S$  be a closed irreducible curve on whose closed points the group acts transitively. Suppose  $\mathcal{J} = \mathcal{I}_S$  away from  $C$  and the nonzero module  $\mathcal{N} = \mathcal{I}_S/\mathcal{J}$  has support exactly  $C$ . Then*

$$\text{Ass}(\mathcal{O}/\mathcal{J}) = \{\eta_S, \eta_C\}.$$

*Proof.* For a finite module over a Noetherian ring, associated primes are finite, minimal points of its support are associated, and the associated primes of a submodule are associated to the ambient module; see [28, Lemma 10.63.3, Lemma 10.63.5, and Proposition 10.63.6]. These assertions apply on a finite affine cover and give the corresponding coherent-sheaf statements. Associated points outside  $C$  consist only of  $\eta_S$ , since the quotient there is the structure sheaf of a reduced irreducible variety. An associated point with closure inside  $C$  is either  $\eta_C$  or a closed point. The finite associated set is preserved by the group action. Transitivity on the infinitely many closed

points of  $C$  excludes the latter possibility. Finally  $\eta_C$  is a minimal point of the support of  $\mathcal{N}$ , hence is associated to  $\mathcal{N}$  and therefore to  $\mathcal{O}/\mathcal{J}$ . The minimal point  $\eta_S$  is associated as well. This gives exactly the two asserted points.  $\square$

The coordinate assertion in Lemma 33.5 is supplied by (33.5). The length-two assertion and the associated-point step of the following proof are, respectively, Lemma 33.6 and Lemma 33.7. We retain the full argument to specify where its nilpotence estimate is used.

*Proof of Theorem 33.1.* We first identify the quadratic support. A hyperplane  $U$  can be totally isotropic for a nonzero Hankel form  $H_\ell$  only if that form has rank at most two: the maximal isotropic dimension of a rank- $r$  form on an  $(n+1)$ -space is  $n+1 - \lceil r/2 \rceil$ . A rank-one Hankel form gives an evaluation hyperplane. A rank-two Hankel form has radical  $H^0(\mathcal{O}(n)(-Z))$  for a length-two divisor  $Z$  on  $\mathbb{P}^1$ . For clarity, this last assertion includes a double point: the proper family of secant lines indexed by length-two divisors covers the rank-at-most-two Hankel variety of Lemma 27.1; on a length-two quotient the rank-two pairing is nondegenerate, so its kernel is exactly the indicated space. Since an isotropic hyperplane contains this radical, its annihilating functional lies on the corresponding secant or tangent line to  $C_n$ . Hence  $|D_2| \subset S_n$ . Conversely a distinct-support secant imposes a nonzero quadratic endpoint relation. Closedness supplies the whole  $S_n$ , proving  $|D_2| = S_n$ .

Outside  $S_n$ , quadratic multiplication is onto; the propagation lemma proves surjectivity in every higher degree. On  $S_n$  the rank lemma gives the two remaining cases of (33.1). Thus  $|D_m| = S_n$  globally. Since  $S_n$  is an irreducible threefold, Lemma 33.5 equates local Krull and tangent dimensions at every point off  $C_n$ . The local rings there are regular. In particular  $\mathcal{J}_m = \mathcal{I}_{S_n}$  away from  $C_n$ .

The coherent quotient  $\mathcal{N}_m = \mathcal{I}_{S_n}/\mathcal{J}_m$  is therefore supported on  $C_n$ , and some power of  $\mathcal{I}_{C_n}$  kills it. This gives the inclusion of  $\mathcal{I}_{S_n}$  in the left side of (33.2). The reverse inclusion holds because  $\mathcal{I}_{S_n}$  is prime and  $S_n$  is not contained in  $C_n$ . This proves the saturation identity.

At an evaluation point  $x$ , the original corank is  $m$ . Schur elimination consequently expresses  $\mathcal{J}_{m,x}$  as the ideal of  $m$ -minors of a matrix whose entries vanish at  $x$ . In the regular local ring  $\mathcal{O}_{\mathbb{P}(V_n^*),x}$ , with maximal ideal  $\mathfrak{m}_x$ , one has

$$\mathcal{J}_{m,x} \subset \mathfrak{m}_x^m. \quad (33.6)$$

Choose coordinates  $h_0, \dots, h_n$  dual to the binary monomials so that  $x = [1 : 0 : \dots : 0]$ , and work on  $h_0 = 1$ . The function

$$F = \det \begin{pmatrix} 1 & h_1 & h_2 \\ h_1 & h_2 & h_3 \\ h_2 & h_3 & h_4 \end{pmatrix}$$

vanishes on  $S_n$ , since every two-evaluation moment sequence has Hankel rank at most two. Its initial form at  $x$  is  $h_2 h_4 - h_3^2$ , a nonzero quadratic. Therefore  $F^j$  has order  $2j$ , and (33.6) implies that its class modulo  $\mathcal{J}_m$  is nonzero whenever  $2j < m$ . Transitivity of  $\mathrm{SL}_2(\mathbb{C})$  on  $C_n$  proves (33.4) at every evaluation point. For  $m \geq 3$  this shows in particular that  $\mathcal{N}_m$  is nonzero with support all of  $C_n$ .

It remains to exclude any additional associated points. The multiplication and its maximal-minor ideal are  $\mathrm{SL}_2(\mathbb{C})$ -equivariant. The associated points of a coherent sheaf form a finite set, stable under automorphisms. Off  $C_n$  the scheme is the smooth irreducible open part of  $S_n$ , so its only associated point there is  $\eta_{S_n}$ . An associated point whose closure is contained in  $C_n$  is either  $\eta_{C_n}$  or a closed point of  $C_n$ . The latter is impossible: transitivity would force every closed point of the curve into this finite set. Thus the only possible embedded associated point is  $\eta_{C_n}$ . It actually occurs because the nonzero submodule  $\mathcal{N}_m$  has support  $C_n$ , whose generic point is a

minimal point of its support and hence an associated point. Associated points of a submodule are associated points of the ambient module. This proves (33.3), and completes the proof.  $\square$

The expected determinantal codimension here is  $\binom{n+m-1}{m} - mn$ , while the actual codimension is  $n - 3$ . For  $m \geq 3$  this is an excess family with a complete associated-point classification. The nilpotency bound is unbounded as  $m$  grows even though the reduced threefold and its unique embedded support do not change. This conclusion is stronger than the two-point-sector calculation: it excludes every other reduced failure component and every other embedded associated subvariety in this entire family. It does not identify the full embedded primary ideal, or assert an all-contact, all- $c$  higher-product classification.

### 34. COMPARISON WITH PRIOR MULTIPLICATION AND FAILURE-LOCUS RESULTS

The use of secants to account for failure of normal generation is classical. It is important here to distinguish the fixed complete series, its varying proper subseries, the Hankel annihilator space, and the scheme on the parameter space of subseries. The following comparisons refer to specific results, not to an inference from the titles of articles.

| Result compared                   | Parameter space and conclusion                                                                                                                    | Use and distinction here                                                                                                                                                   |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [7, Section 1; Theorem 2.1]       | Hankel matrices of indeterminates; determinantal dimensions, normality and rational singularities.                                                | These supply the annihilator-rank geometry. They do not identify the singularities of its varying isotropic-plane image. Our residual Fitting formulas concern that image. |
| [8], [19, Section 2.9]            | A matrix satisfying the expected-grade condition has its maximal-minor resolution.                                                                | We use this resolution after proving the height. The rank competition, exceptional incidences and residual normal derivative are separate arguments.                       |
| [5, Theorem 0.2; Proposition 2.2] | A fixed completely embedded surface; numerical cohomology conditions force failure on finite linear sections, including a higher-product version. | Here the subseries vary, without cross-products. These cited statements do not give the component or residual Fitting formulas on $\text{Gr}(c, V_n)^L$ .                  |
| [14, Theorems 1 and 3]            | Complete very ample curve series; criteria and secant explanations for failure of normal generation.                                              | The ambient complete series on $\mathbb{P}^1$ here is normally generated. The failure comes from varying proper subseries.                                                 |

For the component theorem the additional assertions are the full rank-incidence optimization, the five exceptional isotropic incidences, and descent of the signed families. For the global singularity theorem they are the identification of the polar residual space and the equality of the restricted cotangent Fitting ideals with its multiplication minors. The local node theorem further uses a relative determinant divisor and its normal derivative, not merely the dimensions of rank strata. The cotangent presentation, Schur elimination and determinantal resolution themselves are standard algebraic tools and are not claimed as new constructions.

The closely related paper [4] requires a distinct qualification. The preceding source audit inspected its publisher's first page, which introduces higher-order embedding conditions and varying restriction maps to finite schemes. The remaining pages, containing the theorem statements needed for a complete comparison, were not available in the primary text retrieved for this revision. Accordingly no assertion that its theorems exclude or contain the results here is made. The comparison with this particular paper remains unverified; the table above is not a substitute for it. In particular a difference of parameter-space descriptions on the first page would not by itself establish a priority claim.

The local presentations in Section 26 use classical Schur elimination and the analytic implicit function theorem. The generic determinantal singularity, the quadratic cone  $xz - y^2 = 0$ , and the cubic cusp are not claimed as new singularities. The assertion specific to multiplication is the formula for the joint polar relations, its single-annihilator specialization, and the explicit residual obstruction and realization maps. A theorem-level comparison of these formulations with the failure-locus literature must distinguish an exact nonlinear presentation from the restricted Fitting identities; neither is a substitute for the unresolved comparison with [4].

**34.1. The intrinsic derivative and classical determinantal geometry.** The intrinsic map to  $\text{Hom}(\ker \mu, \text{coker } \mu)$  is the normal derivative to a matrix rank stratum. This interpretation, and the tangent condition  $\psi(\ker \varphi) \subset \text{im } \varphi$ , are classical; see [10, Remark 1.18]. Invertible row and column operations preserve minor ideals [10, Lemma 1.3], and the generic rank-stratum germ is a smooth factor times a smaller determinantal germ [10, Section 1.3.2, equation (13)]. Essential nonsingularity is transversality to that stratum [10, Definition 2.11 and Proposition 2.12]. Fitting ideals are independent of the chosen presentation and commute with base change [25, Tags 07Z8 and 07ZA].

Accordingly, neither the existence of an intrinsic normal derivative nor the generic determinantal normal form is claimed as a new general theorem. Section 26 gives a self-contained application of these tools, identifies their dual obstruction through the actual multiplication polars, and separates the resulting exact nonlinear equations from their first-order Fitting data. The structural hypotheses are explicitly tested there; the generic realization example is not used as a substitute for a natural higher-product classification.

The latter role belongs to Theorem 33.1. It proves a global statement on the natural parameter space  $\mathbb{P}(V_n^*)$ , for every  $n \geq 4$  and  $m \geq 2$ , rather than inferring a classification from a general transversality criterion. The base-point-free pencil argument and rank-two Hankel description are classical ingredients. The additional scheme conclusions proved here are smoothness at every non-evaluation secant, saturation off the evaluation curve, the complete associated-point set for  $m \geq 3$ , and a degree-dependent lower bound on nilpotency. They follow from the polar rank computation, the order of the Schur minors at an evaluation, and equivariance. They are not consequences of the generic matrix normal form alone. This comparison identifies the content requiring a priority assessment; it does not assert that an exhaustive search has established its historical originality.

**34.2. Contact-algebra reduction and the scope of the new classification.** Theorems 23.2–20.2 concern the entire relative generating Grassmannian  $X_{d,2}^\circ$ , including all multiplicity partitions of its length- $d$  divisor. The subseries have codimension  $d - 2$ ;  $d$  is arbitrary. For this family the result is a complete scheme theorem, not only a two-point annihilator sector: the component ideals, primary exponents, absence of embedded points, nilpotent layers, all coranks, and moving-divisor normalization are explicit. The marked length-two divisor in the normalization is recovered from the failure scheme; it is not imposed on the parameter space as a restriction to one preselected obstruction sector.

The determinant of a power basis, the Vandermonde and confluent Vandermonde identities, and the change-of-basis formula for discriminants are elementary classical identities; no priority is claimed for those identities. Their proofs are included to fix all multiplicities, including at nonreduced contacts. The exact multiplication-cokernel transport in Theorem 23.2 is what permits these identities to describe the original maximal-minor scheme after arbitrary base change. The resulting primary decomposition is not inferred from an order-of-vanishing lower bound.

For a precise comparison, [5, Theorem 0.2 and Proposition 2.2] are existence results for deficient finite linear sections of a fixed completely embedded surface. Those statements do not contain the contact-pencil primary formula or the normalization constructed here. This

observation is a comparison of the specified statements, not a claim about all work on failure loci. The theorem pages of [4] have still not been obtained, so its comparison remains explicitly undetermined. The accompanying source audit records the parameter-space, degree, scheme, corank, and associated-prime questions separately; a missing source is not treated as evidence of novelty.

### 35. AN APPLICATION TO CALIBRATED INFORMATION RECOVERY

Let  $J$  be fixed and invertible, let  $H(\theta)_{ij} = \theta_{i+j}$ , and set  $R_\theta = JH(\theta)J^\top$  in its positive moment cone. For a prescribed smooth contact, a normal frame  $Z$  determines  $U = J^\top \text{ran } Z$ . The normal Hessian of the squared  $R_\theta^{-1}$ -distance is the inverse of  $Z^\top R_\theta Z$  (Lemma B.1). Hence this contact gives exactly the restriction of the moment functional to  $U^2$ . Several independent contacts give the dual of (24.1). The information-loss locus is therefore the real positive realization of the failure geometry studied above, subject to the exact contact hypotheses.

Theorem D.1 supplies the positive native realization and global separation conditions. At a real crossing with two real branches, Corollary 31.10 gives the local small-singular-value law  $\varepsilon^a \log(1/\varepsilon)$ . These are applications of the geometry, not assumptions in its proofs. The appendices give the complete likelihood, boundary, score-compression, finite-offset and estimation arguments. In particular they do not identify a computed feature with an independent noisy distance observation.

#### APPENDIX A. CALIBRATED INFORMATION METRICS AND THE GEOMETRIC REDUCTION

Let  $M \subset \mathbb{R}^k$  be a known residual image and  $R > 0$  an inverse information matrix. Consider

$$C_R(b) = \inf_{x \in M} (x - b)^\top R^{-1} (x - b). \quad (\text{A.1})$$

At a smooth point, its full second jet sees only the normal compression of  $R$ . A statistical model may constrain  $R$  enough to remove this ambiguity. The inverse problem is therefore governed both by the geometry of  $M$  and by the linear space of admissible information matrices.

We study the calibrated family

$$R_\theta = JH(\theta)J^\top, \quad H(\theta)_{ab} = \theta_{a+b}, \quad 0 \leq a, b \leq k-1,$$

where  $J$  is fixed and invertible, and  $\theta$  lies in an open positive moment cone of dimension  $p = 2k-1$ . Set  $n = k-1$  and  $V_n = H^0(\mathbb{P}^1, \mathcal{O}(n))$ . Dehomogenization identifies real coefficient vectors with  $\mathcal{P}_n$ . A normal space  $N$  corresponds to  $U = J^\top N$ . The observations specify the moment functional on  $U^2 = \text{span}\{fg : f, g \in U\}$ . Multiple contacts specify a sum of such spaces, without cross-contact products.

The central geometric issue is not only whether a general product map is onto, but how large its entire failure locus is. Put  $q = \binom{c+1}{2}$  and

$$X_L = \text{Gr}(c, V_n)^L, \quad \Delta^{(L)} = \{(U_1, \dots, U_L) : \sum_\nu U_\nu^2 \neq V_{2n}\}.$$

Here codimension refers to complex codimension of the reduced closed set. Theorem 24.1 gives the precise formula. The common-secant condition means that all  $U_\nu$  lie in a hyperplane represented by a point of the secant variety of the evaluation rational normal curve in  $\mathbb{P}(V_n^*)$ . Thus the second term of (24.2) is a concrete geometric obstruction, not a formal expected codimension. For one contact, the common-base-point locus lies properly inside this larger irreducible family and is not itself a component of the failure locus.

The numerical formula has a structural refinement. Write  $a = Lq - p + 1$  and  $b = Lc - 3$ . In the stable range  $c \geq 8$ ,  $k \geq 2c + 1$ , Theorem G.1 proves that the maximal-minor scheme is integral and Cohen–Macaulay when  $a < b$ . Its generic annihilator is a nondegenerate Hankel form. At  $a = b$  the scheme is reduced and Cohen–Macaulay, with precisely one nondegenerate component

and  $2^{L-1}$  signed secant components. If  $b < a$ , the latter are exactly the maximal-dimensional components. They are rational and generically smooth in the determinantal scheme. The signs record independent choices of the two isotropic hyperplanes of a rank-two form, modulo their simultaneous interchange; the common-secant family above is only the all-equal choice.

The change of regime has an intrinsic residual description. At two distinct secant points, write  $U_\nu \cap \text{rad } H_\ell = gK_\nu$ , where  $g$  is their vanishing quadratic. The wall is

$$L\binom{c}{2} = 2k - 5,$$

exactly the saturation threshold for multiplication of the  $(c - 1)$ -planes  $K_\nu$  in the  $(k - 2)$ -dimensional residual space. This identity determines the conormal dimension of a general signed component. The expected-codimension cycle and local conditioning exponents follow from the structural theorem. The earlier all-dimension codimension formula is not restricted by the additional hypotheses of this refinement.

For the synchronized residual image

$$q_A(\eta) = ((a_i^\top \eta)^2)_{i=1}^k, \quad A \in \mathbb{R}^{k \times d},$$

the normal dimension is  $c = k - d$ . Native realizability is an open positive-vector condition. Global separation at a prescribed normal space requires, in addition, that the corresponding tangent plane not split across a nontrivial coordinate partition. This exact criterion is proved in Theorem D.1.

**Theorem A.1** (Sharp native single-contact threshold). *Fix  $d \geq 2$  and any separated root and clock configuration of Section C. For  $k > d$ , there exist full-rank loadings  $A$  with  $q_A(e_1) > 0$  for which the single full second jet  $D^2 C_{R_\theta}(q_A(e_1))$  identifies every  $\theta \in \Omega$  if and only if*

$$\binom{k - d + 1}{2} \geq 2k - 1.$$

*In the sufficient range the assertion holds on a nonempty Zariski-open loading set, for the global distance function. In particular*

$$k_{\min}(d) = d + \left\lceil \frac{3 + \sqrt{16d + 1}}{2} \right\rceil = d + 2\sqrt{d} + O(1). \quad (\text{A.2})$$

*Unrestricted positive matrices at the same contact retain an invisible space of dimension  $d(2k - d + 1)/2$ .*

A contact here is a matrix-valued observation with  $q$  entries, not a scalar datum. The comparison in (A.2) is across calibrated families with varying  $k$ , fixed  $d$ , and arbitrary separated configurations at each  $k$ ; it is not a universal threshold for every information model. Independently designed contacts can use different loadings with the same  $J$ . Several points of one fixed loading are not asserted to vary independently.

**Method and relation to earlier theory.** Hankel duality with polynomial multiplication is classical [21]. The saturated one-space maximal-rank statement is also a consequence of classical postulation of general rational curves [2, 3, 20]. Our proof of Theorem 24.1 instead combines the classical dimensions of Hankel rank strata [7, Sections 1–2] with an isotropic-incidence count, an optimization over all ranks, and an explicit common-secant family. This proves the full codimension formula and the coupled multi-space assertion, not just generic surjectivity. The classical maximal-minor height bound supplies the other matching upper bound [9, Introduction]. These classical inputs are distinguished from the incidence optimization and native fixed-space realization proved here.

Failure of quadratic normality has a substantial antecedent literature. Green–Lazarsfeld [14, Theorems 1 and 3] study complete very ample linear series on curves, giving normal-generation criteria and finite secant configurations explaining failure under degree and cohomology hypotheses. Ballico [5, Theorem 0.2] studies failure cycles for a fixed embedded surface and detects failure on finite linear sections, using a Grassmannian incidence and cohomology. The principle that secants can explain quadratic-normality failure is therefore classical. Here the complete series on  $\mathbb{P}^1$  is fixed and normally generated; the varying objects are its proper  $c$ -dimensional subseries, and for  $L > 1$  the observation uses a direct sum of within-subseries products, without cross-products. Our structural assertion concerns the components and generic scheme structure over this product of Grassmannians. Neither the complete-series degree criterion nor the surface-section failure-cycle theorem supplies this signed component classification or the residual wall. We distinguish these particular conclusions from the classical inputs on Hankel ranks, determinantal resolutions, and cycles [7, 8, 19, 16, 11].

Polynomial product spaces also occur in the study of Gram spectrahedra [23]; the present observation is the dual moment functional on within-contact products. The monomial specialization is a restricted additive-basis problem [27, 17, 18], and can have a strictly worse threshold. Quadratic sensing and phase retrieval concern reconstruction of the input to a quadratic measurement map [1]; our unknown is instead an information metric, and injectivity on all of  $\text{Sym}_d$  is not assumed. Complete-conormal projective duality [12] addresses a different, larger observation category.

The statistical statements likewise keep their experiments separate. The coalesced-root model has a smooth positive extension, but its physical local parameter set is a cone. Its efficient Gaussian quotient and  $N^{-1/4}$  root-amplitude scale are derived explicitly in Section H. For observed Poisson exposures we form normal-compression score statistics directly from the counts. Their local information loss is exactly the loss of rank of the product map. This is a local likelihood statement in a specified target subexperiment, not a claim of global equivalence of all raw data and finitely many distance readings. The finite-sample estimators and computed finite-offset features in the appendices are retained as distinct procedures. Standard constrained and regular asymptotic principles are attributed to [13, 26].

## APPENDIX B. EXACT FIBRES OF PRESCRIBED CONTACT DATA

For  $\theta \in \mathbb{R}^p$  write

$$\ell_\theta \left( \sum_{a=0}^{2n} f_a t^a \right) = \sum_a f_a \theta_a.$$

Let  $\Omega \subset \mathbb{R}^p$  be open with  $R_\theta > 0$ . At each prescribed point  $x_\nu$ , assume that the global image has one embedded smooth germ and that its local minimizing projection computes (A.1). Parameter differentiation below uses analytic germs and a common projection tube on compact metric sets. These conditions are verified for the native loading family in Section D.

Let  $Z_\nu$  have orthonormal columns spanning the Euclidean normal space. Put

$$B_\nu = Z_\nu^\top R_\theta Z_\nu, \quad G_\nu = \frac{1}{2} Z_\nu^\top D^2 C_{R_\theta}(x_\nu) Z_\nu.$$

**Lemma B.1** (Normal block). *One has  $G_\nu = B_\nu^{-1}$ . The full Hessian has tangent kernel and is determined by this normal block.*

*Proof.* The quadratic contact with a manifold equals that with its tangent space  $T$ . At the minimizer of  $(u - v)^\top R^{-1}(u - v)$  over  $v \in T$ , the vector  $R^{-1}(u - v)$  is normal. Hence  $u - v = RZn$ , and normal projection gives  $Bn = Z^\top u$ . The minimum is  $(Z^\top u)^\top B^{-1} Z^\top u$ . A local parametrization and its positive tangent Gram matrix justify the minimizing projection by the implicit function theorem and yield this quadratic term.  $\square$

Set  $U_\nu = J^\top \text{ran } Z_\nu$  in polynomial coordinates, and define

$$\mathcal{U} = \text{im} \left( \bigoplus_{\nu=1}^L \text{Sym}^2 U_\nu \xrightarrow{\text{multiplication}} \mathcal{P}_{2n} \right), \quad \mathcal{L}\theta = (B_\nu(\theta))_\nu.$$

The codomain of  $\mathcal{L}$  has the direct-sum Frobenius norm.

**Theorem B.2** (Exact contact fibre). *The exact second-jet fibre through  $\theta$  is*

$$(\theta + \mathcal{U}^\perp) \cap \Omega.$$

*Its local dimension is  $p - \dim \mathcal{U}$ . Recovery on  $\Omega$  is equivalent to  $\mathcal{U} = \mathcal{P}_{2n}$ , or to injectivity of  $\mathcal{L}$ .*

*Proof.* For normal vectors  $v, w$  with coefficient polynomials  $f_v, f_w$ ,

$$v^\top (R_{\theta'} - R_\theta) w = \ell_{\theta' - \theta}(f_v f_w).$$

Equality of the jets is equality of the positive normal blocks, by Lemma B.1. It is therefore precisely annihilation of  $\mathcal{U}$ . Inversion is a diffeomorphism on positive blocks; openness of  $\Omega$  gives the asserted fibre dimension.  $\square$

For one  $d$ -dimensional germ there are at most  $q = \binom{k-d+1}{2}$  independent products. For unrestricted matrices, fixing the normal–normal block leaves tangent–tangent and mixed blocks free, with total dimension

$$\frac{k(k+1) - (k-d)(k-d+1)}{2} = \frac{d(2k-d+1)}{2}. \quad (\text{B.1})$$

Small perturbations in every such direction preserve positivity.

#### APPENDIX C. THE NATIVE INFORMATION FAMILY

Fix distinct strictly positive rank-one four-cell probability tables  $M_1, M_2$ , roots  $0 < r_1 < \dots < r_k < D$ , and  $m = 2k + 1$  clocks  $D < T_1 < \dots < T_m$ . Use

$$f_1(T) = f_{1,0}(T) \prod_i ((T - r_i - s_i/2)^2 - v_i), \quad p_j = \pi_j M_1 + (1 - \pi_j) M_2,$$

$$\frac{\pi_j}{1 - \pi_j} = \frac{\alpha f_1(T_j)}{(1 - \alpha) f_2(T_j)}.$$

The fixed factors are positive at the clocks. At  $s = v = 0$ , the free coordinates are log mixing odds and the root-sum displacements; the physical retained coordinates satisfy  $v_i \geq 0$ . The likelihood is positive and analytic on a two-sided neighbourhood of this point. Its restriction to  $v \geq 0$ , not that extension's unrestricted estimator, is the physical experiment.

Let  $\lambda_j > 0$  be information weights and set

$$\kappa_j = \pi_j^2 (1 - \pi_j)^2 \sum_{a=1}^4 \frac{(M_{1a} - M_{2a})^2}{p_{ja}}, \quad V_j = (1, -((T_j - r_i)^{-1})_i, -((T_j - r_i)^{-2})_i).$$

The  $k$  simple poles precede the  $k$  double poles. Write

$$\begin{aligned} h(t) &= \prod_i (t - r_i)^2, \quad p_*(t) = \prod_i (t - r_i), \quad P(t) = \prod_j (t - T_j), \\ d_i &= \frac{P(r_i)}{\prod_{a \neq i} (r_i - r_a)^2}, \quad \xi_j = \frac{h(T_j)}{P'(T_j)}, \\ J_{ia} &= [t^a] d_i \frac{p_*(t)}{t - r_i}, \quad \rho_j = \frac{\xi_j^2}{p_*(T_j)^2}. \end{aligned} \quad (\text{C.1})$$

**Proposition C.1** (Profiled quadratic information). *The retained block of the inverse full information, equivalently the inverse nuisance Schur complement, is*

$$R = JH(\theta)J^\top, \quad \theta = \sum_{j=1}^m \frac{\rho_j}{\lambda_j \kappa_j} w_{2n}(T_j), \quad w_s(t) = (1, t, \dots, t^s)^\top. \quad (\text{C.2})$$

The matrix  $J$  is invertible. Independent positive information weights give an open moment cone  $\Omega$  of dimension  $p = 2k - 1$ .

*Proof.* Multiplying a linear combination of the columns of  $V$  by  $h$  gives a polynomial of degree at most  $2k$ . Vanishing at the  $2k + 1$  clocks forces it to be zero; principal parts at the roots and the constant at infinity then force every coefficient to vanish. Thus  $V$  is invertible and the full information is  $V^\top \text{diag}(\lambda_j \kappa_j) V$ .

For arbitrary log-odds increments  $y_j$ , interpolate

$$g(t) = \sum_j h(T_j) y_j \frac{P(t)}{(t - T_j) P'(T_j)}.$$

The retained coefficient at  $r_i$  is  $-g(r_i) / \prod_{a \neq i} (r_i - r_a)^2$ . Thus the retained rows  $Z_0$  of  $V^{-1}$  are

$$(Z_0)_{ij} = \frac{\xi_j d_i}{T_j - r_i} = \frac{\xi_j}{p_*(T_j)} (Jw_n(T_j))_i.$$

Their weighted Gram matrix gives (C.2). Evaluation of the polynomials defining  $J$  at the distinct roots is diagonal with nonzero entries, proving invertibility. The positive moment weights range over an open orthant. The moment matrix has rank  $p$  by Vandermonde independence, so its image is open. Positive atomic weights at distinct clocks give positive definite  $H(\theta)$ .  $\square$

The weights in this proposition need not be unknown likelihood parameters. For a realized fixed allocation they are  $n_j/N$ ; for a limiting allocation they are its limits; a different prespecified  $\lambda$  defines a target design metric. Section I uses a further protocol in which exposures are unknown but their counts are observed. Section H identifies (C.2) as the covariance of the efficient Gaussian quotient of the physical boundary experiment.

#### APPENDIX D. NATIVE REALIZATION AND GLOBAL SEPARATION

All spaces in this section are real. The following result holds for any fixed  $J \in \text{GL}_k(\mathbb{R})$ , not just the interpolation matrix.

**Theorem D.1** (Exact fixed-space realization). *Let  $1 \leq d < k$ ,  $c = k - d$ ,  $U \in \text{Gr}(c, k)$ , and  $T = JU^\perp$ . The space  $U$  is the normal-polynomial space of a local immersed image  $q_A$  at  $q_A(e_1) > 0$  if and only if*

$$T \cap \mathbb{R}_{>0}^k \neq \emptyset. \quad (\text{D.1})$$

There is such a realization with

$$q_A^{-1}(q_A(e_1)) = \{e_1, -e_1\} \quad (\text{D.2})$$

if and only if, in addition,

$$D_s T \neq T \quad \text{for every } s \in \{\pm 1\}^k \setminus \{\mathbf{1}, -\mathbf{1}\}. \quad (\text{D.3})$$

Here  $D_s = \text{diag}(s)$ . On each sign chamber of the full-rank loading domain with  $Ae_1$  nowhere zero, the map

$$A \mapsto U(A) = J^\top (\text{ran}(\text{diag}(Ae_1)A))^\perp$$

is a submersion onto its positive realizable image. That image is nonempty, Euclidean open, and Zariski dense.

*Proof.* Put  $z = Ae_1$ . The tangent space is  $T = \text{ran}(\text{diag}(z)A)$  and contains  $t = z^2 > 0$ . Taking orthogonal complements in  $U = J^\top T^\perp$  gives  $T = JU^\perp$  and proves necessity of (D.1). Conversely choose  $t > 0$  in  $T$ , extend it to a basis  $t, t_2, \dots, t_d$ , and set

$$z_i = \sqrt{t_i}, \quad A = [z, \text{diag}(z)^{-1}t_2, \dots, \text{diag}(z)^{-1}t_d]. \quad (\text{D.4})$$

Then  $\text{diag}(z)A$  has this prescribed basis, and  $q_A(e_1) = t$ .

Equality of coordinatewise squares is equivalent to  $A\eta = D_s z$  for a sign vector  $s$ . Because  $D_s$  commutes with  $\text{diag}(z)$ ,

$$D_s z \in \text{ran } A \iff D_s t \in T. \quad (\text{D.5})$$

If  $D_s T = T$  for a nonconstant  $s$ , the right side holds for every  $t \in T$ ; positivity ensures  $D_s t \neq \pm t$ , giving another preimage. Conversely, under (D.3), each  $T \cap D_s T$  is a proper linear subspace of  $T$ . Their finite union cannot cover the relatively open positive cone in (D.1). Choose  $t$  outside that union and use (D.4). Equation (D.5) leaves only the constant signs, hence exactly the two preimages in (D.2).

For the submersion statement, on a fixed sign chamber the change of frames  $A \mapsto B = \text{diag}(z)A$  is a diffeomorphism onto full-rank frames whose first column is strictly positive. Its inverse has  $z_i = \pm\sqrt{B_{i1}}$  with the fixed signs and  $A_{ij} = B_{ij}/z_i$ . The ordinary frame-to-plane map is a submersion. Orthogonal complement and multiplication by fixed  $J$  preserve that property. Positivity persists under small changes of a frame, and planes containing  $\mathbf{1}$  show nonemptiness. A polynomial vanishing on a nonempty real open subset of an affine Grassmannian chart vanishes identically, proving Zariski density.  $\square$

The obstruction in (D.3) has a concrete interpretation. If  $I$  is the set of positive signs of  $s$ , then  $D_s T = T$  precisely when

$$T = (T \cap \mathbb{R}^I) \oplus (T \cap \mathbb{R}^{I^c}).$$

Thus a prescribed normal space may be locally realizable yet admit no globally separated realization at that same space. Theorem D.1 classifies exactly this distinction.

**Lemma D.2** (Generic global distance). *A nonempty Zariski-open subset of the full-rank loading domain with  $Ae_1$  nowhere zero satisfies (D.2). For each such fixed  $A$ , the global image has one embedded analytic germ at  $b_0 = q_A(e_1)$ . For every compact set  $\mathcal{R} \subset \text{Sym}_k^{>0}$  there is a neighbourhood of  $b_0$  on which the distance to that germ equals the global distance, and the minimizing projection is analytic with uniform bounds for  $R \in \mathcal{R}$ .*

*Proof.* For a nonconstant sign vector, the condition  $D_s Ae_1 \in \text{ran } A$  is algebraic, defined by augmented  $(d+1)$ -minors. It is proper: choose a vector  $z$  with no zero entries and a  $d$ -plane containing  $z$  but not  $D_s z$ , then take a frame with first column  $z$ . Removing the finitely many such proper sets proves the first assertion. The ambient loading space is the irreducible affine space  $\mathbb{R}^{kd}$ ; full rank and nonzero first coordinates specify a Zariski-open domain, so these finite open intersections remain dense.

Full rank gives

$$\|q_A(\eta)\|_2 \geq k^{-1/2} \|A\eta\|_2^2 \geq k^{-1/2} \sigma_{\min}(A)^2 \|\eta\|_2^2,$$

so  $q_A$  is proper. Its derivative is injective at both preimages, and their local image germs coincide by evenness. Properness excludes a remote sheet accumulating at  $b_0$ . The trial image  $b_0$  bounds the distance by  $C\|b - b_0\|^2$ . Uniform positive metric bounds on  $\mathcal{R}$  therefore force all minimizing images into this germ when  $b$  is close enough to  $b_0$ . The positive tangent Hessian and the implicit function theorem give the projection tube. Its radius and derivative constants depend on  $A$ , the local germ and  $\mathcal{R}$ ; no uniformity over all generic loadings is claimed.  $\square$

*Proof of Theorem A.1.* Necessity is the dimension bound in Section B. If it holds and  $d \geq 2$ , then  $c = k - d \geq 5$ . Theorem 24.1 with  $L = 1$  gives a nonempty complex Zariski-open full-product locus. Its defining minors have real coefficients and some minor is a nonzero polynomial on a real affine chart. A nonzero real polynomial cannot vanish on any nonempty real open set. Thus the full-product locus meets the positive realizable image of Theorem D.1.

On pivot charts normal frames are rational functions of  $A$ , so full product rank pulls back to a nonempty Zariski-open subset of the irreducible loading domain. Intersect it with Lemma D.2's dense open set. Theorem B.2 gives recovery from the global distance for every  $\theta \in \Omega$  with the same  $J$ . Writing  $k = d + c$  gives  $c^2 - 3c - 4d + 2 \geq 0$ , whose positive root yields (A.2). Formula (B.1) gives the unrestricted ambiguity.  $\square$

For  $d = 2, 3, 4, 5, 6$ , the optimal  $k$  are 7, 8, 10, 11, 13. At the optimum the full symmetric measurement  $S \mapsto (a_i^\top S a_i)_i$  is necessarily noninjective for  $d \geq 5$ : check  $d = 5, 6$  directly, and use  $k_{\min} \leq 2d < d(d+1)/2$  for  $d \geq 7$ . This is not full-symmetric phase retrieval in disguise.

**Corollary D.3** (Independent native designs and the real secant family). *Fix  $4 \leq c < k$  and a native interpolation matrix  $J$ . For  $L$  independently chosen loading matrices of size  $k \times (k - c)$ , generic globally separated contacts recover the common native moment parameter if and only if  $Lq \geq p$ . The common-secant family  $\Sigma_L$  contains real, positive, globally separated native realizations. At its general such points its loading pullback has real local codimension  $Lc - 3$ .*

*Proof.* Apply the product of the loading submersions and generic global-separation opens to Theorem 24.1, using real nonvanishing of minors as above. For the second assertion choose distinct  $x, y > D$  and  $h = \text{ev}_x + \varepsilon \text{ev}_y$  with  $\varepsilon > 0$ . By (C.1), every coordinate of  $Jw_n(t)$  is negative for  $t > D$ . Thus  $t = -Jh > 0$ . A general  $d$ -plane  $T$  containing  $t$  avoids each of the finitely many vectors  $D_s t$  for nonconstant  $s$ , since none is proportional to  $t$  and  $d < k$ . Such a  $T = JU^\perp$  has  $U \subset \ker h$  and, by (D.4)–(D.5), a globally separated realization. Choose the  $T_\nu$  independently. These choices form real open subsets in the secant incidence. Its generically one-to-one projection is birational in characteristic zero and is an isomorphism on dense open subsets. The loading submersion consequently preserves its codimension there.  $\square$

The exact formula (24.2) is an algebraic statement over  $\mathbb{C}$ . The corollary supplies a real native component of that dimension when the secant term is minimal. It does not infer real points on every complex component in the other regime. It also does not replace the independent-design assumption by multiple arbitrary points on one fixed quadratic image.

## APPENDIX E. RANK STRATA AND QUANTITATIVE INVERSION

Let  $\Delta_r$  be the determinantal scheme on  $\text{Gr}(c, V_n)$  where multiplication has rank at most  $r$ . At a point of rank *exactly*  $r$ , put  $K_U = \ker \mu_U$  and  $Q_U = V_{2n}/U^2$ .

**Proposition E.1** (Infinitesimal rank map). *The tangent space to  $\Delta_r$  is the kernel of*

$$\begin{aligned} \beta_U : \text{Hom}(U, V_n/U) &\longrightarrow \text{Hom}(K_U, Q_U), \\ \beta_U(\phi) \left( \sum_i f_i \odot g_i \right) &= \sum_i \{ \phi(f_i)g_i + f_i\phi(g_i) \} \pmod{U^2}. \end{aligned} \tag{E.1}$$

*If  $\beta_U$  is onto, the scheme is smooth there of codimension  $(q - r)(p - r)$ . The corresponding native contact fibre has dimension  $p - r$ .*

*Proof.* Express nearby planes as graphs and differentiate products. A different lift of  $\phi$  changes the result only by an element of  $U^2$ . Near a nonzero  $r$ -minor, invertible block operations reduce multiplication to an invertible  $r$ -block and a Schur complement. Rank at most  $r$  is equivalent to the vanishing of that complement, whose differential is (E.1). Its kernel is the scheme tangent

space, and surjectivity gives smoothness with the displayed codimension. The fibre assertion is Theorem B.2.  $\square$

The same construction on  $X_L$  uses the direct sum of the graph derivatives. The assertion is about the scheme defined by minors, and does not presuppose its reducedness.

Use coefficient Euclidean norms, the induced symmetric tensor norm on multiplication, direct-sum Frobenius norms on blocks, and the standard Grassmannian Riemannian distance. Let  $\Delta_{\mathbb{R}}$  denote the real failure set. Let  $\gamma$  be the least singular value of an injective  $\mathcal{L}$ .

**Theorem E.2** (Stability and transverse degeneracy). *Suppose  $m_0 I \leq B_\nu \leq M_0 I$  and  $\hat{G}_\nu = G_\nu + E_\nu$ , with  $\max_\nu \|E_\nu\|_{\text{op}} \leq (2M_0)^{-1}$ . Then*

$$\left\| \mathcal{L}^\dagger(\hat{G}_\nu^{-1})_\nu - \theta \right\|_2 \leq \frac{2M_0^2}{\gamma} \left( \sum_\nu \|E_\nu\|_F^2 \right)^{1/2}. \quad (\text{E.2})$$

*The least multiplication singular value is bounded above by a constant times distance to  $\Delta_{\mathbb{R}}$ . Near a real corank-one point at which the corresponding kernel-to-cokernel differential is onto, it is bounded above and below by positive local constants times that distance. The same local comparison holds in loading space for distance to the pulled-back failure set.*

*Proof.* The inverse identity gives  $\hat{G}_\nu^{-1} - B_\nu = -\hat{G}_\nu^{-1} E_\nu B_\nu$ . The assumed bounds make its Frobenius norm at most  $2M_0^2 \|E_\nu\|_F$ . Apply  $\mathcal{L}^\dagger$ , whose norm is  $\gamma^{-1}$ .

In finitely many compact orthonormal frame charts, multiplication is Lipschitz in the plane coordinates. Singular values are Lipschitz in matrix entries and vanish on the real failure set, giving the upper estimate. At corank one, the Schur complement is a row of length  $Lq - p + 1$ . Bounded invertible block operations compare the least singular value with the norm of this row. Surjectivity of its differential gives local coordinates in which the row is a coordinate projection, proving the lower estimate as well. For fixed  $J$ , passage from an orthonormal polynomial frame to  $J^\top Z$  changes norms by bounded positive factors. The loading submersion gives local product coordinates and transfers the distance comparison.  $\square$

No linear distance law is asserted at nontransverse or singular points. Compact design sets disjoint from failure have a positive lower stability constant by continuity; this alone is not a dimension-uniform numerical estimate or a random-loading tail bound.

## APPENDIX F. STRUCTURED COMPRESSION BEYOND HANKEL MATRICES

Let  $\mathcal{K} \subset \text{Sym}_k$  have dimension  $s$  and contain a positive matrix. Write  $\rho(\mathcal{K}, c)$  for the maximum rank of  $X \mapsto X|_{N \times N}$  over  $N \in \text{Gr}(c, k)$ .

**Proposition F.1** (Generic transfer). *For generic full-rank native loadings with  $Ae_1$  nowhere zero, compression has rank  $\rho(\mathcal{K}, c)$  and the global contact fibre in an open positive subset of  $\mathcal{K}$  has dimension  $s - \rho(\mathcal{K}, c)$ .*

*Proof.* Maximal compression rank is a nonempty Zariski-open condition. Apply Theorem D.1 with  $J = I$  and Lemma D.2. Positive block inversion preserves fibres, and rank-nullity gives the dimension.  $\square$

This is a transfer principle, not a computation of  $\rho$  for an unspecified family. The following computation includes multiplicities in a fixed spectral decomposition.

**Theorem F.2** (Block-spectral rank). *Let  $P_1, \dots, P_s$  be nonzero mutually orthogonal real symmetric projections with sum  $I_k$ , and let  $\mathcal{K} = \text{span}\{P_1, \dots, P_s\}$ . Then*

$$\rho(\mathcal{K}, c) = \min\{s, \binom{c+1}{2}\}.$$

For the diagonal family the sharp threshold is  $\binom{c+1}{2} \geq k$ , with least ambient dimension  $d + \lceil (1 + \sqrt{1 + 8d})/2 \rceil$  at fixed  $d = k - c$ . For the family of real symmetric circulant matrices it is  $\binom{c+1}{2} \geq \lfloor k/2 \rfloor + 1$ .

*Proof.* The upper bound is dimensional. Suppose  $c \leq s$ . Choose one unit vector in each  $\text{ran } P_i$ . In their span, a normal frame with rows  $u_i^\top$  gives compressed matrices  $u_i u_i^\top$ . The  $c$  vectors  $e_a$  followed by the  $c(c-1)/2$  vectors  $e_a + e_b$  have linearly independent outer products: first inspect off-diagonal entries, then diagonal entries. Select the first  $s$  when  $s \leq q$ , or retain all  $q$  and add rows when  $s > q$ . Starting with the  $e_a$  gives full column rank. Column-orthonormalization transforms every compressed matrix by the same invertible congruence and therefore preserves linear independence. If  $c > s$ , take a  $c$ -plane containing one unit vector from every  $\text{ran } P_i$  and extend their span. Evaluation on these distinguished vectors separates the  $s$  coefficients. This proves the rank formula in all cases.

For diagonal matrices,  $s = k$  and substitution  $k = d + c$  gives  $c^2 - c - 2d \geq 0$ . A real symmetric circulant matrix is diagonal in the real Fourier decomposition: the zero frequency, the Nyquist frequency when present, and the paired sine-cosine frequencies give  $\lfloor k/2 \rfloor + 1$  orthogonal spectral projections. Its eigenvalue is arbitrary on each such block, so the preceding formula applies.  $\square$

Independent Bernoulli log-odds coordinates with information weights  $\lambda_i$  have inverse information  $\text{diag}((\lambda_i p_i(1 - p_i))^{-1})$ , providing a genuine positive diagonal information family. The block-spectral theorem concerns the stated matrix families and does not claim that every such family is generated by the calibrated polynomial experiment, or compute ranks for all Toeplitz or graph-sparse families.

#### APPENDIX G. COMPONENTS, SCHEME STRUCTURE, AND THE RESIDUAL WALL

Let  $\mathcal{D}_L$  denote the scheme of maximal minors of  $\mu$ , and write

$$e = Lq, \quad a = e - p + 1, \quad b = Lc - 3, \quad m_X = Lc(k - c), \quad \mathcal{E} = \bigoplus_{\nu=1}^L \text{Sym}^2 \mathcal{S}_\nu.$$

Thus  $|\mathcal{D}_L| = \Delta^{(L)}$ . The letters  $a, b$  in this section are codimensions, not loading vectors. Theorem 24.1 holds throughout its stated range. The structural results below assume the additional stable-range conditions

$$c \geq 8, \quad k \geq 2c + 1, \quad e \geq p. \quad (\text{G.1})$$

For the classical input in Lemma 27.1, the precise affine dimension statement is  $\dim R = 2t - 2$  on page 2 of [7]'s arXiv version; set  $t = r + 1$ . The secant identification is its equation (2.0.1). Exact-rank strata are open in their rank-at-most closures; confluent forms remain included. These inputs do not supply the component classification below.

These conditions exclude the small equality cases in the rank optimization and the two families of maximal even-dimensional orthogonal Grassmannians.

For distinct  $x, y \in \mathbb{P}^1$ , choose nonzero representatives of their evaluation functionals and let  $t \neq 0$ . Set

$$h_+ = \text{ev}_x + t \text{ev}_y, \quad h_- = \text{ev}_x - t \text{ev}_y, \quad \ell = \text{ev}_x^{(2n)} - t^2 \text{ev}_y^{(2n)}. \quad (\text{G.2})$$

Rescaling the representatives only reparametrizes this family. For  $\varepsilon \in \{+, -\}^L / \{\varepsilon \sim -\varepsilon\}$ , let  $\Sigma_\varepsilon$  be the closure of the tuples satisfying  $U_\nu \subset \ker h_{\varepsilon_\nu}$ . The all-equal pattern is the common-secant variety  $\Sigma_L$  of Proposition 27.4. Define also

$$\mathfrak{F}_L = \{([\ell], (U_\nu)) : \text{rank } H_\ell = k, H_\ell|_{U_\nu \times U_\nu} = 0\}, \quad \Phi_L = \overline{\text{pr}_{X_L} \mathfrak{F}_L}. \quad (\text{G.3})$$

The closure in this definition is essential: the generic annihilator of  $\Phi_L$  will be nondegenerate, but its specializations need not be.

**Theorem G.1** (Component classification and determinantal scheme). *Assume (G.1).*

(i) *If  $a < b$ , the scheme  $\mathcal{D}_L$  is integral and Cohen–Macaulay of codimension  $a$ , with support  $\Phi_L$ . Its general point has multiplication corank one, a unique projective annihilator of Hankel rank  $k$ , and a smooth determinantal neighbourhood. The projection from  $\mathfrak{F}_L$  is birational onto  $\Phi_L$ .*

(ii) *If  $a = b$ , the scheme  $\mathcal{D}_L$  is reduced and Cohen–Macaulay of pure codimension  $a$ . Its irreducible components are exactly*

$$\Phi_L \quad \text{and} \quad \Sigma_\varepsilon \quad (\varepsilon \in \{+, -\}^L / \{\pm 1\}).$$

*There are  $1 + 2^{L-1}$  components. The general point of each has multiplication corank one and is smooth in the determinantal scheme. The generic annihilator has rank  $k$  on  $\Phi_L$  and rank two on every  $\Sigma_\varepsilon$ .*

(iii) *If  $b < a$ , the maximal-dimensional irreducible components of  $\Delta^{(L)}$  are exactly the  $2^{L-1}$  varieties  $\Sigma_\varepsilon$ , of codimension  $b$ . Their general points have multiplication corank one and are smooth in  $\mathcal{D}_L$ . Every  $\Sigma_\varepsilon$  is rational.*

(iv) *In the expected-codimension range  $a \leq b$ , the fundamental cycle is*

$$[\mathcal{D}_L] = c_a(\mathcal{E}^*) = \left[ \prod_{\nu=1}^L \prod_{1 \leq i \leq j \leq c} (1 + x_{\nu i} + x_{\nu j}) \right]_a \quad \text{in } A^a(X_L), \quad (\text{G.4})$$

where  $x_{\nu i}$  are the formal Chern roots of  $\mathcal{S}_\nu^*$  and the brackets select codimension  $a$ .

Part (iii) classifies maximal-dimensional components, not possible smaller components or embedded structure in the excess range. Parts (i) and (ii), by contrast, concern the entire determinantal scheme. Formula (G.4) uses the classical determinantal cycle formula after the expected codimension and reducedness have been established; the cycle formula itself is not a new general intersection-theoretic principle.

### G.1. The two extremal incidences.

**Lemma G.2** (Strict rank gap). *Under (G.1), every annihilator incidence stratum has dimension strictly less than  $m_X - \min(a, b)$ , except possibly the strata*

$$r = 2, \quad s_1 = \cdots = s_L = 1; \quad r = k, \quad s_1 = \cdots = s_L = c.$$

*Their dimensions are respectively  $m_X - b$  and  $m_X - a$ .*

*Proof.* Use  $D_{r,s}$  and  $h_r$  from Lemmas 27.1 and 27.2. If  $r = k$ , the radical is zero and  $s = c$ . Suppose  $r < k$ . For  $s = 0$ ,  $LD_{r,0} - h_r = (Lc - 2)r + 1 > b$ . For  $1 \leq s \leq c - 2$ , the lower endpoint  $r = 2s$  in the feasible interval gives

$$g(s) = 1 + s(Lc - 4) - Ls(s - 1)/2.$$

Its coefficient before minimizing in  $r$  is  $L(c - s) - 2 \geq 0$ . Moreover,

$$g(1) = b, \quad g(c - 2) - b = (c - 3)\{L(c + 2)/2 - 4\} > 0.$$

Concavity places every integer  $2 \leq s \leq c - 2$  strictly above  $b$ . At  $s = 1$  the coefficient  $L(c - 1) - 2$  is strictly positive, so equality requires  $r = 2$ .

For  $s = c - 1$  and  $L = 1$ , the calculation in Lemma 27.3 gives at least  $a + k - 2c + 1 \geq a + 2$ . For  $s = c - 1$  and  $L \geq 2$ , it gives at least  $b + (c - 2)\{L(c + 1)/2 - 4\} > b$ . For  $s = c$ ,  $r < k$ , it gives at least  $a + 1$ . Thus, for each fixed nonexceptional  $r$  and every feasible  $s$ ,  $LD_{r,s} - h_r > \min(a, b)$ . Averaging this inequality over the  $L$  choices  $s_\nu$  gives  $\sum_\nu D_{r,s_\nu} - h_r > \min(a, b)$ . At  $r = 2$ , replacing any  $s_\nu = 1$  by zero increases this sum. The two displayed dimensions follow from  $D_{2,1} = c$  and  $D_{k,c} = q$ .  $\square$

The proof explains the two ends of the codimension formula: an arbitrary nondegenerate Hankel form imposes  $q$  isotropy equations per contact, whereas a rank-two form leaves a choice between two hyperplanes. Intermediate ranks incur a strict dimension loss in (G.1).

**Lemma G.3** (Signed secant incidences). *The  $2^{L-1}$  varieties  $\Sigma_\varepsilon$  are distinct irreducible rational subvarieties of  $\Delta^{(L)}$ , each of codimension  $b$ . Their incidence parametrizations are birational. A general point of any one of them admits exactly one rank-two projective annihilator.*

*Proof.* A general point  $[h]$  of the secant threefold  $S_n$  has a unique unordered secant decomposition. Indeed, evaluations at four distinct points are linearly independent for  $n \geq 3$ , by interpolation; the corresponding confluent statement follows by Hermite interpolation. Thus two different secant lines cannot meet away from their endpoints, and tangent or limiting cases form a proper closed subset. The parameter space is birational to a projective-line bundle over  $\text{Sym}^2 \mathbb{P}^1 = \mathbb{P}^2$ , hence is rational. Sending  $[h_+]$  to  $[h_-]$  in (G.2) is a rational involution, independent of the ordering of the two evaluation points.

Normalize the sign pattern by  $\varepsilon_1 = +$ . Over the dense open set of distinct secants with nonzero coefficients, the signed incidence is a product of relative Grassmannians, with dimension  $3 + Lc(k-1-c) = m_X - b$ . It is irreducible and rational: its underlying hyperplane bundles are trivial on a dense open subset of the rational base. The functional  $\ell$  in (G.2) annihilates every product, for either sign, so its image closure lies in  $\Delta^{(L)}$ .

Here is the projection calculation explicitly. Fix  $h \in S_n$  in the above open set. In  $\text{Gr}(c, \ker h)$  the exceptional planes admitting another secant hyperplane are the projection of

$$\{(h', U) : h' \in S_n \setminus \{h\}, U \subset \ker h \cap \ker h'\}.$$

Its dimension is at most  $3 + c(k-2-c)$ , strictly smaller than  $c(k-1-c)$  because  $c > 3$ ; if  $c > k-2$  it is empty. Therefore a general first plane determines  $h$  uniquely. This proves generic injectivity and, in characteristic zero, birationality of the signed projection. The recovered  $h$  also determines its partner and the rank-two annihilator. For a general tuple no  $U_\nu$  lies in both partner hyperplanes. Consequently two different normalized sign patterns have different images. The exact-rank-two stratum not represented by two distinct evaluations has smaller dimension; it cannot affect these generic assertions.  $\square$

**Lemma G.4** (The nondegenerate incidence). *The variety  $\mathfrak{F}_L$  is smooth and irreducible of dimension  $m_X - a$ . Its image contains a point whose first plane lies in no secant hyperplane.*

*Proof.* Over the open set of nonsingular Hankel forms, the fibres are products of  $\text{OGr}(c, k)$ . The restriction equations have independent differentials, as in Lemma 27.2. Since  $2c < k$ , the special orthogonal group acts transitively on these isotropic Grassmannians; a Witt basis proves transitivity, and the special orthogonal group over  $\mathbb{C}$  is connected. Thus the fibres are smooth and geometrically irreducible of dimension  $c(k-c) - q$ . The base is a smooth irreducible open subset of  $\mathbb{P}^{p-1}$ , so the relative incidence is smooth and irreducible, with the stated dimension.

Fix one nonsingular form  $H$  and put  $d_H = c(k-c) - q$ . In any hyperplane  $K$  the restricted form has radical of dimension zero or one. In the nondegenerate case its isotropic  $c$ -planes have dimension  $d_H - c$ . In the one-dimensional radical case, the strata avoiding and containing the radical have dimensions respectively

$$d_H - c, \quad d_H - (k - c - 1) \leq d_H - c,$$

when nonempty. These formulas follow by the same choice-of-radical-intersection and lift calculation as Lemma 27.2. Varying  $K$  among secant hyperplanes contributes only three parameters. Their union in  $\text{OGr}(c, k)$  therefore has dimension at most  $d_H - c + 3 < d_H$ . Choose  $U_1$  outside it and choose the other isotropic planes arbitrarily. This proves the last assertion.  $\square$

**G.2. Corank one and the residual multiplication map.** Let  $\mathfrak{J} \subset \mathbb{P}(V_{2n}^*) \times X_L$  be the scheme defined by  $\ell\mu = 0$ .

**Lemma G.5** (Corank-one chart). *Above the open subset on which  $\mu$  has rank at least  $p - 1$ , the projection from  $\mathfrak{J}$  to  $\mathcal{D}_L$  is an isomorphism. At a point of corank one, infinitesimal pairs satisfy*

$$\dot{\ell}(uv) + H_\ell(\dot{u}, v) + H_\ell(u, \dot{v}) = 0 \quad (u, v \in U_\nu, \ 1 \leq \nu \leq L). \quad (\text{G.5})$$

Here  $\dot{U}_\nu \in \text{Hom}(U_\nu, V_n/U_\nu)$  and  $\dot{\ell}$  is taken modulo  $\mathbb{C}\ell$ .

*Proof.* On a nonzero  $(p - 1)$ -minor, invertible row and column operations reduce  $\mu$  to an identity block of size  $p - 1$  and a remaining row of length  $e - p + 1$ . Its entries generate the maximal-minor ideal. The normalized last coordinate of a projective left kernel vector is one; the other coordinates are uniquely eliminated by the identity block. The remaining equations are exactly that row. This proves the scheme isomorphism, not only a bijection of points. Differentiating the product equations gives (G.5).  $\square$

**Lemma G.6** (Generic smoothness of the signed components). *If  $b \leq a$ , a general point of each  $\Sigma_\varepsilon$  has corank one and the scheme  $\mathcal{D}_L$  is smooth there of codimension  $b$ .*

*Proof.* The strict rank gap excludes every intermediate-rank annihilator at a general point of  $\Sigma_\varepsilon$ . If  $b < a$ , the full-rank incidence has smaller dimension and is excluded as well. If  $a = b$  and full-rank annihilators occurred over a dense subset of  $\Sigma_\varepsilon$ , their fibres would have positive dimension: the projective annihilator space already contains its rank-two point, and its full-rank part is open. This would force  $\dim \mathfrak{F}_L > m_X - a$ , a contradiction. Lemma G.3 then implies corank one. Indeed, any further annihilator would give a projective line of annihilators, and hence either a forbidden rank or more than one rank-two point.

For the tangent calculation take the distinct evaluation points in (G.2). Let  $g$  be the binary quadratic vanishing at them. Then

$$E = \text{rad } H_\ell = gV_{n-2}, \quad U_\nu = W_\nu \oplus \mathbb{C}a_\nu, \quad W_\nu = U_\nu \cap E = gK_\nu,$$

where the  $K_\nu$  are independent general  $(c - 1)$ -planes in  $V_{n-2}$ . The inequality determining this regime is precisely

$$a - b = L \binom{c}{2} - (2k - 5) \geq 0. \quad (\text{G.6})$$

Apply Theorem 24.1, already proved in Section 27, with  $k' = k - 2$  and  $c' = c - 1$ . Its hypotheses hold and (G.6) is its saturation condition. Consequently, on a nonempty open subset of the signed incidence,

$$\sum_\nu W_\nu^2 = g^2 V_{2n-4}, \quad \dim \sum_\nu W_\nu^2 = p - 4. \quad (\text{G.7})$$

There is no inductive use of the present structural theorem.

Equation (G.5) on  $W_\nu \times W_\nu$  first requires  $\dot{\ell}$  to annihilate (G.7). Thus  $\dot{\ell}$  has dimension  $(p - 1) - (p - 4) = 3$ . For each such  $\dot{\ell}$ , the remaining equations consist of the  $c - 1$  pairs  $(w, a_\nu)$  and the pair  $(a_\nu, a_\nu)$ . Choose  $b_\nu \in V_n$  with  $H_\ell(a_\nu, b_\nu) = 1$ . The  $b_\nu$ -coordinates of  $\dot{U}_\nu(w)$  and  $\dot{U}_\nu(a_\nu)$  solve these  $c$  equations independently; the last coefficient is two, which is nonzero. Thus the tangent dimension of  $\mathfrak{J}$ , and by Lemma G.5 of  $\mathcal{D}_L$ , is

$$3 + \sum_\nu \{c(k - c) - c\} = m_X - b.$$

The local dimension is at least this number because the point lies on  $\Sigma_\varepsilon$ . Equality of dimension and tangent dimension proves regularity of the local ring and the stated smoothness.  $\square$

Identity (G.6) is a deformation-theoretic interpretation of the change of regime. Saturation of the multiplication map after removing the two evaluation points is exactly the condition under which the rank-two component acquires its predicted smooth normal space. This is stronger than locating a minimum in an incidence count.

### G.3. The whole scheme in expected codimension.

*Proof of Theorem G.1.* Suppose first that  $a \leq b$ . On a product of standard affine Grassmannian charts the coordinate ring is a polynomial ring in  $m_X$  variables over  $\mathbb{C}$ , the matrix of  $\mu$  has size  $p \times e$ , and  $I = I_p(\mu)$  is its maximal-minor ideal. On every chart meeting the failure scheme, Theorem 24.1 gives height  $I \geq a$  and the maximal-minor height inequality gives height  $I \leq e - p + 1 = a$ . Since the ring is regular, grade  $I = a$ . The Eagon–Northcott grade criterion therefore provides a free resolution of  $R/I$  of length  $a$ ; in particular it is Cohen–Macaulay and unmixed of codimension  $a$  [8, 19]. This application uses the grade criterion, not a claim that the entries of  $\mu$  are independent variables.

Every component consequently has dimension  $m_X - a$ . By Lemma G.2, its generic annihilator must have rank  $k$ , or, when  $a = b$ , rank two. In the first case it is dominated by the irreducible incidence  $\mathfrak{F}_L$ , so there is at most one such component. Lemma G.4 supplies a point outside every signed secant variety; purity and the strict rank gap force a full-rank component through the resulting non-secant part of the scheme. Hence this component exists and is  $\Phi_L$ , with dimension  $m_X - a$ .

The full-rank incidence has this same dimension. Its generic fibre is therefore zero-dimensional, but this fibre is the full-rank open subset of the projective annihilator space. It follows that the generic multiplication corank is one. The generic fibre is a single point, and Lemma G.5 identifies a dense open subset of  $\Phi_L$  with an open subset of the smooth variety  $\mathfrak{F}_L$ . This proves birationality and generic scheme smoothness there.

If  $a < b$ , the rank-two incidence has smaller dimension, leaving exactly  $\Phi_L$ . If  $a = b$ , every distinct  $\Sigma_\varepsilon$  has the dimension of a component, and Lemma G.3 identifies all the rank-two components; the confluent rank-two base is a proper lower-dimensional subset. There are no others by the strict rank gap. Lemma G.6 proves generic scheme smoothness on them. A Cohen–Macaulay scheme has no embedded associated primes; if it is generically reduced at every minimal prime, it is reduced. Applying this fact proves (i) and (ii), including integrality in (i).

If  $b < a$ , all maximal-dimensional components have dimension  $m_X - b$ . The full-rank incidence and all the intermediate-rank incidences are too small to dominate one. Lemma G.3 therefore gives exactly the listed components, their codimension and rationality, and Lemma G.6 gives their generic corank and smoothness. This proves (iii).

Finally apply the Thom–Porteous formula to the dual map  $V_{2n}^* \otimes \mathcal{O} \rightarrow \mathcal{E}^*$ , with source rank  $p$ , target rank  $e$ , and rank bound  $p - 1$  [16, 11]. The expected codimension is  $a$ , as verified above, so the formula is  $c_a(\mathcal{E}^*)$ . The roots of  $\text{Sym}^2 \mathcal{S}_\nu^*$  are  $x_{\nu i} + x_{\nu j}$ ,  $i \leq j$ , which gives (G.4). Reducedness shows that in the equality case this is the sum of the component cycles with coefficient one. No excess-range use of the expected-codimension formula is made.  $\square$

### G.4. Real native strata and local conditioning.

**Corollary G.7** (All signed components have positive native points). *For a native interpolation matrix  $J$ , every  $\Sigma_\varepsilon$  has a Zariski-dense set of real points admitting positive, globally separated native realizations. If (G.1) holds and  $b \leq a$ , some such points are smooth points of  $\mathcal{D}_L$  of real local codimension  $b$ . Their pullbacks to loading space have the same local codimension.*

*Proof.* Choose real  $x, y > D$  and sufficiently small  $t > 0$ . Every coordinate of  $-J \text{ev}_x$  is positive, so both  $-Jh_+$  and  $-Jh_-$  are strictly positive. Independently for each contact, choose a general  $(k - c)$ -plane  $T_\nu$  containing  $-Jh_{\varepsilon_\nu}$  and avoiding the vectors obtained from it by nonconstant

coordinate sign changes. The fixed-space construction in Theorem D.1 then gives the required globally separated positive loading and  $U_\nu \subset \ker h_{\varepsilon_\nu}$ . These choices contain a real open subset of the signed incidence. Its birational parametrization is defined over  $\mathbb{R}$  and is an isomorphism on dense open subsets. A nonzero real polynomial cannot vanish on this real open subset, so it meets the generic corank-one and residual-spanning opens of Lemma G.6. The real smooth codimension equals the complex codimension there. The loading submersions of Theorem D.1 transfer the assertion. In the finite sign test,  $s$  and  $-s$  define the same unordered coordinate partition. Positivity specifies a Euclidean-open, Zariski-dense real image, not a positivity-defined Zariski-open set.  $\square$

**Corollary G.8** (Local small-singular-value exponent). *At any real smooth corank-one point furnished by Corollary G.7, there is a loading coordinate neighbourhood, disjoint from the other components, on which the least multiplication singular value  $\gamma_{\text{mult}}$  satisfies*

$$C_1 \text{dist}(A, |\mathcal{D}_L|_{\text{load}}) \leq \gamma_{\text{mult}}(A) \leq C_2 \text{dist}(A, |\mathcal{D}_L|_{\text{load}}). \quad (\text{G.8})$$

*Here the neighbourhood has compact closure in a fixed frame chart and in a fixed full-rank loading sign chamber. On a sufficiently small coordinate product box, a probability law with density bounded above and below by positive constants satisfies*

$$c_1 \epsilon^b \leq \Pr\{\gamma_{\text{mult}} \leq \epsilon\} \leq c_2 \epsilon^b \quad (0 < \epsilon < \epsilon_0). \quad (\text{G.9})$$

*At a real smooth corank-one point of  $\Phi_L$  in the range  $a \leq b$ , the same assertion holds with  $a$  in place of  $b$ .*

*Proof.* Use the Schur-complement row of Lemma G.5. Bounded invertible matrix operations compare its Euclidean norm with  $\gamma_{\text{mult}}$ . At a smooth scheme point of codimension  $b$ , the differentials of its entries span a conormal space of dimension  $b$ . Select  $b$  independent entries as normal coordinates. The whole row vanishes on the failure submanifold; bounded derivatives give the upper distance estimate, and the selected coordinates give the lower estimate. The loading submersion and compact chart bounds preserve these comparisons. Integration in the  $b$  normal coordinates, with bounded Jacobian and density on the product box, gives (G.9). The full-rank case is identical with codimension  $a$ .  $\square$

These are local probability laws near specified smooth strata. They do not assert a global tail for arbitrary random loadings or constants uniform in  $c, k, L$ . In particular, singular intersections and smaller components in the excess range are not covered by this local tube calculation.

**Residual structure and the scope of the classification.** The codimension formula leads to a description beyond the stable component theorem retained in the appendices. Theorem 30.1 removes its dimension restriction: in the original range  $4 \leq c < k$  there are exactly five nonendpoint equality incidences. In each the plane contains the entire radical and is maximal isotropic in the quotient. Orientation monodromy, rather than a choice of one connected orthogonal family, determines their irreducibility.

The common mechanism is Theorem 28.1. For an annihilator  $\ell$ , put  $W_\nu = U_\nu \cap \text{rad } H_\ell$ . The obstruction to solving the infinitesimal plane equations is exactly the restriction of  $\dot{\ell}$  to  $\sum_\nu W_\nu^2$ . This statement holds at every annihilator rank. At rank two it turns singularity of a corank-one point into failure of a smaller multiplication map. At the wall a general point of the residual determinant divisor is an ordinary double crossing between the nondegenerate and signed components (Theorem 31.9), with a logarithmic conditioning law (Corollary 31.10).

The reduced rank-two incidence has a normalization indexed by sign patterns. Its intersections with a common annihilator have a complete relative-Grassmannian description, while its full isotropy fibres carry an explicit square-zero nilradical (Theorem 31.1 and Proposition 31.2). These are different scheme statements. A reduced global scheme need not have reduced fibres. For

higher symmetric powers the analogous labels form dihedral phase orbits, rather than a simple power of the number of hyperplanes (Theorems 32.2 and 32.3).

The all-dimension component theorem is proved from the codimension formula and the residual tangent sequence; it does not rely on the stable theorem retained in the appendices. The full boundary, count-comparison and finite-sample arguments are also retained there, so that their statistical assumptions do not enter the geometric proof.

Our use of secant obstructions is situated in the classical failure-locus literature, including Ballico's higher-order failure-locus paper [4], the complete-series work of Green–Lazarsfeld [14], and the failure-cycle study [5]. The statements proved here specify the varying subseries, the absence of cross-contact products, the exceptional incidences, and the local scheme strata. The accompanying literature audit records which primary texts were checked and which theorem-level comparisons remain unverified. For [4], the first page has now been inspected in addition to the bibliographic record; this is not a completed theorem-by-theorem comparison. Section 34 specifies this distinction.

For clarity, the Hankel shape identification used below is an equality of ideals in the same polynomial ring, not just a dimension analogy. Writing  $S = \mathbb{C}[z_0, \dots, z_{2k-2}]$  and  $H_{u,v} = (z_{i+j})_{0 \leq i < u, 0 \leq j < v}$  for  $u + v = 2k$ , the classical identity recalled in [7, Section 1] is

$$I_{r+1}(H_{k,k}) = I_{r+1}(H_{r+1, 2k-r-1}) \subset S \quad (1 \leq r < k).$$

The latter is a maximal-minor Hankel ideal with quotient of affine dimension  $2r$ . Projectivization gives  $2r - 1$ ; removing the smaller rank locus gives the exact-rank stratum used in the incidence proof.

## APPENDIX H. THE PHYSICAL BOUNDARY EXPERIMENT

The algebraic extension through  $v = 0$  in Section C is useful for differentiation, but it does not remove the physical restriction  $v \geq 0$ . We now identify the actual local experiment. The argument is the finite-cell likelihood expansion underlying constrained Gaussian limits [13, 26]; all normalizations and the synchronized profile are specified here.

Let  $u \in \mathbb{R}^{k+1}$  denote the nuisance log-odds and root-sum coordinates. Observe independent categorical samples with  $n_j/N \rightarrow \lambda_j > 0$ , and let  $\Delta_N = (\Delta_{N,u}, \Delta_{N,v})$  be the score at  $(u, v) = (0, 0)$  divided by  $\sqrt{N}$ . Partition

$$I = V^\top \text{diag}(\lambda_j \kappa_j) V = \begin{pmatrix} I_{uu} & I_{uv} \\ I_{vu} & I_{vv} \end{pmatrix}, \quad Q = I_{vv} - I_{vu} I_{uu}^{-1} I_{uv} = R^{-1}.$$

**Theorem H.1** (Cone likelihood and synchronized profile). *Fix a compact set  $K \subset \mathbb{R}^{k+1} \times \mathbb{R}_{\geq 0}^k$ , with the Euclidean product norm. For  $(a, h) \in K$ , the physical local alternatives  $u = a/\sqrt{N}$ ,  $v = h/\sqrt{N}$  satisfy, uniformly on compact index sets,*

$$\log \frac{dP_{N,a,h}}{dP_{N,0,0}} = (a, h)^\top \Delta_N - \frac{1}{2}(a, h)^\top I (a, h) + o_P(1). \quad (\text{H.1})$$

Here uniformity means that the supremum over  $K$  of the absolute remainder converges to zero in  $P_{N,0,0}$ -probability. Under the corresponding alternatives,

$$Y_N = R(\Delta_{N,v} - I_{vu} I_{uu}^{-1} \Delta_{N,u}) \implies N(h, R). \quad (\text{H.2})$$

After quotienting out unrestricted nuisance translation, this is a Gaussian shift with parameter restricted to the physical cone. For synchronized alternatives  $h = q_A(\zeta)$ , the profiled limiting log likelihood, relative to its value at  $h = 0$ , is

$$\frac{1}{2} Y^\top R^{-1} Y - \frac{1}{2} C_R(Y), \quad C_R(Y) = \inf_{\zeta} (Y - q_A(\zeta))^\top R^{-1} (Y - q_A(\zeta)). \quad (\text{H.3})$$

*The common physical root displacement has scale  $N^{-1/4}\zeta$ .*

*Proof.* The finite set of clock probabilities is strictly positive on a two-sided neighbourhood of the baseline. Its derivatives through order three are bounded on a smaller neighbourhood. At  $u = a/\sqrt{N}$ ,  $v = h/\sqrt{N}$ , expand each cell log probability. Normalization makes the mean linear term vanish, the expectation of the Hessian is minus the score covariance, and the summed cubic remainder is  $O(N^{-1/2})$  on compact index sets. The centred Hessian fluctuation is  $o_P(1)$ . The finite-dimensional independent-group central limit theorem gives  $\Delta_N \Rightarrow N(0, I)$  at the baseline and proves (H.1). Under the local alternatives its mean tends to  $I(a, h)$  and its covariance to  $I$ , either by the same cell calculation or the likelihood change of measure. The efficient score in (H.2) has mean  $Qh$  and covariance  $Q$ , proving the assertion after multiplication by  $R$ .

In the limiting score experiment the nuisance block is independent of the efficient block. Its mean is  $I_{uu}a + I_{uv}h$ , which is an unrestricted translation as  $a$  varies. Discarding this nuisance translation yields (H.2); this is a quotient, not a claim that an unknown nuisance parameter can be simulated without information. Profiling the efficient Gaussian likelihood over  $h = q_A(\zeta)$  and completing the square gives (H.3). The map  $q_A$  is proper for full-rank  $A$ , so the infimum is attained and its optimizing  $\zeta$  are bounded whenever  $Y$  is bounded. Uniform LAN therefore also gives the iterated compact-localization limit of profiled likelihoods: first restrict  $(a, \zeta)$  to a fixed compact set, pass to the limit, and then exhaust the limiting experiment by such sets. No assertion about remote maximizers of an unrestricted finite-sample likelihood is required. Finally  $q_A(N^{-1/4}\zeta) = N^{-1/2}q_A(\zeta)$ , proving the displacement scale.  $\square$

The unrestricted vector  $Y$  is a statistic in the limiting experiment, not the constrained maximum likelihood estimator. The latter is a metric projection onto  $\mathbb{R}_{\geq 0}^k$ , or onto  $q_A(\mathbb{R}^d)$  in the synchronized experiment. At the boundary it need not be normal; for the nonconvex synchronized image the projection can be set-valued. The distance value in (H.3) is nevertheless unambiguous.

**Corollary H.2** (Sharp root-amplitude scale on a physical ray). *Fix  $a_0 \in \mathbb{R}^d \setminus \{0\}$  and  $v_0 = q_A(a_0) \neq 0$ . On the one-sided submodel with physical displacement  $ta_0$ ,  $0 \leq t \leq TN^{-1/4}$ , and nuisance  $u = a/\sqrt{N}$  in a fixed bounded set, the minimax squared risk for estimating  $t$  is of order  $N^{-1/2}$ , for each fixed  $T > 0$ . Thus the amplitude scale is  $N^{-1/4}$ , whereas its squared amplitude is on the  $N^{-1/2}$  parameter scale.*

*Proof.* The variance parameter is  $v = t^2v_0$ . Set  $J_0 = v_0^\top Q v_0 > 0$  and use the baseline efficient score to define

$$\hat{b}_N = \frac{v_0^\top Q Y_N}{\sqrt{N}J_0}, \quad \hat{t}_N = \sqrt{\max\{\hat{b}_N, 0\}}.$$

For completeness, let  $s_{je}$  be the baseline score of cell  $e$  in group  $j$ . There are finitely many such bounded vectors. Taylor expansion of the cell probabilities, rather than convergence in distribution, gives

$$\sup_{(a,h) \in K} \|\mathbb{E}_{a,h} \Delta_N - I_N(a, h)\| \leq C_K N^{-1/2}, \quad \sup_{(a,h) \in K} \|\text{Var}_{a,h} \Delta_N\|_{\text{op}} \leq C_K,$$

where  $I_N$  uses the actual allocations  $n_j/N$  and  $I_N \rightarrow I$ . The first bound sums the bounded second-order probability remainders; the second sums variances of independent bounded scores. The efficient linear map consequently has mean  $h + o(1)$  and bounded covariance, uniformly on  $K$ . With  $h = \sqrt{N}t^2v_0$ , the definition of  $\hat{b}_N$  now gives bias  $o(N^{-1/2})$  and variance  $O(N^{-1})$ , and hence  $\mathbb{E}(\hat{b}_N - t^2)^2 \leq C/N$ . This argument supplies the moment control separately from LAN. For  $b \geq 0$ ,  $(\sqrt{\max\{x, 0\}} - \sqrt{b})^2 \leq |x - b|$ . Cauchy–Schwarz therefore gives the upper bound  $C'/\sqrt{N}$ .

For the lower bound take the two physical alternatives  $t = 0$  and  $t = \tau N^{-1/4}$  with the same nuisance, where  $0 < \tau \leq T$  is sufficiently small. Their likelihood divergence is bounded by a

constant times  $\tau^4$ , by (H.1) or its finite-cell Taylor proof. Their total variation is consequently bounded away from one. The two-point testing bound for squared loss gives a positive constant times the squared separation  $\tau^2 N^{-1/2}$ . The claim concerns this labelled nonnegative ray; it does not assert identifiability of the sign of a general loading vector.  $\square$

## APPENDIX I. DIRECT COUNT-TO-CONTACT EXPERIMENT COMPARISON

Here the observations differ from fixed-count categorical marks. Let the mark laws, hence  $\kappa_j$ , be known, and observe independent exposures

$$N_j \sim \text{Pois}(N\lambda_j), \quad j = 1, \dots, m.$$

This is the known-mark-law subexperiment of an observed-exposure protocol. Fix an interior centre  $\lambda^0 > 0$ . Write  $D$  for the derivative of the native moment functional at that centre:

$$D_{\cdot j} = -\frac{\rho_j}{(\lambda_j^0)^2 \kappa_j} w_{2n}(T_j), \quad \Lambda = \text{diag}(\lambda_j^0), \quad \Sigma = D\Lambda D^\top > 0.$$

The positivity follows from Vandermonde rank. For a local moment index  $h \in \mathbb{R}^p$  in a fixed compact ball, define the least-favourable target submodel

$$\lambda_N(h) = \lambda^0 + N^{-1/2} B_* h, \quad B_* = \Lambda D^\top \Sigma^{-1}. \quad (\text{I.1})$$

It has  $\theta(\lambda_N(h)) = \theta(\lambda^0) + h/\sqrt{N} + O(N^{-1})$ . Its local information is  $\Sigma^{-1}$ , since  $DB_* = I_p$  and  $B_*^\top \Lambda^{-1} B_* = \Sigma^{-1}$ .

Let  $\mathcal{L}$  be the known normal-compression operator, possibly rank deficient, and put  $r = \text{rank } \mathcal{L}$ . Choose an orthonormal coordinate frame  $F$  for its range. Generate independent  $U_j \sim \text{Unif}(-1/2, 1/2)$ , independently of the counts, and retain only

$$T_N = F^\top \mathcal{L} D \frac{N_{\cdot} - N\lambda^0 + U}{\sqrt{N}}. \quad (\text{I.2})$$

The randomization law is part of the protocol; its realized auxiliary values are not retained. This statistic is formed directly by fixed linear weights on the counts. No estimator of  $\theta$  is constructed first. The small randomization prevents lattice encoding in this protocol: without such randomization or a genuine finite-precision coarsening, an exact real-valued linear encoding of an integer vector can be injective even when its real matrix has deficient rank. Weak convergence alone would not exclude that lattice encoding.

For experiments  $\mathcal{A} = (P_h)$  and  $\mathcal{B} = (Q_h)$  on the same parameter set, use

$$\delta(\mathcal{A}, \mathcal{B}) = \inf_K \sup_h \|K P_h - Q_h\|_{\text{TV}}, \quad \Delta(\mathcal{A}, \mathcal{B}) = \max\{\delta(\mathcal{A}, \mathcal{B}), \delta(\mathcal{B}, \mathcal{A})\}.$$

The infimum is over parameter-independent Markov kernels and  $\|P - Q\|_{\text{TV}} = \sup_A |P(A) - Q(A)|$ , equal to one half the  $L^1$  distance for densities. All comparisons below use a fixed compact local parameter set.

**Lemma I.1** (Quantitative jittered Poisson comparison). *Uniformly for  $h$  in a fixed compact set, the distribution of*

$$X_N^\circ = (N_{\cdot} - N\lambda^0 + U)/\sqrt{N}$$

*under (I.1) is within  $CN^{-1/5}$  in total variation of  $N(B_* h, \Lambda)$ . The constant depends on the fixed dimension, positive exposure bounds, and the compact index set. The full jittered vector and the raw counts are exactly equivalent experiments, by independent jittering and rounding. The raw local target experiment is therefore asymptotically equivalent to  $Z \sim N(h, \Sigma)$ .*

*Proof.* We give the density estimate, including the lattice interpolation and tails. In one coordinate write  $K \sim \text{Pois}(N\lambda)$ , with  $0 < \lambda_- \leq \lambda \leq \lambda_+$ , and centre first at its own mean. The density  $f_{N,\lambda}$  of  $(K - N\lambda + U)/\sqrt{N}$  is constant, equal to  $\sqrt{N} \Pr(K = j)$ , on each cell of width  $N^{-1/2}$ . Put  $z_j = (j - N\lambda)/\sqrt{N}$  and let  $\varphi_\lambda$  be the centred normal density with variance  $\lambda$ . Stirling's formula with remainder

$$\log j! = (j + \tfrac{1}{2}) \log j - j + \tfrac{1}{2} \log(2\pi) + r_j, \quad 0 < r_j < 1/(12j),$$

and the Taylor formula for  $\log(1 + u)$  give, uniformly for  $|z_j| \leq 2N^{1/10}$  and all sufficiently large  $N$ ,

$$\left| \log \frac{\sqrt{N} \Pr(K = j)}{\varphi_\lambda(z_j)} \right| \leq C \{N^{-1/2}(1 + |z_j|^3) + N^{-1}(1 + |z_j|^4)\}.$$

Indeed  $j = N\lambda(1 + z_j/(\sqrt{N}\lambda))$ , the quadratic term is  $-z_j^2/(2\lambda)$ , and the remaining logarithmic, cubic, and Stirling terms have the displayed bounds. Within a cell,  $|\log \varphi_\lambda(z) - \log \varphi_\lambda(z_j)| \leq CN^{-1/2}(1 + |z_j|)$ . Consequently on  $|z| \leq R_N = N^{1/10}$  the density ratio differs from one by at most  $CN^{-1/5}$ . Integrating this relative bound costs at most  $CN^{-1/5}$ . Outside this region,

$$\mathbb{E}\{(K - N\lambda + U)/\sqrt{N}\}^2 = \lambda + (12N)^{-1},$$

so Chebyshev's inequality bounds the step-density tail by  $CR_N^{-2}$ ; the same bound holds for the Gaussian tail. Thus

$$\|f_{N,\lambda} - \varphi_\lambda\|_{L^1} \leq CN^{-1/5}. \quad (\text{I.3})$$

Translation to the centre  $N\lambda^0$  changes neither the bound nor its uniformity. The finitely many small  $N$  are absorbed by increasing  $C$ .

For  $m$  independent coordinates, the total variation distance between products is at most the sum of the coordinate distances. This proves the same order for covariance  $\text{diag } \lambda_N(h)$ . To replace it by  $\Lambda$ , differentiate the normal density with respect to each variance on the fixed positive compact interval. The  $L^1$  norms of these derivatives are uniformly bounded (they are the normal expectations of the absolute variance scores). Since  $\|\lambda_N(h) - \lambda^0\| = O(N^{-1/2})$  uniformly, covariance replacement costs  $O(N^{-1/2})$ . This proves the asserted quantitative comparison.

Rounding  $\sqrt{N}X_N^\circ + N\lambda^0$  to the nearest integer recovers each count almost surely, while independent jittering gives the reverse Markov kernel. On arbitrary Gaussian inputs the rounding kernel clips negative integers to zero, so it is defined on the entire observation space. In the limiting normal vector  $X \sim N(B_*h, \Lambda)$ , the statistic  $Z = DX$  has law  $N(h, \Sigma)$  and is sufficient: conditional simulation is

$$X = B_*Z + E, \quad E \sim N(0, \Lambda - \Lambda D^\top \Sigma^{-1} D \Lambda),$$

with  $E$  independent and its law independent of  $h$ . These kernels, the total variation approximation, and contraction of total variation under kernels prove the experiment comparison uniformly on the compact parameter set. In particular this proves the asserted local Le Cam comparison, not merely a central limit theorem.  $\square$

**Theorem I.2** (Known-mark local information retained by contact scores). *With known mark laws and the preassigned centre and target submodel (I.1), the experiment retaining (I.2) is, uniformly on compact local parameter balls, asymptotically equivalent to*

$$T \sim N(F^\top \mathcal{L}h, F^\top \mathcal{L}\Sigma\mathcal{L}^\top F). \quad (\text{I.4})$$

*Its limiting information matrix for  $h$  is*

$$I_{\text{contact}} = \mathcal{L}^\top (\mathcal{L}\Sigma\mathcal{L}^\top)^\dagger \mathcal{L} = \Sigma^{-1/2} P_{\text{ran}(\Sigma^{1/2}\mathcal{L}^\top)} \Sigma^{-1/2}. \quad (\text{I.5})$$

It has rank  $r$  and kernel  $\ker \mathcal{L}$ . The contact-score reduction is locally asymptotically equivalent to the raw target subexperiment if and only if  $\mathcal{L}$  is injective. Hence it retains all local moment information exactly when the within-contact polynomial products span  $V_{2n}$ .

*Proof.* Apply the fixed linear map  $F^\top \mathcal{L} D$  to the total variation approximation in Lemma I.1. Contraction gives convergence in total variation to (I.4), whose covariance is positive definite in its  $r$  coordinates. Identity kernels therefore compare these experiments in both directions with error tending to zero. The usual normal-shift likelihood gives the first expression in (I.5). For  $M = \mathcal{L} \Sigma^{1/2}$ , the identity  $M^\top (M M^\top)^\dagger M = P_{\text{ran } M^\top}$  gives the second. Its rank and kernel follow immediately.

If  $\mathcal{L}$  is injective,  $Z$  is recovered from  $\mathcal{L} Z$  by a left inverse. Thus (I.4) is equivalent to  $N(h, \Sigma)$ , and Lemma I.1 gives equivalence to the raw target subexperiment. Conversely choose distinct  $h_0, h_1$  in a local parameter ball with  $h_1 - h_0 \in \ker \mathcal{L}$ . Their limiting contact distributions coincide, whereas  $N(h_0, \Sigma)$  and  $N(h_1, \Sigma)$  have strictly positive total variation distance. By the triangle inequality, any kernel purporting to reconstruct both raw limit laws has error at least half that distance. The same positive lower obstruction holds asymptotically by the uniform comparisons. The polynomial criterion is Theorem B.2.  $\square$

The uncentred contact block is  $B_\nu(\theta) = \mathcal{L}_\nu \theta$ . Its local fluctuation is precisely the linear type appearing in (I.2). Passing to inverse-Hessian contact scores applies the fixed invertible block transformation  $E_\nu \mapsto -B_{\nu,0}^{-1} E_\nu B_{\nu,0}^{-1}$ , so it preserves the information statement. These are correlated raw-data score statistics, not independently noisy observations of distance values.

The centre  $\lambda^0$  and the local submodel (I.1) are specified before observing the data. The theorem does not claim global sufficiency when both exposure and mark laws are unknown, or that a plug-in estimate of the centre automatically gives the same comparison. It supplies a genuine direct reduction and a complete local information-loss calculation in the stated full-dimensional target experiment. When  $\mathcal{L}$  is injective, exact range-compatible score encoding does not introduce a  $\gamma^{-2}$  statistical variance penalty; independent measurement errors and numerical perturbations have the distinct stability behaviour of Theorem E.2 and Appendix M.

**I.1. Deterministic finite precision and the full exposure tangent model.** The auxiliary randomization above prevents an exact real-valued encoding of the integer count vector. A finite-precision observation provides another, deterministic, way to remove that encoding.

**Theorem I.3** (Deterministic quantized score reduction). *Keep the known-mark target submodel and put  $C = F^\top \mathcal{L} D$ . Suppose  $r \geq 1$ . For a deterministic mesh  $\eta_N > 0$ , retain only*

$$Q_N = \lfloor \eta_N^{-1} C(N - N\lambda^0) / \sqrt{N} \rfloor \in \mathbb{Z}^r,$$

*with the floor taken coordinatewise. If  $\eta_N \rightarrow 0$  and  $\sqrt{N}\eta_N \rightarrow \infty$ , this experiment is asymptotically equivalent to (I.4). More precisely, for  $0 < \eta_N \leq 1$ ,*

$$\Delta(Q_N, \mathcal{T}) \leq C_0 \{N^{-1/5} + (\sqrt{N}\eta_N)^{-1} + \eta_N\}. \quad (\text{I.6})$$

*In particular  $\eta_N = N^{-1/4}$  is admissible. The information and equivalence criterion of Theorem I.2 therefore hold for this deterministic protocol as well.*

*Proof.* Couple the unjittered score  $S_N = C(N - N\lambda^0) / \sqrt{N}$  to  $S_N^\circ = C X_N^\circ$  with the same counts. Their coordinatewise difference is at most  $d_N = C_1 / \sqrt{N}$ , deterministically. The two quantizers can differ only if a coordinate of  $S_N^\circ$  lies within  $d_N$  of the grid  $\eta_N \mathbb{Z}$ . By Lemma I.1, its joint law is within  $CN^{-1/5}$  of the nonsingular normal law in (I.4).

For a univariate normal density  $f$  with variance in a fixed positive interval, the grid-boundary probability is at most  $Cd_N/\eta_N$  if  $d_N \leq \eta_N/2$ . To see this uniformly in its mean, sum the

supremum of  $f$  on each grid interval and use

$$\eta_N \sum_{j \in \mathbb{Z}} \sup_{[j\eta_N, (j+1)\eta_N]} f \leq 1 + \eta_N \int_{\mathbb{R}} |f'(x)| dx.$$

There are at most two boundary pieces per interval. A union bound over the  $r$  coordinates proves  $\Pr\{\lfloor S_N/\eta_N \rfloor \neq \lfloor S_N^\circ/\eta_N \rfloor\} \leq C\{N^{-1/5} + d_N/\eta_N\}$ . When  $d_N > \eta_N/2$  the asserted bound follows by increasing its constant. Quantization contracts total variation, so  $Q_N$  is within the same bound of a quantized Gaussian, uniformly in the local index.

The kernel adding an independent uniform point in the reported grid cell reconstructs the cell-average Gaussian density. Its  $L^1$  distance from the Gaussian is at most  $C_r \eta_N \int \|\nabla f(x)\| dx \leq C \eta_N$ , by the integral mean-value inequality on cubes. The reverse kernel is deterministic quantization. Combining these kernels proves both deficiencies in (I.6). Randomization is used only by this comparison kernel, not by the observed statistic  $Q_N$ . The case  $r = 0$  is the one-point experiment and needs no mesh.  $\square$

An actual finite alphabet may be obtained by an overflow symbol outside a ball of radius  $N^{1/10}$ . Uniform second-moment bounds add only  $O(N^{-1/5})$  to the comparison. Thus infinite digital precision is not hidden in the conclusion.

The least-favourable submodel has a precise relationship to the full exposure tangent experiment, rather than being an intrinsic physical distance observation.

**Proposition I.4** (Full exposure model and nuisance quotient). *For known marks and arbitrary bounded exposure perturbations  $g \in \mathbb{R}^m$ , set  $\lambda_N(g) = \lambda^0 + g/\sqrt{N}$ . The raw count experiment converges in Le Cam distance, uniformly on compact sets, to  $X \sim N(g, \Lambda)$ . In this Gaussian limit the transformation*

$$Z = DX, \quad E = (I_m - B_* D)X$$

*has independent components, with means  $h = Dg$  and  $\xi = (I_m - B_* D)g \in \ker D$ . Quotienting the limit experiment by translations of this unrestricted nuisance gives exactly  $Z \sim N(h, \Sigma)$ ; its contact compression has information (I.5). The least-favourable submodel is the section  $\xi = 0$ .*

*Proof.* The density proof of Lemma I.1 applies unchanged to  $g$  in a compact set; jittering and rounding give both kernels. Since  $DB_* = I_p$ ,  $g = B_* h + \xi$  is a bijective linear parametrization by  $h \in \mathbb{R}^p$  and  $\xi \in \ker D$ . The cross-covariance is

$$D\Lambda(I_m - B_* D)^\top = D\Lambda - \Sigma B_*^\top = 0.$$

The Gaussian components are therefore independent, the covariance of  $Z$  is  $\Sigma$ , and the law of  $E - \xi$  does not depend on either parameter. Translation by  $\ker D$  acts transitively on its nuisance means and leaves  $Z$  fixed. This proves the quotient assertion. It does not supply a kernel simulating an unknown  $\xi$  from  $Z$ , nor assert equivalence to the full experiment with that nuisance retained. The compact-set comparisons are made before this unrestricted Gaussian translation quotient; a compact nuisance set is not silently replaced by a translation group.  $\square$

The centre remains an experimental input. A simple sufficient condition for replacing it by an independent pilot estimate is  $\sqrt{N}\|\hat{\lambda}^0 - \lambda^0\| \rightarrow 0$  in probability, together with a fixed positive compact range for the pilot and continuity of the weights. On events where the scaled error is small, the centring shift is  $o(1)$ , the weights and Gaussian covariances converge, and the same uniform density bounds and mixture kernels apply. A root- $M$  pilot gives this sufficient condition when  $M/N \rightarrow \infty$ . This is a sufficient, not an optimal, pilot regime. An ordinary pilot of size comparable to  $N$  does not meet it: its random centring error must then be included in the limit experiment. Nothing here asserts automatic plug-in sufficiency for unknown marks or an unrecorded pilot.

## APPENDIX J. UNIFORM CONSTANTS AND THE NUISANCE INVARIANT

We record explicit constants for the Poisson step-density estimate in Lemma I.1. They concern the same experiment and do not replace its finite-precision protocol.

**Proposition J.1** (An explicit central-region threshold). *Let  $K \sim \text{Pois}(N\lambda)$ , let  $U \sim \text{Unif}(-1/2, 1/2)$  independently, and suppose  $0 < \lambda_- \leq \lambda \leq \lambda_+$ . Let  $f_{N,\lambda}$  be the density of  $(K - N\lambda + U)/\sqrt{N}$ , and let  $\varphi_\lambda$  be the centred Gaussian density of variance  $\lambda$ . Set*

$$C_0 = 10(1 + \lambda_-^{-1} + \lambda_-^{-2}), \quad N_0 = \left\lceil \max\{1, 2/\lambda_-, (4/\lambda_-)^{5/2}, C_0^5\} \right\rceil.$$

For every integer  $N \geq N_0$ ,

$$\|f_{N,\lambda} - \varphi_\lambda\|_1 \leq (3C_0 + 2\lambda_+ + 1)N^{-1/5}. \quad (\text{J.1})$$

*Proof.* Write  $z_j = (j - N\lambda)/\sqrt{N}$  and  $u_j = z_j/(\sqrt{N}\lambda)$ . If  $|z_j| \leq 2N^{1/10}$  and  $N \geq N_0$ , then  $|u_j| \leq 1/2$  and  $j \geq N\lambda_-/2 \geq 1$ . For  $F(u) = (1+u)\log(1+u) - u$ , Taylor's theorem gives  $|F(u) - u^2/2| \leq (2/3)|u|^3$  on this interval. Stirling's remainder bound  $0 < r_j < 1/(12j)$  therefore yields

$$\left| \log \frac{\sqrt{N} \Pr(K=j)}{\varphi_\lambda(z_j)} \right| \leq \frac{2|z_j|^3}{3\sqrt{N}\lambda_-^2} + \frac{|z_j|}{\sqrt{N}\lambda_-} + \frac{1}{6N\lambda_-}.$$

Within a cell of half-width  $1/(2\sqrt{N})$  the change of the Gaussian log density is at most  $|z_j|/(2\sqrt{N}\lambda_-) + 1/(8N\lambda_-)$ . On  $|z| \leq N^{1/10}$  its cell centre satisfies  $|z_j| \leq 2N^{1/10}$ ; the sum of these bounds is at most  $C_0 N^{-1/5} \leq 1$ . Thus the density ratio differs from one by at most  $3C_0 N^{-1/5}$ , which bounds the central integral. The second moments of the step density and the Gaussian are  $\lambda + 1/(12N)$  and  $\lambda$ . Chebyshev's inequality bounds the sum of their outside integrals by  $(2\lambda_+ + 1)N^{-1/5}$ . Adding proves the result.  $\square$

For the local exposure model, let  $H = \sup_h \|B_* h\|_\infty$  on the prescribed compact index set. Taking also  $N \geq (2H/\min_j \lambda_j^0)^2$  puts every exposure in  $[\min_j \lambda_j^0/2, \max_j \lambda_j^0 + \min_j \lambda_j^0/2]$ . Equation (J.1) then supplies the uniform coordinate constants; product total variation is at most the sum of the coordinate total variations. The covariance replacement and quantization bounds in the preceding appendix remain unchanged.

**Definition J.2** (Nuisance translation invariant). In the limiting Gaussian experiment of Theorem H.1, use the independent nuisance score and efficient observation as coordinates. Let  $\mathbb{R}^{k+1}$  act by translation on the nuisance score coordinate and trivially on the efficient observation  $Y$ . The invariant observation is the measurable maximal invariant of this action, namely  $Y$ .

Thus the phrase “quotienting out nuisance translation” in that theorem means passage to this explicitly defined invariant observation. Independence of the Gaussian blocks follows from their zero cross covariance, and arbitrary nuisance translation removes any information in the first block about the retained parameter for invariant procedures. It is not an equivalence of the full experiment, in which the nuisance itself remains an unknown parameter, to  $Y$  alone.

## APPENDIX K. CLASSICAL MAXIMAL RANK AND MONOMIAL COMPARISON

The proof of the global codimension theorem does not invoke curve postulation. For comparison, the complete classical statement used for a single general space includes the injective range as well.

**Proposition K.1** (Classical rational maximal rank). *For  $4 \leq c \leq n+1$ , a general  $c$ -plane  $U \subset H^0(\mathbb{P}^1, \mathcal{O}(n))$  satisfies*

$$\text{rank}(\text{Sym}^2 U \longrightarrow H^0(\mathbb{P}^1, \mathcal{O}(2n))) = \min\left\{\binom{c+1}{2}, 2n+1\right\}.$$

The full-rank locus is a nonempty Zariski-open set defined over  $\mathbb{R}$ .

*Precise classical reduction.* Apply the maximal-rank theorem in [20, Theorem 1.2] to a general Brill–Noether curve with genus  $g = 0$ , degree  $n$ , ambient projective dimension  $r = c - 1 \geq 3$ , and restriction degree two. Its Brill–Noether number is  $(r + 1)(n - r) \geq 0$ . A general  $c$ -dimensional subseries of  $H^0(\mathbb{P}^1, \mathcal{O}(n))$  is base-point-free and embeds  $\mathbb{P}^1$  in  $\mathbb{P}^r$ : its projection centre avoids the secant and tangent locus of the rational normal curve because  $r \geq 3$ . The space of such subseries is irreducible and parametrizes the general rational Brill–Noether map, up to the choice of a basis and source coordinates. Restriction of quadrics has source  $\text{Sym}^2 U$  and target  $H^0(\mathbb{P}^1, \mathcal{O}(2n))$ , exactly the multiplication map above. This specializes the general theorem to the classical nonspecial rational range of [2, 3]. The complete series is immediate from monomials. Nonvanishing maximal minors in real matrix charts gives the real assertion. This proof attributes, rather than reproves, the classical injective-range theorem.  $\square$

For a monomial space  $U_S = \text{span}\{t^a : a \in S\}$ ,

$$U_S^2 = \mathcal{P}_{2n} \iff S + S = \{0, \dots, 2n\}. \quad (\text{K.1})$$

This is the restricted additive 2-basis problem, distinct from an unrestricted polynomial subspace [27, 17, 18].

**Proposition K.2** (Strict monomial obstruction and constructive bases). *If  $n \geq 3$  and (K.1) holds, then  $\binom{|S|+1}{2} \geq 2n + 2$ . Thus no monomial normal space attains the native optimum at  $(d, k) = (3, 8)$  or  $(5, 11)$ .*

For  $a \geq 2$ ,  $n \geq 2a$ , write  $n = ba + r$ ,  $0 \leq r < a$ . The set

$$S = [0, a - 1] \cup \{a, 2a, \dots, ba\} \cup [n - a + 1, n]$$

has  $S + S = [0, 2n]$  and  $|S| \leq 2a + \lfloor n/a \rfloor - 1$ . In particular such bases exist with  $|S| \leq 2\sqrt{2n} + O(1)$ .

*Proof.* Representing  $0, 1, 2n - 1, 2n$  forces  $0, 1, n - 1, n \in S$ . The distinct unordered pairs  $(0, n)$  and  $(1, n - 1)$  coincide in sum, so at least one of the  $\binom{|S|+1}{2}$  products is redundant. At the displayed native optima the pair count is exactly  $2n + 1$ , respectively 15 and 21, making monomial recovery impossible.

For the construction, adding the multiples  $0, a, \dots, ba$  to  $[0, a - 1]$  covers  $[0, ba + a - 1] \supset [0, n]$ . Adding them to  $[n - a + 1, n]$  covers  $[n - a + 1, 2n - r]$ . Adding the upper interval to itself covers  $[2n - 2a + 2, 2n]$ , which overlaps because  $r \leq a - 1$ . The last multiple belongs to the upper interval, giving the cardinality bound. Choose an integer nearest  $\sqrt{n/2}$  for the asymptotic bound, handling the finitely many smaller values by the full interval.  $\square$

For  $k = 3d + 2$  one has the explicit family  $S = [0, d] \cup [2d + 1, 3d + 1]$ , with complementary exponents  $E = [d + 1, 2d]$ . The three sum intervals are  $[0, 2d]$ ,  $[2d + 1, 4d + 1]$  and  $[4d + 2, 6d + 2]$ . Each column of the interpolation matrix  $J$  has one strict coordinatewise sign, as follows by expanding  $d_i \prod_{a \neq i} (t - r_a)$  with positive roots. In particular a signed  $J e_{d+1}$  is positive. Taking  $T = J \text{span}\{e_a : a \in E\}$  realizes the monomial complement locally by Theorem D.1; generic small loading perturbations give globally separated full-product realizations. The unrestricted fibre dimension for this family is  $5d(d + 1)/2$ . The smaller sets  $\{0, 1, 3, 5, 6\}$  and  $\{0, 1, 2, 5, 8, 11, 12, 13\}$  also satisfy (K.1); no optimal restricted-basis constant is claimed.

For a nonorthonormal normal frame  $W = ZO$ , set  $K = W^\top W$  and  $G = \frac{1}{2} W^\top D^2 C_R W$ . The exact dual readout is

$$KG^{-1}K = W^\top RW. \quad (\text{K.2})$$

Indeed  $K = O^\top O$  and  $G = O^\top B^{-1}O$ , and direct inversion proves the identity. If  $W$  has columns  $J^{-\top} e_a$  for  $a \in S$ , its readout entries are  $\theta_{a+b}$ . Dropping the Gram factors in (K.2) would give an incorrect normalization.

## APPENDIX L. FINITE CALIBRATED SAMPLES AND OBSERVED EXPOSURES

Keep the roots, clocks and calibrated tables fixed and separated. Put  $\delta = M_1 - M_2 \neq 0$ , assume  $\pi_j \in [\varepsilon, 1 - \varepsilon]$  with  $0 < \varepsilon < 1/2$ , and define

$$\kappa(u) = u^2(1-u)^2 \sum_{a=1}^4 \frac{\delta_a^2}{M_{2a} + u\delta_a}, \quad g(u) = 1/\kappa(u).$$

On this interval  $g$  is Lipschitz with a finite constant  $L_g$ . We distinguish a chosen target-design weight  $\lambda$  from the realized sample allocation  $n_j/N$ . The theorem below estimates the metric indexed by the chosen known  $\lambda$ . It estimates the realized normalized sample information only when those weights are chosen as  $n_j/N$ , and the limiting information when  $n_j/N \rightarrow \lambda_j$ .

**Theorem L.1** (Fixed-design categorical recovery). *Observe  $n_j \geq 1$  independent marks at clock  $j$  with law  $p_j$ , independently across groups. For empirical cell frequencies  $\bar{p}_j$ , set*

$$\hat{\pi}_j = \text{proj}_{[\varepsilon, 1-\varepsilon]} \frac{\langle \bar{p}_j - M_2, \delta \rangle}{\|\delta\|_2^2}, \quad W_{\cdot j} = \frac{\rho_j}{\lambda_j} w_{2n}(T_j), \quad \hat{\theta} = W(g(\hat{\pi}_j))_j.$$

Then  $\hat{\theta} \in \Omega$  and

$$\mathbb{E}\|\hat{\theta} - \theta\|_2^2 \leq \frac{\|W\|_{\text{op}}^2 L_g^2}{\|\delta\|_2^2} \sum_j n_j^{-1}. \quad (\text{L.1})$$

On compact interior design sets, the induced estimates of  $R$  and the inverse normal blocks have squared risk  $O(\sum_j n_j^{-1})$ . If  $\mathcal{L}$  is injective, inversion of those induced jets returns  $\hat{\theta}$  exactly.

*Proof.* Before clipping, the estimate of  $\pi_j$  is unbiased. For a one-hot cell vector  $X$ ,  $\text{Var}\langle X, \delta \rangle \leq \|\delta\|_2^2$ . Projection cannot increase distance to the true point in the interval, so  $\mathbb{E}|\hat{\pi}_j - \pi_j|^2 \leq (n_j \|\delta\|_2^2)^{-1}$ . Apply the Lipschitz bound for  $g$  and the operator norm of  $W$ . All estimated moment weights are positive. On compact design sets they lie in a fixed positive box; the moment image is a compact convex subset of  $\Omega$ . The map to  $R$  and positive block inversion are Lipschitz there. Finally the induced inverse jets satisfy  $\hat{G}_\nu^{-1} = \mathcal{L}_\nu \hat{\theta}$ , proving the last assertion.  $\square$

The bound allows the  $\pi_j$  to be coupled by the polynomial model and does not assert a full-dimensional fixed-design likelihood. Its constants need not stay bounded when roots or clocks collide, or probabilities approach the boundary.

**Theorem L.2** (An additional observed-exposure protocol). *Observe independent counts  $N_j \sim \text{Pois}(N\lambda_j)$  and, conditionally,  $N_j$  categorical marks of law  $p_j$ . Suppose  $0 < \lambda_- \leq \lambda_j \leq \lambda_+$ . Clip  $N_j/N$  to this interval, use the preceding clipped mark estimate, and use any value in  $[\varepsilon, 1 - \varepsilon]$  when  $N_j = 0$ . Substitution into (C.2) gives uniform squared risk at most  $K/N$  on fixed compact designs.*

*An open exposure box has an open image in the moment cone. If the mark laws are known and exposures are interior, the plug-in moment estimate has limiting covariance*

$$\Sigma_\theta = \sum_j \frac{\rho_j^2}{\lambda_j^3 \kappa_j^2} w_{2n}(T_j) w_{2n}(T_j)^\top > 0 \quad (\text{L.2})$$

after multiplication of its error by  $\sqrt{N}$ . This is the efficient functional covariance in that known-mark-law subexperiment; its squared-risk order  $N^{-1}$  is locally unimprovable.

*Proof.* Clipping gives  $\mathbb{E}|\hat{\lambda}_j - \lambda_j|^2 \leq \lambda_+/N$ . For  $X \sim \text{Pois}(a)$ ,

$$\mathbb{E}[X^{-1}; X > 0] \leq 2\mathbb{E}[(X+1)^{-1}] = 2(1 - e^{-a})/a, \quad \Pr(X = 0) = e^{-a} \leq 1/a.$$

Conditioning on the counts therefore gives mark-estimation risk  $O(N^{-1})$ . The weight functions  $\rho_j/(\lambda\kappa(u))$  are uniformly Lipschitz on the fixed positive compact rectangle, proving the risk bound.

The exposure-to-moment Jacobian has columns  $-\rho_j w_{2n}(T_j)/(\lambda_j^2 \kappa_j)$  and full row rank. This gives the open-image assertion. The Poisson central limit theorem and the delta method give (L.2). The exposure information per unit  $N$  is  $\text{diag}(1/\lambda_j)$ ; its inverse pushed forward by this Jacobian is exactly the displayed covariance. The regular local information bound for smooth functionals, or the normal target experiment in Lemma I.1, gives efficiency and the lower risk order [26].  $\square$

Counts are genuinely observed in this additional protocol. Unknown exposures cannot be inferred from fixed-count marks alone. Jointly unknown marks and exposures need not have the known-mark-law efficient covariance (L.2).

#### APPENDIX M. COMPUTED FINITE-OFFSET FEATURES AND INDEPENDENT NOISE

At a globally separated informative point  $b_0$ , choose the normal frame  $Z$  and the  $q$  vectors  $v_a$  consisting of its columns and their pairwise sums. Quadratic evaluation is an isomorphism  $\mathcal{Q} : \text{Sym}_c \rightarrow \mathbb{R}^q$ . Define

$$\mu_h(\theta)_a = \frac{1}{2} \{C_{R_\theta}(b_0 + hv_a) + C_{R_\theta}(b_0 - hv_a)\}.$$

Fix a compact convex  $\Theta \subset \Omega$  containing the true parameter and the clipped plug-in estimates. For sufficiently small  $h > 0$  all targets are positive and lie in the common analytic projection tube. Symmetric Taylor expansion, also differentiated in  $\theta$ , gives

$$h^{-2}\mu_h(\vartheta) = \mathcal{Q}((\mathcal{L}\vartheta)^{-1}) + h^2 r_h(\vartheta), \quad \sup_{0 < h \leq h_0} \|r_h\|_{C^1(\Theta)} < \infty. \quad (\text{M.1})$$

**Theorem M.1** (Computed finite-offset encoding and inversion). *For a raw estimator with squared risk at most  $K/N$ , compute  $\hat{\mu}_h = \mu_h(\hat{\theta})$  and set*

$$\tilde{\theta}_h = \mathcal{L}^\dagger [\mathcal{Q}^{-1}(h^{-2}\hat{\mu}_h)]^{-1}.$$

*For small enough  $h$  all inverses exist and*

$$\mathbb{E}\|\tilde{\theta}_h - \theta\|^2 \leq 2K/N + 2C^2 h^4 / \gamma^2.$$

*For a fixed design,  $h_N = o(N^{-1/4})$  makes this output equivalent to the raw estimator at scale  $\sqrt{N}$ . With uniform tube and metric bounds along varying designs, it suffices that  $\sqrt{N}h_N^2/\gamma_N \rightarrow 0$ . A deterministic contact-computation error at most  $\tau_N$  adds parameter error at most  $C'\tau_N/(\gamma_N h_N^2)$ , provided positive inversion remains valid.*

*Proof.* Smooth positive inversion in (M.1) gives

$$F_h(\vartheta) := [\mathcal{Q}^{-1}(h^{-2}\mu_h(\vartheta))]^{-1} = \mathcal{L}\vartheta + h^2 E_h(\vartheta), \quad \|E_h\|_{C^1} \leq C.$$

Thus  $\tilde{\theta}_h = \hat{\theta} + h^2 \mathcal{L}^\dagger E_h(\hat{\theta})$  and its extra error is at most  $Ch^2/\gamma$ . Squaring proves the risk and scaling claims. An error  $\tau_N$  perturbs the argument of inversion by at most  $\|\mathcal{Q}^{-1}\|\tau_N/h_N^2$ ; use the inverse identity in Theorem E.2.  $\square$

This theorem concerns deterministic re-encoding of an already available estimator. It is not an independent observational route or a new minimax rate. Its offset condition controls approximation bias; the leading statistical risk is inherited from the raw estimate. The direct score reduction of Section I does not use this construction.

For comparison, consider the different experiment of independent noisy readings

$$Y_i = \mu_h(\theta) + \sigma \xi_i, \quad \xi_i \sim N(0, I_q), \quad i = 1, \dots, N.$$

For compact convex  $\Theta$  and sufficiently small fixed  $h$ , the exact finite-offset mean satisfies

$$ah^2\|\theta - \theta'\| \leq \|\mu_h(\theta) - \mu_h(\theta')\| \leq bh^2\|\theta - \theta'\|$$

for  $a, b > 0$ . Indeed  $F_h = \mathcal{L}\theta + h^2E_h$  and  $h^2\|DE_h\| \leq \gamma/2$  give a lower bound by the triangle inequality and the mean-value bound for  $E_h$ ; positive inversion and  $\mathcal{Q}$  transfer it to  $\mu_h$ . The upper bound follows directly from (M.1). Fitting the exact mean by least squares consequently gives

$$\sup_{\theta \in \Theta} \mathbb{E}\|\hat{\theta}_N - \theta\|^2 \leq \frac{4q\sigma^2}{Na^2h^4},$$

since the fitted mean is within twice the sample-mean error of the true mean. At an interior parameter the information is  $I_h = \sigma^{-2}D\mu_h^\top D\mu_h > 0$ , the asymptotic covariance is  $I_h^{-1}$ , and the local asymptotic minimax risk is  $\text{tr}(I_h^{-1})/N$  [26]. The least eigenvalue of  $I_h/h^4$  is comparable to  $\gamma^2/\sigma^2$  under the fixed metric bounds, by differentiating (M.1). Gaussian tails and the lower Lipschitz bound justify convergence of local risks. A quadratic-truncated fit at fixed  $h$  would instead generally retain an  $O(h^2)$  bias and is not used. The  $h^{-4}$  variance in this independent-noise experiment must not be transferred to the correlated computed or raw-score statistics.

#### APPENDIX N. COMPLETE CONTACT, DESIGN FIBRES, AND ENDPOINT REPRESENTATIONS

These statements concern different observation categories from a prescribed normal block. We record their mathematical interfaces and proofs so that the one-contact and multi-contact results do not erase the complementary conclusions.

**N.1. Complete conormal ambiguity.** Suppose  $\mathcal{A}_A : \text{Sym}_d \rightarrow \mathbb{R}^k$ ,  $S \mapsto (a_i^\top S a_i)_i$ , is injective. The complete conormal union of the rank-one image is dense in the inverse image, under  $\mathcal{A}_A^*$ , of the singular symmetric matrices: a conormal  $n$  at  $q_A(\eta)$  satisfies  $(\sum_i n_i a_i a_i^\top) \eta = 0$ . Conversely a real singular matrix has a nonzero real kernel vector. For  $d \geq 3$ , no nonzero quadratic form vanishes on all singular symmetric matrices. To see this, restrict a quadratic polynomial to matrices diagonal in any fixed orthogonal basis. Every monomial of degree at most two omits at least one of the  $d$  diagonal variables, so vanishing on every coordinate hyperplane forces the restriction to vanish identically. Every real symmetric matrix is orthogonally diagonalizable, so the polynomial is zero. For  $d = 2$ , direct substitution of rank-one matrices leaves exactly the determinant line. This is the elementary coordinate consequence of classical Veronese duality [12].

Under the surjective map  $\mathcal{A}_A^*$ , pure kernel and mixed quadratic terms must also vanish, since singular symmetric matrices span  $\text{Sym}_d$ . Thus the complete-conormal ambiguity is zero for  $d \geq 3$  and one-dimensional for  $d = 2$ . In the latter case it is generated by  $B_{ii} = 0$ ,  $B_{ij} = \det(a_i, a_j)^2$  and is intersected with the admissible native positive cone. In root coordinates the native linear span consists exactly of symmetric matrices satisfying

$$\frac{(r_j - r_i)X_{ij}}{d_i d_j} - \frac{(r_l - r_i)X_{il}}{d_i d_l} + \frac{(r_l - r_j)X_{jl}}{d_j d_l} = 0 \quad (i < j < l). \quad (\text{N.1})$$

The divided-difference identity for  $(t - r_i)^{-1}(t - r_j)^{-1}$  proves these relations for  $JH(\theta)J^\top$ . Conversely they leave  $k$  free diagonal entries and  $k - 1$  off-diagonal difference coordinates, hence dimension  $2k - 1$ , equal to the Hankel span. Testing  $B$  in (N.1) gives the exact two-parameter transversality criterion. Equality of second jets here is not equality of all higher contact terms.

**N.2. Exact dual-jet query count.** In a deterministic exact-real oracle whose scalar answers are values of the dual normal quadratic forms, full product rank requires exactly  $p = 2k - 1$  scalar queries. Squares span within-contact products by polarization, so select  $p$  independent square functionals for sufficiency. For necessity, follow any adaptive transcript at an interior parameter. Fewer than  $p$  scalar linear answers have a nonzero common kernel. A sufficiently

small perturbation in that kernel remains admissible and reproduces the same answers, hence the same adaptive choices, by induction. This is a query statement about the specified dual oracle, not a statistical sample bound or a count of ordinary scalar distance readings.

**N.3. Reciprocal design fibres.** Put  $a_j = \rho_j / \kappa_j > 0$ , let  $V_m$  be the  $p \times m$  moment matrix, and use reciprocal weights  $\lambda_j = a_j / \omega_j$ . Define on the open moment cone

$$\chi(\theta) = \min_{V_m \omega = \theta, \omega > 0} \sum_j a_j / \omega_j.$$

The zeroth moment bounds every fibre. The objective tends to infinity at its boundary and is strictly convex, so there is a unique minimizer. Its diagonal Hessian  $D_\omega > 0$  and differentiation of the constrained first-order equations give

$$D^2\chi = (V_m D_\omega^{-1} V_m^\top)^{-1} > 0.$$

Indeed  $D_\omega d\omega = V_m^\top d\nu$  and  $V_m d\omega = d\theta$ , while  $\nu$  is the gradient of the value. The implicit function theorem also gives smoothness. Here  $m = p + 2$ , so the affine fibre has dimension two. Every ray from its minimizer meets a higher level once, because the restriction is strictly convex and diverges at the positive-fibre boundary. Its higher levels are smooth circles. Consequently the unit-exposure-budget image is  $\{\theta : \chi(\theta) \leq 1\}$ , with singleton boundary fibres and circular interior fibres. Homogeneity  $\chi(t\theta) = t^{-1}\chi(\theta)$  makes the boundary regular relative to the open cone. Injective contact coordinates transport this geometry without changing the fibres.

**N.4. Endpoint value-graph representations.** For a labelled semialgebraic value graph specified over algebraic coefficients, fix a hidden dimension and a specified normalization of a positive block matrix. Every representation

$$F(x) = \min_{z \geq 0} (x, z)^\top \begin{pmatrix} A & B \\ B^\top & C \end{pmatrix} (x, z)$$

is characterized exactly by positivity, that normalization, and the quantified relations

$$z \geq 0, \quad Cz + B^\top x \geq 0, \quad z_i(Cz + B^\top x)_i = 0,$$

together with the value equality for every labelled  $x$ . Strict convexity in  $z$  makes these Karush–Kuhn–Tucker conditions necessary and sufficient. Real quantifier elimination therefore computes the representation fibre at the fixed hidden dimension [6]. Enumeration over dimensions terminates under a finite-representability promise, yielding the first nonempty normalized fibre; no unconditional finite hidden-dimension bound is implied. This input is a scalar value graph, not an optimizer map as in inverse parametric programming [15, 22].

**N.5. Deterministic local root costs.** The native expansion  $p(u, v) = p_0 + L_p u + D_p v + O((\|u\| + \|v\|)^2)$  gives the information in Proposition C.1. With squared Hellinger distance  $h^2(p, q) = \sum_e (\sqrt{p_e} - \sqrt{q_e})^2$ , profiling the free coordinates yields the quadratic efficient cost after multiplying by  $4/t^2$ . Comparing a physical target variance  $tb$ ,  $b \geq 0$ , with synchronized  $v = tq_A(\eta)$  gives (A.1). Full rank of  $A$  gives coercivity and bounded scaled minimizing sequences, making the Taylor remainder uniform on the relevant local chart. Physical splitting has size  $\sqrt{t}$ . For  $C^2$  nonlinear loadings  $g_i$  with  $g_i(0) = 0$  and  $Dg_i(0) = a_i^\top$ ,

$$g_i(\sqrt{t}\eta)^2 = t(a_i^\top \eta)^2 + O(t^{3/2})$$

uniformly on bounded  $\eta$ . These local statements do not require injectivity of  $\mathcal{A}_A$  on all of  $\text{Sym}_d$ ; global chart selection uses the separated-fibre hypotheses. At  $t = N^{-1/2}$  the same cost is the physical local Gaussian profile of Theorem H.1.

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